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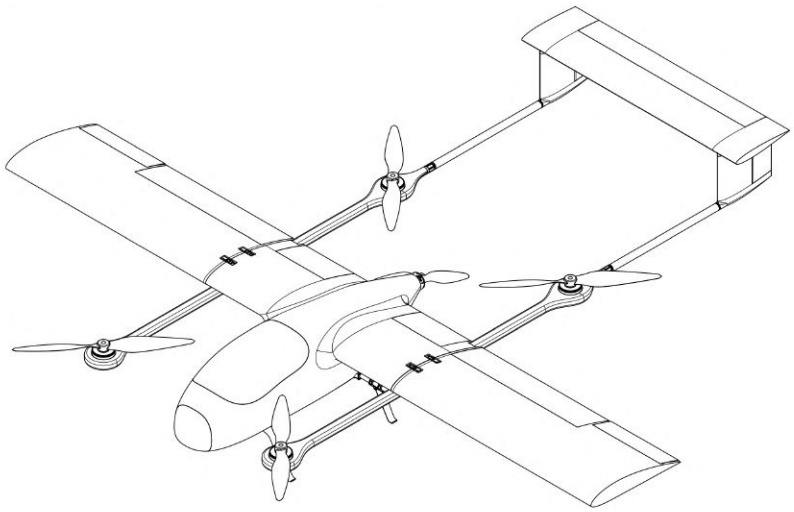
Technical Report

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Executive Summary

Agricultural spraying is a $4.06 billion global industry, yet conventional methods expose farmers to harmful chemicals, offer poor operational productivity, and are increasingly inadequate for modern farming demands. Autonomous agricultural UAVs represent a compelling alternative, with the market exhibiting a compound annual growth rate of 18.5%. A team of eight engineering students was tasked with creating a conceptual design of an autonomous agricultural spraying UAV and developing a supporting business case, responding to a brief that required the problem to be formally defined before design work could begin.

The project was managed using an iterative approach structured around two formal design reviews: a Preliminary Design Review (PDR) and a Critical Design Review (CDR). Centralised General Values document, and regular meeting summaries were used to maintain alignment and traceability throughout the project. A reduced application of the SEBoK System Concept Definition methodology provided a structured path from stakeholder identification and needs elicitation through to a verifiable set of system requirements, ensuring the design addressed the correct problem.

Concept ideation using a morphological chart produced a hybrid VTOL fixed-wing configuration, selected for its range advantage over quadcopters and operational flexibility. The design evolved across three iterations, with PDR driving targeted refinements from iteration 2 to iteration 3. The final configuration was a twin-boom hybrid VTOL fixed-wing UAV with a pusher cruise propeller, a T-tail, and an integrated fuselage payload bay. Key specifications include an MTOW of 140.8 kg, a 47 kg spray payload, a 15-minute flight time, a 5.46 m wingspan, a cruise speed of 25.8 m/s, and an operational productivity of up to 26 ha/h. Unit cost is estimated at approximately $50,000, delivered under a spraying-as-a-service model targeting medium-holder farmers across India.

Requirements compliance was satisfactory, with 65% of needs fully met. Future work includes CFD, structural FEA, spray drift analysis, detailed interface CAD, and physical prototype testing to validate the analytical predictions made during this design phase.

AI Statement

Generative AI was used for grammar checking and rephrasing to improve this report's readability and structure, however, all technical content and figures are original work.

Notes on Figures

Large Excel tables are included as figures to provide evidence of methodology and working. Due to their size, these tables have been reduced for formatting purposes. However, they were exported using the "Print to PDF" function, meaning they remain fully legible when the document is zoomed. The author acknowledges and apologises for any inconvenience caused by the reduced on-page scale and inconsistency in text size.

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Nomenclature

Term	Meaning
Organically farmed crops	Crops farmed without the use of external chemicals
PPE	Personal Protection Equipment
UAV	Unmanned Aerial Vehicle (the aircraft exclusively)
CAGR	Compound Annual Growth Rate
IME	Integrated Mechanical and Electrical
GDBP	Group Design & Business Project
CAD	Computer Assisted Design
TI	Technical Integrator
PM	Project Manager
MTOW	Maximum Take Off Weight
COG	Centre of Gravity
GA	General Assembly
PDR	Preliminary Design Review
CDR	Critical Design Review
SEBoK	Systems Engineering Body of Knowledge
UAS	Unmanned Aerial System
Smallholder Farmer	A farmer that owns < 2 hectares of land
SOI	System of Interest
USP	Unique Selling Point
ID	Identification
VTOL	Vertical Take-Off and Landing
ConOps	Concept of Operations
FMEA	Failure Mode Effects Analysis
Cl	Coefficient of Lift
CAAC	Civil Aviation Administration of China
DGCA	Directorate General of Civil Aviation
EASA	European Union Aviation Safety Agency
PPA	Product Positional Analysis
RF-Transparent	Radio Frequency Transparent
COTS	Commercial Off The Shelf
TBD	Team Bath Drones
MFS	Main Fuselage Structure
WCS	Wings Connection Structure
CFRP	Carbon Fibre Reinforced Polymer
DFMEA	Design Failure Mode Effects Analysis
PFMEA	Process Failure Mode Effects Analysis
Li-Po	Lithium Polymer
CFD	Computational Fluid Dynamics
FEA	Finite Element Analysis

Table 1: Symbols used in the report and their meaning

1 Introduction

1.1 General Overview

Crop spraying, the application of herbicides, pesticides, and fertilisers to agricultural fields, has become an integral part of modern farming, with studies indicating up to a 25% higher yield compared to organically farmed crops[1]. As a result, the agricultural spraying industry is valued at approximately $4.06 billion and is projected to grow to $6.6 billion by 2034[2].

However, conventional pesticide and herbicide application poses significant health risks to farmers when appropriate PPE is not used[3]. This has created demand for more automated and safer spraying solutions. In parallel, the rise of drone technology has accelerated the adoption of agricultural UAVs, with the market experiencing a CAGR of 18.5%[4]. These systems offer improved precision, efficiency, and cost-effectiveness while reducing exposure risks for operators and limiting chemical overuse on crops.

The engineering team, consisting of one IME student, two mechanical engineering students, and five aerospace engineering students, was tasked with developing a business case and conceptual design for an agricultural spraying UAV as part of the GDBP unit. The aim of the project was to produce a credible system-level design that satisfies stakeholder-derived needs and requirements, demonstrates commercial viability, and complies with relevant UAV airworthiness regulations. **Figure 1** presents an overview of the final system described in this report.

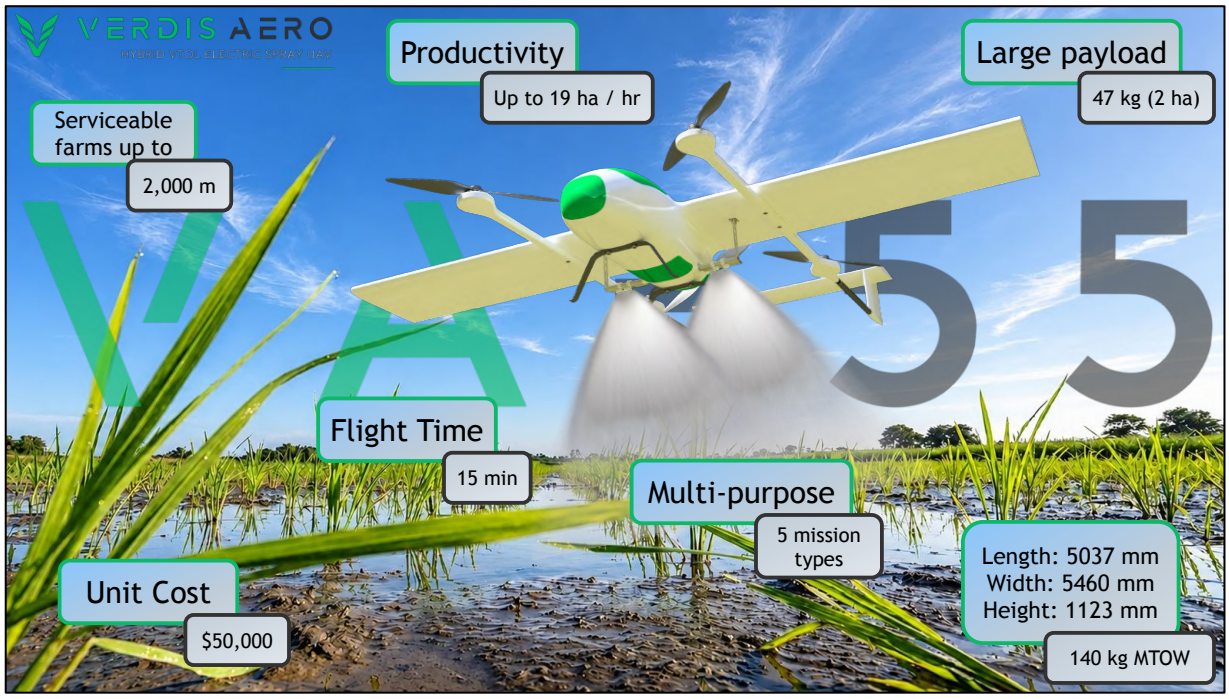


Figure 1: Overview of the UAV with general specifications outlines

1.2 Project Scope and Report Outline

The scope of this project included preliminary detailed design of the UAV, incorporating analytical calculations and early-stage simulation work, the development of a CAD model capturing key system features, a preliminary lifecycle analysis, and a detailed business case for the product. A representation of this scope within the engineering V-model is shown in **Figure 2**.

Full-scale manufacturing, comprehensive lifecycle assessment, advanced high-fidelity modelling and simulation, and physical prototyping and testing were excluded from the scope. These limitations were driven by project time constraints, team size, and deliverable requirements.

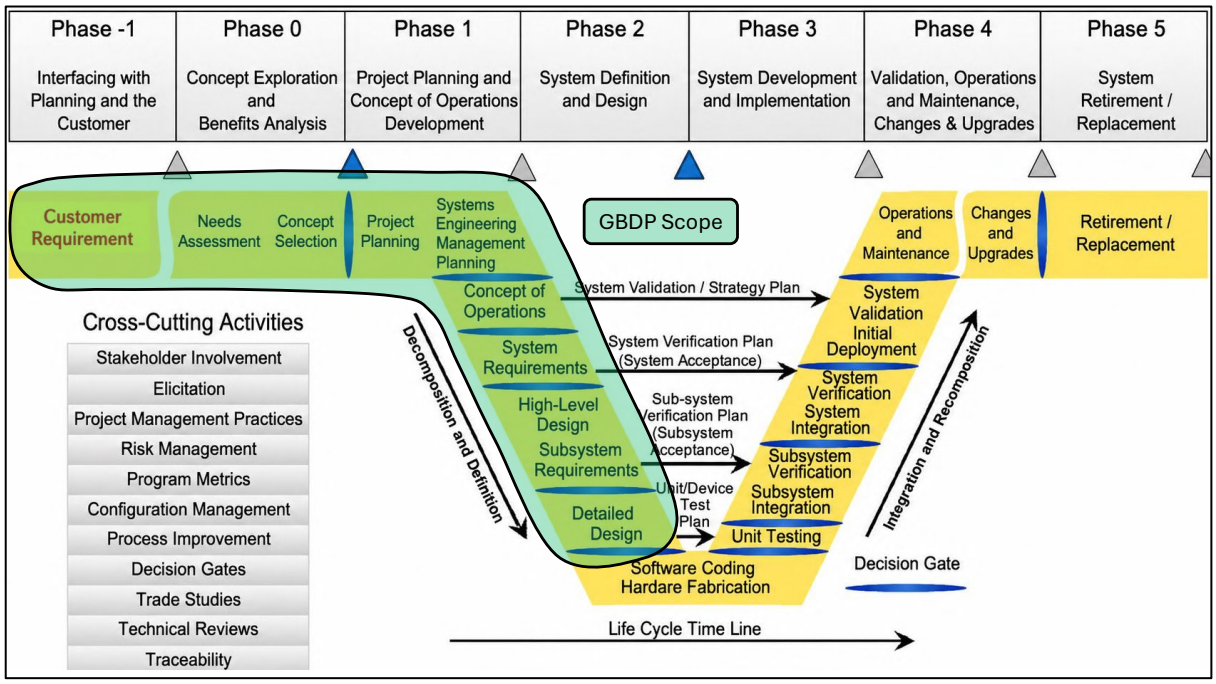


Figure 2: Engineering V diagram highlighting activities that are part of this project's scope

This report focuses on the role of the Technical Integrator (TI) and Project Manager (PM) within the project. Both roles are broad and multidisciplinary in nature, with responsibilities spanning technical coordination, systems integration, and project delivery. The key objectives and responsibilities of each role are summarised in **Table 2**.

Role	Technical Integrator	Project Manager
Responsibilities	- Ensure each sub-system's outputs are consistent with one another - Ensure all sub-systems physically and functionally fit together properly - Develop and maintain a common set of design requirements for each sub-system to meet - Identify and enforce global system constraints (MTOW, COG, Power etc..) - Determine design trade-offs to maximise requirements compliance - Organise the CAD model and design spaces	- Deliver the completed project on time - Develop a project timeline and ensuring it is followed - Enforce the project's scope so the team doesn't get off track - Organise, plan and keep a record of each meeting and design decision - Ensure all the project's resources are used to their maximum potential - Act as a liaison between each team member (communication management) - Ensure work-life balance and good moral across the team

Table 2: Table showing the responsibilities of a TI and PM

The aim of this report is to present the UAV's design process, providing justification and supporting evidence for key requirements and design decisions, alongside a breakdown of the final system supported by a GA drawing. The project management element is also addressed to illustrate how the work was structured and coordinated.

System concept definition is used to identify key stakeholders, risks, and primary design requirements, and to explain how these were derived. The report then examines each design iteration, associated failure modes, and key design trade-offs to show how the system evolved throughout development. Physical subsystem interfaces are also described, with one interface analysed in detail through its failure modes and supporting technical drawings. Finally, the report concludes with a discussion of future work, reflections on the design process, and overall project conclusions.

By the end of this report, the reader should understand how the design requirements were derived, how the system evolved across iterations, how subsystem integration was managed, and to what extent the final design meets its requirements.

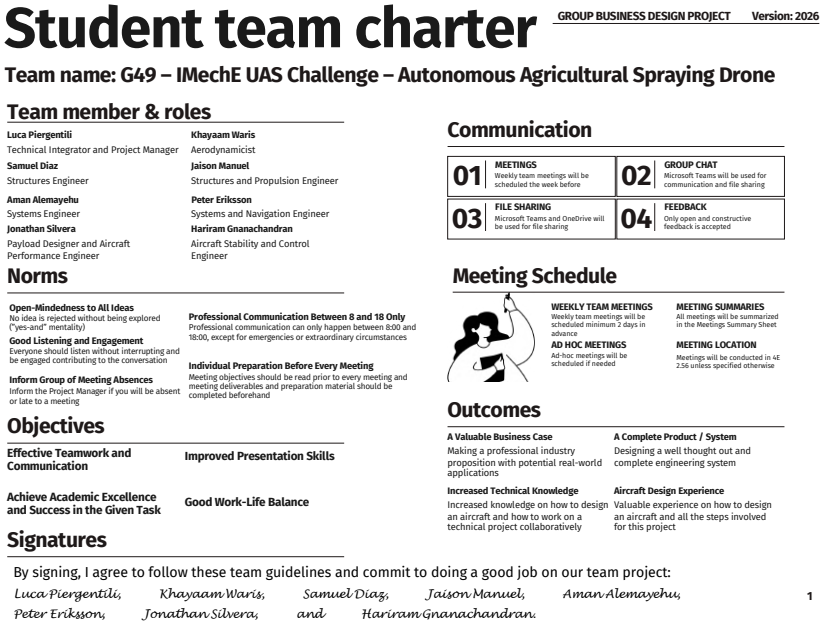
2 Project Initiation

2.1 Project Brief and Objectives and Project Initiation

At the GDBP kick-off meeting, the team was presented with the brief (**Appendix A**) that outlined the core task of designing an agricultural spraying UAV. After analysing the brief, the team was given the expected deliverables for this project:

1. A set of presentations to show the progress of the work that included: a PDR, Mid-Term review, CDR and Final Presentation
2. A shared Business report and an individual Technical report
3. A fully justified design, including a CAD model and GA drawings

Behavioural norms, weekly meetings organisation and academic and personal objectives were then discussed, forming an acceptable code of conduct that was summarized in the Team Charter (**Figure 3**).



Student team charter

GROUP BUSINESS DESIGN PROJECT Version: 2026

Team name: G49 – IMechE UAS Challenge – Autonomous Agricultural Spraying Drone

Team member & roles

Luca Piergentili
Technical Integrator and Project Manager

Khayaam Waris
Aerodynamicist

Samuel Diaz
Structures Engineer

Jaison Manuel
Structures and Propulsion Engineer

Aman Alemayehu
Systems Engineer

Peter Eriksson
Systems and Navigation Engineer

Jonathan Silvera
Payload Designer and Aircraft Performance Engineer

Hariram Gnanachandran
Aircraft Stability and Control Engineer

Norms

Open-Mindedness to All Ideas
No idea is rejected without being explored ("yes-and" mentality)

Good Listening and Engagement
Everyone should listen without interrupting and be engaged contributing to the conversation

Inform Group of Meeting Absences
Inform the Project Manager if you will be absent or late to a meeting

Professional Communication Between 8 and 18 Only
Professional communication can only happen between 8:00 and 18:00, except for emergencies or extraordinary circumstances

Individual Preparation Before Every Meeting
Meeting objectives should be read prior to every meeting and meeting deliverables and preparation material should be completed beforehand

Objectives

Effective Teamwork and Communication

Improved Presentation Skills

Achieve Academic Excellence and Success in the Given Task

Good Work-Life Balance

Signatures

By signing, I agree to follow these team guidelines and commit to doing a good job on our team project:

Luca Piergentili, Khayaam Waris, Samuel Diaz, Jaison Manuel, Aman Alemayehu, Peter Eriksson, Jonathan Silvera, and Hariram Gnanachandran.

Communication

01 MEETINGS Weekly team meetings will be scheduled the week before

02 GROUP CHAT Microsoft Teams will be used for communication and file sharing

03 FILE SHARING Microsoft Teams and OneDrive will be used for file sharing

04 FEEDBACK Only open and constructive feedback is accepted

Meeting Schedule

WEEKLY TEAM MEETINGS
Weekly team meetings will be scheduled minimum 2 days in advance

AD HOC MEETINGS
Ad-hoc meetings will be scheduled if needed

MEETING SUMMARIES
All meetings will be summarized in the Meetings Summary Sheet

MEETING LOCATION
Meetings will be conducted in 4E 2.56 unless specified otherwise

Outcomes

A Valuable Business Case
Making a professional industry proposition with potential real-world applications

A Complete Product / System
Designing a well thought out and complete engineering system

Increased Technical Knowledge
Increased knowledge on how to design an aircraft and how to work on a technical project collaboratively

Aircraft Design Experience
Valuable experience on how to design an aircraft and all the steps involved for this project

1

Figure 3: Team Charter

2.2 Project Schedule

To ensure the project remained on schedule, a Gantt chart was created in Excel (**Figure 4, Appendix B**), based on Professor Lock's Air Taxi GDBP timetable (**Appendix C**) and reviewed by the team. As the brief provided no system requirements, ample time was allocated upfront for requirements generation and market research before design began. Preliminary design (Iterations 1 and 2) was completed pre-PDR, PDR feedback then carried into design refinement (Iteration 3), concluding at CDR and design freeze. A buffer of at least one day before each review allowed time for presentation preparation. Post-CDR time was reserved exclusively for report writing, with exhibition preparation planned following report submission.

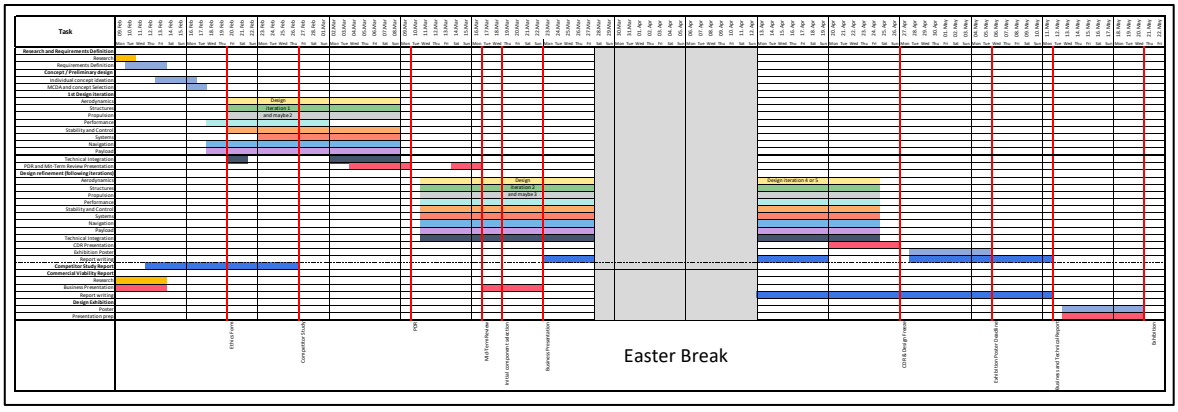


Figure 4: Gantt Chart

Two weeks before CDR, a final project timeline (**Figure 5**, Appendix **B**) was created, incorporating daily tasks and deadlines to ensure all required work was completed on time. Daily objectives and submissions were announced at 09:00 and progress was reviewed at 18:00. If delays occurred, affected deadlines were adjusted accordingly. The

timeline was developed using team input on remaining work and required interdependencies between tasks.

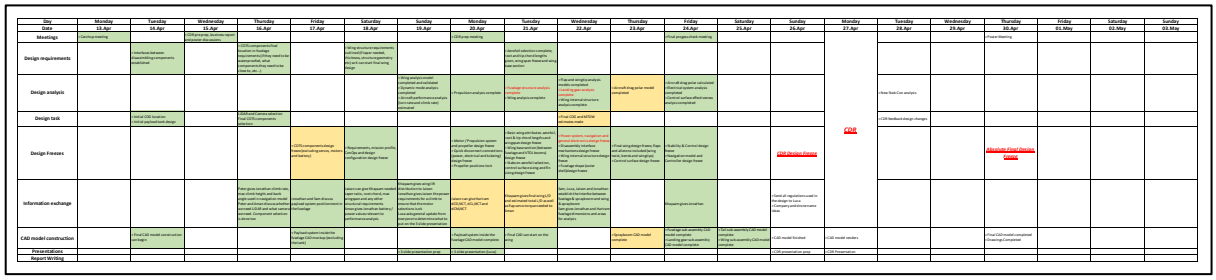


Figure 5: CDR Timeline

2.3 Role Assignment and Team Structure

Team roles were assigned using a Skills Table (**Figures 6** & **7**) created at project kick-off. Each member recorded their engineering-related skills in **Figure 6**, while **Figure 7** mapped the skills required for each role and the most suitable team member. Final roles (**Figure 8**) were assigned based on both skillset and personal preference. No conflicts arose during role allocation, as each member's preferences aligned well with their abilities.

Name	Preferred Role	Course	Skills and knowledge points
Luca	Technical Integrator and willing to help with aero, stab-con and performance	Aerospace Engineering	Good organisation and managing skills (took lead in previous projects), good surface level knowledge in various roles (except avionics / systems), extensive experience with CAD software and led Team 3 in the Team Bath Drones Design and Make activity in the first semester.
Sam	Structures (Aero if any available not already delegated)	Mechanical Engineering	Contributed for sem1 TBD mini project, passionate about CAD, skilled at automating data workflows through Python and Excel. Good at scripting in Python. Experience in MATLAB
Jonathan	Performance / Payload	Aerospace Engineering	Good at using MATLAB and scripting to automate calculations, size aircraft components, assess propulsion and performance trade-offs, support concepts through sketches and CAD, apply core aerospace mathematics related to performance/propulsion/stab and control, analyse engineering data, and extract key requirements from large technical and certification documents.
Khayaam	Aerodynamics	Aerospace Engineering	TBRe lead Rear wing Aerodynamic designer, Ansys CFD experience, adept at CAD Solidworks, inventor, work on mechanical assemblies for TBRe related to suspension and chassis design. FEA experience in Ansys/solidworks for metal parts/assemblies. in depth DFM and GD&T experience OpenFoam experiece ongoing.
Jaison	Structures / Propulsion	Aerospace Engineering	Experienced in MATLAB, FEA, CAD, FMEAs, Python and Excel. Involved with engineering drawings with GD&T and regulation adherence. Expereince in design for manufacture and general manufcaturing
Hariram	Structuress / Stab Con	Aerospace Engineering	Strong with MATLAB, proficient with CAD and FEA in SolidWorks, good surface level knowledge in aerospace roles
Aman	Structures / Systems	Integrated Mechanical and Electrical Engineering	Experience in mechanical design, within Team Bath Roving and design of mechancical integration of electrical systems while using Inventor/Solidworks/FEA. Background in electrical, LTSPICE/PSPICE analysis in ORCAD & coding background (Python,MATLAB,C++) so design and specification of electrical power systems and system integration of mechanical and electrical sub-systems would be sensible for my background for Systems role.
Peter	Systems / Navigation	Mechanical Engineering	Experience with Drones systems design and assembly. Experience with PCB design. Experience with modelling and simulation and Simulink.

Figure 6: Team member skills

Role	Needed Skills	Good Fits for Role	Selected for Role
Aerodynamics	Good understanding of 1st semester Aerodynamics, experience with CFD (Computational Fluid Dynamics) software (e.g., Ansys, OpenFOAM) and prior experience with aerodynamic surface design	Khayaam	Khayaam
Navigation	Experience with MATLAB or other simulation software, knowledge of drone navigation hardware (radar, sensors)	Peter	Peter
Performance	Good understanding of 1st semester Performance and Design, this will be a minor role for this aircraft	Luca, Jonathan, Khayaam, Jaison, Hariram	Jonathan
Propulsion	Good understanding of 1st semester Propulsion, knowledge or interest in drone propulsion	Luca, Jonathan, Khayaam, Jaison, Hariram	Jaison
Payload	Good CAD skills, experience with pipe standards and fluid behaviour	Luca, Sam, Jonathan,	Jonathan
Stability and Control	Good understanding of 1st semester Stability and Control, good MATLAB skills and mathematics skills (ODEs)	Jonathan, Jaison, Hariram	Hariram
Systems	Good coding skills, knowledge in electrical systems and battery technology and understanding of the requirements to make an electrical system safe and well integrated	Aman	Aman
Structures	Good understanding of 1st semester Aerospace Structures or Structural Mechanics, good CAD skills, experience with FEA and good understanding of composites	Sam, Jonathan, Jaison	Sam
Technical Integrator	Planning, organisation and communication skills with previous experience as group leader, good CAD skills, requirements management skills and good surface level knowledge of all roles	Luca	Luca

Figure 7: Skill required for each role

Final Roles	
Luca	Technical Integrator and Project Manager
Sam	Structures
Jonathan	Payload and Performance
Khayaam	Aerodynamics
Jaison	Structures and Propulsion
Hariram	Stability & Control
Aman	Systems
Peter	Navigation and Systems

Figure 8: Final roles

A slight deviation from the suggested team roles was made to better suit the team to this project. Payload and Navigation roles replaced Performance, and the second Structures role. Performance tasks were covered by Aerodynamics, Propulsion and Payload. The TI also functioned as a PM as both TIs and PMs must have a general view of the project.

The team structure consisted of the TI/PM coordinating the project by setting weekly goals, managing inter-team requirements, and organising meetings to discuss design decisions. This reflected a hybrid Agile–Waterfall project management approach, a combination of the structured planning and sequential phases of Waterfall with the flexibility and iterative development practices of Agile[5]. The structure was chosen for its simplicity and effectiveness; however, the TI's decisions were not absolute, as team input and approval were always considered. This encouraged flexibility and avoided an overly dictatorial leadership style.

3 Project Management and Planning

3.1 Communications Management

All online communication was conducted through Microsoft Teams due to its professional communication and file-sharing capabilities. A general chat was used for project-wide communication, while private discussions were conducted through direct messages. The general chat was kept professional to maintain clarity and usability. Team members were encouraged to enable notifications and check Teams daily to minimise response times, which proved effective for most members. The Teams drive was used to centrally store all project documents and work, ensuring accessibility for the entire team.

3.2 Meeting Organisation

All team meetings were conducted in person at the University to improve collaboration and minimise technical difficulties. Meetings were typically held at least twice per week, with the date, time, location, and duration planned in advance to accommodate members' timetables. Each meeting was organised at the end of the previous one, with key details later shared in the general chat. When2meet (**Figure 9**) was used to coordinate availability.

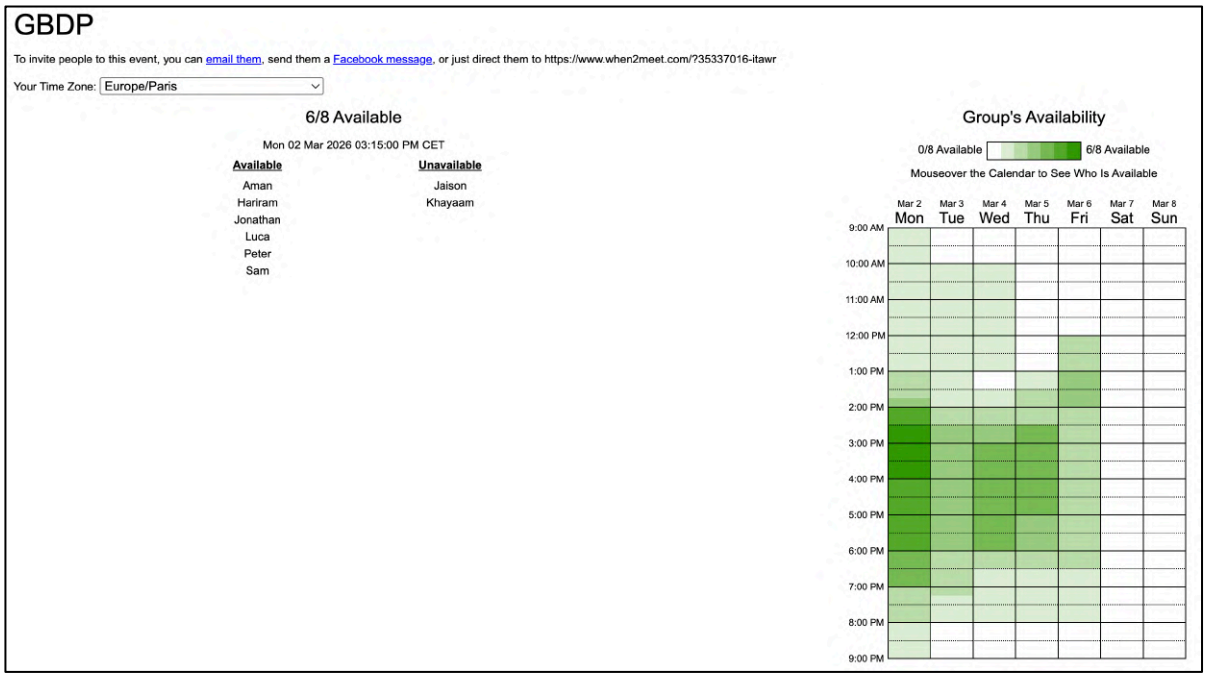
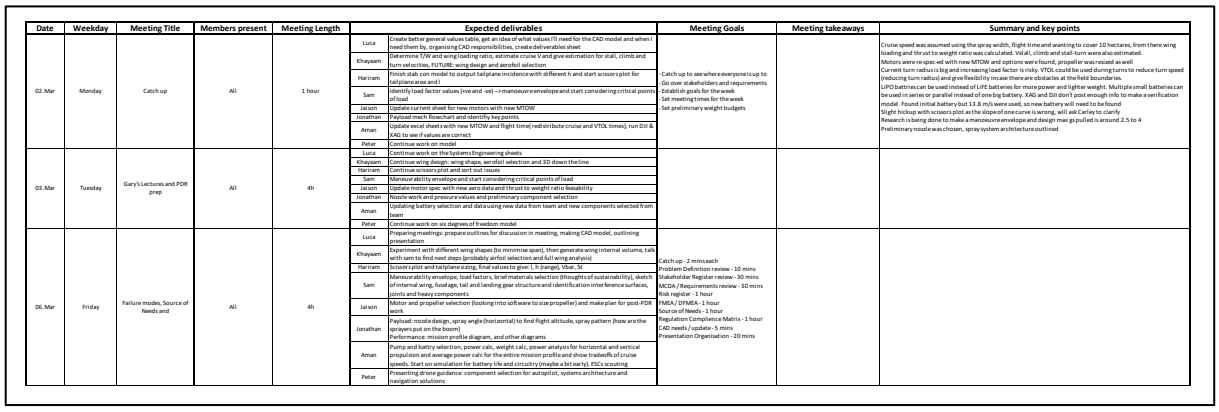


Figure 9: When2meet availability poll

A Meeting Summaries spreadsheet (**Figure 10**) was created to track meetings and support future planning. It recorded expected deliverables, meeting objectives, summaries, key takeaways, attendance, and duration. This ensured absent members could remain updated and design decisions could be tracked.



Date	Weekday	Meeting Title	Members present	Meeting Length	Expected delivrables		Meeting Goals	Meeting takeaways	Summary and key points
02.Mar	Monday	Catch up	All	1 hour	Luca	Create better general values table, get an idea of what values I'll need for the CAD model and when I need them by, organising CAD responsibilities, create deliverables sheet	-Catch up to see where everyone is up to -Go over stakeholders and requirements -Establish goals for the week -Set meeting times for the week -Set preliminary weight budgets		Cruise speed was assumed using the spray width, flight time and wanting to cover 10 hectares, from there wing loading and thrust to weight ratio was calculated. Vstall, climb and stall-turn were also estimated. Motors were re-spec-ed with new MTOW and options were found, propeller was resized as well Current turn radius is big and increasing load factor is risky. VTOL could be used during turns to reduce turn speed (reducing turn radius) and give flexibility incase there are obstacles at the field boundaries. LiPO battries can be used instead of LiFE batteries for more power and lighter weight. Multiple small batteries can be used in series or parallel instead of one big battery. XAG and DJI don't post enough info to make a verification model. Found initial battery but 13.8 m/s were used, so new battery will need to be found Slight hickup with scissors plot as the slope of one curve is wrong, will ask Carley to clarify Research is being done to make a manoeuvre envelope and design max gs pulled is around 2.5 to 4 Preliminary nozzle was chosen, spray system architecture outlined
					Khayaam	Determine T/W and wing loading ratio, estimate cruise V and give estimation for stall, climb and turn velocities, FUTURE: wing design and aerofoil selection			
					Hariram	Finish stab con model to output tailplane incidence with different h and start scissors plot for tailplane area and I			
					Sam	Identify load factor values (+ve and -ve) --> manoeuvre envelope and start considering critical points of load			
					Jaison	Update current sheet for new motors with new MTOW			
					Jonathan	Payload mech flowchart and identify key points			
					Aman	Update excel sheet's with new MTOW and flight time(redistribute cruise and VTOL times), run DJI & XAG to see if values are correct			
					Peter	Continue work on model			
03.Mar	Tuesday	Gary's Lectures and PDR prep	All	4h	Luca	Continue work on the Systems Engineering sheets			
					Khayaam	Continue wing design: wing shape, aerofoil selection and 3D down the line			
					Hariram	Continue scissors plot and sort out issues			
					Sam	Maneuverability envelope and start considering critical points of load			
					Jaison	Update motor spec with new aero data and thrust to weight ratio feasability			
					Jonathan	Nozzle work and pressure values and preliminary component selection			
					Aman	Updating battery selection and data using new data from team and new components selected from team			
					Peter	Continue work on six degrees of freedom model			
06.Mar	Friday	Failure modes, Source of Needs and	All	4h	Luca	Preparing meetings: prepare outlines for discussion in meeting, making CAD model, outlining presentation	-Catch up - 2 mins each Problem Definition review - 10 mins Stakeholder Register review - 30 mins MCDA / Requirements review - 30 mins Risk register - 1 hour FMEA / DFMEA - 1 hour Source of Needs - 1 hour Regulation Compliance Matrix - 1 hour CAD needs / update - 5 mins Presentation Organisation - 20 mins		
					Khayaam	Experiment with different wing shapes (to minimise span), then generate wing internal volume, talk with sam to find next steps (probably airfoil selection and full wing analysis)			
					Hariram	Scissors plot and tailplane sizing, final values to give: l_h (range), Vbar_h			
					Sam	Maneuverability envelope, load factors, brief materials selection (thoughts of sustainability), sketch of internal wing, fuselage, tail and landing gear structure and identification interference surfaces, joints and heavy components			
					Jaison	Motor and propeller selection (looking into software to size propeller) and make plan for post-PDR work			
					Jonathan	Payload: nozzle design, spray angle (horizontal) to find flight altitude, spray pattern (how are the sprayers put on the boom) Performance: mission profile diagram, and other diagrams			
					Aman	Pump and battry selection, power calc, weight calc, power analysis for horizontal and vertical propulsion and average power calc for the entire mission profile and show tradeoffs of cruise speeds. Start on simulation for battery life and circuitry (maybe a bit early), ESCs scouting			
					Peter	Presenting drone guidance: component selection for autopilot, systems architecture and navigation solutions			

Figure 10: Excerpt from Meeting Summaries sheet

Meetings followed a predefined structure based on the meeting goals, beginning with a brief plan and individual progress updates to ensure alignment across the team. Time limits were assigned to each topic to avoid overruns, and meeting summaries and expected deliverables were recorded at the end of each session.

3.3 Change Management and Tracking

As all sub-systems were interdependent, a General Workbook Excel file was created to centralise information relevant to every sub-system, ranging from technical accounts to concept selection. The General Values sheet (**Figures 11** and **12**, Appendix **D**) contained all variables shared across multiple sub-systems. Each variable was assigned an owner, to allow queries to be directed appropriately, and records of creation date, latest revision,

and author were maintained. A connection score was also given to highlight the variable's importance relative to other sub-systems. Updates to variables with a Connection score of 4 or above were communicated through Microsoft Teams.

This system reduced delays associated with requesting information between team members and minimised the risk of inconsistent values being used across sub-systems, which could compromise system integration. It also provided the TI with visibility over overall project progress. When a value was modified, the previous value and justification for the change were recorded in the Archive Values sheet (**Figure 12**), ensuring traceability throughout the design process.

Category	Variable Name	Symbol	Current Value	Units	Connection	Ownership	Notes	Date Added	Last Edited	Edited by
Overall	Maximum Take Off Weight (MTOW)	W	1381.6	N	5	Luca		18.Feb	24.Feb	Luca
	Maximum Take Off Mass (MTOM)	m	140.8	Kg	5	Luca		18.Feb	24.Feb	Luca
	Overall Width	w	5462	mm	3	Luca		18.Feb	16.Mar	Luca
	Overall Length	l	5764	mm	3	Luca		18.Feb	16.Mar	Luca
	Overall Height	h	1211	mm	2	Luca		18.Feb	16.Mar	Luca
	Flight time	t	15	mins	3	Luca		18.Feb	28.Mar	Luca
	Operational radius		4774.5	m	1	Luca		18.Feb	28.Mar	Luca

Figure 11: Excerpt from General Values Sheet

Weight	Definition (Connection)
1	Only impacts the base system
2	Impacts the base system and one other system
3	Impacts the base system and two or three other systems
4	Impacts the base system and four or five other systems
5	Impacts all systems

Figure 12: Value Importance Scale

Category	Variable Name	Symbol	It 1	Source of Value	It 2	Reason for Change	It 3	Reason for Change
Overall	Maximum Take Off Weight (MTOW)	W	1373.4	Initial estimation from competitor research	1031.094	Too heavy, resulted in a 20m wing span	981	Used a simplified number
	Maximum Take Off Mass (MTOM)	m	140	Initial estimation from competitor research	105.1064	Too heavy, resulted in a 20m wing span. Used a payload weight fraction of 0.47 and worked backwards to find MTOW	100	Changed to use a simpler number
	Overall Width	w	5180	CAD model	5180	Wing redesign		
	Overall Length	l	4740	CAD model	4740	Component integration in the fuselage		
	Overall Height	h	1585	CAD model	1585	Drone redesign		
	Flight time		30	Was originally based on competitors but was reduced to allign to payload depletion rate				
	Operational Radius		4000.5	Calculations (08/04/2026)				

Figure 13: Excerpt from Archive Values Sheet

3.4 Decision Making Bias

In group work, design decisions can often become dominated by the team leader or the most confident speaker, limiting exploration of alternative ideas. To mitigate this, all team members were encouraged to present one or more original concepts, supported by research, for every major design decision. During discussions, the PM motivated all members to contribute, and no idea was discarded without appropriate consideration. Regular meetings with supervisors and postgraduate mentors also helped reduce anchoring bias and groupthink. Signifiacnt design decisions were either made through trade-off studies or voting, as seen for Iteration 3's configuration change and the morphological concept selection respectively.

3.5 Project Management Risks

Every project carries a range of risks that may affect successful delivery. The risks associated with this project were identified and compiled into the PM Risk Register

(**Figures 14** and **15**) which was used to support project management decisions. Its construction is outlined in **Appendix E**. The end-of-project time crunch and the Easter break were identified as the most significant risks. Time pressure near submission is an inherent feature of large projects and can lead to reduced work quality; this was mitigated by incorporating schedule buffers into the Gantt chart (**Figure 4**). The Easter break also posed a risk to progress due to reduced access to university facilities, supervisor feedback, and in-person collaboration as team members returned home. This potential slowdown was accounted for in the project timeline to minimise its impact on overall progress.

Category	Risks or Problems	Notes / Description	Impact Description	Probability	Severity	Risk Score	Mitigation	Probability	Impact	Mitigated Risk Score
Timeline Risk	Team member conflicts and Disrespectful behaviour	When team members argue or are at odds with each other	Lack of communication and team work, Independent work with no cooperation and ineffective team meetings due to disagreements between team members	2	4	8	Create a team charter that outlines acceptable and unacceptable behaviours	5	3	3
Communications Risk	Absenses	Absenses during in-person meetings	Ineffective catchups post-meeting with absent members may leave the absentees with incomplete information. Absent team members cannot contribute to design decisions and discussions, therefore delaying design decisions.	5	2	10	Meeting summaries sheet, absent members can join meetings through teams and proper post-meeting catchups	4	4	3
Timeline Risk	Easter Break	2 week break from university	Reduces time availible for project and to recieve help / guidence	5	3	15	Assign work goals / deliverables for after the break and take the break into account when making the gantt chart	5	3	5
Timeline Risk	Scheduling Conflicts	When the team members' individual shcedules clash with each other or group activities	Difficulty in meeting and deadlines planning, absenses during meetings and reduced meeting length	5	2	10	Meeting and deadlines are annouced a week in advance, polls are made to decide meeting time and duration and allowing people to join meetings online	5	4	3
Technical Risk	Delivery of poor work	When a team member's deliverable is not made to a good standard	Slow in design development or research, disagreements between team members and if the work was done to a very poor standard (and is needed to continue with the design), it would have to be redone	3	4	12	Informing academic supervisors of poor work from a team member(s)	2	4	3
Technical Risk	Lack of help and guidance from supervisors or PHD students	When supervisors or PHD students do not provide proper help / guidance needed to continue with the project	More time needed to perform analysis, unclear academic goals leading to inadequate outputs and decreased quality of work	3	3	9	Communication through teams and outlook with the supervisors and PHD students. Scheduled weekly meetings	4	2	5
Communications Risk	Ineffective communication	Team members taking too long to reply, ignoring messages or not understanding the messages' content	Team members lacking information, work getting delayed or not done, absenses at team meetings and conflict between team members	4	2	8	Notifications being switched on for Teams and Outlook and team charter outlining work hours	3	3	3
Timeline Risk	Late for submission deadlines	When work is subitted late for a deadline	Grade being capped at 40%	1	5	5	Adding buffer time to account for delays in the project plan / timeline	4	4	1
Timeline Risk	End-of-project time crunch	End-of-project deliverables overload and increased workload as a result of procrastination	Burnout, missed deadlines, poor quality or plagiarized / rushed work delivered	4	5	20	Week-by-week deliverables to ensure the project stays on time and buffer time for deadlines	5	4	5

Figure 14: Excerpt from Project Management Risk Register

Severity		Probability		Impact		Probability	
1	Minor therefore mitigations are not strictly needed	1	Remote	1	This solution will have a moderate impact on the risk / problem with a moderate success rate, but no impact to its root / cause	1	The solution has little chance to be implemented
2	Small therefore mitigations should be considered vaguely	2	Low	2	This solution will have a positive impact on the risk / problem with a moderate success rate but no impact to its root / cause	2	The solution may be implemented if no risks related to its implementation are identified
3	May need to be considered but has no majour impact on the project so mitigations should be considered	3	Moderate	3	This solution will have a positive impact on the risk / problem but little impact on its root / cause with a moderate success rate	3	The solution will be implemented if no risks related to its implementation are identified
4	Will need to be considered and mitigations should be implemented	4	High	4	This solution will have a positive impact on both the risk / problem and its root / cause with a high success rate	4	The solution will be implemented if no major risks related to its implementation are identified
5	Major and will impact the project in multiple ways and therefore must be mitigated	5	Inevitable	5	This solution will solve the risk / problem with a high success rate	5	The solution will be implemented

Figure 15: Project Management Risk Register Scales Meaning

4 System Concept Definition – 1,409 words

The System Concept Definition process, as defined in the Systems Engineering Body of Knowledge (SEBoK), provides a structured framework for translating a poorly defined problem into a well-scoped system design effort[6]. In its full industrial application, several months and a specialist team are dedicated to this process, for the purposes of this project, a reduced and adapted version of the methodology was followed. Despite this compression, the core sequence of the process was preserved: the problem was formally

defined, stakeholder needs were explicitly captured and requirements, tracing back to their source needs, were generated. The value of adopting a recognised methodology is that it provides a justifiable path from the original brief to a set of specific system requirements, reducing the risk of designing a technically elegant solution to the wrong problem.

4.1 Problem definition

A problem must be fully defined before solution development begins. While the project brief (**Appendix A**) provided the initial starting point, it was intentionally open-ended and contained limited specific requirements. The key takeaways were[7]:

1. The team is required to design a "UAS capable of autonomously planning and dispersing a liquid payload over a set target area"
2. The UAS's mission is to spray crops and spread fertilizer
3. Smallholder farmers are a key market and commercial opportunity

Further research was required as a specific market nor system numerical attributes (range, payload size, etc…) were outlined.

Smallholder rice farmers in India were identified as the primary target market, as India contains 29% of global farms, 90% of which are under 2 hectares in size and predominantly, 60%, grain-based, particularly rice, which requires frequent spraying[8,9]. Large-scale farms and alternative applications were also examined to identify potential secondary and tertiary markets. Based on this analysis, an in-depth problem definition was developed (**Figure 16**, Appendix **F**). The SOI was defined as the agricultural spraying UAV, with the operator, farm, and airspace identified as enabling and receiving systems. The proposed business USP is a modular UAV platform optimised for crop spraying, with additional capability for alternative missions such as crop monitoring and solar panel cleaning.

Market	Who	What	Why	Where	When	Opportunity	Risks	Competition
Primary	Smallholder Farmers	Rice farms need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	Current spraying methods: 1) Spray a little area at the time, i.e. poor operational productivity (hectares sprayed per hour) 2) Are fully manual, labour intensive and not automated 3) Heavily expose the farmer to the harmful products being sprayed and are commonly inappropriatly storaged, therefore bringing health risks to the farmer's family and surrounding community 4) Are the only ones available to farmers as they have limited monetary resources	India, Sub-Saharan Africa, Europe	Short Term	Agricultural UAVs are few in this market therefore competition will be limited. The introduction of crop-spraying UAVs will remove a large burden to the farmer and will reduce their exposure to the dangerous chemicals, prolonging their health and life expectancy.	1) Limited purchasing power: Smallholder rice farmers have very low financial resources. 2) Low technological literacy: Some farmers may struggle to adopt or operate UAV technology.	> Manual (knapsack) spraying > Small tractor spraying
Secondary	Big farm owners + China	Large farm er owners need: 1) a large scale, 2) efficient, 3) automated and 4) productive spraying method	They own large farms (1) and m ust spray all their crops (4). Multiple spraying devices may be needed and therefore, a farmer cannot monitor multiple spraying operations all at once (3). These spraying methods must be cost effective so the farmer doesn't spend too much money (2)	USA, Australia, China, Latin America	Long Term	1) Reduced labour: Less manual effort required for spraying. 2) Improved safety: Lower exposure to harmful chemicals. 3) Higher productivity: Faster and more consistent spraying. 4) Time savings: Farmers can focus on other tasks. 5) Greater efficiency: More precise application uses less chemical. 6) Scalability: Larger farms gain greater efficiency benefits.	1) High competition: Agricultural UAV markets are already saturated.	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Tractor or Boom sprayers
Tertiary	Other agricultural applications (mapping and monitoring) and external applications (solar panel cleaning)	Agricultural applications: large farms require crop surveillance to determine health and growth status, livestock owners require livestock surveillance to track the movement of their animals and large land owners require land mapping to plan future plantations or building sites External applications: solar farms could be sprayed with cleaning agents to clean the solar panels. Spraying of greenhouses, payload transport, monitoring purposes, surveillance and mapping are all mission profiles the drone can accomplish	Using a drone for these tasks may increase productivity and / or efficiency. These alternatives uses provide flexibility to the business and allow for expansion	Worldwide	Parallel to primary business / Long Term	UAVs are well-tailored for these tasks, and the external applications for drones are numerous. This allows for additional profit margins in low seasonal farming demands.	The drone platform was not specifically designed for these use cases and could be less efficient than alternatives. Could add more fixed costs to the company.	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Current methods of performing the task

Figure 16: Problem Definition Table

4.2 Project Stakeholders

To develop system requirements, project stakeholders and their associated needs must first be identified. A stakeholder is defined as: *"an individual or organisation with a right, share, claim, or interest in a system or in its possession of characteristics that meet their needs and expectations"*[10]. Stakeholders and their relationship to the project were identified through brainstorming and by working backwards from the SOI lifecycle. A Stakeholder Register (**Figures 17** and **18**, Appendix **G**) was then developed, capturing

stakeholder identification, their relationship to the SOI, key interests, and a prioritisation score used to determine major stakeholders.

The influence score represents the extent to which a stakeholder can affect the project, while the interest score reflects their level of involvement or concern. Weightings of 1.4 and 0.6 were assigned to the influence and interest scores respectively. Influence was given the greater weighting because it has a more direct impact on the project and its outcomes. The weighted scores were then summed and rounded to the nearest whole number to obtain the overall stakeholder score which informed the development of the Integrated Set of Needs.

The target farmer was assigned a medium influence score because their needs strongly shape the design specifications, although strict compliance with those needs is not mandatory. In contrast, aerospace regulators were given a high influence score, as regulatory compliance is essential to achieve airworthiness, despite their comparatively low direct interest in the project itself.

Category	Stakeholder	ID	Association with the Project	Interest in Project Aspect	Specific Concerns	Influence Score	Interest Score	Total Score
Technical Team	Operations team	T2	This team sets up and operates the drone while it is spraying, they are the service providers	Operational / Safety & Usability	Ease of drone operation, safety from chemical exposure, quick payload and battery reload between missions to minimise downtime, and effective service delivery for customers.	3	4	7
	Mechanics	T3	This team repairs the drone if it gets damaged	Operational / Repairability	Modular design for quick field repairs, reliable availability of spare parts (COTS components), and standardized tooling requirements.	3	4	7
Rest of the Company	Business team	R1	This team identifies the target market, the client needs and outlines a set of needs / requirements that then influence the design	Commercial / Profitability	Cost of Goods Sold (COGS), pricing strategy, route to market (selling to FPOs/CHCs), competitive advantage, and achieving target profit margins.	3	5	7
End users / Customers	Target farmer	U1	This is the primary target customer	Economic / Performance & Yield	Affordability of the spraying service, guaranteeing zero physical damage to the crop, reduced health risks to the farmer and highly effective pest control and increased crop productivity.	3	5	9
	Surrounding Farmers	U3	These have the closest relationship with the farmer and may be a source for more customers. They also provide opportunity for the bulk spraying business model but may also hindure the service	Economic / Performance & Yield and Social / Environmental & Safety	Fear of pesticide "drift" contaminating their own crops or water supply, noise pollution from heavy multi-rotors, and privacy concerns regarding the drone's cameras if they are not part of the bulk operation	4	3	7
Governing Bodies	Local government	G1	Provides guidelines and grants that enable the company to operate the drone and deliver a service. They could also be a source of income for the target customer leading to potentially more sales	Macro-Economic / Strategic	Success of modernization schemes, creating rural tech employment, and improving national food security	4	2	7
	Agricultural regulators	G2	Provides guidelines and grants that enable the company to operate the drone and deliver a service.	Regulatory /Environmental & Chemical Safety	Approval of specific pesticide formulations for Ultra-Low Volume (ULV) aerial spraying and verifying that the drone actively reduces chemical and water waste	5	4	9
	Aerospace / Airworthyness regulators	G3	Provide guidelines and grants that enable the company to operate the drone and deliver a service.	Regulatory / Aviation Safety	Airworthiness type certification, "No Permission, No Takeoff" (NPNT) compliance, strict geo-fencing to prevent flight in restricted zones, and weight limits.	5	2	8
Business Partners	Investors / Stockholders	B3	Provide funding for this business venture	Financial / Growth & Returns	Market capture timeline, intellectual property defensibility, scaling the business model, and overall return on their capital investment.	4	5	9

Figure 17: Excerpt from the Stakeholder Register, showing the most important stakeholders

Influence		Interest		Total Score		Total Score Formula
How much they can influence the project (change the design)		How invested they are to this project		The resulting importance of a stakeholder		**The Influence score was given a weighing of 1.4 and the Interest score was given a weighing of 0.6 for the evaluation of the total score. This was done because a stakeholder's influence on the project is more important than their interest in it. Formula: 1.6*Infl + 0.6*Intr = Tot Sc**
1	Has little to no influence on the project	1	Has little to no interest on the project	1 - 2	Stakeholder is remotely connected to this project	
3	Is considered a driver in design decisions	3	Is interested in the project	3 - 5	Stakeholder is connected to this project	
4	Is a majour driver in design decisions	4	Has a large interest in the project	6 - 8	Stakeholder is an important member of this project	
5	Is the one of the main drivers of design decisions	5	Is solely focused on this project	9 - 10	Stakeholder is a key member of this project	

Figure 18: Scoring Parameters used in Stakeholder scoring

4.3 Source of Needs / Drivers & Constraints

Following stakeholder identification, their needs were identified through brainstorming and market research and the Source of Needs table (**Figure 19**, Appendix **H**) was constructed. These were classified as either Drivers (factors motivating project development) or Constraints (factors limiting the design space)[11,12]. Only stakeholders with an interest score of 4 or above, or an overall score of 7 or above, were considered to focus the analysis on the most relevant parties.

Category	Need or Constraint	ID	Description / Notes	Impact on the project	Related Stakeholder	Source
Drivers	Pesticides needed to grow rice in south-east asia	D1	N/A	Drives the main function of the drone	Target Farmer	https://croplife.org/wp-content/uploads/pdf_files/Case-Study-108-Rice-in-Asia2.pdf
	Herbicides needed to grow crops	D2	N/A	Drives the main function of the drone	Target Farmer	https://ejournals.onlinelibrary.wiley.com/doi/full/10.1002/ps.3598
	Fertilizer needed to increase crop yield	D3	N/A	Drives the main function of the drone	Target Farmer	https://d1wqtxts1xzle7.cloudfront.net/31216237/579-584.doc-libre.pdf?1392254554=&response-content-disposition=inline%3B+filename%3DEffect_of_Nitrogen_and_Phosphorus_Fertil.pdf&Expires=1776078761&Signature=gWBuK3fSDNx-gIVNQE3QcHOy7rsQtTDrCKZ2oMHsQaiwXOiBJMmF4EMUtTED6PGUBvysbuPFFUaOhO9RtqvyJBUTTEWUNadYGsETTYDuBvCTokPQcrvuXbJTX9ReNbBTqEOMbQvz~Qa9dw3vBoBbMAerkE2HFHchaaQ6ubJroF2J9JbKPGCtyaJZAFCHxwfRuk4OJJfPEkgzwG6CCF-pUnzoeKGrfqsfQbOKBbUY6JuXEFsD44HkZtUBJOh8-lCZvTCamQkaWMdDlLqUpFaY6OEyfDDe2ilt-rgiR7JfNG3TJjPUbRpUuLK25eN74rT1L2I5ENOvsmw__&Key-Pair-Id=APKAJJDIFSGSSJBBV4ZA
Constraints	Drone's need for infrastructure independence	C1	The drone must not need any infrastructure on the farm for it to operate	May limit take-off options, charging possibilities, drone autonomy solutions (longer range GPS may have to be used) and set-up processes	Target Farmer	Group Brainstorming and Discussions
	Drone operations being weather resiliant	C2	N/A	The drone may have to possess and IP rating and be capable of flying in difficult weather conditions	Target Farmer, Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Drone needing flexble mission capabilities	C3	Spraying irregular boundaries, obstacle filled fields and varying crop heights	Will add development time and complexity to the drone's software	Target Farmer	Group Brainstorming and Discussions
	Drone noise restrictions	C4	Compliance with noise guidelines set by aerospace regulators	Will add development time and complexity to the drone's hardware	Farmer's Family, Surrounding Farmers, Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Aerospace regulations	C5	N/A	Will add development time and complexity to the drone's hardware and software	Aerospace / Airworthyness regulators	Group Brainstorming and Discussions

Figure 19: Excerpt from the Source of Needs table

4.4 Project Risk Register

SEBoK's methodology includes a project risk analysis step. Unlike failure modes, risks are defined as events or situations that could negatively impact or potentially halt the project. These are summarised in a Risk Register (**Figures 20** and **21**), a management tool used to identify, assess, and track technical and programme uncertainties affecting project objectives. Its construction is outlined in **Appendix I**. While some risks were addressed during the design phase, others can only be mitigated during the project's real-life execution. Risks and mitigations were identified through brainstorming, research, and analysis of the SOI lifecycle, and contributed to the development of the Integrated Set of Needs.

Category	Risk	ID	Notes / Description	Impact Description	Associated Stakeholder	Probability	Severity	Risk Score	Mitigation	Probability	Impact	Mitigated Risk Score	Status	Date identified	Date Closed	Closure Comments
Commercial Risk	Competition	R1	Established ag-drone manufacturers (like DJI or XAG) or local manual labor and tractor services offering cheaper, more familiar alternatives.	Loss of market share, forced price reductions to remain competitive, and difficulty securing long-term service contracts with farmers.	Design / Engineering and Business teams	4	4	16	Offer a strong unique selling point over the competition. Meet the needs of the target market	4	3	5	Open	18.Mar		
Commercial Risk	Lack of consumer demand	R2	Farmers showing skepticism toward drone technology or being unwilling to change their traditional, proven spraying and mapping methods.	Low adoption rates, stagnation in sales or service bookings, and inability to clear the high upfront R&D and manufacturing costs.	Smallholder farmer, Large farm owner, Business team and Operations team	1	5	5	Offer a "Service-as-a-Product" model to lower the entry barrier, allowing farmers to benefit from the tech without the upfront purchase and training costs. Price the product at a loss for introductory schemes for market penetration, have good relations with customer and referral schemes	4	5	1	Open	18.Mar		
Commercial Risk	Lack of funding from external bodies	R3	Insufficient capital secured from venture capitalists, government agricultural grants, or bank loans to support ongoing operations.	Insufficient funds to continue or develop the platform, leading to stalled R&D, inability to scale production, or potential bankruptcy.	Design / Engineering and Business teams, Investors, Local government and farmers financial services	2	5	10	Apply for targeted government agricultural innovation grants and establish early paid pilot programs to demonstrate traction to investors. Investigate in marketing initiatives.	3	3	3	Open	18.Mar		
Commercial Risk	Low profit margin	R4	High manufacturing / development costs for drone components combined with the low price ceiling that cost-conscious farmers are willing to pay.	Entering the market with a massive debt burden, severely restricted cash flow, and an inability to reinvest in future technology upgrades.	Business team, Investors and Competition	2	3	6	Standardise components across the fleet to achieve volume discounts and automate flight planning to reduce the labor hours required per acre. Market diversification	5	4	2	Open	18.Mar		
Technical Risk	Change in regulations, becoming more strict	R5	Aviation authorities (like EASA or FAA) or environmental agencies introduce stricter compliance laws.	Project delays, increased R&D costs to redesign systems, or complete denial of operational authorization.	Smallholder farmers, Airworthiness regulators, Agricultural regulators, Local government and Design / Engineering Team	2	5	10	Design the aircraft to exceed current safety standards (e.g., building in independent flight termination systems) to ensure future compliance via software updates.	3	5	2	Open	18.Mar		
Commercial Risk	High initial setup and manufacturing costs	R6	Large capital expenditure required for tooling, molds, and early-stage production lines.	Entering the market with a large debt, reducing cash flow and lowering net profit margins.	Business teams and Investors	4	4	16	Utilize modular carbon fiber components and 3D printing for early production, only outsourcing heavy tooling after securing minimum order volumes.	4	3	5	Open	18.Mar		
Programme Risk	Poor roads around the target farm	R7	Poor rural infrastructure limits access to fields or target application sites.	Difficulty in reaching the target site with the ground station, leading to delayed missions and reduced operational efficiency.	Farmer, Drone operator / pilot team	1	4	4	Mount the ground control station and charging infrastructure on an off-road capable vehicle allowing it to navigate on muddy and unpaved farm tracks.	2	3	1	Closed	18.Mar	14.04.2026	Ford Transit van was considered as the main transport vehicle for the service
Programme Risk	Unpredictable weather	R8	Heavy rain, high winds, or extreme heat preventing safe flight operations.	Flight cancellations, missed narrow spraying windows, and potential physical damage to the airframe or electronics.	Smallholder farmer, Large farm owner and Operations team	4	5	20	Integrate hyper-local weather telemetry at the ground station and establish clear "Go/No-Go" operational thresholds for wind speed and rain.	4	4	5	Open	18.Mar	27.Apr	Design freeze produced a working drone concept that satisfied all operational requirements
Programme Risk	Scheduling complexity / narrow spraying windows	R9	Short optimal timeframes to spray specific crops combined with multi-client coordination.	Issues with farmer organization and agricultural society organization leading to underutilized assets and loss of revenue.	Smallholder farmer, farmer's neighbour, Business team and Operations team	3	3	9	Develop automated fleet-scheduling software that groups neighboring farms and optimizes routes based on crop maturity and weather windows. Have good relations with the target farmers and offer referral schemes to align their spraying timetable	4	5	2	Open	18.Mar		

Figure 20: Excerpt from the Risks Register, showing the most important risks

Severity Scale		Probability		Occerrance / Year	Impact Scale		Probability	
1	Minor therefore mitigations are not strictly needed	1	Very low	0	1	This solution will have a moderate impact on the risk / problem with a moderate success rate, but no impact to its root / cause	1	The solution has little chance to be implemented
2	Small therefore mitigations should be considered vaguely	2	0.05	18.25	2	This solution will have a positive impact on the risk / problem with a moderate success rate but no impact to its root / cause	2	The solution may be implemented if no risks related to its implementation are identified
3	May need to be considered but has no majour impact on the project so mitigations should be considered	3	0.15	54.75	3	This solution will have a positive impact on the risk / problem but little impact on its root / cause with a moderate success rate	3	The solution will be implemented if no risks related to its implementation are identified
4	Will need to be considered and mitigations should be implemented	4	0.35	127.75	4	This solution will have a positive impact on both the risk / problem and its root / cause with a high success rate	4	The solution will be implemented if no major risks related to its implementation are identified
5	Major and will impact the project in multiple ways and therefore must be mitigated	5	0.5 <	182.5	5	This solution will solve the risk / problem with a high success rate	5	The solution will be implemented

Figure 21: Scoring Parameters used in the Risks Register

4.5 Integrated set of Needs

The Integrated Set of Needs table (**Figures 22** and **23**, **Appendix J**) is the output of the Stakeholder Needs Definition process. It is generated by combining stakeholder needs (drivers and constraints) with the results of the risk and problem definition analyses to form a structured set of system needs (requirements)[11]. These requirements ensure the SOI is designed to address the defined problem. Each requirement must be linked to an associated stakeholder need and risk, and is specified to be singular, measurable, and unambiguous to enable later verification through the compliance matrix. Competitor research, the preliminary business proposition and preliminary ConOps also contributed to the requirements generation.

Each requirement was given an ID and classified as either a "need" or a "want" to distinguish between essential and optional functionality. A prioritisation score was assigned to each requirement and multiplied by the score of the most relevant stakeholder to generate an overall requirement importance score. This ranking was used to ensure consistent and justifiable trade-offs where compliance with conflicting requirements was required.

Requirement Identification			*Requirement Definition*													
Category	**Requirements**	**ID**	**Description / Notes**	**Need or Want**	**Want Classification**	**Related Stackholder(s)**	**Rationale**	**Requirement Source (Needs and Risks)**	**Specific Information Source**	**Associated Risks**	**Stakeholder sore:**	**Matrix Requirement Scores**	**Corrected Requirement Scores**	**Total Score Multiplication**	**Date Added**	**Design Feature / Response**
Operation	The drone's spraying must cover the entire area that has been marked for spraying	CN1	N/A	Need	N/A	Target Farmer and Design Team	If this need is not met, the service will not be carried out correctly	D1, D2, D3, D5, C3, R16	N/A	Increased development time and cost	9	6	8	72	01.Mar	Navigation system
	The drone must be capable of spraying the field or area outlined in the mission plan autonomously	CN11	The drone must spray the field(s) that has been marked for spraying by the farmer or operations team, regardless of obstacles or the field's shape	Need	N/A	Target Farmer, Farmer's neighbours, Farmer's village, Business team	The farmer is paying to have their field sprayed	D7, C3, R1, R16	N/A	Increased development time and cost	9	3	8	72	01.Mar	Navigation system Consideration of irregular fields (non rectangular) is beyond GDBP scope
Regulation	The drone must reduce the farmer's exposure to harmful chemicals during spraying	RN1	The spraying process must be independent from the farmer, spraying drift must be minimised and the correct area must be sprayed	Need	N/A	Target Farmer, Farmer's family, Business team, Insurance, Indian government	Unique selling point of the business	D4, D6, C16, C17	N/A	Increased development time and cost	9	4	7	63	01.Mar	Drone flight and spraying procedures
	The drone and its use case must respect the regional laws of the area where it is operating	RN2	N/A	Need	N/A	Indian government, Farmers financial support services, Indian government, Farmers Union companies, Farmers financial support services, Business team, Insurance	To allow the drone to fly	C17, C22, R29	N/A	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone must not be above 150 kg MTOW	RN3	To remain in the "medium" category according to C.5.#.189	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	The medium category of drones requires less airworthiness checks to be certified and thus would reduce costs	C2, C4, C5, C17, C22, R29	https://easa.europa.eu/document-library/powered-by-easa/publication-data-2021-22-23.pdf	Increased development time and cost and limit in drone capabilities	7	Not scored in the Matrix	8	56	01.Mar	Design spec according to regulations review
	The drone must respect every regulation associated with airworthiness in its category according to EASA	RN4	Specific regulation: EASA's Easy Access Rules for Unmanned Aircraft Systems (Regulations (EU) 2019/947 and 2019/945).	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	Airworthyness and to allow the drone to fly	C2, C4, C5, C17, C22, R29	https://www.easa.europa.eu/en/document-library/easy-access-rules/easy-access-rules-unmanned-aircraft-systems-regulations-eu	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone's V-n envelope must respect guidelines set by EASA	RN5	Specific regulation: CS-VLA, CS-23 and SC-VTOL	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	Airworthyness and to allow the drone to fly	C2, C4, C5, C17, C22, R30	N/A	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone must adhere to the agricultural regulations set by the local government	RN6	N/A	Need	N/A	Agricultural regulators, Business team, Insurance	Unique selling point of the business and to allow the drone to fly	C16, C17, R29	N/A	Increased development time and cost	9	6	8	72	01.Mar	Design spec according to regulations review
Manufacturing and End-of-life	The drone must be built using commercially available and industrially produced materials	MN7	N/A	Need	N/A	Bespoke component manufacturers, Mechanics	Bespoke materials cannot be developed for this drone because of time, cost, complexity and certification	C6, C8	N/A	Increased development time and cost	7	4	8	56	01.Mar	Materials selection during design
	The drone must be assemblable using industry standard tooling and current assembling and manufacturing techniques	MN8	N/A	Need	N/A	Final assembly / manufacturing firm, Mechanics	Bespoke assembly methods / techniques cannot be developed for this drone because of time, cost, complexity and certification	C6, C9	N/A	Increased development time and cost	7	4	8	56	01.Mar	Key considerations during design

Figure 22: Excerpt from the Integrated Set of Needs table, showing the most important needs

Stakeholder Score		***Requirements Score***		***Requirements Total Score***	
Total Score taken from Stakeholder Register		**Value**	**Meaning**	**Value**	**Meaning**
1 - 2	Stakeholder is remotely connected to this project	1	The need has little to no importance regarding the drone's operation and non-compliance is acceptable	1 - 9	This need carries little to no importance to the project
3 - 5	Stakeholder is connected to this project	2	The need has little importance regarding the drone's operation and non-compliance is acceptable	10 - 18	This need carries little importance to the project
6 - 8	Stakeholder is an important member of this project	3	The need has some importance regarding the drone's operation and non-compliance is acceptable but not preferable	19 - 27	This need carries some importance to the project
9 - 10	Stakeholder is a key member of this project	4	The need is somewhat important to the drone's operation and compliance is preferable but not obligatory	28 - 36	This need carries moderate importance to the project
		5	The need is moderatly important to the drone's operation and should be complied with	37 - 45	This need carries moderately high importance to the project
		6	The need is important to the drone's operation and should be complied with	46 - 54	This need carries high importance to the project
		7	The need is very important to the drone's operation and must be complied with	55 - 63	This need carries very high importance to the project
		8	The need is vital to the drone's operation and must be complied with	64 - 72	This need carries vital importance to the project

Figure 23: Scoring Parameters used in the Integrated Set of Needs

4.6 Numerical Requirements Generation

After completing the Integrated Set of Needs, a set of baseline numerical design requirements was established to define the UAV's key specifications. These provided initial targets for early design development. The attributes were derived from competitor product specifications and the highest priority needs that could be expressed numerically. Requirement values were scaled from competitor benchmarks and regulatory constraints to match the UAV's intended mission profile. These were consolidated into the Numerical Requirements table (**Figure 24**, Appendix **K**). When requirements were updated during the project, previous values were retained in an archive to ensure traceability.

Requirement Definition													
Category	**Attribute**	**Requirement**	**Units**	**Importance**	**ID**	**Notes**	**Rationale**	**Source**	**Related Need**	**Date created**	**Created by**	**Last Modified**	**Modified by**
Operation	Operational Productivity (OP)	21 (ideal case) & 18 (normal)	ha/h	High	NR1	How many hectares are sprayed in one hour	The DJI Agras T50 serves a similar market to this UAV, therefore a similar operational productivity is desired	https://dji-ag.com/pages/dji-agras-t50 https://dronespraypro.com/blogs/news/how-many-acres-in-1-day	ON9	13.Feb	Peter	03.Apr	Luca
	Operational Radius	4,000	meters	High	NR2	Dependant on power consumption and battery size, optimised for endurance	The UAV has a fixed-wing advantage over quadcopters and therefore its range can be larger	DJI Agras T25, T25P, T50, T70P, T100 - DJI user manuals	ON10, ON13	13.Feb	Luca	17.Mar	Luca
	Flight time	30	minutes	High	NR3	Dependant on power consumption and battery size, must be high for endurance	The Pelican is fixed-wing agri drone similar to this UAV, therefore flight time should be similar	https://www.flypyka.com/pelican-2 Pyka Pellican range	ON10, ON13	13.Feb	Jaison	25.Feb	Luca
	Payload capacity	47	kg	High	NR4	47 litres of product that can cover approximatly 4.7 acres or 1.9 hectares	The drone must spray two hectares of land in one mission	https://marutdrones.com/ag-365/#faq litres needed to cover one acre	ON10	13.Feb	Jonathan	25.Feb	Luca
	Operational ceiling	1,500	meters	Medium	NR5	The maximum height the drone can operate at	This will increase the number of farms that can be sprayed copared to a drone that can only work at sea level	https://mapsforuspsc.com/largest-rice-producing-states-in-india/ and https://www.researchgate.net/figure/Map-showing-states-and-elevation-zones-of-India_fig1_315835800	ON19	13.Feb	Luca		
	Communication range	5 & Compatible with 4G and sattelite	km	High	NR6	These are dependant on the power output of the hardware (drone & controller) and limits imposed by countries on Operating frequency (check P12 of the Business review document)	This is in line with competitor drones	DJI Agras line of drones	ON2, ON3, ON8, ON11, ON13, ON14, DN3, DN4, DN5	13.Feb	Luca		
Spraying mechanism	Fluid flowrate	9	L/min	High	NR7		This is in line with competitor drones	DJI Agras T25 specs	ON9	13.Feb	Jonathan	25.Feb	Luca

Figure 24: Excerpt from the Requirements table, showing the most important requirements

4.7 Solution Breakdown

The Integrated Set of Needs was used to develop the system concept and business model. The UAV's design and operational features reflect the identified needs, as shown in the Design Feature/Response column of the table. The selected system concept is an autonomous hybrid fixed-wing VTOL UAV with separate flight and transport configurations. This configuration was driven by ON4, ON13, DN3 and aligned with the proposed business USP.

The requirement for autonomy was motivated by ON2, ON3, ON8, farmer productivity demands, and competitive benchmarking. While it was recognised that a hybrid VTOL architecture introduces greater complexity and higher development costs—thereby increasing the breakeven sales volume—the advantages associated with the USP were considered sufficient to justify these trade-offs.

An initial Product System Breakdown diagram (**Figure 25**) was constructed to identify each sub-system. The UAS is composed of 7 sub-systems: Propulsion, Power, Structures, Payload (storage and deployment), GNC, Electronics, and the Ground Station (the aircraft's supporting infrastructure). System interfaces were not yet identified as a drone configuration had not been established.

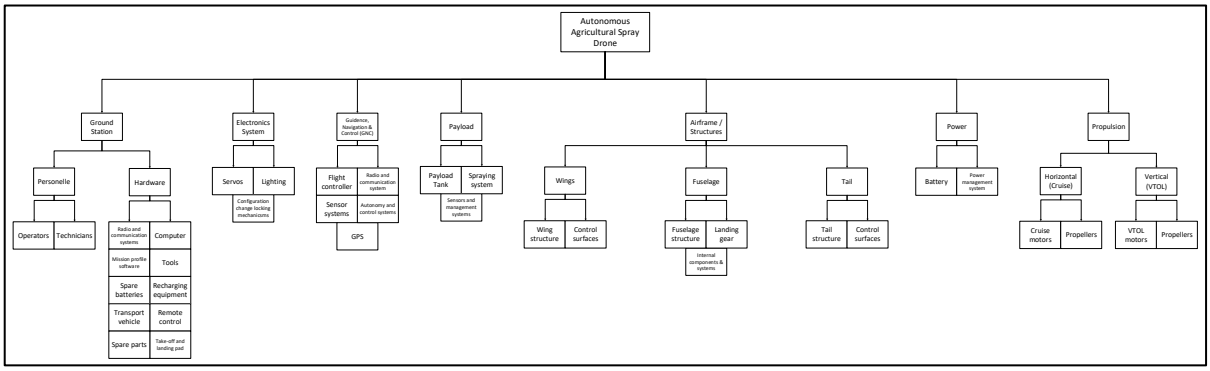


Figure 25: UAV's Product System Breakdown

The selected business model is "spraying as a service", as it provides greater benefits to both the customer and the company compared to a product-only (drone sales) approach. The advantages and disadvantages of this model are summarised in **Table 3**.

Pros	Cons
1. Cheaper for the consumer as they do not have to purchase equipment that they would only use during specific times of the year 2. Maintains the drone's airworthiness as it is not kept in storage for an extended period of time, due to it being used all year round (instead of exclusively during the spraying season) 3. Decreased manufacturing costs for the business as the same fleet is used every year 4. Allows the company to control the platform, ensuring maintenance and operating standards, minimising drone misuse and gathering operational data. 5. Services generate predictable revenue streams 6. Consumer lock-in as customers are likely to use the same service repeatedly 7. Eliminates regulatory liability to end-users	1. Economies of scale would not be present if the drone fleet is small 2. Higher personnel costs as technicians and drone operators are needed to deliver the service 3. Higher training costs as drone operators need to be trained 4. Possibility for complex timetabling and logistics to fit farmers' spraying needs 5. Full operational liability falls on the company 6. Lower short-term income 7. More rigorous regulatory approval

Table 3: Pros and Cons of the spraying as a service model

A ConOps diagram (**Figure 26**) was developed to illustrate the UAV's primary function. A ConOps is a user-focused representation of a proposed system that describes its characteristics from the perspective of the user[13]. It aids in translating abstract requirements into a clear operational scenario that is understandable by the client or supervisor. The UAV's operational concept was derived from requirements ON1–ON4, ON9, ON10, ON14, DN4, DN5, DN7, IN3–IN5, and the proposed business model.

The operation begins with the team conducting a terrain assessment and generating a navigation plan (**A**). The operations team then arrives on-site (**1**), the UAV is then transitioned from its transport configuration to its flight configuration and prepared for flight (**2**). Following pre-flight checks, payload loading and navigation upload (**3**), the UAV performs a vertical take-off (**4**), transitions to forward flight, and completes spraying of the designated area (**5**). It then returns to the launch site, transitions to vertical flight, and lands where it is recharged and refilled, ready for the next mission (**6**).

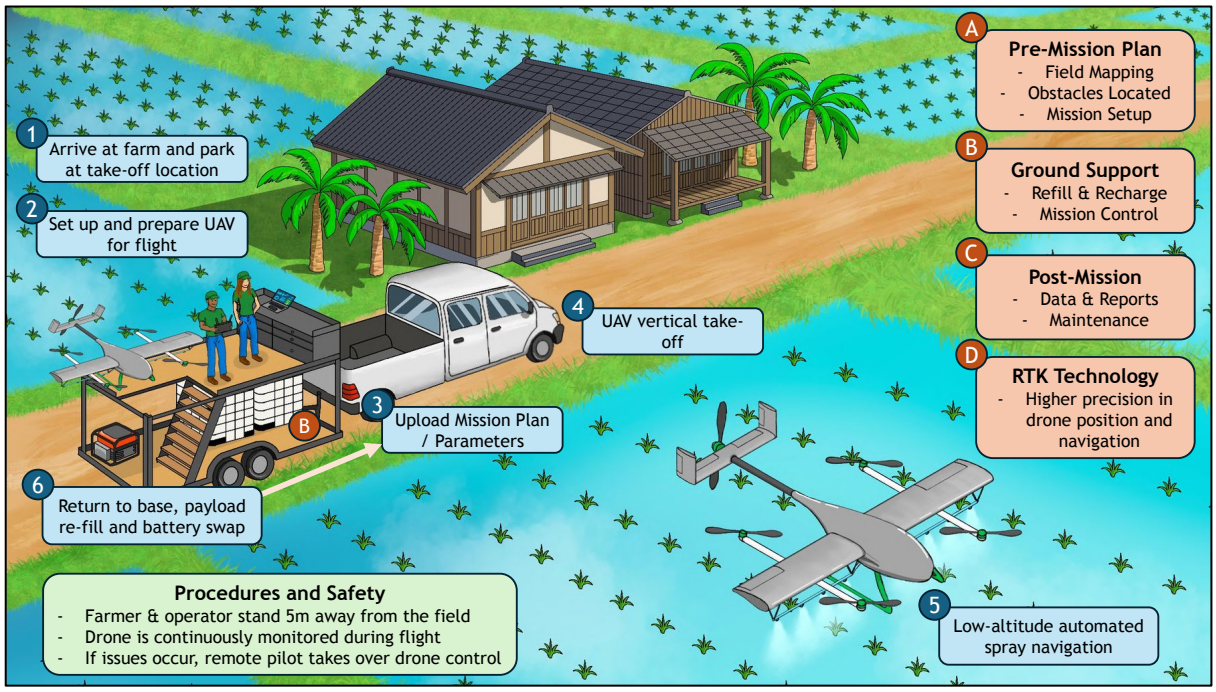


Figure 26: ConOps diagram showing the drone performing a mission

A mission profile diagram (**Figure 27**), that outlines every mission phase, was constructed to show the UAV's operation.

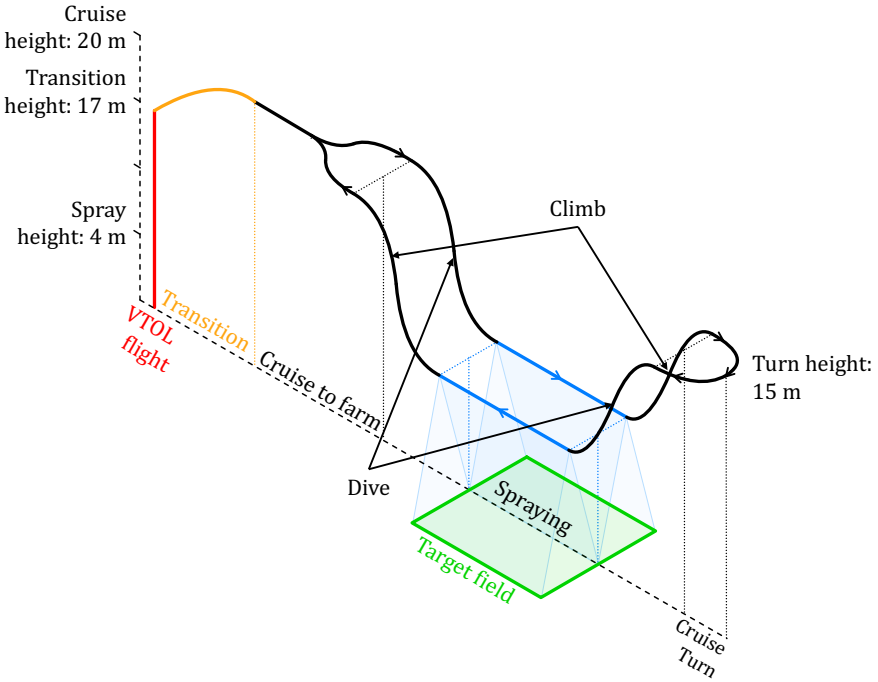


Figure 27: UAV's Mission Profile Diagram
(3D representation chosen for clarity)

5 Solution Execution

This section documents the execution of the design process from initial concept generation through to the final configuration presented at CDR. The design evolved over three iterations, with the PDR serving as a formal checkpoint between the second and third. Each iteration is described through its technical accounts, design challenges, and the methods used to resolve them. Additional aspects of the design process, including weight management, FMEA and CAD development, are also discussed.

5.1 Concept Ideation and Selection

Concept ideation was carried out using a Morphological Chart (**Figure 28**, Appendix **L**), a structured concept generation tool in which the design problem is decomposed into independent functional elements, with feasible solutions generated for each, before a final configuration is formed by combining selected options. This method was chosen as it provided a systematic framework that allowed all team members to contribute ideas across each design element. It also helped prevent premature convergence on a single concept driven by individual preference, ensuring that no viable options were excluded before being fully evaluated.

Each team member was tasked with independently generating one or more concepts for each function, alongside proposing a complete viable UAV configuration. A final brainstorming session was then held to expand and refine the set of concepts. Each functional element concept was presented and discussed, after which voting was conducted to select the most suitable option for each function. The voting process is summarised in **Figure 29**. Members were allocated multiple votes per round to account for the fact that several concepts often had merit, allowing more balanced decision-making without forcing a single choice between competing viable options.

Functions	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5	Idea 6	Idea 7	Idea 8	Idea 9	Idea 10	Idea 11	Idea 12	Idea 13	Idea 14	Idea 15
Configuration			medium-sized Hybrid VTol: 	Picture	Quadcopter										
Propulsion (Horizontal)	Vertical Propeller As seen in Configuration				none	Rotor tilt - rotors tilt forward to provide thrust									
Propulsion (Vertical)	Horizontal Propeller As seen in Configuration				4 Stabilizers	Vertical propellor (see configuration)			In-wing rotors						
Control surfaces	Elevators and ailerons, V-tail -> no rudder.		Ailerons at wing and rudders at tail, no elevators required	Ailerons, Elevators, Rudder, Rotor control during vertical flight	Stabilizers	Rotor speed control - differential thrust manages pitch, roll and yaw control									
Spraying mechanism				Boom across wing trailing edge as seen in idea 2	nozzles on each 4 arms										
Payload location					Centre tank				Held in body moulded tank						

Figure 28: The Morphological Chart

	Category	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea
Idea number	Landing gear 3 votes each	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Description		Fixed Skids	Tripod	Wheels	Retractable	Launch & Catch	No landing gear	Tail sitter								
Votes		8	2	4	5	3	1	1								
Idea number	Propulsion 3 votes each	1	3	7	10	11	9									
Description		Fixed	Tilt	Ducted	In Wing	Tilt Wing	Like the Off-spray									
Votes		8	5	4	3	0	4									
Idea number	Tail 3 votes each	1	2	3	4	7	8	9	10							
Description		V-tail	Inverted V	Twin tail	U-tail	Tail sitter	H-tail	Vertical fin	No tail							
Votes		2	2	3	8	1	5	3	0							
Idea number	Spraying Mechanism 3 votes each	1	2	3	6	7	8	9	10							
Description		Pods	Boom	3 sprayers at the back	Under the vertical rotor	Central Spraying	Two central sprayers	Retractable	Non-wing mounted (on landing gear)							
Votes		3	6	1	3	3	2	1	5							
Idea number	Payload Tank 2 votes each	1	2	3												
Description		In wing	Internal (non removable)	Internal (removable)												
Votes		8	3	5												
Idea number	Power / Fuel 2 votes each	1	2	3	4											
Description		Battery	2-stroke engine	Hydrogen fuel cell	Hybrid system											
Votes		7	2	4	3											
Idea number	Configuration 3 votes each	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Description																
Votes		6	3	0	5	0	2	2	2	0	0	1	0	3	0	0
Idea number	Transport 1 vote each	1	2	3												
Description		Dis-assembly	Folding wings	Telescopic wings												
Votes		6	2	0												

Figure 29: The voting system

A configuration was then chosen to be adapted to integrate the chosen functional element concepts. The final configuration is shown in **Figure 30**.

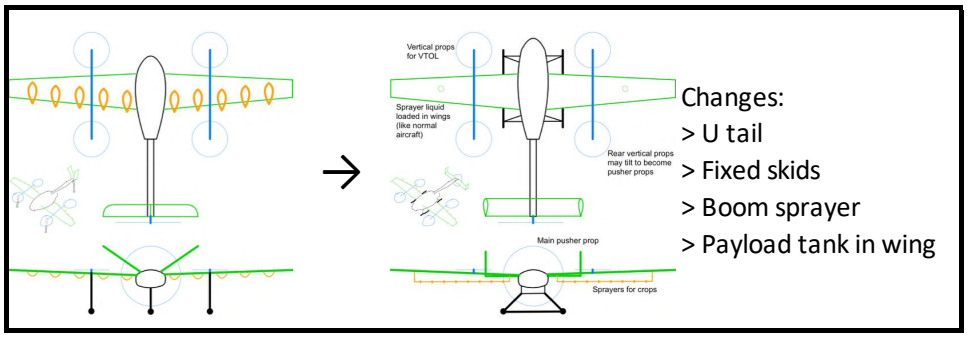


Figure 30: The chosen configuration adapted to fit the chosen function concepts

The selected functional element concepts were primarily chosen for their simplicity and conventional nature. Fixed skids were adopted due to their low mass and structural simplicity, as the aerodynamic benefits of a retractable landing gear were not sufficient to justify the additional weight and complexity. Fixed motors were selected over tilt-rotor systems to simplify the VTOL-to-cruise transition and eliminate the added mechanical complexity and weight of a tilting mechanism. Although this results in the VTOL motors being carried as dead weight during cruise, a configuration that retained the capability of integrating tilt-rotor systems within the VTOL booms was favoured.

A U-tail configuration was chosen due to its favourable lateral stability, clean airflow to the cruise propeller, and well-established conventional design. A spray boom was selected as the payload delivery method as it allows direct access to the nozzles and can be more easily maintained or removed than wing-integrated alternatives. The payload tank was integrated within the wings, following a configuration similar to fuel tanks in commercial aircraft, to utilise otherwise unused internal volume, reduce fuselage size and drag, and minimise sloshing through shorter tank lengths and heights. Battery propulsion was selected based on sustainability considerations and competitor benchmarking.

Configuration 1 was ultimately selected due to its simplicity, compatibility and ease of system integration with the chosen functional element concepts. It also allowed for potential future incorporation of tilt-rotors within the VTOL booms (shown in blue). Additionally, it aligned well with the transport configuration concept, providing clear and

structurally robust disassembly points. Disassembly was preferred over folding or telescopic wing solutions due to reduced structural complexity and fewer performance compromises.

5.2 Iteration 1

After the numerical and qualitative requirements were defined and an initial concept selected, a calculation path (**Figure 31**) was developed to identify the required input parameters and determine the sequence in which values could be estimated or derived. Raymer's Aircraft Design textbook was used to construct the calculation path[14]. This guided the first design iteration by structuring the initial sizing and analysis processes.

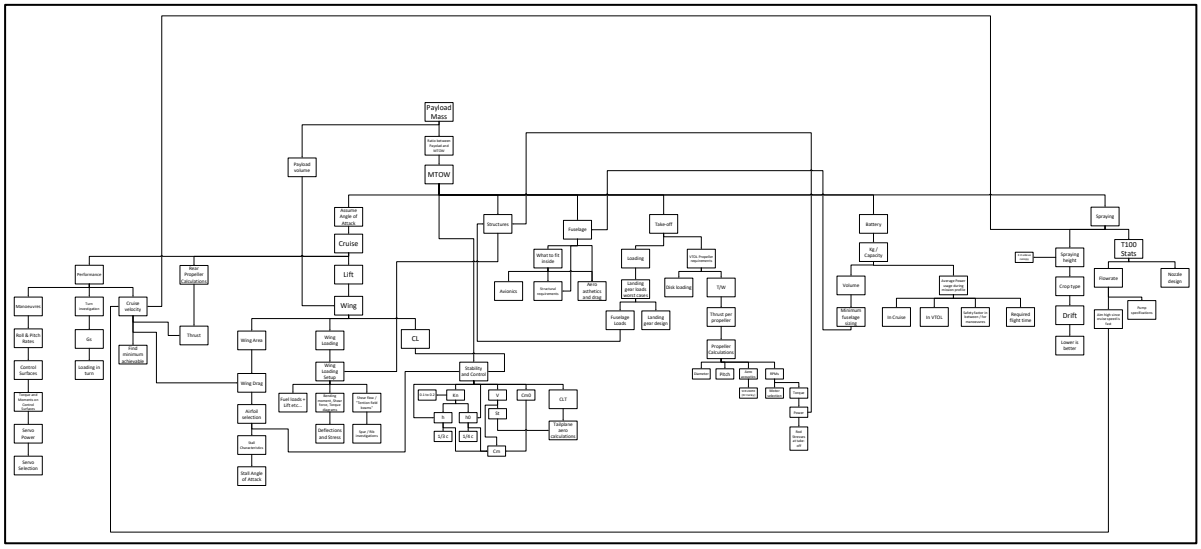


Figure 31: Calculation Path

To determine initial aircraft sizing, a wing platform estimation (**Figure 32**) and constraint diagram (**Figure 33**) were developed. An MTOW of 100 kg (as specified in the requirements) and a cruise speed of 13.8 m/s, in accordance with CAAC UAV regulations (Article 62, translated via Google Translate), were used as initial design inputs under sea-level conditions[15].

The resulting design space was highly constrained, with an insufficient thrust-to-weight ratio that led to an impractically large battery (refer to Aerodynamics and Systems Reports for further details). Additionally, the selected cruise speed was too low to support a feasibly sized aircraft at the specified MTOW. As MTOW was constrained by payload requirements (NR4), and the system needed to satisfy requirement NR11, the cruise speed was revised to 25.8 m/s based on spray width and flight time considerations (refer to Aerodynamics Design Report for further details). This adjustment produced a more viable design space and improved compatibility with higher MTOW configurations, but led to regulatory non-compliance in China.

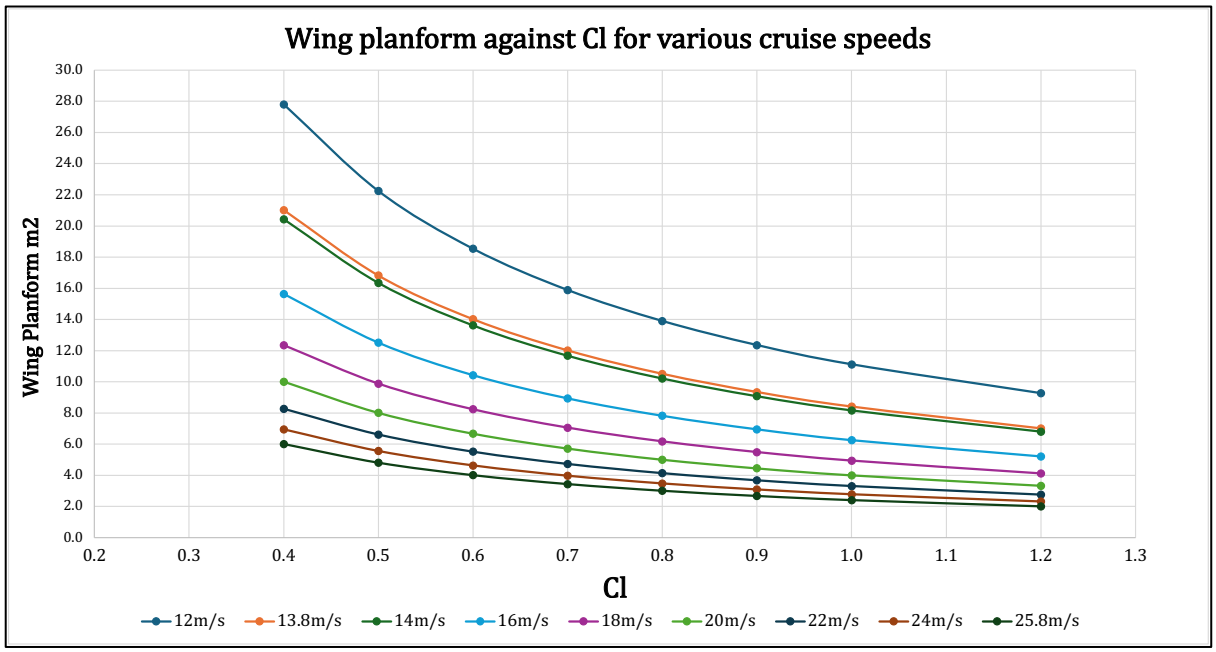


Figure 32 Wing Platform Estimations for Various Cruise Speeds

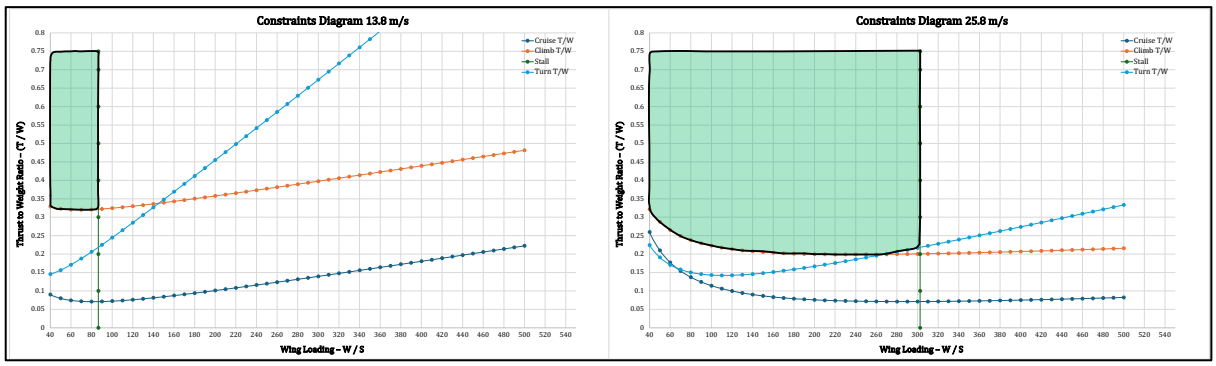


Figure 33: Constraints Diagram (old cruise speed – left, revised cruise speed – right)

Preliminary design was initiated across all sub-systems and initial top-level specifications were established. A Technical Accounts table (**Figure 34**, Appendix **M**) was developed to compare calculated values against requirements and track design compliance. Red indicates non-compliance, green indicates compliance, yellow represents estimated values, and blue denotes uncalculated parameters. As this was the first design iteration, many values were necessarily estimated or undefined, limiting full assessment of requirement compliance.

As MTOW exceeded the initial target, a revised design target of 130 kg was set for Iteration 2.

Category	Attribute	Criteria	Unit	Iteration 1	Verification Method
Operation	Operational Productivity	21 (ideal case) & 18 (normal)	ha / h		
	Max operational radius	4,000	m	6,000	Calculation
	Flight time	30	m	30	Calculation
	Payload Capacity	47	kg	47	Calculation
	Operational Ceiling	1,500	m		
	Communication range	5 & Compatible with 4G and satellite	km		COTS Component Integration
Spraying Mech	Fluid flowrate	9	L/min	19.872	Calculation
Physical	MTOW	100	kg	110	Estimation
	Max MTOW	150	kg	110	Estimation
	Empty weight	53	kg	63	Calculation
	Operational dimentions	L: 3,500 W: 3,000	mm	4500 x 5180	CAD model
	Transporational dimentions	2,765 x 1,640 x 1900 (L x W x H)	mm	2300 x 600 x 600	Estimation
Durability	Operational life	1,000	cycles		
	IP rating	IP55	IP__		
Power	Down time (battery & payload change)	3	min		
	Charging time	8	min		
Other	Regulations compliance				Regulation Checking

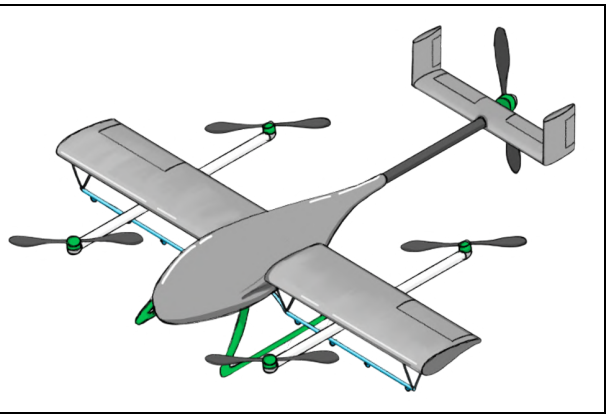


Figure 34: Iteration 1 technical accounts

5.2.1 Initial CAD Model

The CAD model was developed using Autodesk Inventor® Professional 2026. The initial model (**Figure 35**) consisted of cuboids representing the design space allocated to each sub-system (wings, fuselge, tail, etc…). These design spaces were used to define preliminary interfaces between sub-systems, potential disassembly points and enabled initial sizing of the UAV.

The model was parameter-driven via a base component (**Figure 36**), enabling rapid geometric iteration. Changes to key design variables (wingspan, tail moment arm, fuselage length, etc…) could be directly reflected in the CAD model, allowing immediate visualisation of their impact. This approach reduced the risk of conceptual development diverging from physical feasibility by ensuring continuous geometric validation throughout the design process.

The model's parameter-driven architecture was maintained throughout the project, excluding the final CAD model. All parameters were continuously updated in line with the General Values table. This ensured that each design iteration could be rapidly visualised in CAD, enabling consistent physical dimension verification and system-level integration checks.

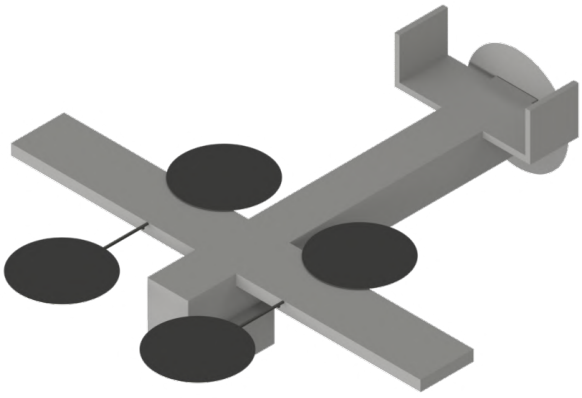


Figure 35: Parameter Driven CAD model

Parameters: Parameters Master Component.ipt

Type Keyword...

Parameter Name	Consumed by	Unit/Type	Equation	Nominal Value	Tolerance	Model Value	Key	Export	Comment
Model Parameters									
d4	Extrusion1	mm	1 mm	1.000000	<Default>	1.000000			
d5	Extrusion1	deg	0.00 deg	0.000000	<Default>	0.000000			
User Parameters									
fuselage_l		mm	4260 mm	d5 is consumed by Extrusion1		4260.000000		☑	Length of the entire aircraft
fuselage_w		mm	500 mm	500.000000		500.000000		☑	
fuselage_h		mm	450 mm	450.000000		450.000000		☑	
chord	vtol_boom_hor_p...	mm	650 mm	650.000000		650.000000		☑	
wingspan		mm	5460 mm	5460.000000		5460.000000		☑	
wing_t	wing_vert_pos	mm	0.2 ul * chord	130.000000		130.000000		☑	
h0	vtol_boom_hor_p...	mm	1.10 ul * chord	715.000000		715.000000		☑	
wing_vert_pos		mm	0 mm + wing_t / 2 ul	65.000000		65.000000		☑	
wing_cent_p	vtol_boom_hor_pos	mm	0.2 ul * chord	130.000000		130.000000		☑	
tailplane_chord	vert_stab_chord, ...	mm	tailplane_a / tailplane_span * 1 m	591.787440		591.787440		☑	
tailplane_t	tail_vert_pos	mm	0.12 ul * tailplane_chord	71.014493		71.014493		☑	
vert_stab_chord	vert_stab_h, vert...	mm	tailplane_chord	591.787440		591.787440		☑	
vert_stab_t		mm	0.1 ul * vert_stab_chord	59.178744		59.178744		☑	
vert_stab_a	vert_stab_h	mm	375 mm	375.000000		375.000000		☑	
tailplane_span	tailplane_chord	mm	vtol_boom_span	1656.000000		1656.000000		☑	
tail_vert_pos		mm	0 mm + tailplane_t / 2 ul	35.507246		35.507246		☑	
lt		mm	3000 mm	3000.000000		3000.000000		☑	
tailplane_cent_p		mm	0.2 ul * tailplane_chord	118.357488		118.357488		☑	
cog	vtol_boom_hor_pos	mm	h0 - 0.29 ul * chord	526.500000		526.500000		☑	
landing_gear_h		mm	400 mm	400.000000		400.000000		☑	
landing_gear_w		mm	750 mm	750.000000		750.000000		☑	
landing_gear_l		mm	800 mm	800.000000		800.000000		☑	
vtol_boom_w		mm	20 mm	20.000000		20.000000		☑	
vtol_boom_h	boom_vert_pos	mm	20 mm	20.000000		20.000000		☑	
vtol_boom_l	vtol_boom_hor_pos	mm	2000 mm	2000.000000		2000.000000		☑	
vtol_boom_hor_pos		mm	vtol_boom_l / 2 ul + h0 - cog + chord / 2 ul - wing_cent_p	1383.500000		1383.500000		☑	The 4 props are centered on the COG
boom_vert_pos		mm	vtol_boom_h / 2 ul	10.000000		10.000000		☑	
vtol_boom_span	tailplane_span	mm	1656 mm	1656.000000		1656.000000		☑	
fuselage_l_boomless		mm	1350 mm	1350.000000		1350.000000		☑	The length of the fuselage "egg"
tail_boom_d		mm	51 mm	51.000000		51.000000		☑	
landing_gear_boom_d		mm	40 mm	40.000000		40.000000		☑	
wing_base_w		mm	1727 mm	1727.000000		1727.000000		☑	
tailplane_base_w		mm	140 mm	140.000000		140.000000		☑	
nozzle_placement		mm	2800 mm	2800.000000		2800.000000		☑	
vert_stab_h		mm	vert_stab_a / vert_stab_chord * 0.5 m	316.836735		316.836735		☑	
tailplane_a	tailplane_chord	mm	980 mm	980.000000		980.000000		☑	
test		mm	30 mm	30.000000		30.000000		☑	

Add Numeric | Update | Purge Unused | Import from XML | Reset Tolerance | << Less

Link | Immediate Update | Export to XML | Done

Figure 36: Base component and Parameters list

Once initial sizing was completed, the CAD model was refined with representative geometry, including a fuselage shape, landing gear, tail boom, and aerodynamic surfaces to better reflect the evolving design.

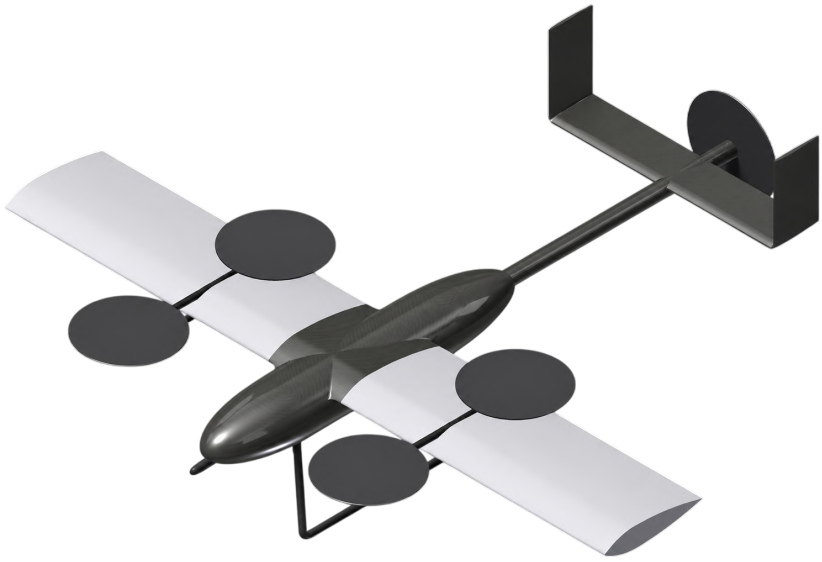


Figure 37: Improved CAD Model

5.3 FMEA

Before Iteration 2 began, a system-level FMEA (**Figures 38** and **39**) was conducted to identify the UAV's top-level failure modes and evaluate potential design, testing, and process-based mitigations. Its construction is outlined in **Appendix N**. Through brainstorming, 13 system-level failure modes were identified.

Faulty obstacle detection and circuitry failure were identified as the most critical failure modes, as they could result in aircraft or crop damage, injury to the client, and are difficult to detect before occurrence. India's humid operating environment may further increase the likelihood of such failures. Design mitigations are limited, as these failures are primarily associated with COTS components, making careful component selection essential. Static obstacles can be identified during pre-mission mapping to reduce collision risk; however, this does not protect against moving hazards such as birds, reinforcing the need for continuous operator supervision during flight operations.

Structural failure modes were assigned low occurrence ratings due to regulatory requirements mandating high structural safety factors in aircraft design. As the UAV is not intended to perform aggressive manoeuvres, it is unlikely to experience the high load factors typically associated with structural failure, and these failure modes were therefore considered less critical.

Full Aircraft Failure Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Mitigation through Analysis and Design	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Testing	Resp.	SEVERITY (1 - 10)	OCCURENCE (1 - 10)	DETECTION (1 - 10)	RPN = S x O x D	IMPLEMENTATION	EFFECTIVENESS	MRPN (Mitigated RPN) RPN = (S x O x D) + (I x E)
When could the failure mode occur?	What sub system is the failure mode part of	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or feature to go wrong? (How could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate this failure mode?			What test can be performed to mitigate this failure mode?	Who is responsible for making sure the actions are completed?						
During flight or during manoeuvres & gusts	Wing	Critical wing structure failure	Aircraft, field or crop damage and service not delivered	10	Assembly / manufacturing error, fatigue, inadequate pre-flight checks and unpredictable weather	1	Pre-flight, pre and post assembly / manufacture and QC checks, weather forecasts, inspections and certification documentation	7	70	Safety factor Structural redundancy in the wings FEA analysis g sensors VTOL system	7	4	Simulated failure test (wings are detached mid flight and the aircraft must recover in VTOL by itself)	Design, manufacturing and operations teams	10	1	7	70	7	4	28
During landing	Landing gear	Structural failure of landing gear	Aircraft, field or crop damage and service delays	3	Fatigue, hard impact and manufacturing defects	2	Pre-flight, pre and post assembly / manufacture and QC checks, inspections and certification documentation	2	12	Safety factor Structural redundancy in fuselage FEA analysis g sensors Removable & swappable sprayboom	7	4	Hard landing test Fatigue testing	Design, manufacturing and operations teams	3	2	2	12	7	4	5
During flight or during manoeuvres & gusts and landing	Fuselage and Tail	Structural failure of fuselage or Tail	Aircraft, field or crop damage and service not delivered	10	Assembly / manufacturing error, fatigue, inadequate pre-flight checks and unpredictable weather	1	Pre-flight, pre and post assembly / manufacture and QC checks, weather forecasts, inspections and certification documentation	4	40	Safety factor Structural redundancy in the tail FEA analysis g sensors VTOL system	7	4	Simulated failure test (tail is detached mid flight and the aircraft must recover in VTOL by itself)	Design, manufacturing and operations teams	10	1	4	40	7	4	16
During flight or during manoeuvres & gusts and landing	Battery	Power loss	Crop damage, electrical fire and service not delivered	10	Thermal degredation, overcharging, incorrect battery usage and handling and short circuitig	4	Flight log per battery, cell voltage checks, smart charging / discharging, proper storage, correct handling procedures and pre-flight checks	3	120	Battery management / protection systems (BMS) Thermal simulation for battery degradation Battery lifecycle analysis Smart battery system Circuit redundancy Waterproofing	9	4	Accelerate mission testing Lifecycle testing	Design and operations teams	10	4	3	120	9	4	48
During flight or during manoeuvres & gusts and landing	Sensors	Faulty obstacle detection	Damage to crop, aircraft and property damage, harm to client, unfulfilled service	10	faulty hardware, power loss to system, dirty/covered sensors, Flight controller software issues	3	Pre flight checks, redundancy within sensor system	6	180	Simulation of obstacle detection code accuracy Robustness through software Rendundancy within system	8	4	Sub-system bench testing Full navigation testing System validation testing	Design team	10	3	6	180	8	4	72
During flight or during manoeuvres & gusts and landing	Control Surfaces	Faulty / loss of actuation	Service not delivered, low efficiency in service	5	Faulty hardware, linkage failure, power loss. structural failure	5	Pre flight checks, QC at factory, documentation and inspection logs	1	25	Redundancy VTOL system	7	2	Fatigue testing Simulated failure test (placing the aircraft in an unstable manoeuvre where it must recover by itself)	Design, manufacturing and operations teams	5	5	1	25	7	2	5
During flight or during manoeuvres & gusts and landing	Circuitry	Circuitry failure	Aircraft, field or crop damage and service not delivered	10	Faulty wiring, short circuit, leakage, disconnected interfaces and overspec power inputs	3	Manufacturing regulation, QC checks, pre-flight checks, documentation and inspection logs	6	180	Thorough PCB model and circuitry mode. Appropriate wire and component selection Built-in failsafes and circuit redundancies	8	4	Segmented and full sub-system bench testing	Design, manufacturing and operations teams	10	3	6	180	8	4	72

Figure 38: Excerpt from System Level FMEA

Severity Scale

Effect	Criteria: Severity of Effect	Ranking
Hazardous - Without Warning	May expose client to loss, harm or major disruption - failure will occur **without** warning	10
Hazardous - With Warning	May expose client to loss, harm or major disruption - failure **with** warning	9
Very High	Major disruption of service involving client interaction, resulting in either associate re-work or inconvenience to client	8
High	Minor disruption of service involving client interaction and resulting in either associate re-work or inconvenience to client	7
Moderate	Major disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to client	6
Low	Minor disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	5
Very Low	Minor disruption of service involving client interaction that does not result in either associate re-work or inconvenience to clients	4
Minor	Minor disruption of service not involving client interaction and does not result in either associate re-work or inconvenience to clients	3
Very Minor	No disruption of service noticed by the client in any capacity and does not result in either associate re-work or inconvenience to clients	2
None	No Effect	1

Occurance Scale

Probability of Failure	Time Period	Per Item Failure Rates	Ranking
Very High: Failure is almost inevitable	More than once per day	>= 1 in 2	10
	Once every 3-4 days	1 in 3	9
High: Generally associated with processes similar to previous processes that have often	Once every week	1 in 8	8
Moderate: Generally associated with processes similar to previous processes which have occasional failures, but not in major proportions	Once every 3 months	1 in 80	7
	Once every 6 months	1 in 400	6
	Once a year	1 in 800	5
Low: Isolated failures associated with similar processes	Once every 1 - 3 years	1 in 1,500	4
Very Low: Only isolated failures associated with almost identical processes	Once every 3 - 6 years	1 in 3,000	3
Remote: Failure is unlikely. No failures associated with almost identical processes	Once Every 7+ Years	1 in 6000	2
			1

Detection Scale

Detection	Criteria: Likelihood the existence of a defect will be detected by process controls before next or subsequent process, -OR- before exposure to a client.	Ranking
Almost Impossible	No known controls available to detect failure mode	10
Very Remote	Very remote likelihood current controls will detect failure mode	9
Remote	Remote likelihood current controls will detect failure mode	8
Very Low	Very low likelihood current controls will detect failure mode	7
Low	Low likelihood current controls will detect failure mode	6
Moderate	Moderate likelihood current controls will detect failure mode	5
Moderately High	Moderately high likelihood current controls will detect failure mode	4
High	High likelihood current controls will detect failure mode	3
Very High	Very high likelihood current controls will detect failure mode	2
Almost Certain	Current controls almost certain to detect the failure mode. Reliable detection controls are known with similar processes.	1

Can only be detected in flight

Can be detected during take-off

Can be detected on the ground / during setup

Implementation Scale

Implementation	Criteria: Likelihood the implementation of the design feature or testing process as a form of mitigation for the failure mode	Ranking
Certain	Will be implemented	10
Very High	Has a very high chance of implementation	9
High	Has a high chance of implementation	8
Moderately High	Has a moderately high chance of implementation	7
Moderate	Has a moderate chance of being implemented	6
Low	Has a low chance of implementation	5
Very Low	Has a very low chance of implementation	4
Remote	Has a remote chance of implementation	3
Very Remote	Has a very remote chance of implementation	2
Impossible	Will not be implemented	1

Effectiveness Scale

Effectiveness	Criteria: Effectiveness of the design feature or testing process in solving or mitigating the failure mode	Ranking
Completely Ineffective	The mitigation will have no effect on the failure mode	10
Partially Ineffective	The mitigation will have very little effect on the failure mode	9
Very Low	The mitigation will reduces the chances of the failure mode occuring substantially	8
Low	The mitigation will moderately reduce the chances of the failure mode occuring	7
Moderately Low	The mitigation will reduce the chances of the failure mode occuring	6
Moderate	The mitigation will partly solve the problem	5
Moderately High	The mitigation will greatly reduce the chances of the failure mode occuring	4
High	The mitigation will reduces the chances of the failure mode occuring substantially	3
Very High	The mitigation will solve the problem almost in its entirety	2
Complete	The mitigation will solve the problem	1

Figure 39: Scoring Parameters used in the FMEA

5.4 Airworthiness and Regulations

An airworthiness mind map (**Figure 40**) was developed to identify the key elements associated with airworthiness and determine which sub-systems required compliance considerations. Relevant regulatory authorities were also identified for each country in which the service could operate.

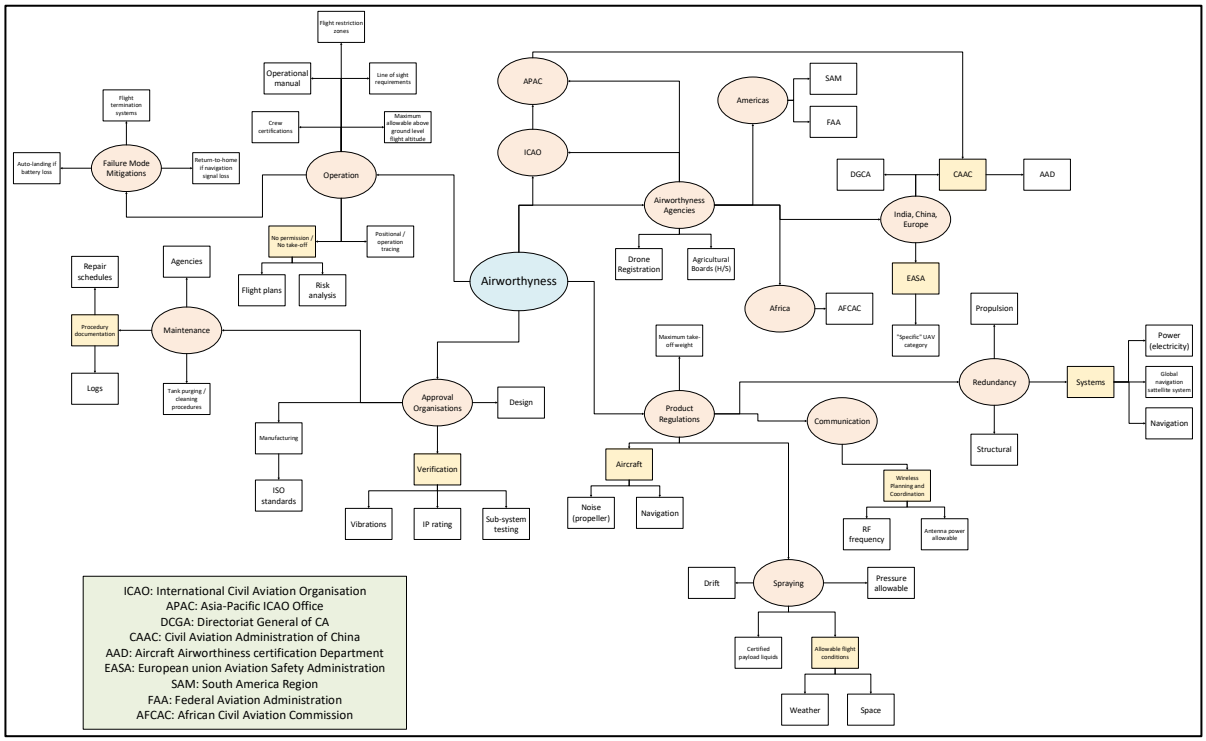


Figure 40: Airworthiness Mind Map

As India was selected as the initial target market, DGCA weight category regulations (GSR 589 Part II) were used to define the UAV's MTOW limit[16]. A 50 kg design margin below the regulatory limit was initially maintained to account for expected weight growth during development; however, following initial aircraft sizing, the design margin was reduced to 20 kg.

Globally, UAV regulations are still relatively immature and often lack detailed guidance. As a result, EASA regulations 2019/947 and 2019/945 were adopted as the primary design reference due to their comparatively comprehensive and stringent nature[17]. Many provisions within these regulations are risk-assessment based, meaning compliance depends on the outcome of a formal operational risk analysis. As such assessments were outside the scope of this project, conventional airworthiness regulations were additionally referenced for design areas not explicitly covered by the UAV-specific framework. Chinese regulations were initially considered; however, regulatory constraints and strong market competition ultimately excluded China as a target market. The regulations applied to each sub-system are summarised in **Table 4**.

Regulatory Body	Parent Country	Regulation	Who used it	What was it used for
EASA	European Union	2019/947	Whole Team	Used as general regulations for UAV design
		2019/945	Whole Team	Used as general regulations for UAV design
		SC-19	Aman (Systems)	For general qualitative regulations of electrical systems and components, accounting for safety, maintenance and handling.
		SC-E-19		
		CS-25.23		
IEC	International Organisation	IEC60529	Aman (Systems)	Specific qualitative regulations for lithium-based batteries (mainly for industrial applications but its a close alternative).
EASA	European Union	CS-VLA	Sam and Jaison (Structures)	Used for guidance on loads and load cases as well as construction of the V-n (manoeuvre) envelope using limit load factors
		CS-23		
		SC-VTOL		
AFWAL	United States of America	MIL-F-8785C	Hariram (Stab & Con)	Used for validating the dynamic stability of aircraft
IARC	International Organisation	N/A	Jaison (Propulsion)	Used to understand emission impact of engine alternative propulsion (Specifically that they classify diesel engine exhaust as a human carcinogen)
DGCA	India	G.S.R. 589	Jonathan (Payload) Luca (TI)	Used to understand agricultural spraying guidelines and design the payload system (Part V) Used to fix the aircraft's MTOW (Part II)
CAA	United Kingdom	CAP1861	Peter (Navigation)	Used to determine what avionics hardware is needed for Beyond Visual Line of Sight (BVLOS)

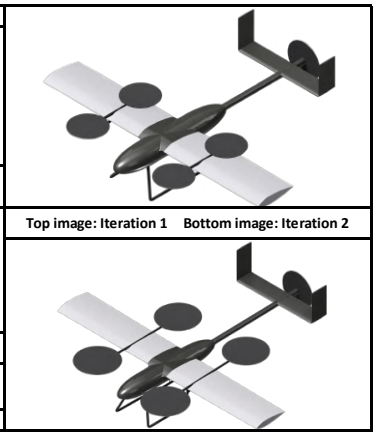
Table 4: List of Airworthiness Regulations followed by each Sub-System

5.5 Iteration 2 and PDR

No major configuration changes occurred between Iteration 1 and Iteration 2; instead, the design was refined and the UAV's theoretical performance capabilities were further developed. This progression is reflected in the updated technical accounts for the iteration (**Figure 41**, Appendix **M**). The newly derived parameters demonstrated compliance with the established requirements, maintaining a high overall level of design compliance.

The CAD model was also updated to reflect the revised iteration. The landing gear and VTOL booms, which were originally centred on the COG, were relocated to improve ground stability and reduce the risk of rotor interference with the wings.

Category	Attribute	Criteria	Unit	Iteration 1	Verification Method	Iteration 2	Verification Method
Operation	Operational Productivity	21 (ideal case) & 18 (normal)	ha / h			55.728	Calculation
	Max operational radius	4,000	m	6,000	Calculation	4,000.5	Calculation
	Flight time	30	m	30	Calculation	15	Battery Spec
	Payload Capacity	47	kg	47	Calculation	47	Payload Spec
	Operational Ceiling	1,500	m				
	Communication range	5 & Compatible with 4G and satellite	km		COTS Component Integration		COTS Component Integration
Spraying Mech	Fluid flowrate	9	L/min	19.872	Calculation	18.48	Calculation
Physical	MTOW	130	kg	110	Calculation and Estimation (previous target was 100 kg)	124	Calculation and Estimation
	Max MTOW	150	kg	110	Estimation	124	
	Empty weight	83	kg	63	Calculation	77	Calculation
	Operational dimentions	L: 3,500 W: 3,000	mm	4500 x 4800	CAD model	4739 x 5180	CAD model
	Transporational dimentions	2946 x 1784 x 1886 (L x W x H)	mm	2300 x 600 x 600	Estimation	2,700 x 1,810 x 720	CAD model Old target: 2,765 x 1,640 x 1900
Durability	Operational life	1,000	cycles			500	Battery Spec
	IP rating	IP55	IP__				
Power	Down time (battery & payload change)	3	min				
	Charging time	8	min				
Other	Regulations compliance				Regulation Checking		Regulation Checking



Top image: Iteration 1 Bottom image: Iteration 2

Figure 41: Iteration 1 and 2 Technical Accounts

A PPA was conducted for the UAV's wingspan, cruise speed, and MTOW using data from Janes aircraft databases (**Figures 42 and 43**)[18]. This comparison was performed to validate the scaling of the design and identify any potential analytical inconsistencies. The UAV's relationships between MTOW and cruise velocity, and between wingspan and cruise velocity, were generally consistent with comparable aircraft within its class, although the cruise speed was comparatively lower. This was considered acceptable due to the UAV's specialised agricultural spraying mission, which does not require high cruise velocities.

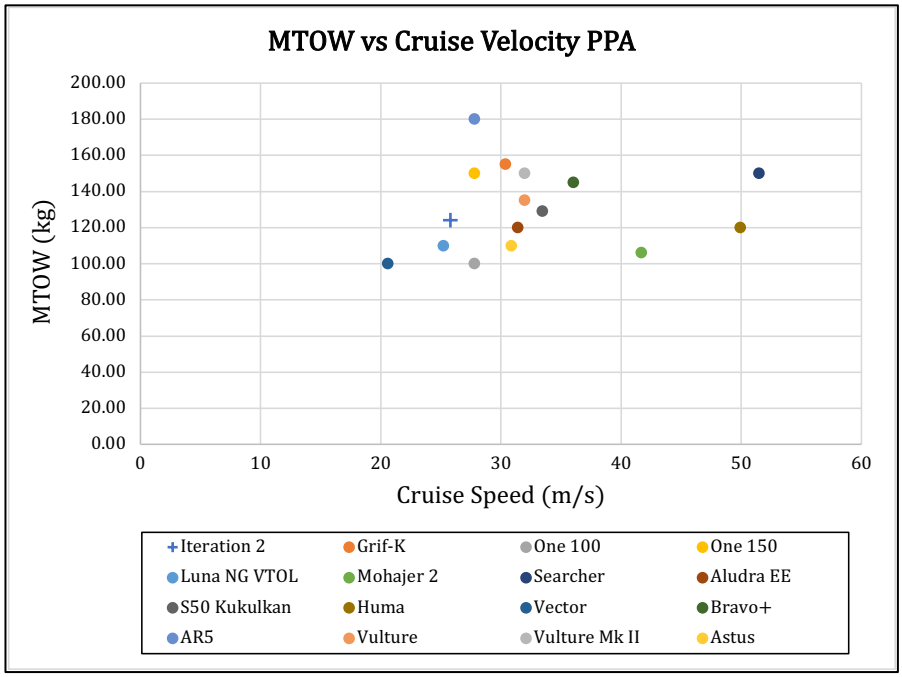


Figure 42: MTOW vs Cruise Velocity Product Position Analysis (Iteration 2's data was used)

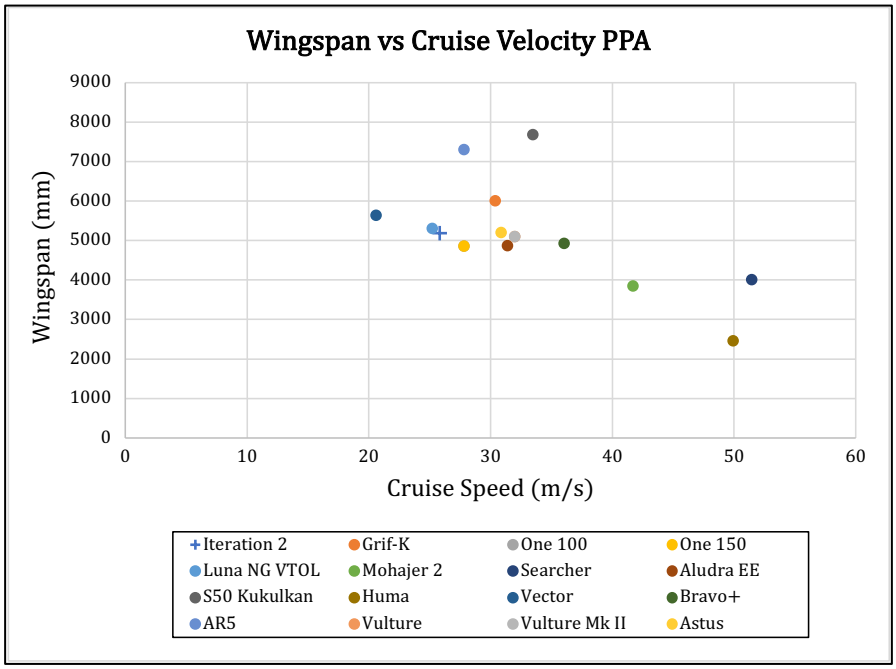


Figure 43: Wingspan vs Cruise Velocity Product Position Analysis (Iteration 2's data was used)

This iteration was presented at PDR, where the primary feedback focused on the underdeveloped lift-to-drag ratio and aerodynamic efficiency analysis. Additional concerns were raised regarding spray tank integration, fluid slosh dynamics, and propeller wash interaction with the spray system, all of which were identified as priorities for detailed design.

The RF transparency limitations of carbon fibre for sensors and avionics were also discussed, leading to consideration of RF-transparent fuselage regions during detailed fuselage development. Finally, the placement of the cruise propeller and payload tanks was questioned, with these concerns ultimately driving significant configuration changes in Iteration 3, discussed in later sections.

5.5.1 Flight Time Change

The initial flight time target was set to 30 minutes to align with the capabilities of the Pyka Pelican, another fixed-wing agricultural aircraft[19]. However, payload system calculations showed that the payload would be depleted in approximately 4.5 minutes of active spraying. Assuming that spraying would account for roughly half of the total mission duration (a conservative estimate), the effective mission time was estimated at only 9 minutes.

To reduce unnecessary battery mass, the flight time requirement was therefore reduced to 15 minutes while still maintaining a substantial energy reserve for operations involving distant fields. VTOL operation time was also reduced, as calculations estimated a maximum cumulative VTOL duration of approximately 80 seconds per mission (accounting for a signal loss scenario). Applying a safety factor of 1.5 increased the design requirement to 2 minutes of VTOL endurance. The trade-offs associated with this decision are summarised in **Figure 43**.

Sub-system	Component	Old Position	Why it was put there	What problems it caused (Why was it changed)	New proposed location	Why was it put there	What downsides are there
Aircraft Mission Profile	Flight Time	30 mins	> Competitor research and initial spec	> The battery needed for this flight time was extremely heavy > All the payload would be depleated in 5 minutes of spraying time (straight flight over fields, excluding turns) so 25 minutes of cruise were excessive and not needed	15 mins	> Productivity analysis and rate of payload depletion rate meant the drone would be in the air significantly longer than necessary. > Reduced flight time allows for a more effective use of time in-field which can help bring battery capacity down. > Reduced battery weight allows for increased weight budgets for other sub-systems	> Flight time is less competitive comapred to other agri-drones in the market in the same weight class > Survay missions flight time is reduced
Aircraft Mission Profile	VTOL time	3 mins	> Allocated time after flight time change	> The battery needed for this flight time was heavy	2 mins	> Ascent rate of 2.5 m/s meant a cruise altitude of 15 m was reached in 6 seconds --> 10 total time to reach cruise accounting for transition to cruise (horizontal) flight > Total VTOL time (in seconds) needed during flight = 10 x 2 + 60 (if signal loss) = 80 s --> safety factor of 1.5 (returning to base with 20% battery) --> 2 minutes	> Less safety margin for VTOL flight

Figure 44: Flight Time Change Trade-offs summary

5.6 Weight Budgets

Weight budgets, defined as allocated mass limits for each sub-system, were used to control weight growth and maintain the UAV's MTOW below the design limit. Initial estimates were derived from selected COTS components and structural mass approximations based on engineering judgement, informing the weight distributions used during Iterations 1 and 2.

Following the PDR, a detailed weight table was developed (**Figure 45**). The first formal weight budget was generated using methods described in UAV design literature by Gundlach and Sadraey, using the target MTOW as the reference[20, 21]. As these methods were theoretical and intended for different UAV classes, a second budget was produced using the TBD 2026 CDR weight breakdown (**Appendix O**). Component masses were

converted into percentages of empty weight and scaled to the target empty weight of the UAV. Empty weight was used instead of MTOW because the TBD UAV and the current iteration had different empty-weight fractions (59% and 66% respectively).

As both budgets were proportional to the target MTOW, final weight budgets were recalculated using the latest estimated MTOW values to assess whether proportional allocations (**Figure 46**, Appendix **N**) were being maintained as the design evolved.

At CDR, the MTOW target was not achieved due to the ambitious empty weight fraction requirement. Nevertheless, the proportional weight budgets were partially satisfied, despite neither estimation method perfectly representing the UAV's final weight distribution. The budget derived from the TBD dataset showed the closest overall agreement with the resulting distribution, while the first method more accurately predicted propulsion and avionics masses.

Sub system	Item	Budget 1 (kg) (130 kg MTOW)	Budget 1 (kg) (Actual MTOW)	Budget 2 (kg) (83 kg MEW)	Budget 2 (kg) (Actual MEW)	PDR Estimate (kg)	Weight (kg)	Notes
Payload	Payload	44.20	47.88	47	47	47	47.00	Taken from requirements table
Structures	Full Aircraft structures	32.50	35.21	48.97	55.36	30	42	Budget 1 & 2 taken from Gundlach's Designing Unmanned Aircraft Systems and Sadraey's Unmanned Aircraft Design and budget 3 & 4 made useing TBD's 2026 drone CDR weight breakdown
	Fuselage Structure	10.40	11.27	18.96	21.43		14.00	Estimate made useing TBD's 2026 drone CDR weight breakdown
	Wings Attachment Structure	3.90	4.22	7.71	8.71		5.23	Estimate made useing TBD's 2026 drone CDR weight breakdown
	Wings Structure	7.80	8.45	15.41	17.42		10.47	Estimate made useing TBD's 2026 drone CDR weight breakdown
	Tailplne Structure	3.90	4.22	1.87	2.11		5.23	Estimate made useing TBD's 2026 drone CDR weight breakdown
	VTOL Booms	1.30	1.41	1.63	1.84		1.70	Estimate made useing TBD's 2026 drone CDR weight breakdown
	Tail Boom	1.30	1.41	0.81	0.92		1.31	Estimate made useing TBD's 2026 drone CDR weight breakdown
	Landing Gear	3.90	4.22	2.58	2.92		3.83	Estimate made useing TBD's 2026 drone CDR weight breakdown
Propulsion	Propulsion system	13.00	14.08	11.92	13.47	9	16.77	Budget 1 & 2 taken from Gundlach's Designing Unmanned Aircraft Systems and Sadraey's Unmanned Aircraft Design and budget 3 & 4 made useing TBD's 2026 drone CDR weight breakdown
Payload	Payload system	6.50	7.04	4.88	5.52	1	8.65	Budget 1 & 2 taken from Gundlach's Designing Unmanned Aircraft Systems and Sadraey's Unmanned Aircraft Design and budget 3 & 4 made useing TBD's 2026 drone CDR weight breakdown
Power / General Electronics	Power system	32.50	35.21	11.34	12.82	19	25.78	Budget 1 & 2 taken from Gundlach's Designing Unmanned Aircraft Systems and Sadraey's Unmanned Aircraft Design and budget 3 & 4 made useing TBD's 2026 drone CDR weight breakdown
Navigation	Navigation & Sensors system	1.30	1.41	5.89	6.66	4	0.86	Budget 1 & 2 taken from Gundlach's Designing Unmanned Aircraft Systems and Sadraey's Unmanned Aircraft Design and budget 3 & 4 made useing TBD's 2026 drone CDR weight breakdown
Total MTOW:		**130.00**	**140.83**	**130.00**	**140.83**	**110.00**	**140.83**	**Assumed Maximum Take-off Wieight (MTOW) and Manufacturer's Empty Weight (MEW) for calculations**
Empty Weight:		**85.80**	**92.95**	**83.00**	**93.83**	**63.00**	**93.83**	
Buffer:		**0.00**	**-10.83**	**0.00**	**-10.83**	**20.00**	**-10.83**	**130**
Regulation Buffer (MTOW 150kg) :		**20.00**	**9.17**	**20.00**	**9.17**	**40.00**	**9.17**	**83**

Figure 45: Weight Budgets Table

Sub system	Item	Budget 1 % of MTOW	Budget 3 % of MEW	Weight % of MTOW
Payload	Payload	34.00	N/A	33.37
Structures	Full Aircraft structures	25.00	59.00	66.63
	Fuselage Structure	8.00	22.84	9.94
	Wings Attachment Structure	3.00	9.28	3.72
	Wings Structure	6.00	18.57	7.43
	Tailplne Structure	3.00	2.25	3.72
	VTOL Booms	1.00	1.96	1.21
	Tail Boom	1.00	0.98	0.93
	Landing Gear	3.00	3.11	2.72
Propulsion	Propulsion system	10.00	14.36	11.91
Payload	Payload system	5.00	5.88	6.14
Power / General Electronics	Power system	25.00	13.67	18.31
Navigation	Navigation & Sensors system	1.00	7.09	0.61

Figure 46: Proportional Weight Budgets Table

A PPA (**Figures 47** & **48**) was conducted using data from Janes aircraft databases to compare the UAV's empty weight fraction, MTOW, and wingspan against similar aircraft[18]. The results indicated that the UAV's size and overall weight characteristics were consistent with aircraft in the same class. Although the empty weight fraction was comparatively low, this aligned with the UAV's mission profile, which prioritised payload capacity over long-range endurance, unlike many comparable aircraft optimised for extended missions.

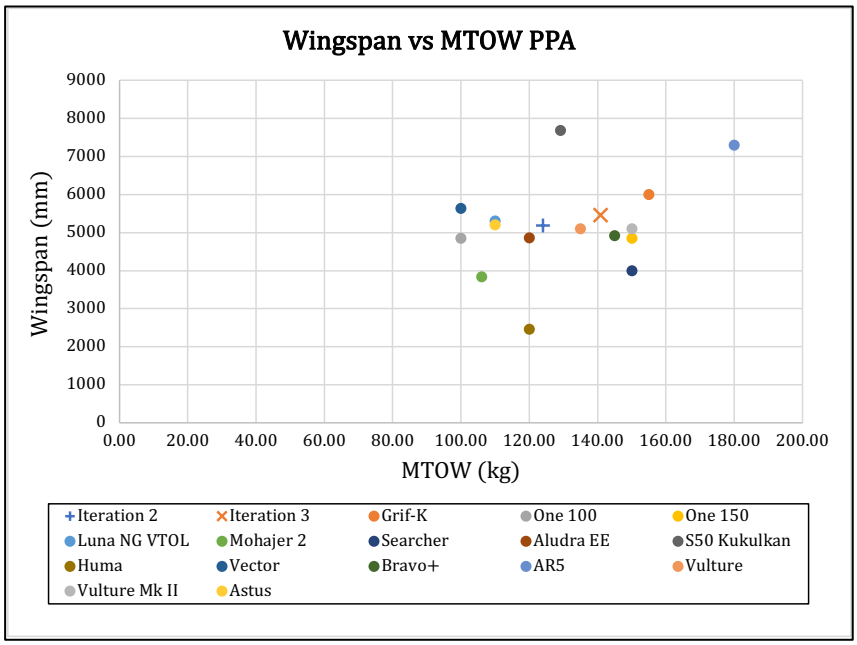


Figure 47: Empty Weight Fraction vs MTOW Product Position Analysis
(Both Iterations 2 and 3 are shown)

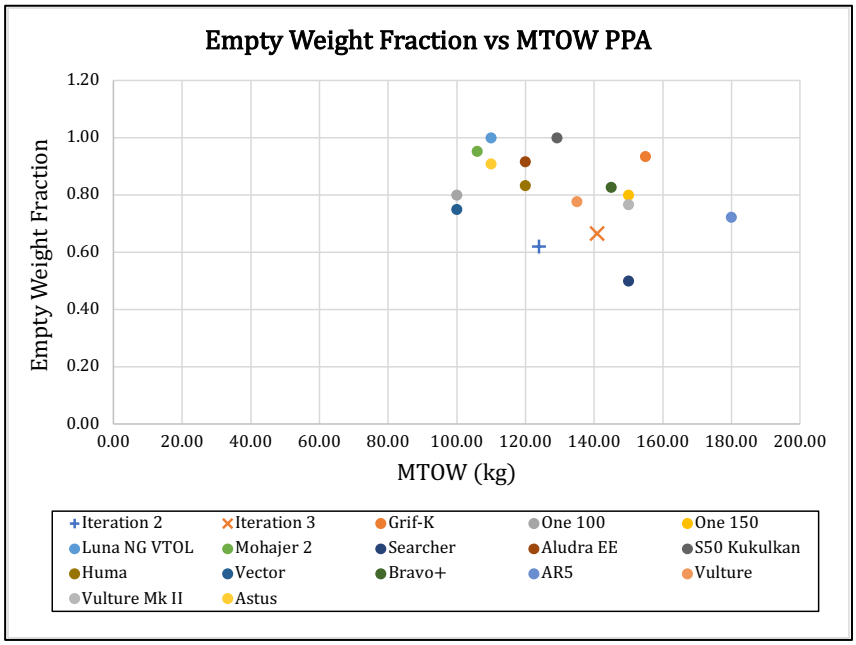


Figure 48: Wingspan vs MTOW Product Position Analysis
(Both Iterations 2 and 3 are shown)

5.7 Iterations 3

Iteration 3 was the final design iteration presented in this project and represents a configuration suitable for CDR-level maturity. The design evolved from Iteration 2 through changes to the configuration and functionality of several sub-systems. By this stage, all major sub-systems had been fully defined, enabling calculation of the UAV's final specifications and completion of the technical accounts using data from the final configuration (**Figure 49**, Appendix **M**).

A high level of requirement compliance was maintained throughout this iteration; however, the previously identified non-compliant attributes remained unresolved.

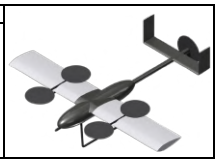
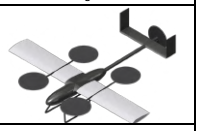
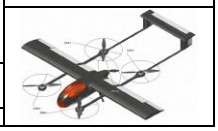
Category	Attribute	Criteria	Unit	Iteration 1	Verification Method	Iteration 2	Verification Method	Iteration 3	Verification Method	
Operation	Operational Productivity	21 (ideal case) & 18 (normal)	ha / h			55.728	Calculation	26 (ideal case) & 19-15 (normal)	Calculation and Simulation	
	Max operational radius	4,000	m	6,000	Calculation	4,000.5	Calculation	4547.25	Calculation	
	Flight time	30	m	30	Calculation	15	Battery Spec	15	Battery Spec	
	Payload Capacity	47	kg	47	Calculation	47	Payload Spec	47	Payload Spec	
	Operational Ceiling	1,500	m					2,000	Calculation	
	Communication range	5 & Compatible with 4G and satellite	km		COTS Component Integration		COTS Component Integration	Yes	COTS Component Integration	
Spraying Mech	Fluid flowrate	9	L/min	19.872	Calculation	18.48	Calculation	16.72	Calculation	Top Image: Iteration 1 Bottom Image: Iteration 2
Physical	MTOW	130	kg	110	Calculation and Estimation (previous target was 100 kg)	124	Calculation and Estimation	140.8329	Calculation	
	Max MTOW	150	kg	110	Estimation	124		140.8329		
	Empty weight	83	kg	63	Calculation	77	Calculation	93.8329	Calculation	
	Operational dimentions	L: 3,500 W: 3,000	mm	4500 x 4800	CAD model	4739 x 5180	CAD model	4701 x 5460	CAD model	
	Transporational dimentions	2946 x 1784 x 1886 (L x W x H)	mm	2300 x 600 x 600	Estimation	2,700 x 1,810 x 720	CAD model Old target: 2,765 x 1,640 x 1900	2730 x 1,729 x 840	CAD model	
Durability	Operational life	1,000	cycles			500	Battery Spec	500 for the battery, more for the rest	Battery Spec	Top Image: Iteration 2 Bottom Image: Iteration 3
	IP rating	IP55	IP__					IP35 (VTOL motors), rest is inside sealed fuselage	COTS Component Spec	
Power	Down time (battery & payload change)	3	min					3 (5 worst case)	Simulation	
	Charging time	8	min					Undefineable		
Other	Regulations compliance				Regulation Checking		Regulation Checking		Regulation Checking	

Figure 49: Iteration 1, 2 and 3 Technical Accounts

An updated product system breakdown (**Figure 50**) was also developed to reflect the revised composition and organisation of the UAV's sub-systems.

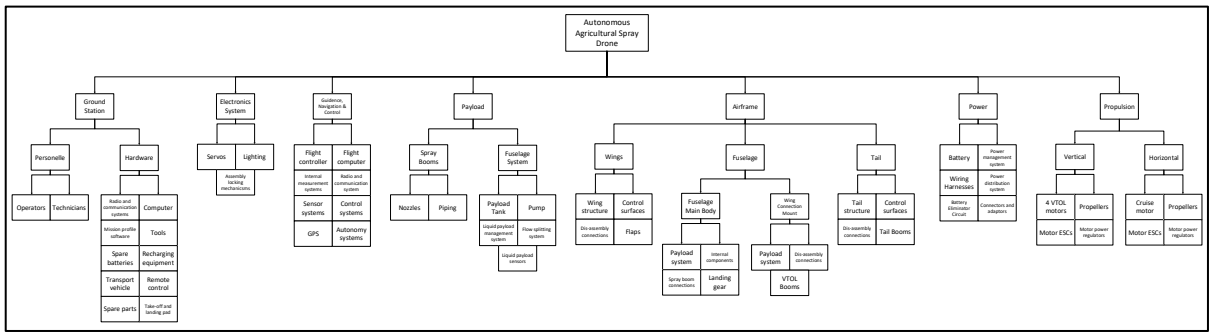


Figure 50: Revised Product System Breakdown

Figure 51 compares the UAV with similar agricultural drones, including the only fixed-wing spraying UAV currently on the market; all other competitors are multirotor platforms. Compared to quadcopters, the UAV is larger and heavier, which is expected due to its fixed-wing configuration. However, this provides advantages in range and endurance, supporting alternative mission profiles such as mapping and crop monitoring.

Operational productivity figures published by manufacturers can be optimistic; for example, the DJI Agras T50 advertises 21 ha/h, while independent studies report values closer to 17 ha/h[22]. To account for this uncertainty, optimistic, realistic, and pessimistic productivity estimates were generated for the UAV using the navigation model (refer to Navigation Report). The optimistic estimate exceeded most competing quadcopters, while the realistic estimate remained broadly competitive. These performance characteristics are largely attributable to the UAV's fixed-wing architecture.

Although the Pyka Pelican operates in a different class and cannot be directly compared, its relative capabilities align proportionally with the proposed UAV. Overall, the design achieved performance specifications competitive with aircraft in its category.

Specs	Units	VA-55	DJI T40	DJI T50	XAG P100 pro	Hylio Ares	Pyka Pelican
Type	N/A	Hybrid Fixed Wing	Quadcopter	Quadcopter	Quadcopter	Quadcopter	Fixed Wing
Dimensions	mm x mm L x W	4701 x 5460	2800 x 3150	2800 x 3085	2868 x 2927	1702 x 1702	6000 x 11500
MTOW	kg	141	90	92	96	105.6	635
Payload Capacity	kg	47	50	40	50	50	300
Payload volume	l	55	70	75	40	50	300
Unit price	$	48,000	15,000	13,000	19,000	15,000	550,000
Operational Productivity	ha / hr	26 (hi) 19 (mid) 15 (low)	21.3	21	19	28	105
Spray Width	m	4 to 11, 6 nominal	11	4 to 11	5 to 10	12	20
Max operational radius	m	4547	2000	2000	2000	Not stated	Not stated
Flight time (full payload)	min	15	7	7	6	8	35 + 10 reserve
Flight time (empty payload)	min	44	18	16.5	18	20	Not stated
Cruise speed	m/s	25.8	7	10	13.8	13.3	33.4
Fluid flowrate	L / min	16.72	12	16	22	26	Not stated
IP Rating	IP__	IP35	IP67	IP67	IPX6K	IP55	Not stated
Source	N/A	Designer spec	T40 Webpage, specs and User Manual	T50 Webpage, specs and User Manual	XAG P100 Webpage, specs and User Manual	Hylio Ares Specs	Pyka Pelican Webpage
Link	N/A	N/A	https://dl.djicdn.com/downloads/t40_t20p/20230720/T40_T20P_User_Manual_v1.4_EN.pdf	https://dl.djicdn.com/downloads/t50_t25/20250109/T50_T25_User_Manual_v1.0_EN.pdf	https://www.xa.com/en/p100pro/p100prospecs User manual: https://drive.google.com/file/d/1y3tSoa8nPHKkPNCODO4rymOEGsqnlKWy/view	https://cdn.prod.website-files.com/5d3b17e74e117222 2cd74eed/68e3f98a06e8f765 8cae3c01_ARES%20(HYL-150)%20Spec%20Sheet%20Metric%205.3.pdf	https://www.flypyka.com/pelican-2

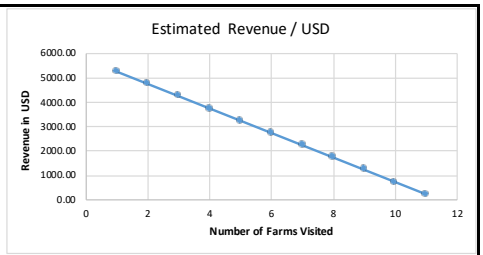
Figure 51: Spec comparison between the UAV and other drones on the market

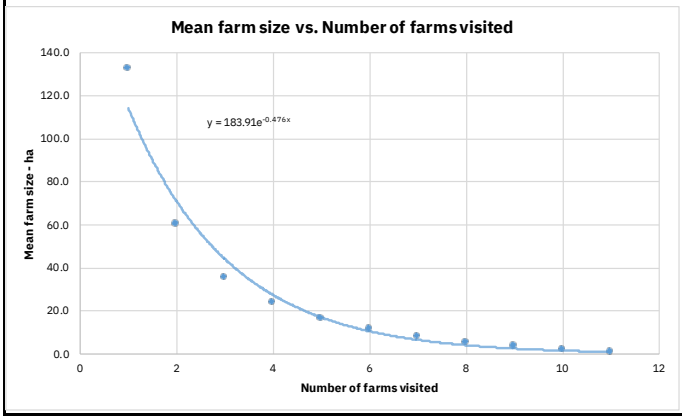
CDR feedback was positive and did not lead to any design changes, thus the planned design freeze was respected.

5.7.1 Problem Definition Change

Calculation of the final UAV specifications enabled preliminary productivity, operating costs, and revenue analysis. Using an estimated daily spraying capacity of 121.6 ha, assuming 3 minutes of downtime between missions and active spraying during 55% of flight time, a Farms Visited and Revenue model was developed (**Figure 52**). The model also accounted for transportation costs and a 30-minute transit time between farms to provide a more realistic estimate of operational revenue.

Number of farms visited per day	Spray time /h	Total hectares covered	Mean farm size / ha	Estimated Revenue / USD
1	8.67	132.8	132.8	5252.11
2	7.84	120.1	60.1	4749.31
3	7.01	107.4	35.8	4246.52
4	6.18	94.7	23.7	3743.72
5	5.35	82.0	16.4	3240.92
6	4.52	69.3	11.5	2738.12
7	3.69	56.5	8.1	2235.33
8	2.86	43.8	5.5	1732.53
9	2.03	31.1	3.5	1229.73
10	1.2	18.4	1.8	726.94
11	0.37	5.7	0.5	224.14





Effective productivity 15.3 ha / h

Figure 52: Farms Visited & Revenue model

The model showed that servicing larger hectarage farms was operationally more efficient, as proportionally less time was spent travelling and preparing the UAV between missions. Combined with additional market research, this indicated that the current UAV design was better suited to a different market segment, leading to an update of the problem definition (**Figure 53**, Appendix **F**).

While the stakeholders, stakeholder needs, and primary operating region remained unchanged, the primary target market shifted towards medium-holder farms, with smallholder farms in bulk becoming the secondary market. Further discussion of this market transition is provided in the Business Viability Report.

	Market	Who	What	Why	Where	When	Opportunity	Risks	Competition
Revised Problem Definition	Primary	Mediumholder (50 to 120 hectares) broad acre crop farms	Corn, wheat, soy-bean, rice and grains need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	These crops have height uniformity, canopy can be easily penetrated by spraying particulates, and the high volume needed for these farms make it ideal for drone-for drone application. This is also the most common crop type farmed globally and therefore will have the highest amount of customer farmers. 30 to 120 hectare farms were selected as a target as they are the ideal size range for the drone.	India, South America, Sub-Saharan Africa	Short Term	They utilise a high volume of herbicides and pesticides, application can be made more efficient with drones. Crops may get damaged with other methods of spraying (tractor and manual), therefore higher crop yields may be aquired with drone spraying. Middle market not directly addressed	1) Limited purchasing power for the farmer 2) High development cost and time 3) Competition from other drone platforms and current spraying method alternatives	> Other spraying / agriculture drone platforms (DJI Agras, XAG Agri) > Tractor or Boom sprayers
	Secondary	Smallholder broad acre crop farmers (in bulk), largeholder broad acre crop farmer	Corn, wheat, soy-bean, rice and grains need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	These crops have height uniformity, canopy can be easily penetrated by spraying particulates, and the high volume needed for these farms make it ideal for drone-for drone application. This is also the most common crop type farmed globally and therefore will have the highest amount of customer farmers. Servicing smallholders in bulk will provide a cost effective and safer alternative to labour intensive manual spraying. Given the extended range of the drone, large farms can also be serviced.	Smallholders: India, Sub-Saharan Africa Largeholders: Australia, USA, South America	Short to Medium Term	Smallholders: Agricultural UAVs are few in this market therefore competition will be limited. The introduction of crop spraying UAVs will remove a large burden to the farmer and will reduce their exposure to the dangerous chemicals, prolonging their health and life expectancy. Largeholders: Large farms require an autonomous and high-output spraying platform	1) Limited smallholder purchasing power 2) Low technological literacy for smallholders 3) Competition from other drone platforms and current spraying method alternatives 4) Saturated largeholder market	> Manual (knapsack) spraying > Small tractor spraying --- > Other spraying / agriculture drone platforms (DJI Agras, XAG Agri) > Tractor or Boom sprayers
	Tertiary	All other crop types and farm sizes	Any other crop that requires spraying	This is the primary mission profile for the drone, although it is optimised for broad acre farms, it can still be used for other crop types	Worldwide	Long Term	Most crop types require spraying, automating this process would lead to lower farmer fatigue and higher crop yield	The drone platform was not specifically designed for these crop types and therefore other competitor products tailored specifically to a given crop type will be more convenient to the farmer	> Manual (knapsack) spraying > Tractor or Boom spraying > Other spraying / agriculture drone platforms (DJI Agras, XAG Agri)
	Quaternary	Other agricultural applications (mapping and monitoring) and external applications (solar panel cleaning)	Agricultural applications: large farms require crop surveillance to determine health and growth status, livestock owners require livestock surveillance to track the movement of their animals and large land owners require land mapping to plan future plantations or building sites External applications: solar farms could be sprayed with cleaning agents to clean the solar panels. Spraying of greenhouses, payload transport, monitoring purposes, surveillance and mapping are all mission profiles the drone can accomplish	Using a drone for these tasks may increase productivity and / or efficiency. These alternative uses provide flexibility to the business and allow for expansion	Worldwide	Parallel to primary business / Long Term	UAVs are well-tailored for these tasks, and the external applications for drones are numerous. This allows for additional profit margins in low seasonal farming demands.	The drone platform was not specifically designed for these use cases and could be less efficient than alternatives. Could add more fixed costs to the company.	> Other spraying / agriculture drone platforms (DJI Agras, XAG Agri) > Current methods of performing the task

Figure 53: Updated Problem Definition

5.7.2 Configuration Change and Payload Tank Re-Positioning

The rear-mounted cruise motor configuration raised concerns during PDR due to the potential structural issues associated with vibrations, gyroscopic loading, and aeroelastic effects. Furthermore, the large moment arm relative to the COG was expected to reduce flight stability, while routing power cables through the tailplane would introduce aerodynamic and integration inefficiencies. The influence of propeller wash on spray drift was also found to be less significant than initially anticipated.

As a result, a trade study was conducted to evaluate alternative propulsion layouts. Evaluation requirements were defined and weighted according to importance, with each concept scored from 1 to 5 against each criterion. The weighted scores were then summed to identify the most suitable configuration, as summarised in **Figure 54**.

Cruise motor location	Evaluation Criteria	Weight	Nose-Mounted	Rear Pusher (current)	Dorsal Pylon	Wing-Mounted	Twin-Boom Pusher
	Spray Performance	20%	2	5	5	1	4
	Aerodynamic/Transition Efficiency	20%	5	4	3	5	3
	CG & Stability	15%	5	3	1	4	4
	Structural Simplicity	20%	5	2	2	3	5
	Repairability and Safety	5%	2	4	5	3	4
	Sensor / Nose Clearance	10%	1	5	5	4	5
	Vibration Magnitude	10%	4	3	4	2	2
	Weighted Total	100%	3.75	3.65	3.3	3.15	3.9

Figure 54: Propeller Repositioning Trade Study

The selected solution relocated the cruise motor to the rear of the fuselage, concentrating the majority of aircraft systems within the fuselage and eliminating the need for additional structural reinforcement in the tail. To accommodate the new propulsion layout, the aircraft configuration was changed to a twin-boom arrangement (Configuration Concept 4, **Figure 28**), as the original central boom conflicted with the revised motor placement.

The VTOL booms were extended to support the tail structure, which was initially redesigned into an inverted U-tail configuration. However, as tailplane span requirements exceeded the maximum allowable span imposed by the transport configuration

constraints, the design evolved into a twin-fin T-tail arrangement. The progression of these configuration changes is illustrated in **Figure 55**.

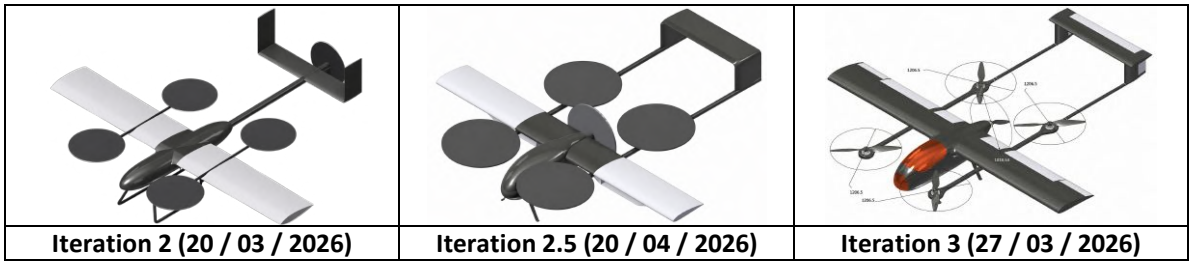


Figure 55: Overall Configuration Changes

The positioning and integration of the payload tank were also reconsidered due to the complexity introduced at the wing–fuselage interface, constraints imposed by the wing's internal structure, and stability concerns during banked manoeuvres. Relocating the tank also eliminated the need for a central fuselage reservoir, reducing overall system mass.

The payload tank was moved into the fuselage, simplifying system integration at the expense of increased fuselage size and associated aerodynamic penalties. Although a removable tank system was initially explored, a waterproof integrated payload bay was ultimately selected. This solution allowed the bay to function both as a spray reservoir and as a cargo compartment for alternative payload, further supporting the company's USP. Integrating the payload bay directly into the aircraft structure also reduced weight compared to using a standalone removable tank. The outcomes of both the cruise motor placement and payload repositioning trade studies are summarised in **Figure 56**.

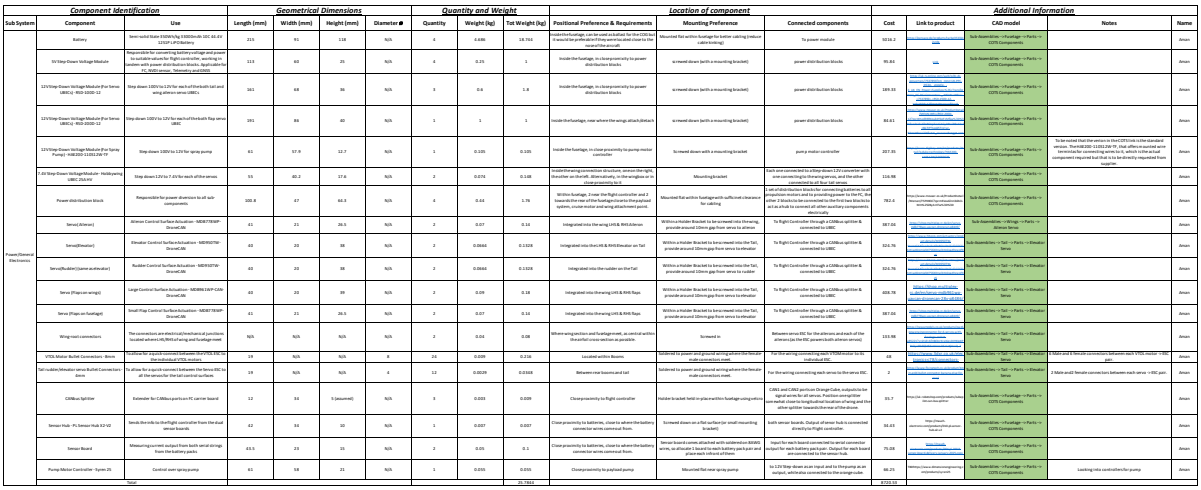
Sub-system	Component	Old Position	Why it was put there	What problems it caused (Why was it changed)	New proposed location	Why was it put there	What downsides are there	Research sources
Propulsion	Cruise Motor	At the back of the tail	> To limit propwash effects on the spray particles as the propwash was going to increase the likelyhood and extent of spray drift	> Large structural requirements for the aircraft's tail, tail boom and fuselage connection point because of the motor's weight (cantilever beam), vibrations, large gyroscopic effects (due to the propeller spinning and large moment arm inducing a large yaw) and aeroelasticity (boom flex) > The location of the motor would have pushed the COG aft (therefore counter weighting would have been needed) > Dirty air would have been ingested by the propiller (reducing efficiency) > Higher possibility for tail strike > Aerodynamic inefficiencies due to cable routing in tailplane	At the back of the fuselage, the configuration was then changed to a twin-boom tail, upside-down U tail	> Improved structural simplicity (as the motor is now mounted on the main structural element of the aircraft) > Avoids spray interference considerations (compared to other proposed cruise motor locations) > Sensor clearance for front of fuselage (compared to other proposed cruise motor locations) > Reduced vibration compared to previous configuration	> Spray might be induced into propeller flow as the motor is now located closer to the boom compared to the prevous configuration > Aerodynamically less efficient as the air is more disturbed (because of the proximity to the wings and fuselage) compared to the previous configuration > Cooling will now need serious consideration as the motor will be more integrated into the fuselage and therefore less exposed to oncoming air > Unknown vibration effects due to disturbed / dirty flow compared to tractor propeller configuration	https://airplaneacademy.co.nz/pusher-vs-puller-propeller-aircraft-compared/
Payload	Payload tank	In the wings	> To use the empty space that will be in the wings structure > To limit payload slosh > Reduce fuselage size	> Structural complexity in the wings that would lead to increased weight in the wings and wingbox > Added payload system complexity and weight > Increased interface complexity (actubing, electrical, power and mechanical connections will be needed between the wings and fuselage > Potential stability issues due to the payload moving around the wing during a turn	Inside the fuselage	> Ease of modularity: to allow the drone to be re-configured for different mission profiles as it would be easier to remove the payload tank if it is located in the fuselage > Simplicity of design: removes design complexity from the structure of the wings and the interface between the wing and fuselage > Simplified payload system design as a single tank is considered and failure modes that cause leakages are reduced > Weight and space saving, instead of having two tanks, the drone has one and therefore it removes they system in charge of making sure the same amount is taken from both tanks (central reservoir) > Easier to clean the tank (the tank needs to be cleaned from mission to mission) > Gravity feed system is simpler: less variables in the payload feed system (as the wing payload would have relied on gravity feed) leading to more consistent spraying > A thinner aerofoil can be used and there are less restrictions on wing design > Simpler system to maintain, repair and inspect > Multiple tanks may lead to inbalances during flight > Increased flexibility in COG placement as the payload tank will be able to be moved	> Space in the wing is not used (that internal volume is no longer being used for anything) > Higher chances of slosh (as bigger tanks have a higher chance for more sloshing) > Bigger fuselage needed to house the payload tank and therefore more drag, more power needed and less wing space (unless wingspan is increased) > Potential structural weak points in the fuselage (as a lot of space is taken up by the tank) > The pump will therefore need to provide all the negative pressure as gravity will no longer help feed the system	Brainstorming
Payload	Payload tank	Separate component	> To allow for the payload tank to be removable to ease washing	> Added extra weight to the fuselage > For the tank to be removable, large hatches would need to be present on the fuselage and there would need to be a lot of clearance in the fuselage structure to allow for the tank to be taken out	Integrated into the fuselage structure	> Reduced weight at the tank is now integrated into the structure > When empty, the tank can be used as a cargo bay area, allowing the drone to serve different missions	> Increased structural complexity > Decreased repairability	Brainstorming

Figure 56: Trade-off Studies Summary

5.7.3 Fuselage Planning, COG Management

A COTS component table (**Figure 57**, Appendix **P**) was developed to record the dimensions, function, mass, cost, and fuselage positioning requirements of each component. Each item was represented as a simplified cuboid and positioned within the fuselage 3D design space. Component placement was determined by positional constraints and relationships with connected systems.

Maintenance access, cooling requirements, and the need for RF-transparent material regions were also considered during integration, informing the placement of fuselage hatches and internal airflow passages.



Component Identification			Geometrical Dimensions				Quantity and Weight			Location of component			Additional Information				
Sub System	Component	Use	Length (mm)	Width (mm)	Height (mm)	Diameter ⌀	Quantity	Weight (kg)	Tot Weight (kg)	Positional Preference & Requirements	Mounting Preference	Connected components	Cost	Link to product	CAD model	Notes	Name
Power/Electronics Electronics	Battery	Semi-solid State 6100Mah,6g 6 6000mAh 10C 44.4V 12S2P LiPO Battery	215	91	118	N/A	4	6.696	18.764	Inside the fuselage, can be used as ballast for the COG but it would be preferable if they were located close to the nose of the aircraft	Mounted flat within fuselage for better cabling (reduce cable kinking)	To power module	1218.2		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
5V Step-Down Voltage Module	Responsible for converting battery voltage and power to suitable values for flight controller, sensing in modern aircraft, motor and servos, flight data file PC, RGB Sensor, Telemetry and GNSS	113	60	25	N/A	4	0.25	1	Inside the fuselage, in close proximity to power distribution blocks	screwed down (with a mounting bracket)	power distribution blocks	95.84		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
12 V Step-Down Voltage Module (For Servos UBEC) : RIOI-10030-12	Step down 100V to 12V for each of the both tail and wing aileron servos UBEC)	161	68	36	N/A	3	0.6	1.8	Inside the fuselage, in a close proximity to power distribution blocks	screwed down (with a mounting bracket)	power distribution blocks	189.33		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
12V Step-Down Voltage Module (For Servos UBEC) : RIOI-10030-12	Step down 100V to 12V for each of the both flap servos UBEC	191	86	40	N/A	1	1	1	Inside the fuselage, near where the wings attach (Attach	screwed down (with a mounting bracket)	power distribution blocks	84.61		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
12V Step-Down Voltage Module (For Spray Pump) : HAB200-1101ZW-TP	Step down 100V to 12V for spray pump	61	57.9	12.7	N/A	1	0.105	0.105	Inside the fuselage, in close proximity to pump motor controller	Screwed down with a mounting bracket	pump motor controller	207.35		Sub-Assemblies --> Fuselage --> Parts --> COTS Components	To be noted that the servos in the COTS list is the standard version. The HAB200-1101ZW-TP, that offers mounted wire terminals for connecting wires to it, which is the actual component required but that is to be directly requested from supplier.	
7-6V Step-Down Voltage Module - Hobbywing UBEC 25A HV	Step down 12V to 7.4V for each of the servos	55	40.2	17.6	N/A	2	0.074	0.148	Inside the wing connection between servos (for right, therefore on the left, 90 mm directly, or the angles are in close proximity to it	Mounting bracket	Each one connected to a Step-down 12V connector with one connecting to the wing servos, and the other connected to all four tail servos	116.98		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
Power distribution block	Responsible for power diversion to all sub components	100.8	47	66.3	N/A	4	0.44	1.76	Within fuselage, 2 near the flight controller and 2 towards the rear of the fuselage (near to the payload system, cruise motor and wing attachment point)	Mounted flat within fuselage with sufficient clearance for cabling	2 sets of distribution blocks for connecting batteries to propulsion motors and to powering power to the FC, the other 2 blocks to be connected to the first two blocks to act as a hub to connect all other auxiliary components electrically	782.4		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
Servos (Ailerons)	Aileron Control Surface Actuation - MEW270SBP-Drone Lab	41	21	26.5	N/A	2	0.07	0.14	Integrated into the wing LHS & RHS Aileron	Within a Holder Bracket to be screwed into the wing, provide around 10mm gap from servo to aileron	To Flight Controller through a CANbus splitter & connected to UBEC	667.66		Sub-Assemblies --> Wings --> Parts --> Aileron Servo		Aman
Servo(Elevator)	Elevator Control Surface Actuation - MEW270SBP-Drone Lab	40	20	38	N/A	2	0.0666	0.1326	Integrated into the LHS & RHS Elevator on Tail	Within a Holder Bracket to be screwed into the Tail, provide around 10mm gap from servo to elevator	To Flight Controller through a CANbus splitter & connected to UBEC	322.76		Sub-Assemblies --> Tail --> Parts --> Elevator Servo		Aman
Servo(Rudder) (same as elevator)	Rudder Control Surface Actuation - MEW270SBP-Drone Lab	40	20	38	N/A	2	0.0666	0.1326	Integrated into the rudder on the Tail	Within a Holder Bracket to be screwed into the Tail, provide around 10mm gap from servo to rudder	To Flight Controller through a CANbus splitter & connected to UBEC	322.76		Sub-Assemblies --> Tail --> Parts --> Elevator Servo		Aman
Servo (Flaps on wings)	Large Control Surface Actuation - MG996 SWP-Cube Drone Lab	40	20	39	N/A	2	0.09	0.18	Integrated into the wing LHS & RHS flaps	Within a Holder Bracket to be screwed into the Tail, provide around 10mm gap from servo to elevator	To Flight Controller through a CANbus splitter & connected to UBEC	403.78		Sub-Assemblies --> Tail --> Parts --> Elevator Servo		Aman
Servo (Flaps on fuselage)	Small Flap Control Surface Actuation - MEW270SBP-Drone Lab	41	21	26.5	N/A	2	0.07	0.14	Integrated into the wing LHS & RHS flaps	Within a Holder Bracket to be screwed into the Tail, provide around 10mm gap from servo to elevator	To Flight Controller through a CANbus splitter & connected to UBEC	667.66		Sub-Assemblies --> Tail --> Parts --> Elevator Servo		Aman
Wing root connectors	The connectors are electrical/mechanical junctions located where LHS/RHS wing and fuselage meet	N/A	N/A	N/A	N/A	2	0.04	0.08	Where wing section and fuselage meet, as central within the airfoil cross-section as possible.	Screwed in	Between servos ESC for the ailerons and each of the ailerons (as the ESC powers both ailerons servos)	113.98		Sub-Assemblies --> Tail --> Parts --> Elevator Servo		Aman
XT60L Motor Bullet Connectors - 8mm	To allow for a quick connect between the XT60L ESC to the individual XT60 motors	19	N/A	N/A	8	24	0.009	0.216	Located within Booms	Soldered to power and ground wiring where the female male connectors meet.	For the wiring connecting each XT60M motor to its individual ESC	68		Sub-Assemblies --> Tail --> Parts --> Elevator Servo	3 Male and 3 female connectors between each XT60L motor --> ESC pair.	Aman
Tail-Audible/Emotion servo Bullet Connectors - 4mm	To allow for a quick connect between the Servo ESC to all the servos for the tail control surfaces	19	N/A	N/A	4	12	0.0039	0.0468	Between rear booms and tail	Soldered to power and ground wiring where the female male connectors meet.	For the wiring connecting each servo to the servo ESC	2		Sub-Assemblies --> Tail --> Parts --> Elevator Servo	2 Male and 2 female connectors between each servo --> ESC pair.	Aman
CANbus Splitter	Extender for CANbus ports on FC carrier board	12	34	5 (assumed)	N/A	3	0.033	0.099	Close proximity to flight controller	Holder bracket held in place within fuselage using velcro	CAN1 and CAN2 pin from Orange Cube, outputs to for signal wires for all servos. Position one right beneath/at close to long bundled location arriving and the other splitter towards the rear of the drone	31.7		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
Sensor Hub - PL Sensor Hub V2-V2	Sends the info to the flight controller from the dual sensor boards	42	34	10	N/A	1	0.027	0.027	Close proximity to batteries, close to where the battery connector wires come out from.	Screwed down on a flat surface (or small mounting bracket)	both sensor boards. Output of sensor hub is connected directly to Flight controller.	34.83		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
Sensor Board	Measuring current output from both servo strings from the battery packs	69.5	20	15	N/A	2	0.05	0.1	Close proximity to batteries, close to where the battery connector wires come out from.	Sensor board comes attached with soldered on BMMS wires, so allocate 1 board to each battery pack pair and place each inboard of them	Input for each board connected to servo connector output for each battery pack pair. Output for each board are connected to the sensor hub.	71.98		Sub-Assemblies --> Fuselage --> Parts --> COTS Components		Aman
Pump Motor Controller - tyrom 25	Control over spray pump	61	58	21	N/A	1	0.055	0.055	Close proximity to by payload pump	Mounted flat near spray pump	5v 12V Step-down as an input and to the pump as an output, with the pump connected to the orange cube	66.25		Sub-Assemblies --> Fuselage --> Parts --> COTS Components	Looking into controllers for pump	Aman
	Total								21.7843			8703.53				

Figure 57: Excerpt from COTS component list

A COG management sheet (**Figure 58**, Appendix **Q**) was developed to support mass distribution and balance calculations. Initially, Stability and Control worked with an assumed COG, referenced from the fuselage nose datum. However, as the final fuselage geometry had not yet been defined, the datum was relocated to the wing leading edge and the assumed COG position recalculated accordingly.

Once the target COG location had been established, the COTS components and payload tank were repositioned along the longitudinal axis to achieve a COG close to the target while maintaining a practical fuselage length. Early calculations considered only components heavier than 1 kg, as smaller items were unlikely to produce significant moments. A gradient format was applied to the moment column to highlight components with the greatest influence on COG location. The battery system was positioned forward to counterbalance the tail assembly, while the payload tank was placed near the wing leading edge to minimise payload-induced moment variation.

After an acceptable COG estimate was achieved, the remaining components were included and stability calculations were repeated for both loaded and unloaded payload conditions. The fuselage geometry was then designed around the internal layout, with lofting performed using the method described in Raymer's Aircraft Design textbook (refer to the Structures Design Report)[14]. Structural COG and mass estimates were incorporated into the COG calculations as design matured.

A final COG calculation was completed once the design was frozen, one day prior to CDR. The resulting COG differed only slightly from previous estimates, requiring minor adjustments to the tailplane and fin sizing to maintain stability. As these changes had negligible influence on overall balance, the configuration was finalised for CDR.

Component Identification			***Datum: Wing Leading Edge***		
Sub-system	**Item**	**Weight (kg)**	**Distance from DATUM (mm)**	**h**	**Moment**
Structures	Fuselage structure	14.00	-154.609	-0.184	-2.577
	Wing attatchment points and VTOL booms structure	5.23	210.000	0.250	1.308
	Wing structure	10.47	210.000	0.250	2.617
	Tailplane Structure	5.23	3209.958	3.821	19.999
	VTOL booms	1.70	359.338	0.428	0.728
	Horizontal tail booms	1.31	2552.062	3.038	3.975
41.7715	Landing gear	3.83	15.570	0.019	0.071
	Sum of Weights	**140.83**	**Sum of Moments**		**5.527**
	Centre of Gravity	**0.039**	**Centre of Gravity (mm)**	**32.97**	
	Centre of Gravity (empty payload)	**0.151**	**Centre of Gravity (no payload) (mm)**	**126.92**	
	Centre of Gravity (ground condition: no battery & payload)	**0.431**	**Centre of Gravity (ground condition: no battery & payload) (mm)**	**361.94**	
	h0	**0.25**	**h0 (mm)**	**210.00**	

Figure 58: Excerpt from COG Management Table

A COG Excursion diagram (**Figure 59**) was developed to illustrate how the COG shifts throughout the mission profile relative to the forward and aft limits, confirming that it remains within acceptable bounds under all operating conditions.

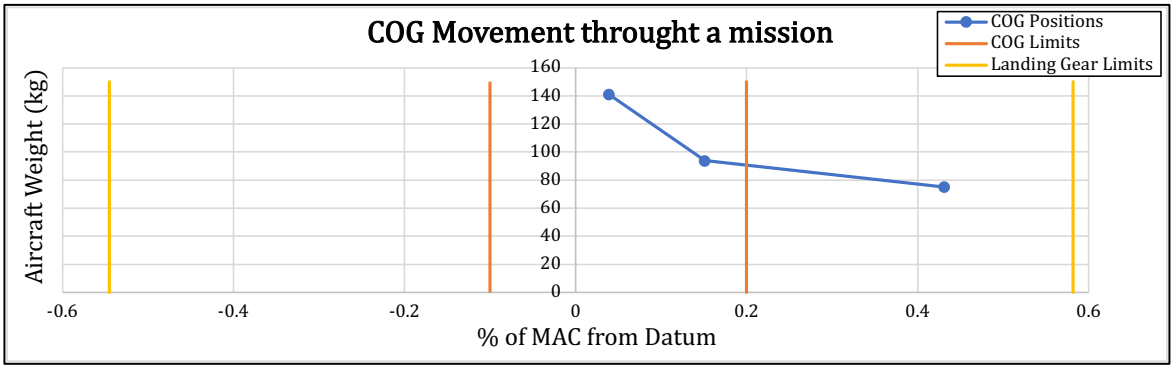


Figure 59: COG Excursion diagram
(forward and aft COG limits were determined by Stability and Control)

5.7.4 Interface Planning

As disassembly is a key requirement, interface design was considered from the outset, as reflected by the dedicated column in the morphological chart (**Figure 28**). Once the sub-system designs reached sufficient maturity, an interface map (**Figure 60**) was developed.

The fuselage was divided into the Main Fuselage Structure (MFS) and the Wing Connection Structure (WCS), with the double-headed arrow indicating full structural integration between these two sub-assemblies. The MFS contains all internal components, including the payload system and landing gear, while all other major aircraft elements interface through the WCS.

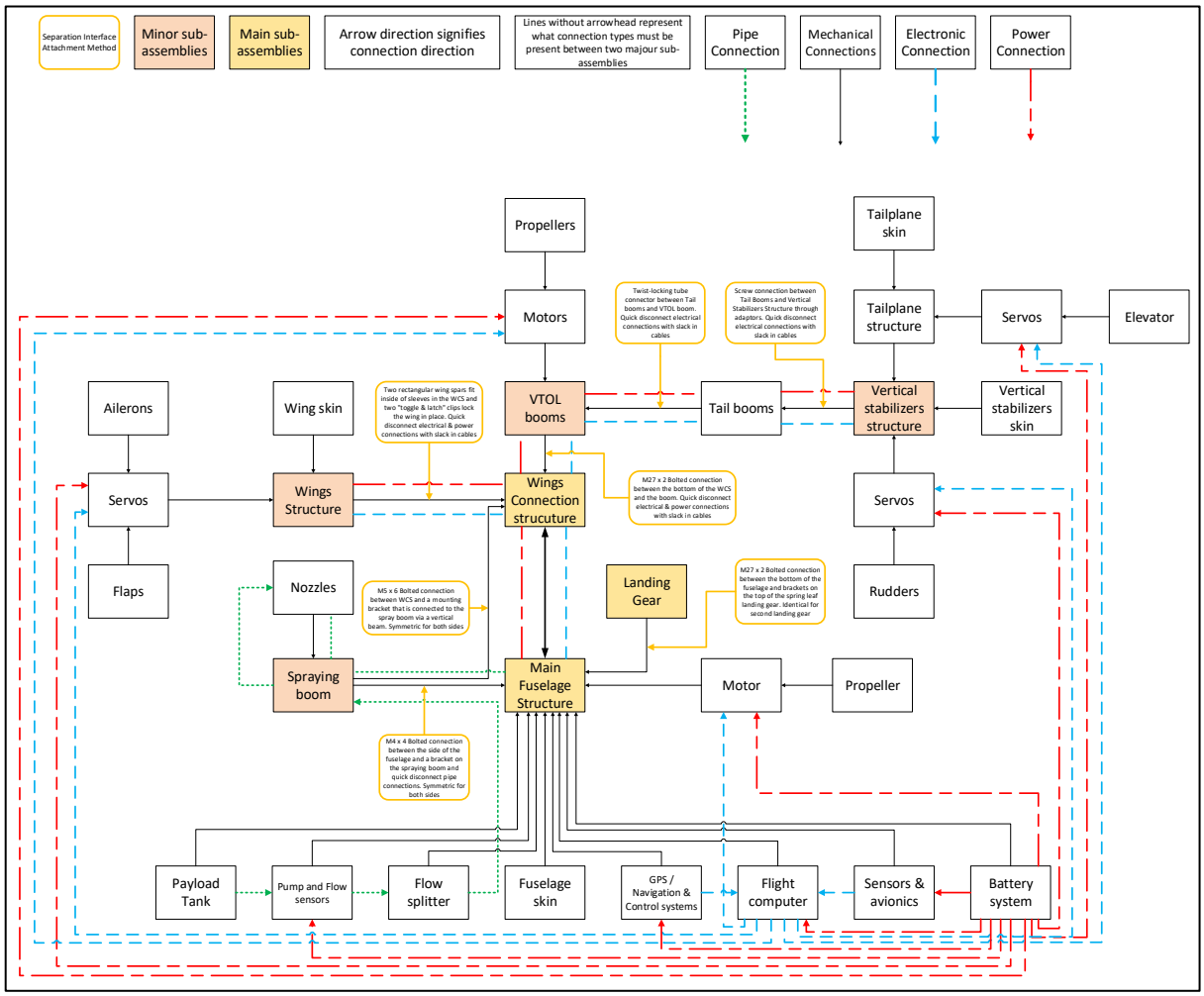


Figure 60: Interface map

Requirement IN5 from the Integrated Set of Needs specified that physical interfaces between sub-systems should be toolless to minimise assembly and disassembly time during transport. This requirement was only partially met, with four of seven interfaces requiring mechanical fasteners.

The spray boom uses fasteners to connect to the fuselage due to spatial constraints that prevented the use of a toolless interface; however, a quick-connect fluid coupling was incorporated within the boom itself. The VTOL booms and landing gear also use fasteners, as they provide higher structural security, a compact integration profile, and simpler interfacing with the MFS and WCS compared to toolless alternatives.

In contrast, the wings and tail were designed as toolless interfaces. The wings used a sliding spar system into the WCS to form a cantilever load path, secured with a locking clip to prevent axial movement. This concept was inspired by the wing interface of the MH620 Heavy-lift VTOL drone[23]. The tail booms attach to the tailplane via threaded adaptors, while the tail assembly uses proprietary twist-lock tube connectors to join the tail and VTOL booms. This arrangement improves aerodynamic cleanliness compared to the MH620-inspired baseline while still allowing internal cable routing and maintaining a toolless operation.

Quick-connect electrical plugs were used across all power and signal connections, with hollow structural members providing internal routing space for wiring. All interface

concepts were reviewed with the structures team to confirm load path integrity before being consolidated into the final interface diagram.

5.7.5 Tail Interface and DFMEA

The interface between the tail sub-assembly and the VTOL booms is the most complex in the system, involving multiple elements to achieve a toolless connection while remaining compatible with COTS CFRP tubular sections. Several alternative concepts, including clips, press-fits, and bolted joints, were evaluated; however, the final design was selected for its operational simplicity, compliance with requirement IN5, and improved aerodynamic efficiency. **Figure 61** (**Appendix R**) present an exploded view, parts list, and connection architecture for this interface.

Since threads cannot be reliably formed in CFRP and press-fit solutions risk loosening over time, titanium adaptor components were introduced to bridge the two sub-assemblies. Titanium was selected due to its high specific strength and compatibility with CFRP, avoiding the galvanic corrosion issues associated with aluminium and carbon composites interfaces[24]. The titanium adaptors are bonded to the CFRP tubes using a structural epoxy adhesive, providing a durable and lightweight joint solution.

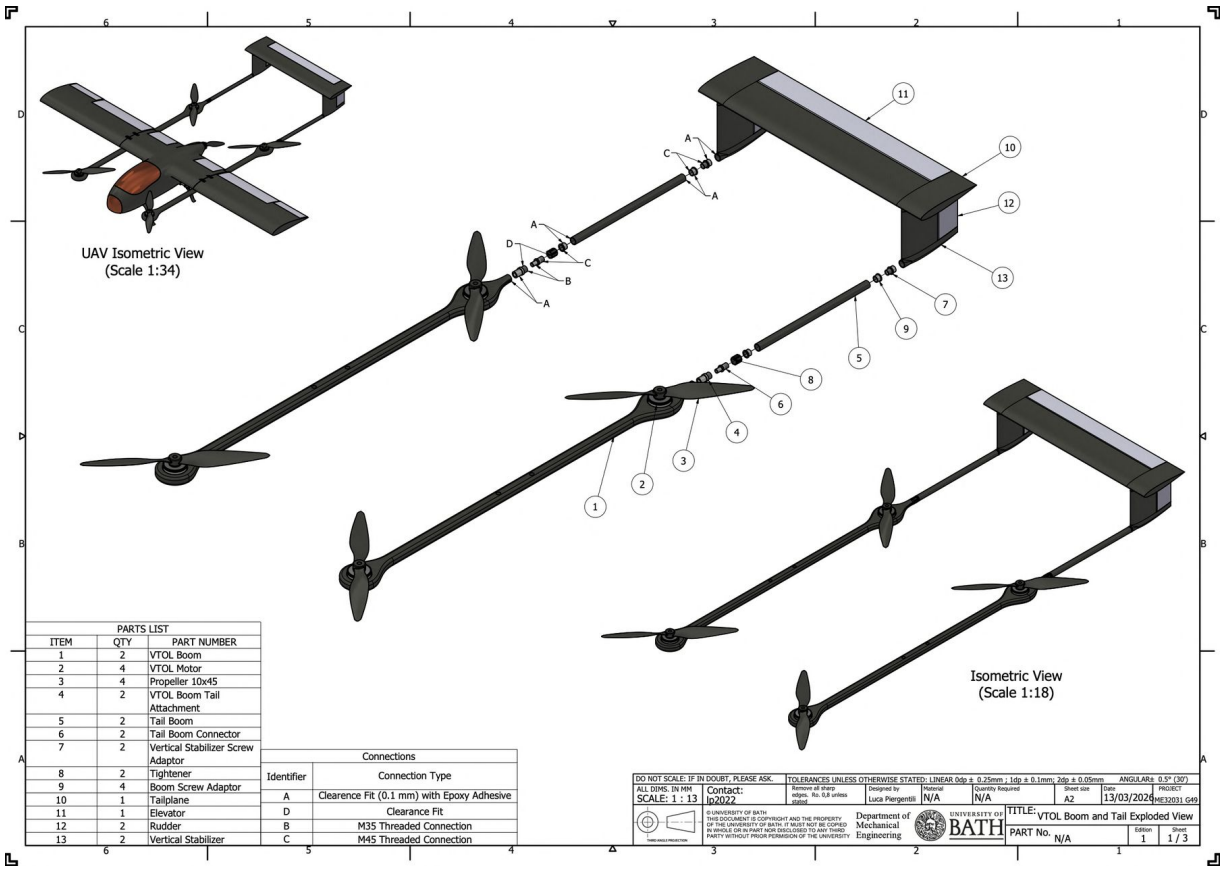


Figure 61: Tail and VTOL Booms interface exploded view

A technical drawing and detailed assembly procedure (**Figures 62** and **63**, **Appendix R**) were produced to document the interface design and its operation. A 0.2 mm diameter clearance between the titanium adaptors and CFRP tubes was introduced to provide sufficient bond-line thickness for the epoxy adhesive, ensuring adequate surface area for a reliable structural joint.

The threaded holes were designed with additional depth beyond the thread engagement length to ensure full thread contact of the mating components, improving connection security and load transfer reliability.

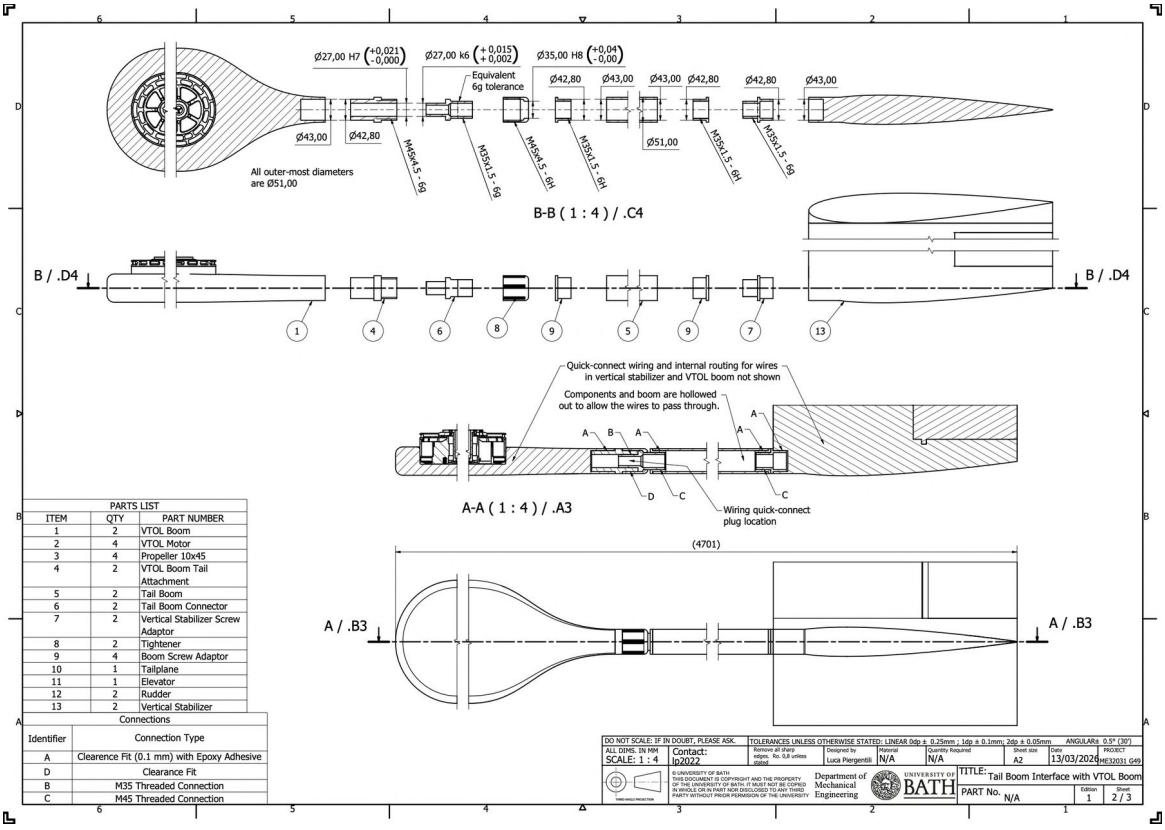


Figure 62: Tail and VTOL Booms interface technical drawing

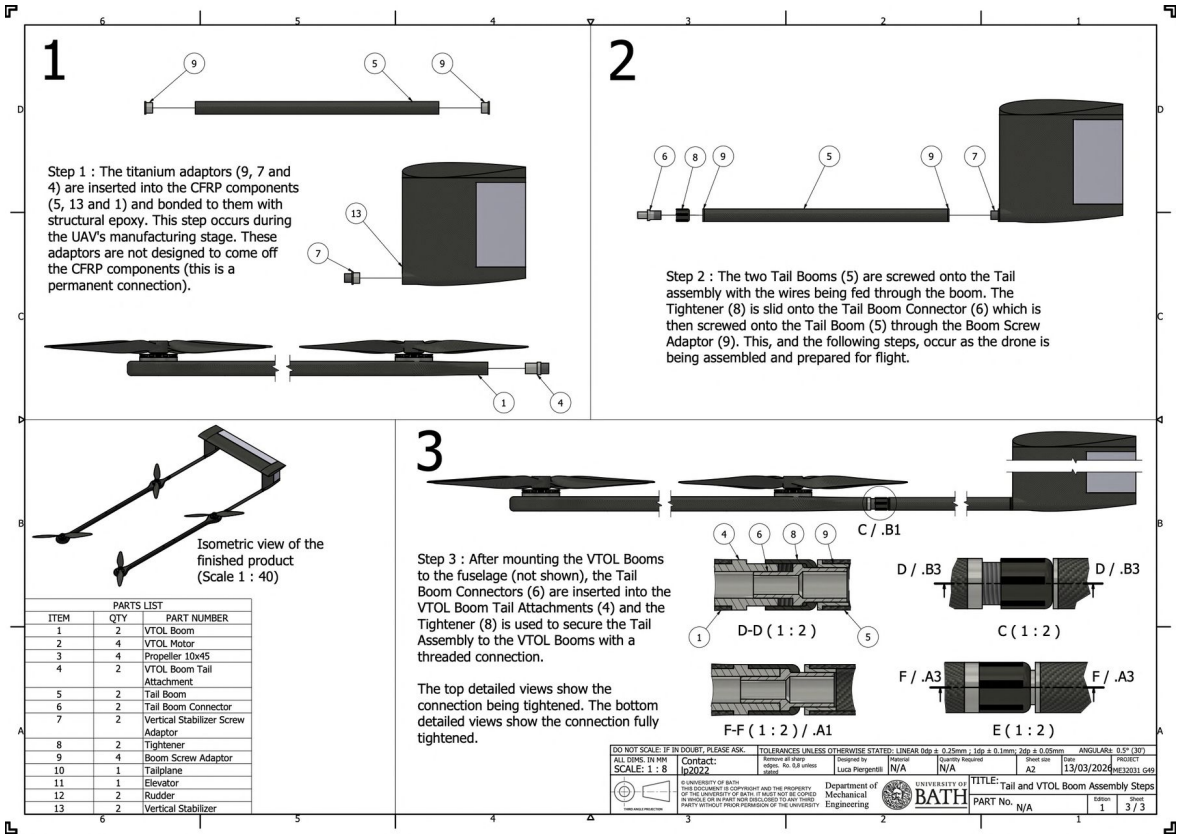


Figure 63: Tail and VTOL Booms interface assembly steps

A DFMEA (**Figure 64**, **Appendix N**), detailing potential failure modes for each component, was developed for this interface. Its construction is outlined in **Appendix N**. The analysis was completed after the first design iteration of the interface to identify critical failure points. Identified failure modes were assessed and, where possible, mitigations were incorporated into the second design iteration.

The most critical remaining failure mode is CFRP cracking, as carbon fibre composites exhibit limited elastic deformation before sudden fracture. As a result, the adaptor–CFRP interfaces remain vulnerable to impact loading even with applied safety factors. To mitigate this risk, a high-clearance landing gear was designed, with appropriate placement defined relative to all COG conditions (**Figure 59**), to minimise the likelihood of tail strikes during landing and ground operations.

VTOL Boom & Tail Interface Design Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Mitigation through Analysis and Design	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Testing	Resp.	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN = S X O X D	IMPLEMENTATION	EFFECTIVENESS	MRPN Mitigated RPN (MRPN) = S X O X E / 10)
When could the failure mode occur?	What sub-system is the failure mode part of	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		How causes the step, change or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate the failure mode?			What test can be performed to mitigate this failure mode?	Who is responsible for making sure the actions are completed?							
During flight or during manoeuvres & gusts	VTOL Boom, Tail Boom and Tail	Epoxy adhesive failing	Structural failure in the Tail (the Tail disassembles)	9	Incorrect application of epoxy during manufacturing, incorrect adhesive used, insufficient surface area for epoxy to contact, product used after its rated service life	3	QC checks, manufacturing instructions, adhesive check, pre-flight checks	2	54	Gap added between interfaces Correct epoxy spec-ed	10	9	Full structures test	Design, manufacturing and operations teams	9	3	2	54	10	9	49
During flight or during manoeuvres & gusts	VTOL Boom	VTOL Boom (1) cracking or failing at interface with VTOL Boom Tail Attachment (4)	Structural failure in the Tail (the Tail detaches from the rest of the aircraft)	8	Incorrect manufacturing (of CFRP component and / or adaptors), prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor Deep insertion length in CFRP component	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78
During flight or during manoeuvres & gusts	VTOL Boom and Tail Boom	Threaded connection between Tightener (8) and VTOL Boom Tail Attachment (4) failing	Tail is no loger secured to the VTOL boom with a threaded connection and may fall out	5	Fatigue, prior damage, rough handling, poor manufacturing	2	Pre-flight checks, QC checks, inspection schedule & logs	4	40	Over-spec-ed threads and interference fit (H7 - k6) between the VTOL Boom Tail Attachment and the Tail Boom Connector	6	7	Full structures test and full assembly test	Design, manufacturing and operations teams	5	2	4	40	6	7	28
During flight or during manoeuvres & gusts	VTOL Boom and Tail Boom	Tail Boom Connector (6) sliding out of the VTOL Boom Tail Attachment (4)	The Tail detaches from the rest of the aircraft	7	Incorrect tolerances between two interfaces, threaded connection between Tightener and VTOL Boom Tail Attachment failing	2	Pre-flight checks, QC checks	2	28	H7 - k6 tolerances between the two components	6	9	Full structures test and full assembly test	Design, manufacturing and operations teams	7	2	2	28	6	9	25
During drone assembly	Tail Boom	Tightener (8) gets jammed and no longer slides over the Tail Boom Connector (6)	The Tail cannot be properly secured onto the VTOL Booms	2	Incorrect tolerances between two interfaces	3	Pre-flight checks, QC checks, inspection schedule & logs	1	6	H8 tolerance on Tightener's inner hole. Lubrification	8	9	Full assembly test	Design, manufacturing and operations teams	2	3	1	6	8	9	5
During flight or during manoeuvres & gusts	Tail Boom	Threaded connection between Tail Boom Connector (6) and Boom Screw Adaptor (9) failing	The Tail detaches from the rest of the aircraft	7	Fatigue, prior damage, rough handling, poor manufacturing	2	Pre-flight checks, QC checks, inspection schedule & logs	4	56	Over-spec-ed threads	7	7	Full structures test and full assembly test	Design, manufacturing and operations teams	7	2	4	56	7	7	39
During flight or during manoeuvres & gusts	Tail Boom	Tail Boom (5) cracking or failing at interface with Boom Screw Adaptors (9)	Structural failure in the Tail (the Tail disassembles or detaches from the rest of the aircraft)	8	Incorrect manufacturing (of CFRP component and / or adaptors), prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor Deep insertion length in CFRP component	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78

Figure 64: Excerpt from the Tail and VTOL Booms Interface DFMEA
(the same scoring system as the FMEA is used here)

5.7.6 PFMEA

As UAV assembly is a critical step in system operation, a PFMEA (**Figure 65**, **Appendix N**) was developed to identify potential failures in the assembly processes. Its construction is outlined in **Appendix N**. Most identified failure modes were attributed to operator or technician error, primarily arising from carelessness or insufficient training. The main mitigation for these risks is improved training procedures and a stringent recruitment process.

The engineering team would be directly involved in the training of the first operators, as they possess the required technical knowledge, after which responsibility would transition to senior operators and technicians. A detailed training programme is beyond the scope of this project.

Most design-based mitigations focused on improving assembly simplicity; however, due to project time constraints, these design-for-assembly improvements were not fully implemented but were identified as priorities for further detailed design.

Drone Assembly Process Failure Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Process Mitigation	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Analysis and Design	Resp.	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN = S * O * D	IMPLEMENTATION	EFFECTIVENESS	MRPN (Mitigated RPN) = S x (O x E / 10)
When could the failure mode occur?	What sub-system is the failure mode part of	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate this failure mode?			What design feature can be implemented to mitigate this failure mode?	Who is responsible for making sure the actions are completed?							
During drone assembly	Entire drone assembly	Incorrect handling of drone components	Broken components, higher likelyhood of structural failure modes occurring	8	Operator & Technician carelessness, Poor training	4	Component handling instructions and procedures, Technician training, strict hiring processes	4	128	Technician training	10	7	Higher structural redundancy and component sturdiness to avoid damage	Design team, Technicians and Operators	8	4	4	128	10	7	90
During drone assembly	Entire drone assembly	Over-tightened screws	Cracks in CFRP components, damage to screw or mounting	7	Incorrect tightening of screws	5	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	6	210	Specific screw torque value set for every screw	8	7	N/A	Design team, Technicians and Operators	7	5	6	210	8	7	147
During drone assembly	Entire drone assembly	Under-tightened screws	Higher likelyhood of components detaching	8	Incorrect tightening of screws	3	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	4	96	Specific screw torque value set for every screw	8	7	N/A	Design team, Technicians and Operators	8	3	4	96	8	7	67
During drone assembly	Entire drone assembly	Not all screws are installed	Higher likelyhood of components detaching	9	Operator & Technician carelessness, Poor training	3	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	27	Detailed assembly instructions	10	7	N/A	Design team, Technicians and Operators	9	3	1	27	10	7	19
During drone assembly	Entire drone assembly	Un-mounted components (not all components are mounted)	Drone cannot fly	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	N/A	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Entire drone assembly	Components mounted in the wrong location	Drone cannot fly	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	Specific interfaces for each component	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Entire drone assembly	Components mounted in the wrong orientation	Drone cannot fly or inability to carry out the mission	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	Specific interfaces for each component	Design team, Technicians and Operators	10	1	1	10	10	7	7

Figure 65: Excerpt from the Assembly Process PFMEA
(the same scoring system as the FMEA is used here)

5.7.7 Requirements Compliance

The final iteration's technical accounts (**Figure 49**) summarise system compliance with the numerical requirements, while a compliance matrix (**Figure 66**, **Appendix S**) was produced to verify adherence to the Integrated Set of Needs. For each requirement, a verification method, expected outcome, and rationale were defined for both the design and testing phases to provide a complete verification framework.

Design-phase verification methods primarily consisted of analyses already performed within this project; however, six verification methods were identified as future work or outside the project scope. Ensuring accuracy in these predictive methods is critical to avoid discrepancies between expected and real-world behaviour that could otherwise necessitate redesign after testing, thereby increasing development costs. Testing-phase verification predominantly relies on full sub-system or system-level trials, as these provide the most representative real-world performance data.

Requirement Identification			***Complience and Verification***							
Category	**Requirements**	**ID**	**Design Phase Verification Technique**	**Expected outcome**	**Rationale**	**Development / Test Phase Verification Technique**	**Expected outcome**	**Rationale**	**Compliance Status**	**Rationale for Non-Complience**
Operation	The drone's spraying must cover the entire area that has been marked for spraying	ON1	Spraying and Navigation model	Theoretical mission profile and operational productivity	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Partial Compliance	Irregular fields were not considered during this design analysis, future work will involve consideration of irregular and curved field shapes
	The drone must be capable of autonomous flight	ON2	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be able to perform its spraying function autonomously	ON3	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of VTOL	ON4	Propulsion analysis and calculation	Theoretical drone performace metrics	Only availible verification technique at this stage	Propulsion test and Full operations test	Real-life drone performance capabilities	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of flying under rain or perturbed (non sunny and with wind) weather conditions	ON5	Structural analysis and calculation	Theoretical structures capabilities	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Full waterproofing analysis was not completed, future work would involve waterproofing design for the fuselage and o-ring design for relevant interphaes
	The drone must be able to withstand the structural loads outlined in the flight envelope specific to its category	ON6	Structural analysis and calculation	Theoretical structures capabilities	Only availible verification technique at this stage	Structural testing	Real-life drone structures behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The payload system must be accessible in order for leak checks to be performed	ON7	Payload design	Compliance or non-compliance depending on payload design outcomes and decisions	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Full waterproofing and leak analysis was not completed, future work would involve waterproofing design for the fuselage, water-tight payload connections and o-ring design for relevant interphaes
	If signal / communication is lost between the drone and the operator, the drone must automatically reposition itself to regain the connection or have a RTH (return-to-home) function	ON8	COTS components spec and Navigation model	System compliance or non-compliance depending on the COTS component specs and built-in function in navigation code	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Beyond GDBP scope	N/A
	The drone operation must have a higher operational productivity than 0.4 to 0.7 hectares per day --> 0.04 to 0.07 hectares / hour	ON9	Spraying and Navigation model and theoretical operational productivity calculations	Theoretical mission profile and operational productivity	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must carry enough payload to spray 2 hectares in one mission	ON10	Structural analysis and calculation	Theoretical structures capabilities	Only availible verification technique at this stage	Payload tank design and Full operations test	Real-life payload system behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of spraying the field or area outlined in the mission plan autonomously	ON11	Spraying and Navigation model with emphasis on spray drift	Theoretical mission profile and operational productivity	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must have accurate obstacle detection	ON12	COTS components spec and Navigation model	System compliance or non-compliance depending on the COTS component specs and built-in function in navigation code	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must follow a flight path that avoids obstacles, no-fly zones and terrain with autonomous operation	ON13	6-degree of freedom model simulation	Simulated behaviour of the drone.	Only availible verification technique at this stage	Flight navigation test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Navigation model only includes a field drawn by the user, no real life mapping, and therefore no-fly zones, were included in the model. Future work would be to import the model into a world map and include no-fly zones in the map, coding the software to avoid them
	The drone must have an operational radius of 4km	ON14	Propulsion and battery analysis and calculation	Theoretical drone perfomace metrics and range	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must have 4G, GPS and sattellite connectivity	ON15	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only availible verification technique at this stage	Off-the-shelf component search and testing	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone spraying systems must be compatible with existing spraying products	ON16	Payload design and material selection	Compliance or non-compliance depending on payload design and material selection from theoretical materials analysis	Only availible verification technique at this stage	Sprayer test and Full operations test	Real-life payload system behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone's battery system must be rechargeable on site (at the farm)	ON17	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only availible verification technique at this stage	Battery recharge test	Real-life battery behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone battery system must be rechargable using existing infrastructure	ON18	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only availible verification technique at this stage	Off-the-shelf component search and testing	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone must be capable of servicing farms located at 4921 ft (1,500 m) above sea level	ON19	Propulsion analysis and calculation	Theoretical drone performace metrics	Only availible verification technique at this stage	High altitude Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone's structure must be capable of surviving a crash landing with the drone travelling at the its cruise velocity	ON20	Structural analysis and calculation	Theoretical structures capabilities	Only availible verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Crash analysis was not performed on the drone, future work would involve full crash analysis after every phase of flight

Figure 66: Excerpt from the Compliance matrix

Design and testing phases verification techniques were also outlined for the numerical requirements, as seen in **Figure 67, Appendix S**.

Requirement Definition						*Complience and Verification*							
Category	Attribute	Requirement	Units	Importance	ID	Design Phase Verification Technique	Rationale	Expected outcome	Development / Test Phase Verification Technique	Rationale	Expected outcome	Iteration 3 Compliance	Rationale For Non-Compliance
Operation	Operational Productivity (OP)	21 (ideal case) & 18 (normal)	ha / h	High	NR1	Calculation	This is the only verification method availible at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Operational Radius	4,000	meters	High	NR2	Calculation	This is the only verification method availible at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Flight time	30	minutes	High	NR3	Calculation and battery spec	This is the only verification method availible at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Non-Compliance	A flight time of 30 minutes was extremely over-specced for the mission. See Flight Time Change Trade-Off Summary for more info
	Payload capacity	47	kg	High	NR4	Defined from requirement	This is the only verification method availible at this stage	Theoretical value	Physical drone construction	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone specifications	Compliance	N/A
	Operational ceiling	1,500	meters	Medium	NR5	Motors spec and Calculation	This is the only verification method availible at this stage	Theoretical value	High altitude full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Communication range	5 & Compatible with 4G and satellite	km	High	NR6	Avionics spec	This is the only verification method availible at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
Spraying mechanism	Fluid flowrate	9	L/min	High	NR7	Calculation	This is the only verification method availible at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A

Figure 67: Excerpt from the Numerical Compliance matrix

Overall, requirement compliance was satisfactory, as shown in **Figure 68**. A small number of requirements require future work to achieve full compliance, while others were defined as out of scope for this project. Only one requirement, DN1, was not met; however, this non-compliance was justified within the design constraints and trade-offs made during development.

Integrated Set of Needs Compliance Summary		
Number of Needs	**Percentage of Total Needs**	**Explenation**
37	65	Percent of needs complied with
3	5	Percent of needs partially complied with
9	16	Percent of needs require future work to be complied with
1	2	Percent of needs not complied with
7	12	Percent of needs are beyond the scope of GDBP
57	100	57 Needs total, divided in 6 categories

Numerical Requirements Compliance Summary		
Number of Requirements	**Percentage of Total Needs**	**Explenation**
9	60	Percent of needs complied with
2	13	Percent of needs partially complied with
1	7	Percent of needs require future work to be complied with
3	20	Percent of needs not complied with
15	100	15 Needs total, divided in 5 categories

Figure 68: Compliance summary table

5.7.8 Final CAD Model and GA Drawings

The parametric construction approach was discontinued for the final CAD model, as Inventor® Professional 2026 has limitations when managing parameters linked to complex geometry. This did not impact the design process, as the main geometric features had already been frozen at this stage.

The final CAD model (**Figure 69**, **Appendix** T) represents the external geometry of the UAV. Structural components that required detailed analysis were fully modelled and integrated into the assembly, as shown in **Figure 70** (**Appendix T**). Material assignments within the CAD model were selected primarily for visual clarity and representation purposes rather than for final engineering specification.

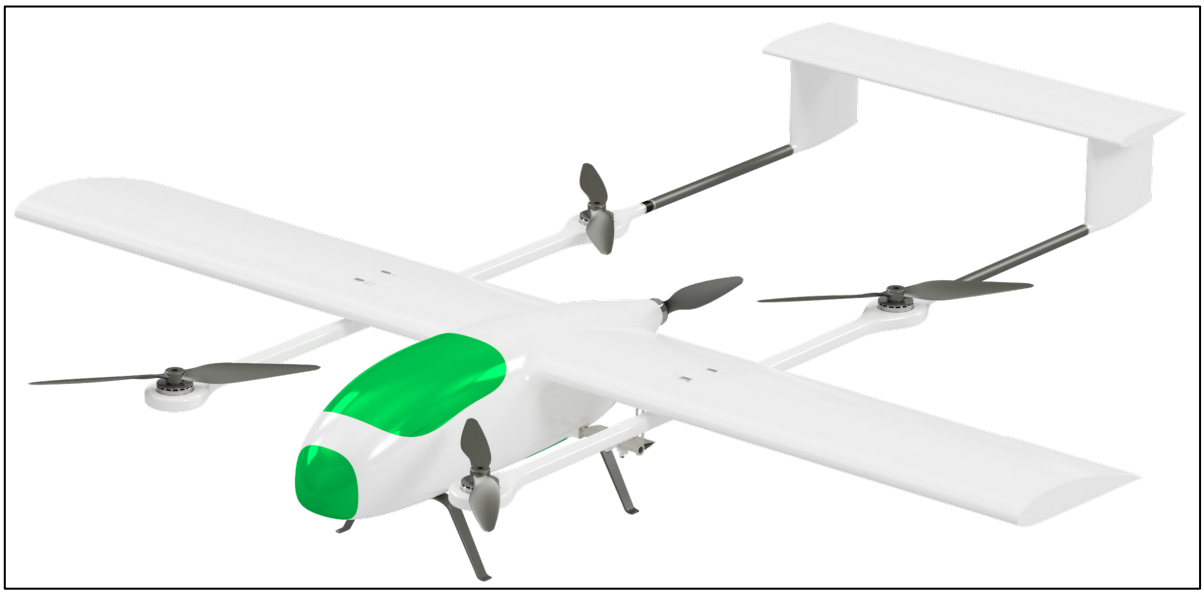


Figure 69: CAD Model render
(operations paint scheme)

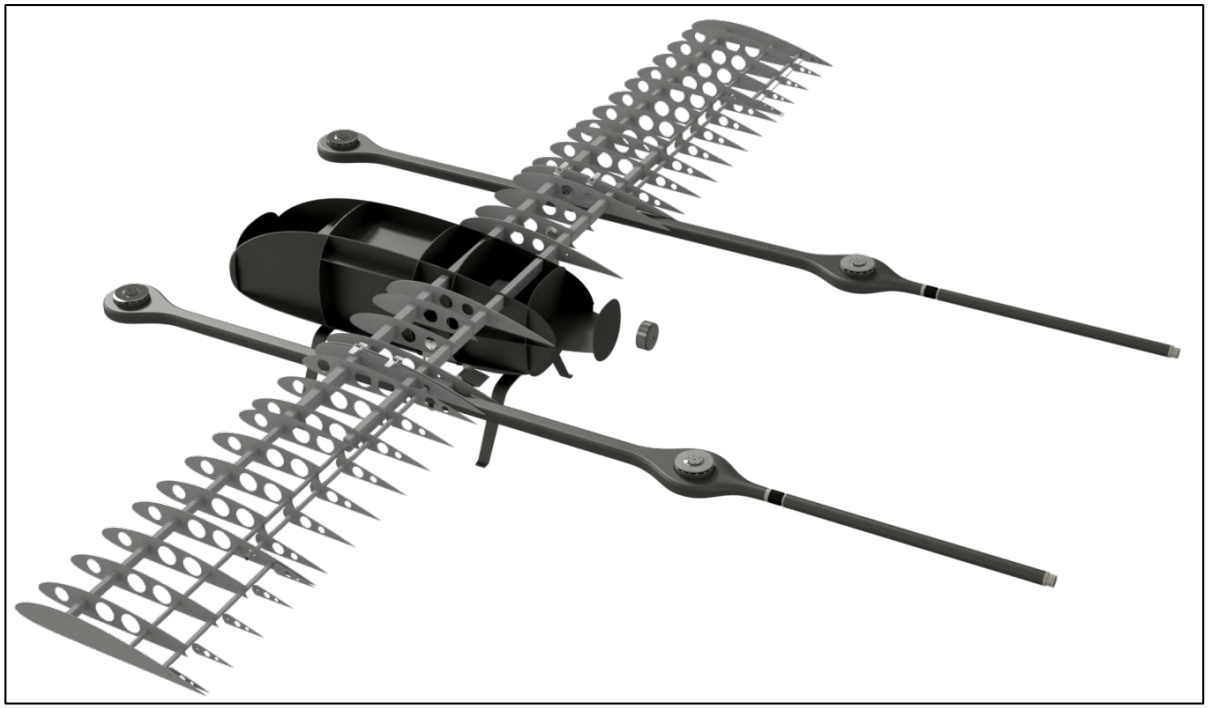


Figure 70: Internal Structures render
(structures chosen for in-depth analysis)

A GA drawing (**Figures 71–73**, **Appendix T**) was produced to define the overall dimensions of the UAV. An exploded view was also included to illustrate the disassembly process for transport. A separate sheet detailing the dimensions of the disassembled components was created to verify compliance with requirement NR11 related to transport constraints.

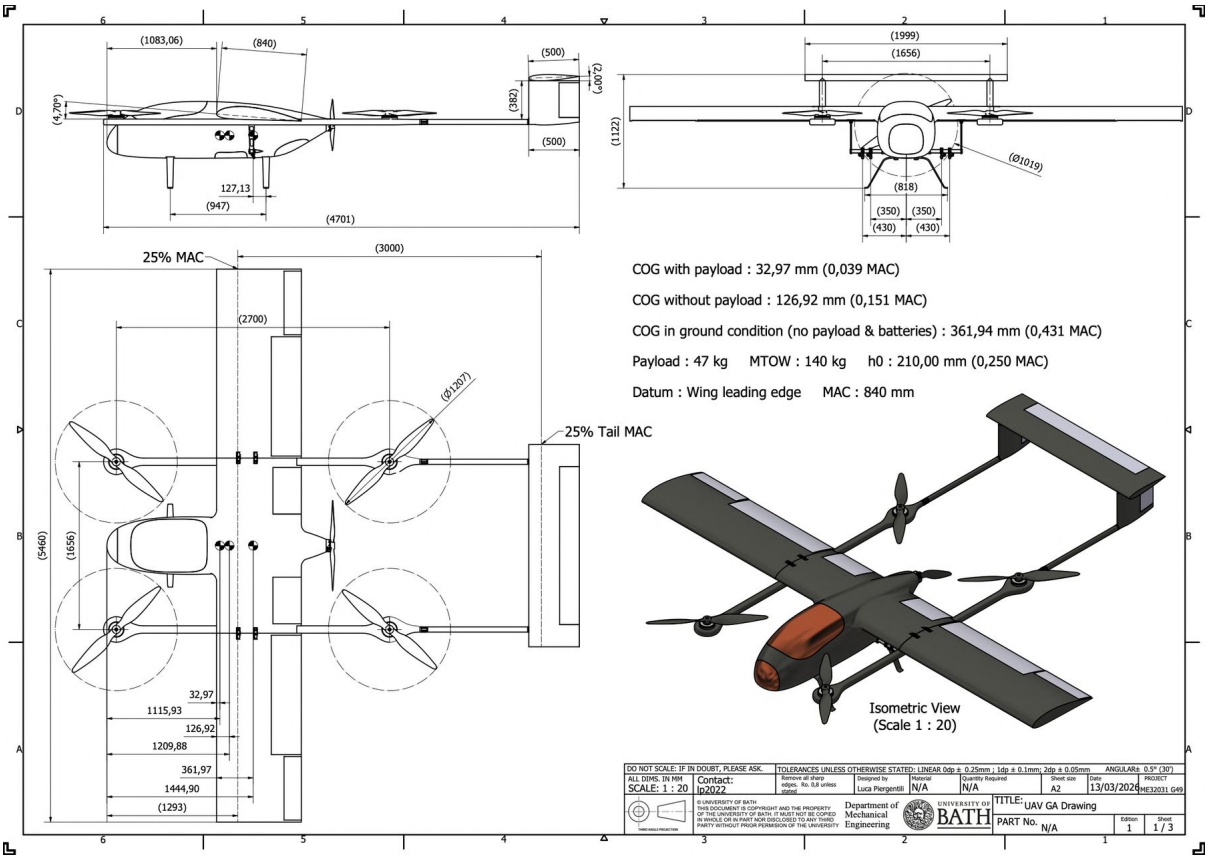


Figure 71: GA Drawing

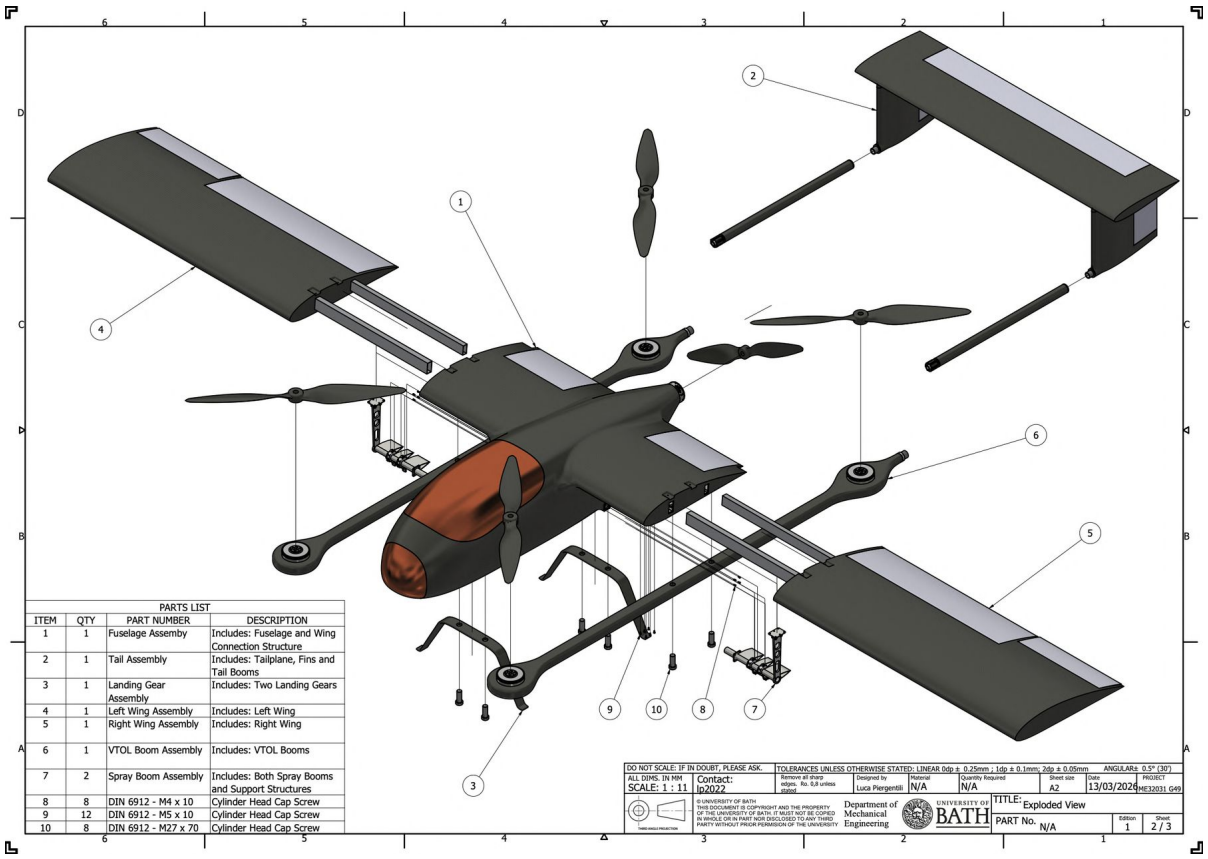


Figure 72: Exploded View

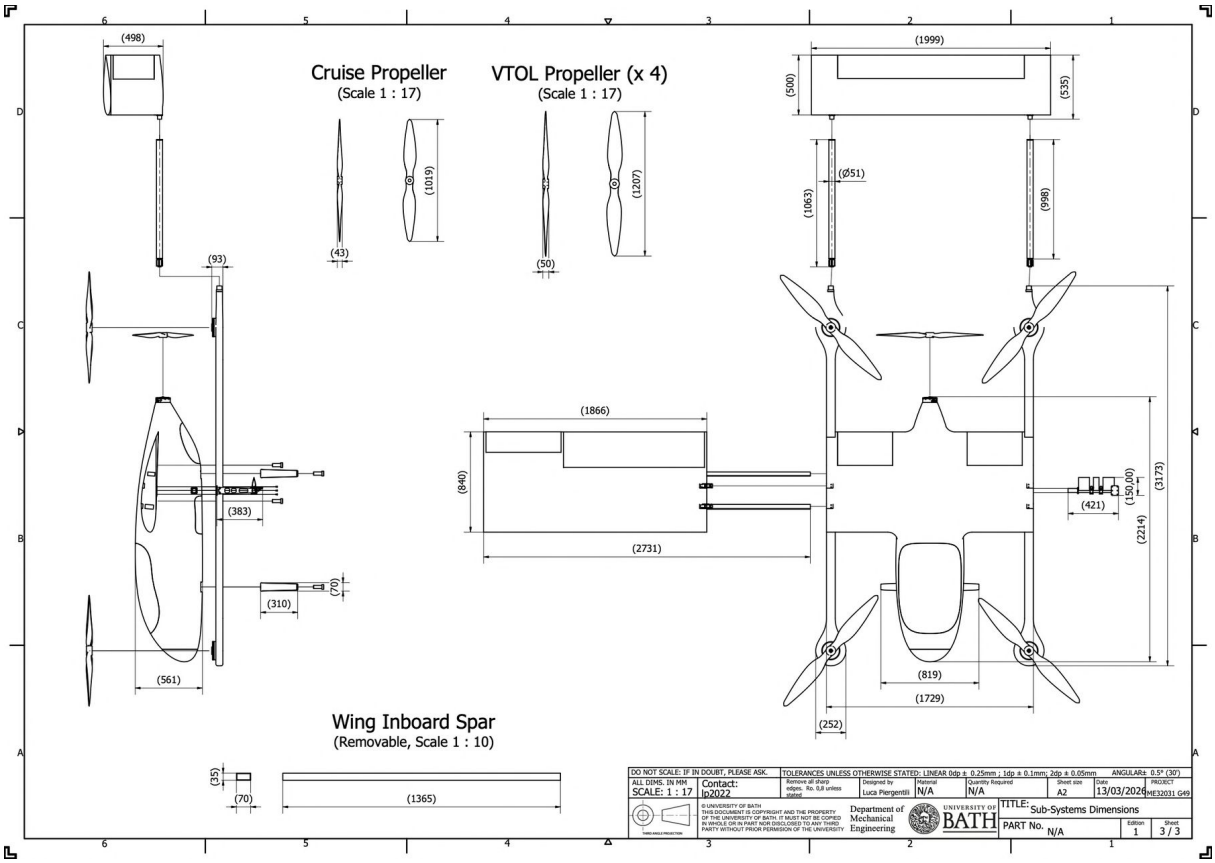


Figure 73: Transportational Dimensions

5.8 Solution Lifecycle and Sustainability

5.8.1 Solution Lifecycle

A preliminary life cycle assessment (**Figure 74**) was conducted for the UAV. Primary structures manufacturing (VTOL booms, tail, wings, fuselage structure, etc.) would take place in India through external manufacturing suppliers, who would also be responsible for raw material procurement. Sub-system assembly, COTS validation, integration, and full aircraft testing would be performed in-house to ensure system-level functionality and compliance.

Following verification, the UAV enters operational service. Maintenance intervals vary by sub-system: the spray tank and nozzles require daily inspection and cleaning, while the internal payload system is inspected weekly[25]. The airframe and primary structure require maintenance after 200 flight hours or six months of operation, with additional repairs performed following any non-catastrophic incidents[26].

As the system is based on COTS components, avionics, propulsion, and battery systems can be upgraded provided new technologies are physically and functionally compatible with the existing architecture. Any upgrades would require full system-level testing to ensure continued airworthiness, but this modularity supports future-proofing and maintains competitiveness within the agricultural UAV market.

At end-of-life, COTS components are disposed of or recycled according to manufacturer guidance. Recycling of CFRP structural components would be attempted where feasible; however, due to the UAV's operating environment and the difficulty of CFRP recycling, full

airframe disposal may be required. The modular architecture reduces waste, as individual sub-systems can be replaced independently and reused across other platforms. For example, if the fuselage is damaged, the wings, booms, and tail assemblies may be retained as spare parts or reused on other aircraft, extending component service life and reducing overall material waste.

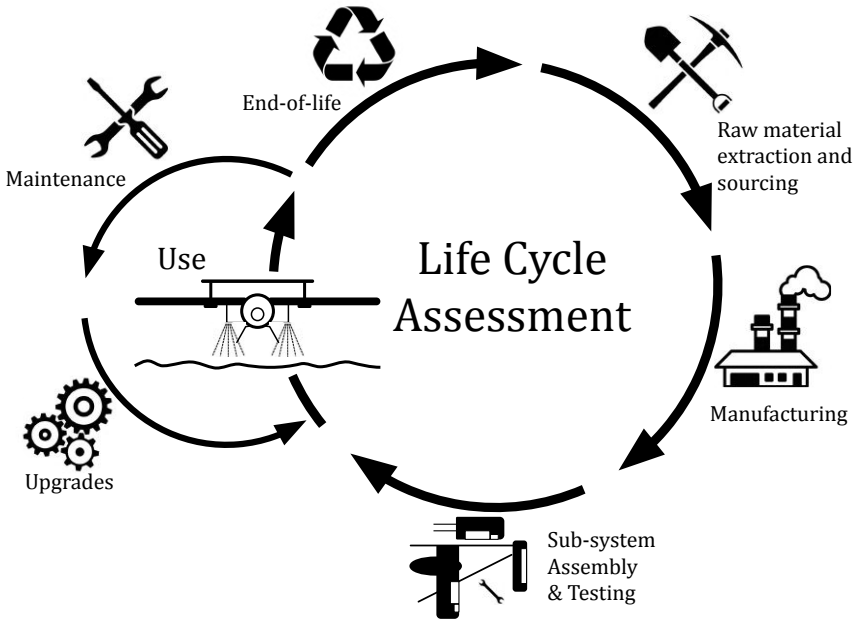


Figure 74: Solution Lifecycle

5.8.2 Sustainability Considerations

To maximise sustainability, the UAV was designed around an electric powertrain, despite many comparable systems in its class relying on piston engines[18]. The modular architecture further reduces environmental impact by enabling component reuse and minimising material waste, as previously discussed. In addition, the UAV's intended role is to replace less efficient and more polluting spraying methods, such as diesel-powered tractors, making it inherently more sustainable at the system level.

However, the use of CFRP structures and Li-Po batteries limits overall sustainability performance. These materials were selected due to current technological constraints in lightweight structures and energy storage, rather than ideal end-of-life recyclability or environmental impact.

6 Future Work

The project timeframe limited the overall scope of analysis and design that could be performed. As a result, several areas require further work, including detailed spray drift analysis, CFD and FEA studies, disassembly interface structural validation, navigation optimisation, flight dynamics modelling, and thermal simulations.

Further development is also needed for fuselage internal layout and hatch design, along with fully defined CAD-level interfaces, to confirm that all sub-systems can be properly integrated within the available design space. Systems integration across avionics, payload control, and actuation systems must also be completed to ensure full electrical and software interoperability.

In addition, design for manufacture and assembly practices should be further implemented, including full definition of joining methods (adhesive vs mechanical), reduction of geometric complexity, and selection of readily available materials that do not require specialised tooling. Design for safety and durability considerations were addressed to meet regulatory requirements. A full lifecycle analysis should also be conducted to incorporate sustainability, recyclability, logistics, and supply chain constraints into future design decisions.

Finally, physical prototyping and testing represent the logical next step following detailed design, enabling validation of system performance, integration, and compliance with requirements and regulations.

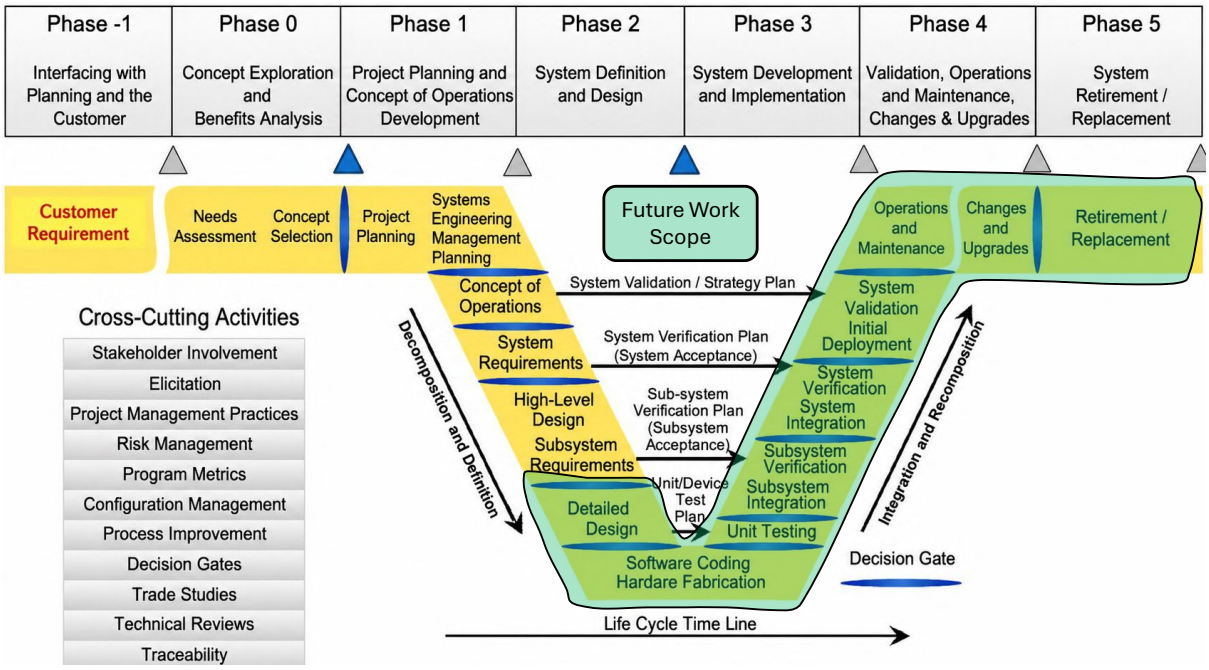


Figure 75: Representation of Future Work on the Engineering V Diagram

7 Reflections

Overall, this project was successfully organised and managed, with the methods used appropriately documented throughout this report. Areas for improvement include greater conciseness when communicating within the team, better time management during meetings, and adopting a more relaxed approach at times where appropriate. Despite this, the team collaborated effectively and was collectively satisfied with the final outcomes.

The technical aspect was also successful, however, improvements could have been made in the early requirement generation phase, as some values were later revised and additional requirements were introduced during the project. In addition, more in-depth analysis at the concept stage could have been carried out to better define and validate the optimal target market.

This project provided early insight into aircraft design processes, as well as broader understanding of industry-level design practices. The SEBoK methodology proved valuable in structuring a detailed design specification, while tools such as the ConOps, mission profile, FMEA, and considerations of airworthiness, sub-system integration, and design trade-offs significantly improved the depth and quality of the design.

The project also developed leadership and teamwork skills, highlighting the importance of soft skills alongside technical competence. The responsibilities of the role were clearly understood and effectively carried out, contributing meaningfully to the overall success of the project.

8 Conclusions

8.1 Project Management Conclusions

The goals and responsibilities outlined in **Table 2** were achieved and generally well maintained throughout the project. The project was delivered on time, with the schedule mostly adhered to and overall scope control maintained, although initial design work was delayed by the Competitor Study. The Easter break was not taken as a holiday, as design work continued throughout this period. Communication between team members was effective, morale remained positive, and a good work–life balance was maintained. All deliverables were submitted ahead of deadlines, and role allocation proved effective, with all members fulfilling their responsibilities. The CAD model was completed after CDR due to the extended development timeline. Meetings were organised with clear objectives that were consistently met, and the meeting summaries provided effective traceability of decisions and design progress.

However, meetings initially tended to overrun, although time management improved significantly as the project progressed. Change management and tracking were generally satisfactory, although not all values were consistently archived and a small number of major changes were not communicated to the entire team; despite this, no significant conflicts or delays resulted. Decision-making bias was largely avoided through structured

discussion of design alternatives and evidence-based final decisions. Project risks (**Figure 14**) and team conflicts were successfully mitigated and resolved where they arose.

8.2 Technical Conclusions

8.2.1 System Level Conclusions

The project's technical objectives outlined in the introduction were successfully achieved. The engineered design solution was deemed satisfactory, with preliminary analysis indicating it is capable of fulfilling its intended mission. Relevant airworthiness regulations were researched and incorporated into the design process.

A credible conceptual design of an agricultural spraying UAV was developed, balancing requirement compliance, regulatory constraints, and commercial viability within a coherent business case. Design decisions were supported by analytical calculations and preliminary simulations, with alternative solutions evaluated where appropriate. A CAD model capturing key system features was produced, alongside detailed engineering drawings.

The concept selection and overall design process followed a structured, iterative approach, with each iteration improving upon the previous one. System-level, design, and process failure modes were identified, with appropriate mitigations implemented through both design and procedural measures. Both PDR and CDR were successfully completed, and the design was formally frozen on schedule.

8.2.2 Technical Integration Conclusions

The TI goals and responsibilities outlined in **Table 2** were also achieved and consistently maintained. The SEBoK systems engineering methodology was followed throughout, enabling the problem, stakeholders, drivers, constraints and risks to be identified and a clear set of qualitative and numerical requirements clearly defined.

Sub-systems were successfully integrated both functionally and physically, while key top-level constraints such as MTOW, overall dimensions, and COG positioning were respected. Design trade-offs were effectively managed to maximise compliance with requirements and regulatory constraints.

All disassembly interfaces were systematically considered, with appropriate solutions defined for each and a preliminary system lifecycle analysis was also completed.

8.2.3 Technical Report Conclusions

This report clearly explains the UAV's design evolution, each requirement, key design decision, and stage of the design process were documented. The progression through design iterations, including associated trade-offs and failure modes, was also detailed, with particular focus given to the Tail–VTOL boom interface. Future work was identified and discussed.

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10 Appendix

Appendix A: Project Brief

G49: IMechE UAS challenge - Autonomous Drone for Agricultural Use

Source: Team Bath Drone

Competition: IMechE UAS challenge

Academic supervisor: Sam Bull

S.C.Bull@bath.ac.uk

Title: IMechE UAS challenge - Autonomous Drone for Agricultural Use

Description: "The team is required to design, build and demonstrate a UAS capable of autonomously planning and dispensing a liquid payload over a set target area. Drones are proving vital for farms in agriculture-dependent economies, saving time, cost and removing humans from contact with harmful chemicals - spraying crops, spreading fertilizer, transporting produce, monitoring livestock locations, mapping field topography and measuring crop health. The tasks for 8 students to undertake will be Technical Integrator, Aerodynamics, Stability and Control, Structures, Propulsion, Systems, Navigation (some tasks may require more than one student).

UAS design requires system level analysis to balance the competing requirements of all technical disciplines.

The business/societal impact is the potential to revolutionise farming in developing countries.

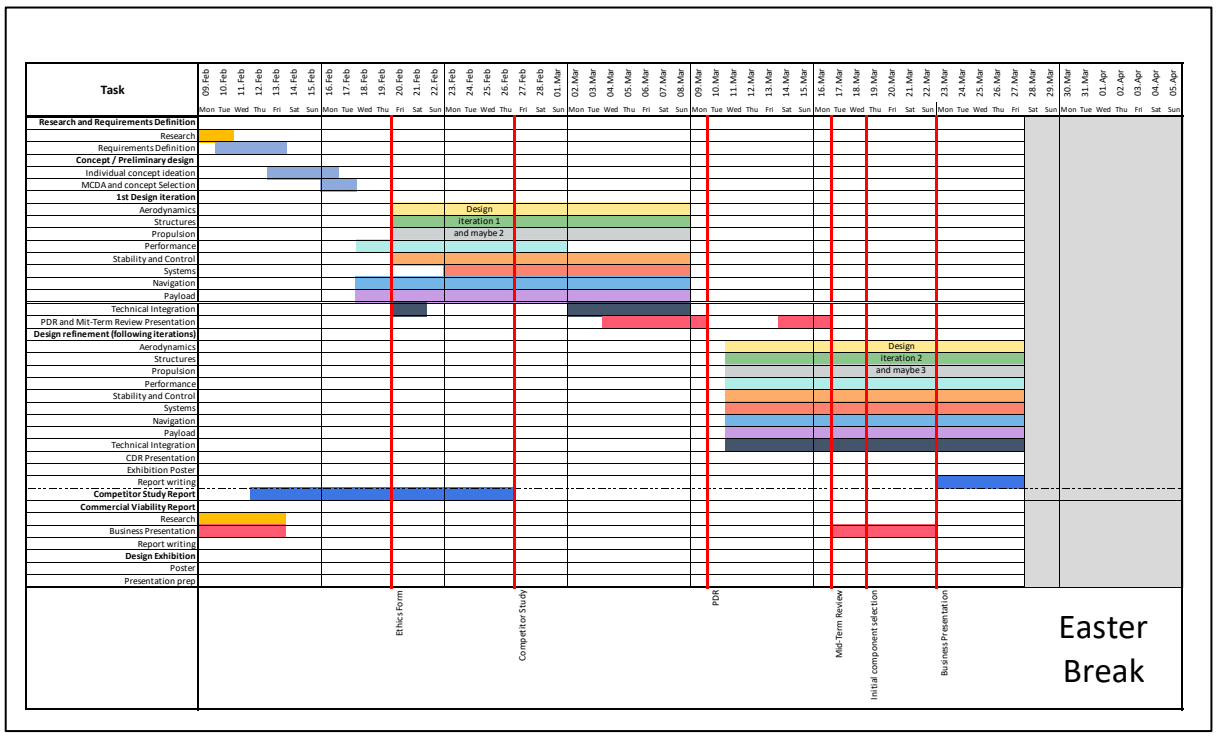
Technical challenges: The design of an autonomous UAS presents a huge technical challenge, blending and balancing the separate disciplines. This years brief involves the dispensing of a liquid payload, which poses a very interesting control challenge for the team. Innovation is encouraged in the initial scoping for a platform configuration - and sustainable choices in materials/systems will be prioritised.

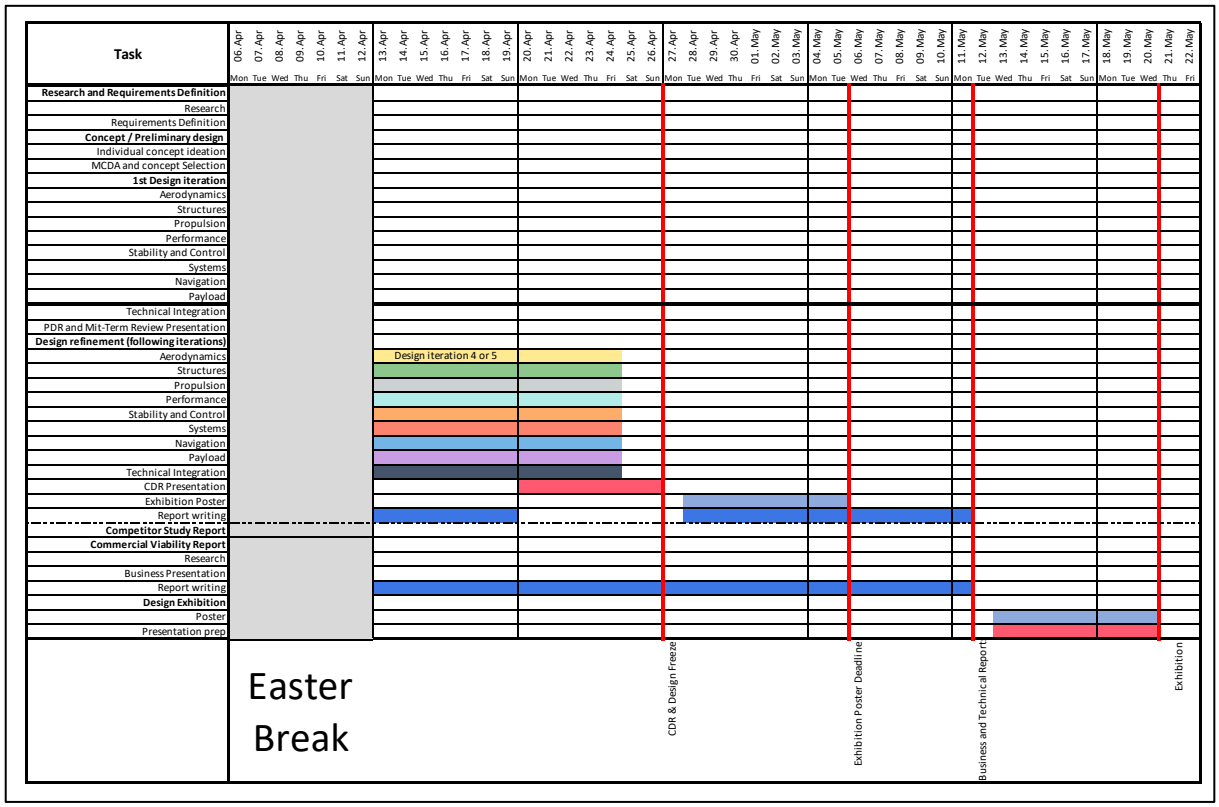
Commercial opportunities: "Drones save farmers time and money. They reduce the need for smallholders – people who farm less than 5 acres (2 hectares), which account for 85% of farms globally – to do dangerous and tiring manual spraying and spreading work on their own farms. They also remove the need to hire workers to do the same.

By eliminating some of the last remaining physically demanding work in farming, drones may also help make agriculture more attractive to rural youth, who are often disillusioned with the drudgery of traditional farming. In addition, drones create new skilled employment opportunities in rural areas for pilots, many of whom are young people."

How will this project benefit your team's competition goals? The team will be following a brief that is very similar to the IMechE UAS challenge. This GBDP will create a pipeline of expertise to support Team Bath Drones and provide the foundation for the core team next year.

Appendix B: Gantt Chart and CDR Timetable





Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date	13.Apr	14.Apr	15.Apr	16.Apr	17.Apr	18.Apr	19.Apr
Meetings	> Catchup meeting		> CDR pre-prep, business report and poster discussions				
Design requirements		> Interfaces between disassimbling components established		> COTS components final location in fuselage requirements (if they need to be waterproofed, what components they need to be close to, etc...)		> Wing structure requirements outlined (if taper needed, thickness, structure geometry etc) so K can start final wing design	
Design analysis							> Wing analysis model completed and validated > Dynamic mode analysis completed > Aircraft performance analysis (turn rate and climb rate) estimated
Design task		> Initial COG location > Initial payload tank design		LiDAR and Camera selection Final COTS components selection			
Design Freezes					> COTS components design freeze (excluding servos, motors and battery)	> Requirements, mission profile, ConOps and design configuration design freeze	
Information exchange				Peter gives Jonathan climb rate, max climb height and bank angle used in navigation model Peter and Aman discuss whether we need LiDAR and what camera we need. Component selection is done too	Jonathan and Sam discuss payload system positionment in the fuselage	Jaison can give Khayaam needed taper ratio, root chord, max wingspan and any other structural requirements Aman gives Jonathan battery / power values relevant to performance analysis	Khayaam gives wing lift distribution to Jaison Jonathan gives Jaison the power requirements for a climb to ensure that the motor selections is ok Luca asks general update from everyone to determine what to put on the 3 slide presentation
CAD model construction		> Final CAD model construction can begin			> Payload system inside the fuselage CAD mockup (excluding the tank)		
Presentations							> 3 slide presentation prep
Report Writing							

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date	20.Apr	21.Apr	22.Apr	23.Apr	24.Apr	25.Apr	26.Apr
Meetings	> CDR prep meeting				> Final progess check meeting		
Design requirements		> Aerofoil selection complete, root and tip chord lengths given, wing span freeze and wing base section					
Design analysis	> Propulsion analysis complete	> Fuselage structure analysis complete > Wing analysis complete	> Flap and wingtip analysis models completed > Landing gear analysis complete > Wing internal structure analysis complete	> Aircraft drag polar model completed	> Aircraft drag polar calculated > Electrical system analysis completed > Control surface effectiveness analysis completed		
Design task			> Final COG and MTOW estimates made				
Design Freezes	> Motor / Propulsion system and propeller design freeze > Quick disconnect connections (power, electrical and tubeing) design freeze > Propeller positions lock	> Basic wing attributes: aerofoil, root & tip chord lengths and wingspan design freeze > Wing base section (between fuselage and VTOL booms) design freeze > Stabcon aerofoil selection, control surface sizing and fin sizing design freeze	> Power system, navigation and general electronics design freeze > Disassembly interface mechanisms design freeze > Wing internal structure design freeze > Fuselage shape (outer shell)design freeze	> Final wing design freeze, flaps and ailerons included (wing twist, bends and wingtips) > Control surface design freeze	> Stability & Control design freeze > Navigation model and Controller design freeze		***CDR Design Freeze***
Information exchange	Jaison can give Hariram dCD/dCT, dCL/dCT and dCM/dCT	Khayaam gives final wing L/D and estimated total L/D as well as flap servo torque needed to Aman	Sam, Luca, Jaison and Jonathan establish the Interfce between fuselage & sprayboom and wing & sprayboom Sam gives Jonathan and Hariram fuselage dimensions and areas for analysis		Khayaam gives Jonathan		> Send all regulations used in the design to Luca > Company and drone name ideas
CAD model construction	> Payload system inside the fuselage CAD model complete	> Final CAD can start on the wing		> Sprayboom CAD model complete	> Fuselage sub-assembly CAD model complete > Landing gear sub-assembly CAD model complete	> Tail sub-assembly CAD model complete > Wing sub-assembly CAD model complete	> CAD model finished
Presentations	> 3 slide presentation (Luca)						> CDR presentation prep
Report Writing							

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date	27.Apr	28.Apr	29.Apr	30.Apr	01.May	02.May	03.May
Meetings				> Poster Meeting			
Design requirements							
Design analysis		> New Stab-Con analysis					
Design task	***CDR***	> CDR feedback design changes					
Design Freezes				***Absolute Final Design Freeze***			
Information exchange							
CAD model construction	> CAD model renders			> Final CAD model completed > Drawings Completed			
Presentations	> CDR Presentation						
Report Writing							

Appendix C: Air Taxi GDBP Timetable

Week	AM (0930-1230)	PM (1415-1615)
1 (02/2) Monday	Introduction and Overview of Systems Engineering 1015 Start this day only	On-line Launch with Sora Aviation includes technical design tutorial Part 1
1 (03/2) Tuesday	Introduction to Design Framework	Cockpit / Cabin Design from Vertical Aerospace 1415 - online
2 (09/2) Monday	Preliminary Design, Creativity and Innovation	Sora Aviation Business tutorial 1415 - online
2 (10/2) Tuesday	Presentation from UAM Team 2025 1115 Start this day only	No Activity
3 (17/2) Tuesday	Airworthiness, Systems Requirements and Flight Operations	Safety
4 (25/2) Wednesday	**Design Integration, iteration and interfaces**	**Aftermarket**
4 (27/2) **Friday**		**Competitors Study due 4pm Friday**
5 (03/3) Tuesday	Design for X	PDR Prep
6 (10/3) Tuesday	**PDR**	No activity

Appendix D: General Values and Archive Values

Category	Variable Name	Symbol	Current Value	Units	Connection	Ownership	Notes	Date Added	Last Edited	Edited by
Overall	Maximum Take Off Weight (MTOW)	W	1381.6	N	5	Luca		18.Feb	24.Feb	Luca
	Maximum Take Off Mass (MTOM)	m	140.8	Kg	5	Luca		18.Feb	24.Feb	Luca
	Overall Width	w	5462	mm	3	Luca		18.Feb	16.Mar	Luca
	Overall Length	l	5764	mm	3	Luca		18.Feb	16.Mar	Luca
	Overall Height	h	1211	mm	2	Luca		18.Feb	16.Mar	Luca
	Flight time	t	15	mins	3	Luca		18.Feb	28.Mar	Luca
	Operational radius		4774.5	m	1	Luca		18.Feb	28.Mar	Luca
Aerodynamics	Lift Coefficient	cL	0.7	N/A	2	Khayaam		01.Mar	01.Mar	Khayaam
	Stall lift Coef	CL_max	1.8		2	Khayaam		01.Mar	01.Mar	Khayaam
	Wing Area	S	3.35	m2	2	Khayaam		01.Mar	01.Mar	Khayaam
	Wing Loading	W/S	280	N/m2	2	Khayaam		08.Mar	08.Mar	Khayaam
	Wingspan	b	5.46	m	3	Khayaam		01.Mar	08.Mar	Khayaam
	Chord length	c	0.84	m	5	Khayaam		01.Mar	08.Mar	Khayaam
	Cruise Velocity	V_c	25.8	m/s	4	Khayaam		18.Feb	24.Feb	Khayaam
	Aspect ratio	AR	6.5		3	Khayaam		25.Feb	25.Feb	Khayaam
	Flap Chord		0.294		3	Khayaam		13.Apr	13.Apr	Khayaam
	Flap Span		3.276		3	Khayaam		13.Apr	13.Apr	Khayaam
	L/D Cruise	T/W			2	Khayaam		13.Apr	08.Mar	Khayaam
	Wing Volume	WV	93.52	L	3	Khayaam	Safety factor of 0.7 was applied to account for some of the internal volume taken up by structures	12.Mar	13.Mar	Luca
	Stall Velocity	V_s	16.1	m/s	2	Khayaam		13.Apr	01.Mar	Khayaam
Stability and Control	Zero-Lift Pitch Moment Coefficient	CM0	-0.082		2	Hariram		05.Mar	05.Mar	Hariram
	COG location with Payload	h	0.039		4	Hariram		05.Mar	14.Apr	Luca
	COG location location without Payload	h	0.151		4	Hariram		18.Feb	14.Apr	Luca
	Non-dimensional Aerodynamic Centre Location	h0	0.25		4	Hariram		05.Mar	08.Mar	Hariram
	h0 - h	h0 - h	0.211		4	Hariram	h taken as the average between the two values	03.Apr	08.Mar	Luca
	Lift-Curve Slope	a	4.85		2	Hariram		05.Mar	05.Mar	Hariram
	Tailplane Lift Curve Slope	a1	4.27		2	Hariram		05.Mar	05.Mar	Hariram
	Control Surface Lift Curve Slope	a2	3.12		2	Hariram		08.Mar	11.Mar	Hariram
	Downwash Gradient	de/da	0.39		2	Hariram		05.Mar	05.Mar	Hariram
	Tail Volume Coefficient	Vh	0.777		1	Hariram		05.Mar	08.Mar	Hariram
	Horizontal Stabiliser Area	ST	0.999	m2	3	Hariram		05.Mar	11.Mar	Hariram
	Horizontal Moment Arm	l_t	3.0	m	3	Hariram		05.Mar	05.Mar	Hariram
	Fin Volume Coefficient	Vv	0.031		1	Hariram		08.Mar	11.Mar	Hariram
	Vertical Stabiliser Area	SF	0.388	m2	3	Hariram		08.Mar	11.Mar	Hariram
	Vertical Moment Arm	l_b	3.0	m	3	Hariram		08.Mar	08.Mar	Hariram
	Horizontal Stabiliser Span	b_h	1.999	m	3	Hariram		08.Mar	21.Apr	Hariram
	Horizontal Stabiliser Chord Length	c_h	0.5	m	3	Hariram		08.Mar	21.Apr	Hariram
	Vertical Stabiliser Span	b_v	0.763	m	3	Hariram		08.Mar	21.Apr	Hariram
	Vertical Stabiliser Chord Length	c_v	0.5	m	3	Hariram		08.Mar	21.Apr	Hariram
	Aileron Span	b_a	1.26	m	2	Hariram		21.Apr	21.Apr	Hariram
	Aileron Span / Wing Span	b_a/b	0.230769231		2	Hariram		08.Mar	21.Apr	Hariram
	Aileron Chord Length	c_a	0.168	m	2	Hariram		08.Mar	21.Apr	Hariram
	Aileron Chord / Wing Chord	c_a/c	0.2		2	Hariram		21.Apr	21.Apr	Hariram
	Aileron Planform Area	S_a	0.21168	m2	2	Hariram		08.Mar	21.Apr	Hariram
	Elevator Span	b_e	1.56	m	2	Hariram		08.Mar	21.Apr	Hariram
	Elevator Span / Tailplane Span	b_e/b_h	0.780390195		2	Hariram		21.Apr	21.Apr	Hariram
	Elevator Chord Length	c_e	0.187	m	2	Hariram		08.Mar	21.Apr	Hariram
	Elevator Chord / Tailplane Chord	c_e / c_h	0.374		2	Hariram		21.Apr	21.Apr	Hariram
	Elevator Planform Area	S_e	0.292	m2	2	Hariram		08.Mar	21.Apr	Hariram
	Rudder Span	b_r	0.686	m	2	Hariram		08.Mar	21.Apr	Hariram
	Rudder Span / Fin Span	b_r / b_v	0.899082569		2	Hariram		21.Apr	21.Apr	Hariram
	Rudder Chord Length	c_r	0.191	m	2	Hariram		08.Mar	11.Mar	Hariram
	Rudder Chord / Fin Chord	c_r / c_v	0.382		2	Hariram		21.Apr	21.Apr	Hariram
	Rudder Planform Area	S_r	0.131026	m2	2	Hariram		08.Mar	11.Mar	Hariram
Performance	Minimum Turn Radius	r	20	m	3	Jonathan		20.Feb	20.Feb	Jonathan
	Maximum Load Factor	n	4	g	4	Jonathan		20.Feb	20.Feb	Jonathan
Structures	Mass of Fuselage Structure	M_f	x	kg	4	Sam		22.Apr	22.Apr	Sam
	Mass of Wing Structure	M_w	15.71	kg	4	Sam		22.Apr	22.Apr	Sam
	Maximum Deflection at Wing Tip	delta_d_tip	x	mm	1	Sam		22.Apr	22.Apr	Sam
	Von Misses Stress	sigma_vm	x	MPa	1	Sam		22.Apr	22.Apr	Sam
	Ultimate load factor	n_ult	5.7	-	1	Sam		08.Mar	08.Mar	Sam
Propulsion	Vertical Propellor Diameter	D_v	47	inch	1	Jaison		25.Feb	10.Apr	Jaison
	Horizontal Propellor Diameter	D_h	35	inch	1	Jaison		25.Feb	21.Apr	Jaison
	Vertical Thrust To Weight Ratio	TWR	2		3	Jaison		25.Feb	21.Apr	Jaison
	Vertical Motor Power	P_v	6	kW	2	Jaison		25.Feb	10.Apr	Jaison
	Horizontal Motor Power	P_h	5.6	kW	2	Jaison		25.Feb	10.Apr	Jaison
Payload	Spray Width (assumed)	w	6	m	5	Jonathan		25.Feb	25.Feb	Jonathan
	C (litres of payload per hectare) (Assumed)	c	15	L/h	3	Jonathan		25.Feb	25.Feb	Jonathan
	Calculated nozzle rate	x	4.18	L/min	2	Jonathan		25.Feb	25.Feb	Jonathan
	Payload mass	mp	47	kg	5	Jonathan	Pyka pellican ratio between payload and MTOW used to get to this value	25.Feb	25.Feb	Jonathan
	Payload Litrage		47	L	4			02.Apr	02.Apr	Jonathan
	Cruise height		2.1	m	2	Jonathan		02.Mar	02.Mar	Jonathan
	Nozzle spacing		0.08/0.24/0.08	m	2		Distance between leftmost nozzle and nearest nozzle/distance between central nozzles/Distance between rightmost nozzle and nearest nozzle	02.Apr	02.Apr	Jonathan
	Total Spray time		4.554263566	mins	2			06.Mar	06.Mar	Jonathan
	Spray area covered per run		2.72835	hectare	2			02.Apr	02.Apr	Jonathan
	Operational spray range		4547.25	m	2			02.Apr	02.Apr	Jonathan
	Operational productivity		26 (ideal case) & 19-15 (normal)	ha/h	2		Assuming a strip with width 6 m is being sprayed	05.Mar	25.Apr	Luca
Systems	shut off Pressure(max)	p(shut off)	100	psi/Pa	2	Aman/Jonathan	Assumed max limit based on average shut-off limits imposed on commercial pumps	29.Mar	29.Mar	Aman
	nozzle Pressure(operating max)	p(nozzle)	50	psi/Pa	2	Aman/Jonathan		08.Mar	18.Mar	Aman
	Battery weight(per battery)	m(per battery)	4.686	kg	4	Aman	Updated to reflect new flight time	08.Mar	09.Apr	Aman
	Battery weight(total)	m(battery_total)	18.744	kg	4	Aman	Updated to reflect new flight time	08.Mar	09.Apr	Aman
	Battery Length (per battery)	l(battery)	215	mm	3	Aman	Updated to reflect new flight time	08.Mar	09.Apr	Aman
	Battery width (per battery)	w(battery)	91	mm	3	Aman	Updated to reflect new flight time	08.Mar	09.Apr	Aman
	Battery Height (per battery)	h(battery)	118	mm	3	Aman	Updated to reflect new flight time	08.Mar	09.Apr	Aman
	Battery Quantity		4		2	Aman	Updated to reflect new flight time	10.Mar	03.Apr	Aman
	Effective Battery Voltage		88.8	V	2	Aman	44.4V is the nominal voltage for each COTS pack. 88.8V is for te=he total 4-pack config. in 2452P	24.Apr	24.Apr	Aman
	Effective Battery Capacity		66	Ah	2	Aman	Updated to reflect new flight time. Value based from COTS battery. Calculated capacity = 51.3	10.Mar	09.Apr	Aman
	Battery Configuration		2452P		2	Aman	Battery packs to be in a 2 series, 2 parallel configuration	10.Mar	09.Apr	Aman
	Continuous C-rate		5.58		2	Aman	Calculated from calculated capacity. Based on COTS battery, 66Ah, value = 4.33	24.Apr	24.Apr	Aman
	Peak C-rate		8.26		2	Aman	Calculated from calculated capacity. Based on COTS battery capacity, 66Ah, value = 6.24. Maximum stated COTS C-rate = 10	24.Apr	24.Apr	Aman
	VTOL Hover Max Power		37.7	kW	1	Aman		24.Apr	24.Apr	Aman
	VTOL Climb Max Power		49.3	kW	1	Aman		24.Apr	24.Apr	Aman
	Cruise Max Power		5.18	kW	1	Aman	Accounts for pump use, if not accounted for: Power = 4.93kW	24.Apr	24.Apr	Aman
	Cruise Climb Max Power		12.9	kW	1	Aman		24.Apr	24.Apr	Aman
	Battery recharge time		Undefineable	mins	1	Aman	N/A	24.Apr	24.Apr	Aman
	Battery lifecycle		500	cycles	1	Aman	500 charge cycles listed per battery. Limited by battery technology used (LiPO)	28.Mar	24.Apr	Luca
	Max VTOL Climb Height		20	m	2	Aman		24.Apr	24.Apr	Aman
	VTOL Climb Velocity	v_VTOL_climb	2	m/s	2	Aman		24.Apr	24.Apr	Aman
	VTOL Total Flight Time	t	2	mins	2	Aman		28.Mar	24.Apr	Aman
	VTOL Total climb time	t	26	s	2	Aman	Half of value is representative for ascend and descend separately	24.Apr	24.Apr	Aman
	VTOL Total Hover time	t	94	s	2	Aman	Half of value is representative for ascend and descend separately	24.Apr	24.Apr	Aman
	Cruise Flight Time	t	13	mins	2	Aman	Total flight time change based on drone productivity assessment.	28.Mar	24.Apr	Aman

Category	Variable Name	Symbol	It 1	Source of Value	It 2	Reason for Change	It 3	Reason for Change	It 4	Reason for Change	It 5	Reason for Change
Overall	Maximum Take Off Weight (MTOW)	W	1373.4	Initial estimation from competitor research	1031.094	Too heavy, resulted in a 20m wing span	981	Used a simplified number				
	Maximum Take Off Mass (MTOM)	m	140	Initial estimation from competitor research	105.1064	Too heavy, resulted in a 20m wing span Used a payload weight fraction of 0.47 and worked backwards to find MTOW	100	Changed to use a simpler number				
	Overall Width	w	5180	CAD model	5180	Wing redesign						
	Overall Length	l	4740	CAD model	4740	Component integration in the fuselage						
	Overall Height	h	1585	CAD model	1585	Drone redesign						
	Flight time		30	Was originally based on competitors but was reduced to allign to payload depletion rate								
	Operational Radius		4000.5	Calculations (08/04/2026)								
Aerodynamics	Lift Coefficient	cL										
	Wing Area	S					3.35					
	Wing Loading	WS					280					
	Wingspan	b					5.18	Wing re-design				
	Chord length	c					0.65	Wing re-design				
	Cruise Velocity	V	13.8	Chinese regulation constraint (CAAC: Interim Regulations on the Flight Management of Unmanned Aerial Vehicles, article 62, paragraph 8)	xxx	Was too slow to fly with the given MTOW. Found using Aerodynamics calculations and assumptions	25.8					
	Aspect ratio	AR					8					
	T/W horizontal	T/W										
Stability and Control	Zero-Lift Pitch Moment Coefficient	CM0										
	COG location with Payload	h	0.33-0.52	Error spotted in model where h0 was set to 0.25 rather than 0.25 + (distance from nose to wing / c)			1.43-1.57	Vertical and horizontal chord lengths to be same	-0.26 to -0.17	Actual COG was calculated and datum changed		
	COG location location without Payload	h										
	Non-dimensional Aerodynamic Centre Location	h0	0.25				1.78	Vertical and horizontal chord lengths to be same	0.16	Actual COG was calculated and datum changed		
	h0 - h	h0 - h										
	Lift-Curve Slope	a							4.32			
	Control-Surface Lift Curve Slope	a1							3.36			
	Control Surface Lift Curve Slope	a2										
	Downwash Gradient	dε/dα							0.36			
	Tail Volume Coefficient	Vh			0.92	Sign error spotted in fin sizing model	0.998	Vertical and horizontal chord lengths to be same	1.3			
	Horizontal Stabiliser Area	ST	0.98	Error spotted in model where h0 was set to 0.25 rather than 0.25 + (distance from nose to wing / c)	1.72	Sign error spotted in fin sizing model	0.724	Vertical and horizontal chord lengths to be same	0.944		1.03	
	Horizontal Moment Arm	l_c					3.0	Vertical and horizontal chord lengths to be same	3.0		3.0	
	Fin Volume Coefficient	Vv			0.174	Sign error spotted in fin sizing model	0.114	Vertical and horizontal chord lengths to be same	0.100		0.032	
	Vertical Stabiliser Area	SF			1.62	Sign error spotted in fin sizing model	0.659	Vertical and horizontal chord lengths to be same	0.578		0.269	
	Vertical Moment Arm	l_b					3.0	Vertical and horizontal chord lengths to be same	3.0		3.0	
	Horizontal Stabiliser Span	b_h					1.702	Vertical and horizontal chord lengths to be same	1.792		2.267	
	Horizontal Stabiliser Chord Length	c_h					0.425	Vertical and horizontal chord lengths to be same	0.527		0.453	
	Vertical Stabiliser Span	b_v					0.993	Vertical and horizontal chord lengths to be same	1.076		0.733	
	Vertical Stabiliser Chord Length	c_v					0.662	Vertical and horizontal chord lengths to be same	0.538		0.367	
	Aileron Span	b_a							2.072		1.255	
	Aileron Chord Length	c_a							0.130		0.168	
	Aileron Planform Area	S_a							0.268		0.211	
	Elevator Span	b_e					1.532	Vertical and horizontal chord lengths to be same	1.613		2.041	
	Elevator Chord Length	c_e					0.160	Vertical and horizontal chord lengths to be same	0.198		0.170	
	Elevator Planform Area	S_e					0.244	Vertical and horizontal chord lengths to be same	0.319		0.347	
	Rudder Span	b_r					0.894	Vertical and horizontal chord lengths to be same	0.968		0.660	
	Rudder Chord Length	c_r					0.248	Vertical and horizontal chord lengths to be same	0.202		0.138	
	Rudder Planform Area	S_r					0.222	Vertical and horizontal chord lengths to be same	0.195		0.091	
Performance	Minimum Turn Radius	r										
	Maximum Load Factor	n										
Structures	Mass of Fuselage Structure	M_f										
	Mass of Wing Structure	M_w										
	Maximum Deflection at Wing Tip	delta_d_tip										
	Von Misses Stress	sigma_vm										
Propulsion	Vertical Propellor Diameter		42		42		47	Changed to reflect new MTOW				
	Horzontal Propellor Diameter		42		29	Changed to reflect new I/O	40	Changed to reflect new MTOW	35	Changed to reflect propellor design output		
	Vertical Thrust To Weight Ratio	TWR	2		2		2					
	Vertical Motor Power	P_v	4.58		4.58		6	Changed to reflect new MTOW				
	Horizontal Motor Power	P_h	10.7		4.91	Changed to reflect new I/O	5.6	Changed to reflect new MTOW				
Payload	Spray Width (assumed)	w										
	C (litres of payload per hectare) (Assumed)	c										
	Calculated nozzle rate	x										
	Payload mass	mp	66.1	Number found using Pyca Pelican payload weight ratio from MTOW	49.4	Initial payload weight was too heavy. Used marutitonee's litres per acres value to find litrage needed for 2 hectares approximately	47	Used a payload weight fraction of 0.47 from MTOW				
Systems	VTOL flight time	s	30		27		3					
	Cruise Flight time	s	15		3		12					
	Battery weight (total)	m	N/A	Based on original cruise speed, and initial motor selection, batteries were above MTOW	17kg	Updated motor & propellor selection, but battery type not compatible with new motors	15.15	Updated motor & Propellor selection, with updated battery type to be compatible with motors				
	Battery Length (per battery)	l(battery)	N/A	Based on original cruise speed, and initial motor selection, batteries were above MTOW		Updated motor & propellor selection, but battery type not compatible with new motors		Updated motor & Propellor selection, with updated battery type to be compatible with motors				
	Battery width (per battery)	w(battery)	N/A	Based on original cruise speed, and initial motor selection, batteries were above MTOW		Updated motor & propellor selection, but battery type not compatible with new motors		Updated motor & Propellor selection, with updated battery type to be compatible with motors				
	Battery Height (per battery)	h(battery)	N/A	Based on original cruise speed, and initial motor selection, batteries were above MTOW		Updated motor & propellor selection, but battery type not compatible with new motors		Updated motor & Propellor selection, with updated battery type to be compatible with motors				

Weight	Definition (Connection)
1	Only impacts the base system
2	Impacts the base system and one other system
3	Impacts the base system and two or three other systems
4	Impacts the base system and four or five other systems
5	Impacts all systems

Appendix E: Project Management Risk Register

Category	Risks or Problems	Notes / Description	Impact Description	Probability	Severity	Risk Score	Mitigation	Probability	Impact	Mitigated Risk Score
Timeline Risk	Team member conflicts and Disrespectful behaviour	When team members argue or are at odds with each other	Lack of communication and team work, Independent work with no cooperation and ineffective team meetings due to disagreements between team members	2	4	8	Create a team charter that outlines acceptable and unacceptable behaviours	5	3	3
Communications Risk	Absenses	Absenses during in-person meetings	Ineffective catchups post-meeting with absent members may leave the absentees with incomplete information. Absent team members cannot contribute to design decisions and discussions, therefore delaying design decisions.	5	2	10	Meeting summaries sheet, absent members can join meetings through teams and proper post-meeting catchups	4	4	3
Timeline Risk	Easter Break	2 week break from university	Reduces time availible for project and to recieve help / guidence	5	3	15	Assign work goals / deliverables for after the break and take the break into account when making the gantt chart	5	3	5
Timeline Risk	Scheduling Conflicts	When the team members' individual shcedules clash with each other or group activities	Difficulty in meeting and deadlines planning, absenses during meetings and reduced meeting length	5	2	10	Meeting and deadlines are annouced a week in advance, polls are made to decide meeting time and duration and allowing people to join meetings online	5	4	3
Technical Risk	Delivery of poor work	When a team member's deliverable is not made to a good standard	Slow in design development or research, disagreements between team members and if the work was done to a very poor standard (and is needed to continue with the design), it would have to be redone	3	4	12	Informing academic supervisors of poor work from a team member(s)	2	4	3
Technical Risk	Lack of help and guidance from supervisors or PHD students	When supervisors or PHD students do not provide proper help / guidance needed to continue with the project	More time needed to perform analysis, unclear academic goals leading to inadequate outputs and decreased quality of work	3	3	9	Communication through teams and outlook with the supervisors and PHD students. Scheduled weekly meetings	4	2	5
Communications Risk	Ineffective communication	Team members taking too long to reply, ignoring messages or not understanding the messages' content	Team members lacking information, work getting delayed or not done, absenses at team meetings and conflict between team members	4	2	8	Notifications being switched on for Teams and Outlook and team charter outlining work hours	3	3	3
Timeline Risk	Late for submission deadlines	When work is subitted late for a deadline	Grade being capped at 40%	1	5	5	Adding buffer time to account for delays in the project plan / timeline	4	4	1
Timeline Risk	End-of-project time crunch	End-of-project deliverables overload and increased workload as a result of procrastination	Burnout, missed deadlines, poor quality or plagiarized / rushed work delivered	4	5	20	Week-by-week deliverables to ensure the project stays on time and buffer time for deadlines	5	4	5
Communications Risk	Lack of knowledge in assigned role	When a team member has little to no knowledge regarding their assigned role	Delay in design or decreased quality in analysis due to the learning curve, more help needed during the project and other team members having to fill in to help, reducing their capability to do their own work	2	5	10	Proper role allocation based on prior knowledge, skills and interests	5	4	3
Technical Risk	Poor system integration	Subsystems (e.g., aero, structures, payload) are developed in isolation and may not fit together physically or functionally	Sub-systmems will need to be re-designed	2	4	8	Shared values, weekly team meetings, good communication between sub-system teams and integration coordination	4	5	2
Technical Risk	Groupthink	When the entire team agrees on one idea without considering others or when ideas are not shut down to avoid team member conflicts	An overly complex, unachievable design is chosen or obvious flaws persist in the design without being pointed out	3	4	12	Effective disucssion during design meetings where all ideas are explored and evaluated. Frequent meetings with supervisors to review design ideas and to check progess	4	3	4
Technical Risk	Over-designing or doing work that is out of GDBP's scope	When complex designs are persued or very in depth analysis is performed on sub-systems	Failure to deliver a complete, functioning core design by the deadline or severe delays in the work, leading to some aspects being rushed	2	5	10	PM and academic supervisors enforcing the project's scope and outlining what work would be too in depth for GDBP	3	5	2
Communications Risk	Uneven workload	When work is unevenly distributed among the 8 team members	Conflict in the team as some team members will be overworked, delay in design and a higher potential for poor quality work being delivered	3	3	9	The role responsabilities are outlined at the start of the project, deliverables for each team member are discussed as a group and each team member ensures that everyone is doing their work correctly.	4	3	3
Technical Risk	Failure to complete task	Failure to design a system that complies with the assigned requirements	A higher potential for a lower grade in the project and team demotivation that could lead to further delays or problems	1	4	4	Consistently asking for feedback from supervisors and PHD students, PDR and CDR	5	4	1
Timeline Risk	Unavailible rooms or spaces to meet	As there are many GDBP teams this year, a suitable space to work together may be difficult to find	Meetings having to be online which would make cooperation and discussions harder	4	2	8	Booking rooms in advance or finding alternative places to meet on and off campus	4	5	2
Technical Risk	Un-updated general values	When the sheet containing the general values of the aircraft (wing span, MTOW, payload capacity etc...) is not updated when a new iteration on the number is made	Individual group members work using non-up-to-date numbers, leading to incohearent and un-integratable sub-systems	3	3	9	Clear communication when a change occurs and good engineering practice	3	3	3
Communications Risk	Bad leadership	When the project manager does not provide goals or a timeline for the team or is completely absent	Unclear project objectives, delay, decrease in output quality and potential project failure	1	5	5	Assigned roles, secondary group leader and academic supervisor supervision, with them taking action if a lack of leadership occurrs.	4	5	1

Severity		Probability		Impact		Probability	
1	Minor therefore mitigations are not strictly needed	1	Remote	1	This solution will have a moderate impact on the risk / problem with a moderate success rate, but no impact to its root / cause	1	The solution has little chance to be implemented
2	Small therefore mitigations should be considered vaguely	2	Low	2	This solution will have a positive impact on the risk / problem with a moderate success rate but no impact to its root / cause	2	The solution may be implemented if no risks related to its implementation are identified
3	May need to be considered but has no majour impact on the project so mitigations should be considered	3	Moderate	3	This solution will have a positive impact on the risk / problem but little impact on its root / cause with a moderate success rate	3	The solution will be implemented if no risks related to its implementation are identified
4	Will need to be considered and mitigations should be implemented	4	High	4	This solution will have a positive impact on both the risk / problem and its root / cause with a high success rate	4	The solution will be implemented if no major risks related to its implementation are identified
5	Major and will impact the project in multiple ways and therefore must be mitigated	5	Inevitable	5	This solution will solve the risk / problem with a high success rate	5	The solution will be implemented

The PM Risk Register was constructed similarly to the SEBoK Risk Register. Each risk was identified through brainstorming. A name, description and impact description was given to each risk to fully define it. Probability (how likely it is for the risk to occur) and severity (how severe the risk's impact is) scores were assigned to each risk and were multiplied to obtain an overall risk score. Appropriate mitigation measures were then identified and given probability and impact (how successful the mitigation is) scores. The mitigate risk score was calculated by dividing the overall risk score by the mitigation's impact score.

Appendix F: Problem Definition

	Market	Who	What	Why	Where	When	Opportunity	Risks	Competition
Initial Problem Definition	Primary	Smallholder Farmers	Rice farms need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	Current spraying methods: 1) Spray a little area at the time, i. e. poor operational productivity (hectares sprayed per hour) 2) Are fully manual, labour intensive and not automated 3) Heavily expose the farmer to the harmful products being sprayed and are commonly inappropriatly stranged, therefore bringing health risks to the farmer's family and surrounding community 4) Are the only ones available to farmers as they have limited monetary resources	India, Sub-Saharan Africa, Europe	Short Term	Agricultural UAVs are few in this market therefore competition will be limited. The introduction of crop spraying UAVs will remove a large burden to the farmer and will reduce their exposure to the dangerous chemicals, prolonging their health and life expectancy.	1) Limited purchasing power: Smallholder rice farmers have very low financial resources. 2) Low technological literacy: Some farmers may struggle to adopt or operate a UAV technology.	> Manual (knapsack) spraying > Small tractor spraying
	Secondary	Big farm owners + China	Large farmer owners need: 1) a large scale, 2) efficient, 3) automated and 4) productive spraying method	They own large farms (1) and must spray all their crops (4). Multiple spraying devices may be needed and therefore, a farmer cannot monitor multiple spraying operations all at once (2). These spraying methods must be cost effective so the farmer doesn't spend too much money (2)	USA, Australia, China, Latin America	Long Term	1) Reduced labour: Less manual effort required for spraying. 2) Improved safety: Lower exposure to harmful chemicals. 3) Higher productivity: Faster and more consistent spraying. 4) Time savings: Farmers can focus on other tasks. 5) Greater efficiency: More precise application uses less chemical. 6) Scalability: Larger farms gain greater efficiency benefits.	1) High competition: Agricultural UAV markets are already saturated.	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Tractor or Boom sprayers
	Tertiary	Other agricultural applications (mapping and monitoring) and external applications (solar panel cleaning)	Agricultural applications: large farms require crop surveillance to determine health and growth status, livestock owners require livestock surveillance to track the movement of their animals and large land owners require land mapping to plan future plantations or building sites External applications: solar farms could be sprayed with cleaning agents to clean the solar panels. Spraying of greenhouses, payload transport, monitoring purposes, surveillance and mapping are all mission profiles the drone can accomplish	Using a drone for these tasks may increase productivity and / or efficiency. These alternatives uses provide flexibility to the business and allow for expansion	Worldwide	Parallel to primary business / Long Term	UAVs are well-tailored for these tasks, and the external applications for drones are numerous. This allows for additional profit margins in low seasonal farming demands.	The drone platform was not specifically designed for these use cases and could be less efficient than alternatives. Could add more fixed costs to the company.	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Current methods of performing the task

	For the Consumer	For the Company	What is being designed?	
Service Model Unique Selling Point	Speed of spraying Lower cost as our company can offer bundles of multiple services in one go	Less equipment is needed to perform multiple mission types as our drone is modular and can be adapted to perform multiple mission profiles	One sentence summary	A drone capable of spraying crops autonomously (with an initial operator input), optimressed for medium sized farms (30 to 120 ha) with the possibility of being adapted for different mission profiles (such as mapping, surveillance and other spraying applications).
Sales Model Unique Selling Point	The drone is modular and can perform multiple mission profiles	The drone is modular and can perform multiple mission profiles and therefore may be more attractive		

	Market	Who	What	Why	Where	When	Opportunity	Risks	Competition
Revised Problem Definition	Primary	Mediumholder (30 to 120 hectares) broad acre crop farms	Corn, wheat, soy-bean, rice and grains need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	These crops have height uniformity, canopy can be easily penetrated by spraying particulates, and the high volume needed for these farms make it ideal for drone for drone application. This is also the most common crop type farmed globally and therefore will have the highest amount of customer farmers. 30 to 120 hectare farms were selected as a target as they are the ideal size range for the drone.	India, South America, Sub-Saharan Africa	Short Term	They utilise a high volume of herbicides and pesticides, application can be made more efficient with drones. Crops may get damaged with other methods of spraying (tractor and manual), therefore higher crop yields may be aquired with drone spraying. Middle market not directly addressed	1) Limited purchasing power for the farmer 2) High development cost and time 3) Competition from other drone platforms and current spraying method alternatives	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Tractor or Boom sprayers
	Secondary	Smallholder broad acre crop farmers (in bulk), largeholder broad acre crop farmer	Corn, wheat, soy-bean, rice and grains need to be sprayed with herbicides, pesticides and fertilisers periodically during the season.	These crops have height uniformity, canopy can be easily penetrated by spraying particulates, and the high volume needed for these farms make it ideal for drone for drone application. This is also the most common crop type farmed globally and therefore will have the highest amount of customer farmers. Servicing smallholders in bulk will provide a cost effective and safer alternative to labour intensive manual spraying. Given the extended range of the drone, large farms can also be serviced.	Smallholders: India, Sub-Saharan Africa Largeholders: Australia, USA, South America	Short to Medium Term	Smallholders: Agricultural UAVs are few in this market therefore competition will be limited. The introduction of crop spraying UAVs will remove a large burden to the farmer and will reduce their exposure to the dangerous chemicals, prolonging their health and life expectancy. Largeholders: Large farms require an autonomous and high output spraying platform	1) Limited smallholder purchasing power 2) Low technological literacy for smallholders 3) Competition from other drone platforms and current spraying method alternatives 4) Saturated largeholder market	> Manual (knapsack) spraying > Small tractor spraying > Tractor or Boom sprayers > Other spraying / agriculture drone platforms (Dji Agras, XAG Agri)
	Tertiary	All other crop types and farm sizes	Any other crop that requires spraying	This is the primary mission profile for the drone, although it is optimised for broad acre farms, it can still be used for other crop types	Worldwide	Long Term	Most crop types require spraying, automating this process would lead to lower farmer fatigue and higher crop yeild	The drone platform was not specifically designed for these crop types and therefore other competitor products explored specifically to a given crop type will be more convenient to the farmer	> Manual (knapsack) spraying > Tractor or Boom sprayers > Other spraying / agriculture drone platforms (Dji Agras, XAG Agri)
	Quaternary	Other agricultural applications (mapping and monitoring) and external applications (solar panel cleaning)	Agricultural applications: large farms require crop surveillance to determine health and growth status, livestock owners require livestock surveillance to track the movement of their animals and large land owners require land mapping to plan future plantations or building sites External applications: solar farms could be sprayed with cleaning agents to clean the solar panels. Spraying of greenhouses, payload transport, monitoring purposes, surveillance and mapping are all mission profiles the drone can accomplish	Using a drone for these tasks may increase productivity and / or efficiency. These alternatives uses provide flexibility to the business and allow for expansion	Worldwide	Parallel to primary business / Long Term	UAVs are well-tailored for these tasks, and the external applications for drones are numerous. This allows for additional profit margins in low seasonal farming demands.	The drone platform was not specifically designed for these use cases and could be less efficient than alternatives. Could add more fixed costs to the company.	> Other spraying / agriculture drone platforms (Dji Agras, XAG Agri) > Current methods of performing the task

Appendix G: Project Stakeholders

Category	Stakeholder	ID	Association with the Project	Interest in Project Aspect	Specific Concerns	Influence Score	Interest Score	Total Score
Technical Team	Design / Engineering team	T1	This team designs and tests the drone	Technical / System Integration	Feasibility of the architecture, optimizing payload-to-weight ratios, balancing battery life with spray capacity, and integrating components flawlessly.	1	5	4
	Operations team	T2	This team sets up and operates the drone while it is spraying, they are the service providers	Operational / Safety & Usability	Ease of drone operation, safety from chemical exposure, quick payload and battery reload between missions to minimise downtime, and effective service delivery for customers.	3	4	7
	Mechanics	T3	This team repairs the drone if it gets damaged	Operational / Repairability	Modular design for quick field repairs, reliable availability of spare parts (COTS components), and standardized tooling requirements.	3	4	7
	End-of-life disposal agencies	T4	This team oversees the disposal of the drone at the end of its life	Environmental / Recycling	Safe handling of hazardous materials, end-of-life processes for batteries, recovering precious metals, and neutralizing chemical-contaminated payload tanks	2	2	4
Manufacturing	Off-the-shelf component suppliers	M1	These companies supply the off-the-shelf components used in the drone	Supply Chain / Commercial	Clear technical specifications, consistent volume forecasting, lead times, and timely payments from the drone manufacturer.	1	1	2
	Bespoke component manufacturers	M2	This company manufactures the custom / biskpoke components for the drone	Technical Specs / Tolerances	Feasibility of custom designs (e.g., proprietary atomizing nozzles), production scalability, and intellectual property (IP) protection.	2	2	4
	Final assembly / manufacturing team	M3	This company assembles the drone	Production / Quality Assurance	Manufacturability of the drone design (Design for Manufacturing), assembly line safety, testing bottlenecks, and labor costs.	2	1	3
Rest of the Company	Business team	R1	This team identifies the target market, the client needs and outlines a set of needs / requirements that then influence the design	Commercial / Profitability	Cost of Goods Sold (COGS), pricing strategy, route to market (selling to FPOs/CHCs), competitive advantage, and achieving target profit margins.	3	5	7
	Other company employees	R2	These teams run all other company functions that do not include: technical design and business related tasks	Organizational / Job Security	Company stability, workplace safety, internal resource allocation, and how the new drone project impacts their existing workloads.	1	3	3
End users / Customers	Target farmer	U1	This is the primary target customer	Economic / Performance & Yield	Affordability of the spraying service, guaranteeing zero physical damage to the crop, reduced health risks to the farmer and highly effective pest control and increased crop productivity.	3	5	9
	Farmer's family	U2	These either work with the farmer or influence the farmer's decisions	Social / Health & Livelihood	Reduced exposure to toxic pesticides for the farmer (health benefits), improved household income, and potential job displacement if they previously relied on manual spraying wages.	1	4	4
	Surrounding Farmers	U3	These have the closest relationship with the farmer and may be a source for more customers. They also provide opportunity for the bulk spraying business model but may also hindure the service	Economic / Performance & Yield and Social / Environmental & Safety	Fear of pesticide "drift" contaminating their own crops or water supply, noise pollution from heavy multi-rotors, and privacy concerns regarding the drone's cameras if they are not part of the bulk operation	4	3	7
	Farmer's village	U4	These have a close relationship with the farmer and may be a source for more customers. They also provide opportunity for the bulk spraying business model but may also hindure the service	Economic / Performance & Yield and Social / Environmental & Safety	Fear of noise pollution from heavy multi-rotors and privacy concerns regarding the drone's cameras if they are not part of the bulk operation	2	4	5
	Consumers for Non-Spraying Drone Services	U5	This is the secondary target customer	Task Automation	Livestock or crop monitoring, landscape mapping, solar panel washing and other external drone applications	2	3	5
Governing Bodies	Local government	G1	Provides guidelines and grants that enable the company to operate the drone and deliver a service. They could also be a source of income for the target customer leading to potentially more sales	Macro-Economic / Strategic	Success of modernization schemes, creating rural tech employment, and improving national food security	4	2	7
	Agricultural regulators	G2	Provides guidelines and grants that enable the company to operate the drone and deliver a service.	Regulatory /Environmental & Chemical Safety	Approval of specific pesticide formulations for Ultra-Low Volume (ULV) aerial spraying and verifying that the drone actively reduces chemical and water waste	5	4	9
	Aerospace / Airworthyness regulators	G3	Provide guidelines and grants that enable the company to operate the drone and deliver a service.	Regulatory / Aviation Safety	Airworthiness type certifiion, "No Permission, No Takeoff" (NPNT) compliance, strict geo-fencing to prevent flight in restricted zones, and weight limits.	5	2	8
Business Partners	Farmers Union companies	B1	Help the company reach more customers	Economic / Farmer's Welfare	Increased walefare and productivity for farmers and farming modernisation	3	3	6
	Farmers financial support services	B2	Help the company reach more customers	Financial / Risk & Repayment	The economic viability of the drone as a financed asset, the likelihood of the farmer/FPO defaulting on loans, and proof of increased farm revenue to justify credit risk.	3	3	6
	Investors / Stockholders	B3	Provide funding for this business venture	Financial / Growth & Returns	Market capture timeline, intellectual property defensibility, scaling the business model, and overall return on their capital investment.	4	5	9
Utilities Providers	Manufacturers of sprayed products	P1	Provide the product that is sprayed onto the fields	Technical / Chemical Compatibility	Ensuring their specific liquid or nano-fertilizers do not clog the spray nozzles, and that the drone's downwash provides correct canopy penetration.	3	3	6
	Connectivity providers	P2	Provide the 4G and / or satellite connection that ensures drone navigation	Technical / Network Infrastructure	Bandwidth requirements for telemetry, managing network handovers during flight, and ensuring stable 4G/5G coverage in remote rural areas without signal drops	2	1	3
	Electricity providers	P3	Provide the electricity used to recharge the drone's batteries	Operational / Energy Grid	The impact of localized grid load spikes caused by charging multiple high-capacity drone batteries in rural areas with fragile electrical infrastructure.	1	1	2
	Logistics for transport	P4	Provide the means to transport parts to manufacturers and our drone to farmers to perform the service	Operational / Supply Chain & Distribution	Form factor and foldability so the drone fits into standard small commercial vehicles; vibration and shock resistance to survive deeply rutted, unpaved rural roads; and strict hazardous materials (HazMat) compliance for safely storing and transporting large, high-capacity lithium-ion batteries without fire risks	2	1	3
Other	Insurance	O1	Provide finicial security in the event of mishaps / accidents	Financial / Risk Mitigation	Liability coverage for crashes or property damage, crop damage claims from chemical drift, and the availability of black-box flight logs to investigate incidents.	1	1	2
	Competitors	O2	Provides an alternative to our service	Market Dominance / Competition & Technical Differentiation	Cost-to-Payload Ratio: They worry about new entrants undercutting their price-per-liter capacity. Service Accessibility: They are concerned with "field-repairable" designs that bypass their centralized (and often slow) authorized repair networks. Regulatory Compliance: They monitor for technical shortcuts (like speed-cap bypasses) that could lead to unfair advantages or safety scandals affecting the whole industry	3	2	5
	3rd party product testers	O3	Provide external review of the system and therefore may bring positive publicity	Technical / Validation	Having unbiased, standardized methodologies to rigorously test the drone's spray uniformity, battery endurance, and safety claims before public release.	1	2	3

Influence		Interest		Total Score		Total Score Formula
How much they can influence the project (change the design)		How invested they are to this project		The resulting importance of a stakeholder		**The Influence score was given a weighing of 1.4 and the Interest score was given a weighing of 0.6 for the evaluation of the total score. This was done because a stakeholder's influence on the project is more important than their interest in it. Formula: 1.6*Infl + 0.6*Intr = Tot Sc**
1	Has little to no influence on the project	1	Has little to no interest on the project	1 - 2	Stakeholder is remotely connected to this project	
3	Is considered a driver in design decisions	3	Is interested in the project	3 - 5	Stakeholder is connected to this project	
4	Is a majour driver in design decisions	4	Has a large interest in the project	6 - 8	Stakeholder is an important member of this project	
5	Is the one of the main drivers of design decisions	5	Is solely focused on this project	9 - 10	Stakeholder is a key member of this project	

Appendix H: Source of Needs / Drivers & Constraints

Category	Need or Constraint	ID	Description / Notes	Impact on the project	Related Stakeholder	Source
Drivers	Pesticides needed to grow rice in south-east asia	D1	N/A	Drives the main function of the drone	Target Farmer	https://rrojelife.org/wp-content/uploads/pdf_files/Case-Study-108-Rice-in-Asia2.pdf
	Herbicides needed to grow crops	D2	N/A	Drives the main function of the drone	Target Farmer	https://scijournals.onlinelibrary.wiley.com/doi/full/10.1002/ps.3598
	Fertilizer needed to increase crop yield	D3	N/A	Drives the main function of the drone	Target Farmer	https://d1wqtxts1xzle7.cloudfront.net/31216237/579-584.doc-libre.pdf?1392254554=&response-content-disposition=inline%3B+filename%3DEffect_of_Nitrogen_and_Phosphorus_Fertil.pdf&Expires=1727607876&Signature=gWBuK3h5DXx-gVkNCF3rQ4sQy7rvsTpTvCY27qAN4qGsiwXTpLMzvEAFMJtcTZtQSFGrBuydvJRFFiaz9eYpDkuCy1BuTTEWUJNpJYUCpETTYDuBy8acTCqLP0LryvJMJTTMN4sN9dTTaEfDkMkDvy-QpNbaeb4xBMAer6rkZ2HFHUhaaQ6uB2oF2-YbJWPkDCtgzAZAFCHJwfiBuk4OUPRAgowG6CCF-pAnoHGrfaxJOb0KBOuYkiuXEJEu34HkZotUB1OhB-tCfx7CzmOiuWMxHJaUeBoFaYKGfcxfDOe2Ib-rpJ87zlNGSTJrPUbNaUaUcKZSwN74rt1iZISENOcsmw__&Key-Pair-Id=APKAIUCHFSGGSLRBV4ZA	
	Manual spraying is operationally inefficient and a better alternative is needed	D4	N/A	Drives the main function of the drone and will give an operational productivity target	Target Farmer	https://www.gflprayer.com/How-Many-Acres-Can-A-Sprayer-Spray-id41906936.html
	Manual spraying is labour intensive	D5	N/A	Drives the main function of the drone	Target Farmer	Group Brainstorming and Discussions
	Current spraying techniques bring health risks to farmers, therefore an alternative is needed	D6	N/A	Drives the main function of the drone and will set a spray drift limit	Target Farmer, Farmer's Family	https://doi.org/10.4103/ijoem.ijoem_121_18
	More automation in farming brings benefits	D7	N/A	Drone will be designed to be autonomous	Target Farmer	https://www.plugandplaytechcenter.com/insights/how-automation-transforming-farming-industry
	Increased use of Agricultural drones in the farming industry	D8	N/A	A USP (unique feature or better specs) will have to be developed for the UAS to gain market share from the competition.	Target Farmer	https://www.fortunebusinessinsights.com/agriculture-drones-market-102588#:~:text=KEY%20MARKET%20INSIGHTS,Download%20Free%20Sample
	Need to increase agricultural output	D9	N/A	Drives the main function of the drone	Local Government	https://www.spglobal.com/en/research-insights/special-reports/india-forward/making-agriculture-more-efficient-for-food-security-and-sustainable-growth
	A valuable business proposition and availible markets	D10	The drone needs to have a business justification	Informs the requirements and specs of the drone and will provide external funding	Business team, Investors / Stockholders	Group Brainstorming and Discussions
	Governments push to make farming more attractive to a new generation	D11	Governments givning incentives to increase the farming workforce	May provide subsidies to the company	Local government, Design / Engineering team, Business team	https://www.researchgate.net/publication/387836211_Initiatives_to_Attract_and_Retain_Youth_in_Farming
Constraints	Drone's need for infrastructure independence	C1	The drone must not need any infrastructure on the farm for it to operate	May limit take-off options, charging possibilities, drone autonomy solutions (longer range GPS may have to be used) and set-up processes	Target Farmer	Group Brainstorming and Discussions
	Drone operations being weather resiliant	C2	N/A	The drone may have to possess and IP rating and be capable of flying in difficult weather conditions	Target Farmer, Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Drone needing flexible mission capabilities	C3	Spraying irregular boundaries, obstacle filled fields and varying crop heights	Will add development time and complexity to the drone's software	Target Farmer	Group Brainstorming and Discussions
	Drone noise restrictions	C4	Compliance with noise guidelines set by aerospace regulators	Will add development time and complexity to the drone's hardware	Farmer's Family, Surrounding Farmers, Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Aerospace regulations	C5	N/A	Will add development time and complexity to the drone's hardware and software	Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Limited company budget	C6	N/A	May limit the design complexity and overall specs of the drone	Design / Engineering team, Business team	Group Brainstorming and Discussions
	Farmers prefer simple and intuitive equipment to use	C7	N/A	The drone platform will need to be engineered to be intuitive to use	Target Farmer	Group Brainstorming and Discussions
	Use current availible technology	C8	No new bespoke technology (sensors, motors or any electrical system) can be developed for this drone, only COTS components can be used	May limit design capabilities but will serve as a cost control method as developing new components is expensive	Design / Engineering team	Group Brainstorming and Discussions
	Use current manufacturing techniques	C9	No new materials or manufacturing techniques can be developed for this drone	May limit design capabilities but will serve as a cost control method as developing new materials or manufacturing processes is expensive	Design / Engineering team, Mechanics	Group Brainstorming and Discussions
	Limited farmer monetary resources	C10	Farmers may not have a enough money to afford a drone	The "spraying as a service" model may be adopted	Target Farmer	https://www.action.org/growing-resilient-futures-for-india-s-small-scale-farmers/
	Drone pilots prefer drones that can be piloted / operated intuitively	C11	N/A	Will add development time and complexity to the drone's hardware and software	Operations team	Group Brainstorming and Discussions
	Time efficiency with processes	C12	The farmers want the spraying operation to be efficient, i.e. for the setup to not take too long so multiple fields can be sprayed. Simple to use systems are preferred	If the "spraying as a service" model is chosen, down time in between missions will have to be limited	Target Farmer, Operations team	Group Brainstorming and Discussions
	Limited development time	C13	Only 12 weeks were given for this project	Simpliciy may become a main driver of this project	Design / Engineering team	Group Brainstorming and Discussions
	Small design team with limited experience	C14	The team only consists of 8 engineering students	Simpliciy may become a main driver of this project	Design / Engineering team	Group Brainstorming and Discussions
	Drone repairability	C15	Repairability becomes easier if the system can be dis-assembled easily and components are accessible	Will add development time and complexity to the drone's hardware and limit certain design solutions	Operations team, Mechanics	Group Brainstorming and Discussions
	Spray drift regulations	C16	Spray drift must be limited to ensure safety and no contamination of nearby plantations	Will impact the payload spraying design	Agricultural regulators, Surrounding Farmers, Farmer's village	https://www.epa.nsw.gov.au/Your-environment/Pesticides/preventing-pesticide-misuse/reducing-conflict-avoiding-spray-drift#:~:text=You%20can%20also%20*%20Monitor%20weather%20conditions,on%20131.%29%20to%20report%20a%20possible
	Regulation compliance	C17	The drone cannot be or operate illegally	This will limit what markets can be accessed or what permits must be obtained to spray in a given country	Local government, Agricultural regulators, Aerospace / Airworthyness regulators	https://www.hse.gov.uk/pesticides/using-pesticides/general/aerial-spraying.htm#:~:text=To%20use%20pesticides%20by%20aerial%20in%20the%20air%20a%20trial%20*%20Conducting%20efficacy%20data%20trials
	Existing spraying products compatibility	C18	Spray products already exist so the payload system needs to be engineered around them	Will limit the design possibilities of the payload system as liquid viscosity may have an effect	Design / Engineering team, Target Farmer	Group Brainstorming and Discussions
	Crop Spraying windows	C19	Crop spraying only occurs during specific times of the season	Will lead to complex timetabling when multiple farmers will need to be serviced. This may lead to a larger drone fleet needed and therefore manufacturing costs should be reduced, this will impact the design	Design / Engineering team, Target Farmer	https://twigachemicals.com/wp-content/uploads/2022/03/RICE-spray-programm-1.pdf https://biothai.org/about-the-rice-we-eat/
	Spraying product formulae and chemistry	C20	The spraying products may be corrosive to some materials and cannot be contaminated with other chemicals or their effectiveness may reduce	Will limit what materials can be used for the spray tank and the areas around the loading port(s)	Design / Engineering team	https://content.ampp.org/corrosion/article-abstract/6/5/137/27079/Corrosion-of-Metals-by-Insecticidal-Solutions?redirectedFrom=fulltext
	Sustainability	C21	Sustainability is a main focus on modern engineering projects and efforts must be made to minimise enviromental impact	May limit what materials, processes and power generation systems are used on the drone	Design / Engineering team, Local governement	Group Brainstorming and Discussions
	Drone operations safety	C22	The drone system needs to be safe for it to be allowed to operate	Will determine if the drone can be operated of if more development time is needed	Local government, Agricultural regulators, Aerospace / Airworthyness regulators	Group Brainstorming and Discussions
	Farm location	C23	Some farms may be located in remote areas or at high altitudes	The drone will need to be able to operate at high altitudes and a transport solution will need to be determined to move the drone from farm to farm	Design / Engineering team	https://mapsforupsc.com/largest-rice-producing-states-in-india/ and https://www.researchgate.net/figure/Map-showing-states-and-elevation-zones-of-India_fig1_315835800

Appendix I: Risk Register

Category	Risk	ID	Notes / Description	Impact Description	Associated Stakeholder	Probability	Severity	Risk Score	Mitigation	Probability	Impact	Mitigated Risk Score	Status	Date identified	Date Closed	Closure Comments
Commercial Risk	Competition	R1	Established ag-drone manufacturers (like DJI or XAG) or local manual labor and tractor services offering cheaper, more familiar alternatives.	Loss of market share, forced price reductions to remain competitive, and difficulty securing long-term service contracts with farmers.	Design / Engineering and Business teams	4	4	16	Offer a strong unique selling point over the competition. Meet the needs of the target market	4	3	5	Open	18.Mar		
Commercial Risk	Lack of consumer demand	R2	Farmers showing skepticism toward drone technology or being unwilling to change their traditional, proven spraying and mapping methods.	Low adoption rates, stagnation in sales or service bookings, and inability to clear the high upfront R&D and manufacturing costs.	Smallholder farmer, Large farm owner, Business team and Operations team	1	5	5	Offer a "Service-as-a-Product" model to lower the entry barrier, allowing farmers to benefit from the tech without the upfront purchase and training costs. Price the product at a loss for introductory schemes for market penetration, have good relations with customer and referral schemes	4	5	1	Open	18.Mar		
Commercial Risk	Lack of funding from external bodies	R3	Insufficient capital secured from venture capitalists, government agricultural grants, or bank loans to support ongoing operations.	Insufficient funds to continue or develop the platform, leading to stalled R&D, inability to scale production, or potential bankruptcy.	Design / Engineering and Business teams, Investors, Local government and farmers financial services	2	5	10	Apply for targeted government agricultural innovation grants and establish early paid pilot programs to demonstrate traction to investors. Investigate in marketing initiatives.	3	3	3	Open	18.Mar		
Commercial Risk	Low profit margin	R4	High manufacturing / development costs for drone components combined with the low price ceiling that cost-conscious farmers are willing to pay.	Entering the market with a massive debt burden, severely restricted cash flow, and an inability to reinvest in future technology upgrades.	Business team, Investors and Competition	2	3	6	Standardize components across the fleet to achieve volume discounts and automate flight planning to reduce the labor hours required per acre. Market diversification	5	4	2	Open	18.Mar		
Technical Risk	Change in regulations, becoming more strict	R5	Aviation authorities (like EASA or FAA) or environmental agencies introduce stricter compliance laws.	Project delays, increased R&D costs to redesign systems, or complete denial of operational authorization.	Smallholder farmers, Airworthiness regulators, Agricultural regulators, Local government and Design / Engineering Team	2	5	10	Design the aircraft to exceed current safety standards (e.g., building in independent flight termination systems) to ensure future compliance via software updates.	3	5	2	Open	18.Mar		
Commercial Risk	High initial setup and manufacturing costs	R6	Large capital expenditure required for tooling, molds, and early-stage production lines.	Entering the market with a large debt, reducing cash flow and lowering net profit margins.	Business teams and Investors	4	4	16	Utilize modular carbon fiber components and 3D printing for early production, only outsourcing heavy tooling after securing minimum order volumes.	4	3	5	Open	18.Mar		
Programme Risk	Poor roads around the target farm	R7	Poor rural infrastructure limits access to fields or target application sites.	Difficulty in reaching the target site with the ground station, leading to delayed missions and reduced operational efficiency.	Farmer, Drone operator / pilot team	1	4	4	Mount the ground control station and charging infrastructure on an off-road-capable vehicle allowing it to navigate on muddy and unpaved farm tracks.	2	3	1	Closed	18.Mar	14.04.2026	Ford Transit van was considered as the main transport vehicle for the service
Programme Risk	Unpredictable weather	R8	Heavy rain, high winds, or extreme heat preventing safe flight operations.	Flight cancellations, missed narrow spraying windows, and potential physical damage to the airframe or electronics.	Smallholder farmer, Large farm owner and Operations team	4	5	20	Integrate hyper-local weather telemetry at the ground station and establish clear "Go/No-Go" operational thresholds for wind speed and rain.	4	4	5	Open	18.Mar	27.Apr	Design freeze produced a working drone concept that satisfied all operational requirements
Programme Risk	Scheduling complexity / narrow spraying windows	R9	Short optimal timeframes to spray specific crops combined with multi-client coordination.	Issues with farmer organization and agricultural society organization leading to underutilized assets and loss of revenue.	Smallholder farmer, farmer's neighbour, Business team and Operations team	3	3	9	Develop automated fleet-scheduling software that groups neighboring farms and optimizes routes based on crop maturity and weather windows. Have good relations with the target farmers and offer referral schemes to align their spraying timetable	4	5	2	Open	18.Mar		
Programme Risk	Skilled labour shortage	R10	Difficulty finding or retaining qualified drone pilots, engineers, and agronomists.	All design, operation, assembly, and/or business team leaves the company, halting development and flight operations.	Design / Engineering team, Operations team, Technicians Mechanics, Business teams and other company employees	2	4	8	Design the ground station UI to be highly automated and intuitive, reducing the pilot's cognitive workload and specialized training requirements. Provide appropriate training	5	3	3	Open	18.Mar		
Programme Risk	Supply chain issues	R11	Bottlenecks in sourcing specialised drone components, carbon fiber, or batteries.	Parts not delivered or delayed or delivered out of spec or with defects, causing severe production timeline slips.	Off-the-shelf component suppliers, Bespoke component manufacturers and Final assembly / manufacturing firm	2	4	8	Source standardized, off-the-shelf industrial parts where possible and maintain relationships with at least two distinct suppliers for critical components.	4	4	2	Open	18.Mar		
Technical Risk	Bad signal	R12	High interference or lack of satellite/cellular coverage in remote agricultural areas.	Difficulty in mapping and communication, leading to lost telemetry and increased risk of flyaways.	Farmer, Drone operator / pilot team	3	3	9	Equip the drone with high-gain directional antennas and a dual-link command system that combines long-range RF with 4G/5G cellular fallover.	3	3	3	Closed	18.Mar	27.Apr	RTK technology along with moxable antennae will be set around the field, ensuring full coverage
Technical Risk	Bad charging infrastructure	R13	Rural areas lacking high-voltage grids to rapidly recharge high-capacity drone batteries.	No available electrical power outlets, leading to extended downtime between battery swaps and lower daily acreage coverage.	Farmer, Drone operator / pilot team	5	1	5	Use a mobile support trailer equipped with a high-capacity generator or a bank of field-swappable batteries that charge overnight at a central hub.	2	4	1	Open	18.Mar		
Technical Risk	Missing take-off and landing ground clearance	R14	Uneven terrain, high crops, or muddy ground preventing safe VTOL operations.	Lack of space to safely takeoff/land, risking prop strikes and damage to the airframe or payload.	Farmer, Drone operator / pilot team	1	5	5	Utilize a raised, portable folding platform or a heavy-duty rubber landing mat to keep propellers clear of high crops and mud.	4	2	3	Closed	18.Mar	14.Apr	Ford Transit can operate as a take-off and landing pad
Technical Risk	Obstacle filled field	R15	Fields containing power lines, trees, fences, or towers.	High probability of mid-air collisions, requiring advanced obstacle avoidance systems which increase hardware costs.	Drone operator / pilot team	2	3	6	Implement obstacle avoidance systems, adequate navigation software and mandate a pre-flight mapping pass to mark power lines and trees.	5	5	1	Open	18.Mar		
Technical Risk	Irregular field shape / complex geographical topology	R16	Steep slopes or non-uniform boundaries making automated flight pathing difficult.	Inefficient battery usage, uneven chemical application, and complex manual piloting overrides required.	Smallholder farmer, Drone operator / pilot and Design / Engineering Team	5	2	10	Irregular fields and topology will be considered during the development of the navigation algorithms and utilisation of 3D terrain-following radar and AI flight-path software that automatically adjusts altitude to maintain a constant height above the crop canopy.	4	4	3	Open	18.Mar		
Technical Risk	Current technology barrier	R17	Desired flight times, payload capacities, or autonomous features exceeding existing commercial tech.	Drone requires not yet developed or matured technology to function/operate, stalling release.	Design / Engineering team	1	5	5	Consideration of current technological limits and having an apropiate Technology Readyness Level	2	3	2	Closed	18.Mar	27.Apr	Design freeze produced a working drone concept, with availible COTS components being used
Programme Risk	Delay in hardware / software development	R18	Unforeseen bugs, structural failures during testing, MTOW becoming too great or complex coding tasks.	Engineering team takes too long to design drone, allowing competitors to capture market share first.	Design / Engineering team	3	4	12	Use "Hardware-in-the-Loop" simulation to run hardware stress testing and software debugging in parallel before physical prototypes are built.	1	4	3	Open	18.Mar		
Programme Risk	Certification and regulatory delay or denial	R19	Failing to achieve EASA Specific category authorization or CS/CE labeling on time.	Inability to legally operate or sell the product, leading to dead capital and zero revenue.	Local government, Agricultural regulators, Aerospace regulators (DGCA and EASA) and Design / Engineering Team	3	5	15	Engage with the National Aviation Authority (NAA) at the beginning of the design phase to align documentation with SORA standards from day one.	2	4	4	Open	18.Mar		
Technical Risk	Inappropriate choice / specing of commercial off-the-shelf components	R20	Selecting parts that are too heavy, underpowered, or incompatible with the system.	System failure, degraded flight performance, and wasted capital on unusable stock.	Design / Engineering Team	2	5	10	Build a small-scale, inexpensive flying prototype to prove the mathematical modeling, COTS components sourcing and propulsion specs before purchasing full-sized industrial parts.	4	5	2	Closed	18.Mar	27.Apr	Design freeze produced a working drone concept, with availible COTS components being used
Technical Risk	Discrepancy between predicted / theoretical behaviour and real life behaviour	R21	CAD modeling and stress analysis failing to account for real-world environmental wear.	Unexpected structural failures, weight creep, or poor lift generation during actual flight testing.	Design / Engineering team, Operations team, Technicians Mechanics	2	4	8	Enforce a strict margin of safety in CAD stress testing (e.g., testing to 1.5x or 2x expected load) and have assumptions derived from prior research based on more realistic conditions.	5	5	2	Open	18.Mar		
Technical Risk	Unsustainable lifecycle	R22	Quick degradation of expensive components like high-discharge lithium batteries.	Product lifecycle is inappropriate for its use case / product is "pushed too hard," leading to high client churn.	Design / Engineering Team, Operations Team	1	3	3	Implement an active battery management system (BMS) to prevent over-discharging and design high-wear parts like spray nozzles to be easily replaceable. Implement a safety factor and clear operational window.	5	5	1	Open	18.Mar		
Commercial Risk	Sustainability issues	R23	Excessive power consumption or reliance on non-recyclable composite materials.	Our product fails to be sustainable, limiting access to green grants and harming brand reputation.	Design / Engineering team	3	1	3	Evaluate recyclable thermoplastics or bio-composites for non-structural farings and prioritize efficient electric propulsion over gas-powered systems.	5	5	1	Open	18.Mar		
Commercial Risk	Elevated running costs	R24	Higher overheads than projected for operations, transport, and staffing.	Maintenance, battery replacement, transports, wages, etc., eating heavily into profit margins.	Business team, Investors	3	3	9	Design for maximum reliability to extend time between overhauls and train local operators for basic field repairs to avoid shipping delays.	4	4	2	Open	18.Mar		
Technical Risk	Poor spraying	R25	Uneven chemical distribution, drift due to wind, or incorrect droplet size.	Crop damage or ineffective pest control, causing customer complaints and potential legal liability.	Design / Engineering team	2	5	10	Research and design into spraing mechanics and proper prototype field testing	1	2	5	Open	18.Mar		
Technical Risk	Operating in unpreferable environmental conditions	R26	Demanding the drone fly in dusty, hot, or highly humid conditions typical of farm environments.	Still able to operate in various conditions with wind, heat, dust, humidity, light rain, but accelerates component wear.	Design / Engineering team	4	1	4	Rate all critical internal electronics to IP55 (dust-tight and water-resistant) and use conformal coating on all circuit boards to protect against humidity.	3	1	4	Open	18.Mar		
Programme Risk	Rework after failed tests	R27	Prototypes failing physical drop, stress, or flight tests.	Requiring reworks on early stages of manufacturing/testing forcing redesign or rebuild and increasing R&D burn rate.	Design / Engineering team, Operations team, Technicians	4	4	16	Adequate planning buffer time and established countermeasures such as modular design	4	4	4	Open	18.Mar		
Commercial Risk	Seasonal demand volatility	R28	Agriculture being highly dependent on specific growing and harvesting seasons.	Revenue disruption due to seasonal requirements, causing severe cash-flow issues in the off-season.	Operations team, Business team	5	2	10	Pivot the drone's use case in the off-season for tasks or other external applications, i.e. market diversification.	4	2	5	Open	18.Mar		
Technical Risk	Regulatory non-compliance	R29	Airworthyness failed, flying outside of approved parameters or using uncertified radio frequencies.	Heavy government fines, grounding of the fleet, or total revocation of operating licenses.	Design / Engineering team	1	5	5	Hard-code flight boundary limits directly into the flight controller to physically prevent the drone from flying outside authorized operational volumes. Examine and follow all regulations this drone category	3	3	2	Closed	18.Mar	27.Apr	Current design (frozen at CDR) is fully compliant with regulations
Technical Risk	Incapable of VTOL at 1000 m	R30	Air density drops at high altitudes, reducing rotor thrust and lift capacity.	Inability to service farms located in higher altitude or mountainous regions, narrowing the target market.	Design / Engineering Team, Operations Team	1	4	4	Utilize oversized, high-torque rotors or variable pitch propellers to ensure the drone maintains an adequate lift-to-weight ratio in thinner mountain air.	3	4	1	Closed	18.Mar	14.Apr	VTOL motors and power was oversped-ed in the design stage, granting a max operating ceiling of over 1,500 m
Commercial Risk	Service is priced over the target	R31	High manufacturing and operation costs force the end-user price up.	Farmers default to manual labor or tractor spraying, leading to extremely low product adoption rates.	Business team and Target Farmer	3	4	12	Minimize total part count in the airframe design to reduce machining and assembly labor costs, bringing down the end-user price.	4	4	3	Open	18.Mar		
Technical Risk	Gaps in current regulation	R32	Operating in a legal "gray area" where drone laws have not yet been written for large chemical ag-drones.	Sudden forced grounding of the fleet when new laws are abruptly introduced to close the gaps.	Design / Engineering team	2	3	6	Voluntarily design the operation to meet the highest foreseeable safety standards (like SAIL IV) to insulate the business from sudden rule shifts.	4	4	2	Open	18.Mar		
Technical Risk	Bottleneck in pilot training	R33	Complex flight control or heavy training requirements limiting the available pilot pool.	Inability to scale the service side of the business to match drone production output.	Operations team	2	2	4	Partner with existing accredited aviation training schools to handle the ground school and practical STS hours on our behalf.	5	5	1	Open	18.Mar		
Technical Risk	System unreliability	R34	Intermittent hardware glitches, sensor dropouts, or loose physical connections.	Mid-mission failures and forced emergency landings, damaging client confidence.	Design / Engineering team, Operations team, Technicians	2	3	6	Implement redundant flight controllers and sensors so the drone can land safely even if a primary component fails.	4	4	2	Open	18.Mar		
Technical Risk	Weight creep	R35	Adding sensors and structural reinforcements that push the drone over its MTOW limit.	Drastically reduced flight times, loss of VTOL capability, or forced downgrading of the payload tank size.	Design / Engineering team	5	3	15	Enforce a strict "weight budget" for every engineering sub-team and require physical weighing of every part before adding it to the master CAD assembly.	5	5	3	Open	18.Mar		
Technical Risk	Higher than expected maintenance	R36	Moving parts or spray nozzles wearing out much faster than lab testing predicted.	Increased operational downtime and lower overall profitability for the service.	Design / Engineering team, Operations team, Technicians	2	3	6	Build predictive maintenance algorithms that track motor vibrations and flight hours to flag components for replacement before they fail.	4	5	1	Open	18.Mar		
Technical Risk	Defects in hardware and software	R37	Factory assembly errors or unhandled exceptions in the flight control software.	High risk of system crashes, dangerous physical failures, and costly product recalls.	Design / Engineering Team, Operations Team	2	4	8	Adopt a rigorous aerospace-grade Quality Management System (QMS) for assembly and peer-reviewed unit testing for all new flight code.	4	2	4	Open	18.Mar		
Technical Risk	Mission incident	R38	A physical crash or severe malfunction during active operation.	An accident occurs during a mission that damages the crops or injures the farmer or operators. Leads to loss of trust in the system	Target Farmer, Design / Engineering and Business Teams	1	5	5	Ensure system reliability and failure mode mitigation	3	2	3	Open	18.Mar		
Programme Risk	Drone misuse	R39	Operators ignoring safety protocols or flying in restricted airspace without permission.	Severe liability claims, legal battles, and massive damage to the company's public image.	Operations team, Business team	1	4	4	Implement a digital ignition key or biometric operator login on the ground control station so only authorized, trained pilots can fly. Implement a black-box system in the drone that records all sensor values and command imputs during a mission and stores them if an incident happens	4	3	1	Open	18.Mar		
Technical Risk	Incorrect assumptions during modeling	R40	Underestimating the effect of sloshing liquid payloads on center-of-gravity, aerodynamic effects and stability & control characteristics.	Unstable flight characteristics, unpredictable roll rates, and increased risk of mid-air flips.	Design / Engineering team	3	3	9	Rigorous flight testing during the prototype phase of the project	5	2	5	Open	18.Mar		
Commercial Risk	Unit cost too high	R41	Over-engineered components or expensive composites driving up the base cost of one unit.	Low volume of sales, failing to hit the economies of scale needed to turn a profit.	Business team and Target Farmer	4	4	16	Design the airframe to minimize part count and assembly complexity, focusing on multi-functional structural elements.	2	3	5	Open	18.Mar		
Programme Risk	Constraints in supply chain	R42	Tariffs, trade embargoes, or raw material scarcity affecting production.	Massive manufacturing delays and forced pivots to inferior alternative components.	Design / Engineering team	3	3	9	Source standardized components from multiple regions and maintain a buffer stock of critical parts to mitigate trade or shipping delays.	4	3	3	Open	18.Mar		
Programme Risk	Increase in insurance cost	R43	High claim rates in the drone industry pushing up liability premiums.	Squeezed profit margins for both the manufacturer and the service operators.	Business team, Insurance	2	2	4	Install continuous black-box telemetry logging, demonstrating transparent, safe flight data to insurers can help negotiate lower premiums.	5	4	1	Open	18.Mar		
Technical Risk	Subsystem accessibility is poor	R44	Burial of vital electronics deep inside the frame, requiring full teardowns for basic repairs.	Poor access to components will make the system difficult to maintain, extending mission downtime.	Operations team, Technicians	2	3	6	Utilize quick-release fasteners and hinged panels for the avionics bay so technicians can reach core controllers without full teardowns.	4	4	2	Open	18.Mar		
Programme Risk	Extended down-time between missions	R45	Slow battery charging or complex physical refilling of the spray payload.	Lower daily acreage coverage, making the drone less competitive than a standard tractor.	Operations team, Target Farmer	2	3	6	Implement a "slide-in" battery tray system for 30-second swaps and use high-flow, quick-connect valves for instant chemical refilling.	5	4	2	Open	18.Mar		
Programme Risk	Bird strike hazard	R46	Local wildlife attacking or colliding with the drone during low-altitude passes.	Physical damage to the airframe, destroyed propellers, and potential loss of the airframe.	Operations team	3	2	6	Paint the drone in highly visible, contrasting colors and install strobe lights to deter birds during low-altitude passes.	4	2	3	Open	18.Mar		
Commercial Risk	Stakeholder needs not met	R47	Failing to deliver the yield increases or cost savings promised to the farmers.	Failure to secure contract renewals, leading to brand death in close-knit farming communities.	Target Farmer, Design / Engineering and Business Teams	2	5	10	Clear definition of stakeholder needs and requirements	5	3	3	Closed	18.Mar	27.Apr	Current design (frozen at CDR) meets all identified stakeholder needs
Technical Risk	Poor system integration	R48	Subsystems (e.g., aero, structures, payload) are developed in isolation, and may not fit together physically and software-wise when they are integrated	Sub-systems will need to be re-designed	Design / Engineering team	2	4	8	Shared values, weekly team meetings, good communication between sub-system teams and integration coordination	4	4	2	Closed	18.Mar	27.Apr	Current design (frozen at CDR) features fully integrated sub-systems

Severity Scale		Probability		Occerrance / Year	Impact Scale		Probability	
1	Minor therefore mitigations are not strictly needed	1	Very low	0	1	This solution will have a moderate impact on the risk / problem with a moderate success rate, but no impact to its root / cause	1	The solution has little chance to be implemented
2	Small therefore mitigations should be considered vaguely	2	0.05	18.25	2	This solution will have a positive impact on the risk / problem with a moderate success rate but no impact to its root / cause	2	The solution may be implemented if no risks related to its implementation are identified
3	May need to be considered but has no majour impact on the project so mitigations should be considered	3	0.15	54.75	3	This solution will have a positive impact on the risk / problem but little impact on its root / cause with a moderate success rate	3	The solution will be implemented if no risks related to its implementation are identified
4	Will need to be considered and mitigations should be implemented	4	0.35	127.75	4	This solution will have a positive impact on both the risk / problem and its root / cause with a high success rate	4	The solution will be implemented if no major risks related to its implementation are identified
5	Major and will impact the project in multiple ways and therefore must be mitigated	5	0.5 <	182.5	5	This solution will solve the risk / problem with a high success rate	5	The solution will be implemented

The Risk Register defines each risk in terms of its description, impact and affected stakeholders. Similarly to the PM risk register, a probability and severity score was given to each risk, and the two were multiplied to obtain the overall risk score. Mitigations for each risk were outlined and given a probability and impact score that was used to calculate the mitigated risk score by dividing the overall risk score by the impact score.

Appendix J: Integrated Set of Needs

Requirement Identification						Requirement Definition										
Category	Requirements	ID	Description / Notes	Need or Want	Want Classification	Related Stackholder(s)	Rationale	Requirement Source (Needs and Risks)	Specific Information Source	Associated Risks	Stakeholder sore:	Matrix Requirement Scores	Corrected Requirement Scores	Total Score Multiplication	Date Added	Design Feature / Response
Operation	The drone's spraying must cover the entire area that has been marked for spraying	ON1	N/A	Need	N/A	Target Farmer and Design Team	If this need is not met, the service will not be carried out correctly	D1, D2, D3, D9, C3, R16	N/A	Increased development time and cost	9	6	8	72	01.Mar	Navigation system
	The drone must be capable of autonomous flight	ON2	N/A	Need	N/A	Target Farmer, Consumers for Non-Spraying Drone Services, Operations Team and Design Team	Reduces operator workload, competition is autonomous, repeatability and mission will be carried out more optimally	D7, R1	N/A	Increased development time and cost	9	Not scored in the Matrix	6	54	24.Mar	Avionics and Navigation systems
	The drone must be able to perform its spraying function autonomously	ON3	The drone autonomously knows when to spray and when to stop spraying	Need	N/A	Target Farmer and Design Team	Reduces operator workload, competition is autonomous, repeatability and mission will be carried out more optimally	D7, R1	N/A	Increased development time and cost	9	Not scored in the Matrix	6	54	24.Mar	Avionics and Navigation systems
	The drone must be capabled VTOL	ON4	N/A	Want	High	Operations Team and Design Team	Many farms may have limited options, space or infrastructure for proper runway	C1	N/A	Increased development time and cost	7	Not scored in the Matrix	7	49	24.Mar	VTOL propulsion system
	The drone must be capable of flying under rain or perturbed (non sunny and with wind) weather conditions	ON5	Detailed gusts requirements will be developed by the Structures team	Want	High	Consumers for Non-Spraying Drone Services and Design Team	Weather can be unpredictable and this capability offers flexibility	C2, R8	N/A	Increased development time and cost	5	Not scored in the Matrix	7	35	24.Mar	Fuselage waterproofing, IP rated external component selection and over-spec-ed structures
	The drone must be able to withstand the structural loads outlined in the flight envelope specific to its cicategory	ON6	N/A	Need	N/A	Design Team	Needed for airworthyness	C2	N/A	Increased development time and cost	4	Not scored in the Matrix	7	28	24.Mar	Over-spec-ed structures
	The payload system must be accessible in order for leak checks to be performed	ON7	N/A	Want	High	Operations Team and Design Team	Necessary for pre-flight checks	C16	N/A	Increased development time and cost	7	Not scored in the Matrix	5	35	24.Mar	Openable hatches in the fuselage that grant access to the internals
	If signal / communication is lost between the drone and the operator, the drone must automatically reposition itself to regain the connection or have a RTH (return-to-home) function	ON8	N/A	Want	Medium	Operations Team and Design Team	This insures the safety of the drone and anyone in the vicinity	C5, R1	EASA and DGCA requirement	Increased development time and cost	7	Not scored in the Matrix	6	42	24.Mar	Avionics and Navigation systems
	The drone operation must have a higher operational productivity than 0.4 to 0.7 hectares per day --> 0.66 to 0.07 hectares / hour	ON9	The ha / h is based on a 10 hour day and the operational productivity is based on manual spraying	Need	N/A	Target Farmer, Farmer's family, Business team, Competitors	The drone's operational productivity must be higher than the current methods in order for the business to be competitive	D4, D5, D9, C12, R1	https://www.ellanyan.co.in/how-much-does-can-it-operate-50-faq-id41990606.html	Increased development time and cost	9	3	5	45	01.Mar	Navigation system, cruise speed and payload spraying systems spec-ed to achieve a given operational productivity
	The drone must carry enough payload to spray 2 hectares in one mission	ON10	N/A	Need	N/A	Target Farmer	The business model also aims to target Smallholder Farmers --> max farm area area: 2 ha so it would be beneficial to spray an entire farm in one mission	R1, Business Proposition	N/A	Increased development time and cost	9	3	3	27	01.Mar	Payload tank size
	The drone must be capable of spraying the field or area outlined in the mission plan autonomously	ON11	The drone must spray the field(s) that has been marked for spraying by the farmer or operations team, regardless of obstacles or the field's shape	Need	N/A	Target Farmer, Farmer's neighbours, Farmer's village, Business team	The farmer is paying to have their field sprayed	D7, C3, R1, R16	N/A	Increased development time and cost	9	3	8	72	01.Mar	Navigation system Consideration of irregular fields (non rectangular) is beyond GDMP scope
	The drone must have accurate obstacle detection	ON12	N/A	Need	N/A	Design Team	To avoid damage to the drone or the obstacle	C5, R15	N/A	Increased development time and cost	4	Not scored in the Matrix	7	28	16.Apr	Avionics system
	The drone must follow a flight path that avoids obstacles, no-fly zones and terrain with autonomous operation	ON13	The drone doesn't crash into people, obstacles or the ground and doesn't fly in restricted areas	Need	N/A	Target Farmer, Farmer's neighbours, Farmer's village, Indian government, Aerospace regulators, Insurance	It would be undesirable if the drone crashed while flying, both for the safety of the drone and any obstacles. No-fly zones need to be avoided because of regulations	D7, R15	N/A	Increased development time and cost	9	4	6	54	01.Mar	Avionics and Navigation systems
	The drone must have an operational radius of 4km	ON14	Double the competition	Need	N/A	Target Farmer, Business team	Non all fields are located next to a possible take-off zone, so the drone my need to cruise to a field before spraying it. The drone will also need to cruise to cover the edges of a large field.	R1, Business proposition	DJI Agras T25, T25P, T50, T70P, T100 - DJI user manuals	Increased development time and cost	9	3	4	36	01.Mar	Power specifications and appropiate battery size
	The drone must have 4G, GPS and sattellite connectivity	ON15	N/A	Need	N/A	Operations team	Requirement mirrors competitors and ensures connectivity at all times during a mission.	C1, R12	N/A	Increased development time and cost	7	4	6	42	01.Mar	Avionics system
	The drone spraying systems must be compatible with existing spraying products	ON16	Compatible with many or with few different products	Need	N/A	Manufacturers of sprayed products	The drone cannot perform its mission if it cannot spray the payload	C18, C20	N/A	Increased development time and cost	6	5	6	36	01.Mar	Payload tank and payload spraying system design
	The drone's battery system must be rechargeable on site (at the farm)	ON17	Battery exchange time outlined in requirements table	Want	High	Operations team	Avoids having to carry a lot of spare batteies and increases the locations to which the service can be delivered	C1, R13	N/A	Increased development time and cost	4	Not scored in the Matrix	5	20	01.Mar	Battery specing
	The drone battery system must be rechargable using existing infrastructure	ON18	N/A	Need	N/A	Electricity providers	Bespoke charging infrastructure cannot be developed for this drone because of time, cost, complexity and government permission	C1	N/A	Increased development time and cost	2	5	3	6	01.Mar	Battery specing
	The drone must be capable of servicing farms located at 4921 ft (1,500 m) above sea level	ON19	The is the drone's service ceiling	Need	N/A	Target Farmer and Design Team	This will increase the number of farms that can be serviced compared to a drone that can only operate at sea level	C23	https://mesaforyou.com/blogs/drone-products/drone-flight-altitude and https://www.researchgate.net/figure/Map-showing-the-size-and-elevation-zones-of-India_fig1_311605864	Increased development time and cost	9	Not scored in the Matrix	6	54	16.Apr	Over-spec-ed motors
	The drone's structure must be capable of surviving a crash landing with the drone travelling at the its cruise velocity	ON20	N/A	Need	N/A	Operations Team and Design Team	This avoids a complete rebuild of the drone should a crashland occur and all internal components would be better protected, avoiding other failures	C5	N/A	Increased development time and cost	7	Not scored in the Matrix	6	42	06.Apr	Over-spec-ed structures
Design	The payload tank must be removable	DN1	N/A	Want	High	Operations Team and Design Team	This allows a reduced weight when the drone is performing non spraying missions and facilitates washing	C19, C20, R44	N/A	Increased development time and cost	7	Not scored in the Matrix	5	35	24.Mar	N/A
	The payload tank must be shaped in such a way that fascilitates cleaning	DN2	N/A	Want	Medium	Operations Team and Design Team	Reduces down time between two missions that require different payload liquids	C18, C20	N/A	Increased development time and cost	7	Not scored in the Matrix	3	21	24.Mar	Payload tank sloped bottom and large access hatch
	The drone must have a modular design that allows for different mission profiles to be performed	DN3	N/A	Need	N/A	Operations Team and Design Team	Increases product competitiveness and allows business expansion	R28, Business proposition and ConOps concept	N/A	Increased development time and cost and no specialisation in one mission profile	7	Not scored in the Matrix	5	35	24.Mar	Fixed wing design, removable spray booms and modular sensor systems and payload system
	The drone must be disassemblable or have given configuration for transport	DN4	N/A	Need	N/A	Operations Team and Design Team	To ease transport	C23, R1, Business proposition and ConOps concept	N/A	Increased development time and cost	7	Not scored in the Matrix	6	42	24.Mar	Drone configuration
	In its transportational configuration, the drone must fit inside of a medium sized van	DN5	N/A	Need	N/A	Operations Team and Design Team	The drone must be able to be transported from farm to farm as flying it from farm to farm would be impractical. This also facilitates shipping and eliminate the need for a custom vehicle to transport the drone	C23, R7, Business proposition and ConOps concept	N/A	Increased development time and cost	7	Not scored in the Matrix	6	42	24.Mar	Drone configuration dimensions consideration
	Internal components should be easily accessable	DN6	N/A	Want	Medium	Design Team and Mechanics	To fascilitate repairs, modifications or upgrades	C15, R44	N/A	Increased development time and cost	7	Not scored in the Matrix	4	28	06.Apr	Openable hatches in the fuselage that grant access to the internals
	The disassembled drone (or in its transport configuration) must have designated carrying points	DN7	N/A	Need	N/A	Operations Team and Design Team	This will ease the movement of the drone from the van to the take-off zone	Business proposition and ConOps concept	N/A	Increased development time and cost	7	Not scored in the Matrix	6	42	10.Apr	Screw-on handles for the fuselage or ergonomic grab-on points for transport
Business	The drone service must be cheaper than 725 ₹ / hectare	BN1	This is the average cost of manual spraying, therefore our product must match that	Need	N/A	Target farmer, Business team, Competitors	The drone's spraying cost per hectare must be lower than the current methods in order for this business to be competitive	D4, D10, C10	N/A	Compromises made to the design, increased development time and cost or reduced profitability of the business	9	3	2	18	01.Mar	Costing analysis
	It must be easy for the farmer to contact the company to order the service	BN2	N/A	Want	Medium	Target Farmer	Eliminates friction between the farmer and the service	D10, C19, R9	N/A	Higher call center costs	9	1	1	9	01.Mar	Beyond GDMP scope
	The company must have enough drone capacity to service multiple farmers	BN3	N/A	Want	Medium	Target Farmer, Business team	Scalability	D10, C19, R9	N/A	Lack of demand	9	2	2	18	01.Mar	Beyond GDMP scope
	The company must be able to provide the service within 1 week of the farmer contacting the company	BN4	N/A	Want	Medium	Target Farmer	Spraying times during the season are very specific and crops cannot be sprayed late	D10, C19, R9	N/A	Lack of demand	9	1	3	27	01.Mar	Beyond GDMP scope
	The drone must have documentation that proves its spraying effectiveness	BN5	For marketing and to make sure the farmer know that this service is legit	Want	Medium	Target Farmer, Farmer's Union companies, Farmers financial support services, Business team, 3rd party product testers	Marketing and sales	Marketing	N/A	Increased development time and cost	9	1	3	27	01.Mar	Beyond GDMP scope
Regulation	The drone must reduce the farmer's exposure to harmful chemicals during spraying	RN1	The spraying process must be independent from the farmer, spraying drift must be minimised and the correct area must be sprayed	Need	N/A	Target Farmer, Farmer's family, Business team, Insurance, Indian government	Unique selling point of the business	D4, D6, C16, C17	N/A	Increased development time and cost	9	4	7	63	01.Mar	Drone flight and spraying procedures
	The drone and its use case must respect the regional laws of the area where it is operating	RN2	N/A	Need	N/A	Indian government, Farmers financial support services, Indian government, Farmers Union companies, Farmers financial support services, Business team, Insurance	To allow the drone to fly	C17, C22, R29	N/A	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone must not be above 150 kg MTOW	RN3	To remain in the "medium" category according to G.S.R 589	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	The medium category of drones requires less airworthiness checks to be certified and thus would reduce costs	C2, C4, C5, C17, C22, R29	https://easyjet.gov.np/files/Publications/2023-03/easa-uas-publication-2023-01-en.pdf	Increased development time and cost and limit in drone capabilities	7	Not scored in the Matrix	8	56	01.Mar	Design spec according to regulations review
	The drone must respect every regulation associated with airworthiness in its category according to EASA	RN4	Specific regulation: EASA's Easy Access Rules for Unmanned Aircraft Systems (Regulations (EU) 2019/947 and 2019/945).	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	Airworthyness and to allow the drone to fly	C2, C4, C5, C17, C22, R29	https://www.easa.europa.eu/en/domains/civil-drones-rpas/easy-access-rules-easy-access-rules-unmanned-aircraft-systems-regulations-eu	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone's V-n envelope must respect guidelines set by EASA	RN5	Specific regulation: CS-VLA, CS-23 and SC-VTOL	Need	N/A	Aerospace regulators, Business team, Insurance, Design team	Airworthyness and to allow the drone to fly	C2, C4, C5, C17, C22, R30	N/A	Increased development time and cost	7	6	8	56	01.Mar	Design spec according to regulations review
	The drone must adhere to the agricultural regulations set by the local government	RN6	N/A	Need	N/A	Agricultural regulators, Business team, Insurance	Unique selling point of the business and to allow the drone to fly	C16, C17, R29	N/A	Increased development time and cost	9	6	8	72	01.Mar	Design spec according to regulations review
	When operting, the drone should respect the noise contraints established by the local government	RN7	N/A	Need	N/A	Target Farmer, Farmer's family, Farmer's neighbours, Farmer's village, Indian government, Aerospace regulators	To allow the drone to fly	C4, C17, C22, R29	N/A	Increased development time and cost	9	5	6	54	01.Mar	Preliminary noise analysis (in depth propeller noise analysis is out of scope)
	The drone must be safe for any person located within a 30m radius of the area of operation	RN8	N/A	Need	N/A	Indian government, Farmers Union companies, Farmers financial support services Indian government, Farmers Union companies, Farmers financial support services, Business team, Insurance	To allow the drone to fly and spray	C16, C22	N/A	Increased development time and cost	7	6	7	49	01.Mar	Drone flight and spraying procedures
	The spraying system must comply with the relevant regulatory bodies that regulate pesticide, herbicide and fertilizer dispersion over crops or a field	RN9	For example: USA: EPA and state agriculture departments, Brazil: MAPA (ministry of agriculture), Australia: APVMA and state EPAs	Need	N/A	Farmer's neighbours, Indian government, Agricultural regulators, Business team	To allow the drone to spray	C16, C17, R29	N/A	Increased development time and cost	7	5	7	49	01.Mar	Regulations review
Drone-Operator Interaction	The drone must be intuitive to pilot / control	IN1	Exceptional skill must not be required to pilot the drone	Want	High	Operations team, Aerospace regulators	Makes it easier to find drone pilots / operators	D11, C7, C11, R10, R33	N/A	Increased development time and cost	7	2	3	21	01.Mar	Beyond GDMP scope
	The mission setup process must be straight forward	IN2	The mission plan UI must be user friendly and simple to understand	Want	High	Operations team, Target Farmer	This will reduce downtime and allow the farmer to assist in mission planning	C7, C11, R10	N/A	Increased development time and cost	9	Not scored in the Matrix	3	27	13.Apr	Beyond GDMP scope
	Down-time between two drone missions must be limited	IN3	The number of operations between two drone missions must be limited or be fast and simple to execute. It would be preffredble is multiple operations could happen at once	Want	High	Operations team and Target Farmer	This is so good operational productivity can be achieved, multiple farms or big fields can be serviced in the same day and the farmer's time is not wasted	C7, C12, R45	N/A	Increased development time and cost	9	Not scored in the Matrix	5	45	16.Apr	Payload refill and battery swap operations
	The battery swap and payload loading must be tool-less	IN4	N/A	Want	Medium	Operations team	To minimise down-time	C7, C12	N/A	Increased development time and cost	7	2	2	14	24.Mar	Tool-less fastening mechanisms for the wings and tail boom (VTOL boom and landing gear will need tools)
	Transition operation from drone transport configuration to flight configuration must be tool-less	IN5	N/A	Want	Medium	Operations Team and Design Team	This will speed up the drone preparations for flight process	C12	N/A	Increased development time and cost	7	Not scored in the Matrix	2	14	16.Apr	A latch locking mechanisms for all the fuselage access hatches (3D modelling is out of GDMP scope)
	The drone must have the necessary equipment to be remote controlled by a pilot	IN6	N/A	Need	N/A	Operations team	So a pilot can take over in case of an emergency	C5	N/A	Increased development time and cost	7	Not scored in the Matrix	7	49	24.Mar	Avionics system
	The drone's operations must be monitorable from a distance by the operator	IN7	N/A	Need	N/A	Operations team	So the operator can monitor the drone at all times and take over if something goes wrong	D11, C5	N/A	Increased development time and cost	4	Not scored in the Matrix	6	24	01.Mar	Beyond GDMP scope
	The drone must possess a camera that is able to give the operator the drone's POV	IN8	N/A	Want	Medium	Operations Team and Design Team	So the operator can monitor the drone at all times and take over if something goes wrong and it provides the farmer with a visual of the drone's POV	D11, C5	N/A	Increased development time and cost	7	Not scored in the Matrix	4	28	06.Apr	Avionics and Navigation systems
Manufacturing and End-of-life	Material selection, contruction / assembly method and battery technology must be compatible with end-of-life processes	MN1	N/A	Want	Medium	End-of-life disposal agencies	Sustainability	C21	N/A	Increased development time and cost	4	2	2	8	01.Mar	Considerations during design, in-depth analysis is beyond GDMP scope
	Off-the-shelf components must be used in the design with little to no alterations	MN2	No modifications to off-the-shelf components need for integration with the drone	Want	Low	Component suppliers, Mechanics	Minimises cost and assembly time	C8, C8, C14	N/A	Increased development time and cost	7	2	3	21	01.Mar	COTS component selection and integration
	The drone's avionics and sensor systems need to be made from COTS components	MN3	No new or bispoke sensors	Want	High	Design Team	This will reduce development costs	C8, C8, C14, R17	N/A	Increased development time and cost and availible technologies may limit drone capabilities	4	Not scored in the Matrix	6	24	16.Apr	COTS component selection and integration
	The drone's power system needs to be made from COTS components	MN4	No new power systems (batteries or generators)	Want	High	Design Team	This will reduce development costs	C8, C8, C14, R17	N/A	Increased development time and cost and availible technologies may limit drone capabilities	4	Not scored in the Matrix	6	24	16.Apr	COTS component selection and integration
	The drone's propulsion system needs to be made from COTS components	MN5	No new motors or engines	Want	High	Design Team	This will reduce development costs	C8, C8, C14, R17	N/A	Increased development time and cost and availible technologies may limit drone capabilities	4	Not scored in the Matrix	6	24	16.Apr	COTS component selection and integration
	Bespoke components must be manufacturable with existing industrial manufacturing methods	MN6	No new tools must be invented for this project and cutting-edge manufacturing techniques cannot be used	Need	N/A	Bespoke component manufacturers	Bespoke manufacturing methods cannot be developed for this drone because of time, cost, complexity and certification	C8, C8, C14, R17	N/A	Increased development time and cost	4	4	8	32	01.Mar	Key considerations during design
	The drone must be built using commercially availible and industrially produced materials	MN7	N/A	Need	N/A	Bespoke component manufacturers, Mechanics	Bespoke materials cannot be developed for this drone because of time, cost, complexity and certification	C8, C8	N/A	Increased development time and cost	7	4	8	56	01.Mar	Materials selection during design
	The drone must be assemblable using industry standard tooling and current assembling and manufacturing techniques	MN8	N/A	Need	N/A	Final assembly / manufacturing firm, Mechanics	Bespoke assembly methods / techniques cannot be developed for this drone because of time, cost, complexity and certification	C8, C9	N/A	Increased development time and cost	7	4	8	56	01.Mar	Key considerations during design

Stakeholder Score		***Requirements Score***		***Requirements Total Score***	
Total Score taken from Stakeholder Register		**Value**	**Meaning**	**Value**	**Meaning**
1 - 2	Stakeholder is remotely connected to this project	1	The need has little to no importance regarding the drone's operation and non-compliance is acceptable	1 - 9	This need carries little to no importance to the project
3 - 5	Stakeholder is connected to this project	2	The need has little importance regarding the drone's operation and non-compliance is acceptable	10 - 18	This need carries little importance to the project
6 - 8	Stakeholder is an important member of this project	3	The need has some importance regarding the drone's operation and non-compliance is acceptable but not preferable	19 - 27	This need carries some importance to the project
9 - 10	Stakeholder is a key member of this project	4	The need is somewhat important to the drone's operation and compliance is preferable but not obligatory	28 - 36	This need carries moderate importance to the project
		5	The need is moderatly important to the drone's operation and should be complied with	37 - 45	This need carries moderately high importance to the project
		6	The need is important to the drone's operation and should be complied with	46 - 54	This need carries high importance to the project
		7	The need is very important to the drone's operation and must be complied with	55 - 63	This need carries very high importance to the project
		8	The need is vital to the drone's operation and must be complied with	64 - 72	This need carries vital importance to the project

Appendix K: Numerical Requirements

Requirement Definition												
Category	**Attribute**	**Requirement**	**Units**	**Importance**	**ID**	**Notes**	**Rationale**	**Source**	**Related Need**	**Date created**	**Created by**	**Last Modified**	**Modified by**
Operation	Operational Productivity (OP)	21 (ideal case) & 18 (normal)	ha / h	High	NR1	How many hectares are sprayed in one hour	The DJI Agras T50 serves a similar market to this UAV, therefore a similar operational productivity is desired	https://dji-ag.com/jaapa/dji-agras-t50 https://dronespraypro.com/blogs/news/how-many-acres-in-1-day	ON9	13.Feb	Peter	03.Apr	Luca
	Operational Radius	4,000	meters	High	NR2	Dependant on power consumption and battery size, optimised for endurance	The UAV has a fixed-wing advantage over quadcopters and therefore its range can be larger	DJI Agras T25, T25P, T50, T70P, T100 - DJI user manuals	ON10, ON13	13.Feb	Luca	17.Mar	Luca
	Flight time	30	minutes	High	NR3	Dependant on power consumption and battery size, must be high for endurance	The Pelican is fixed-wing agri drone similar to this UAV, therefore flight time should be similar	https://www.flypyka.com/pelican-2 Pyka Pellican range	ON10, ON13	13.Feb	Jaison	25.Feb	Luca
	Payload capacity	47	kg	High	NR4	47 litres of product that can cover approximatly 4.7 acres or 1.9 hectares	The drone must spray two hectares of land in one mission	https://marutdrones.com/ag-365/#faq litres needed to cover one acre	ON10	13.Feb	Jonathan	25.Feb	Luca
	Operational ceiling	1,500	meters	Medium	NR5	The maximum height the drone can operate at	This will increase the number of farms that can be sprayed copared to a drone that can only work at sea level	https://mapsforupsc.com/largest-rice-producing-states-in-india/ and https://www.researchgate.net/figure/Map-showing-states-and-elevation-zones-of-India_fig1_315835800	ON19	13.Feb	Luca		
	Communication range	5 & Compatible with 4G and sattelite	km	High	NR6	These are dependant on the power output of the hardware (drone & controller) and limits imposed by countries on Operating frequency (check P12 of the Business review document)	This is in line with competitor drones	DJI Agras line of drones	ON2, ON3, ON8, ON11, ON13, ON14, DN3, DN4, DN5	13.Feb	Luca		
Spraying mechanism	Fluid flowrate	9	L/min	High	NR7		This is in line with competitor drones	DJI Agras T25 specs	ON9	13.Feb	Jonathan	25.Feb	Luca
Physical	MTOW	130 Absolute Max: 150 kg	kg	High	NR8	Maximum set by DGCA regulation: G.S.R 589 Part II	The Pelican's payload mass fraction was used to arrive at this MTOW	https://ajgaantta.gov.in/WriteReadData/2021/229221.pdf	ON10, RN3	13.Feb	Khayaam	17.Apr	Luca
	Empty weight	83 Absolute Max: 103 kg	kg	Medium	NR9	47% of the aircraft's MTOW is payload, 53% is empty weight	The Pelican's payload mass fraction was used to arrive at this empty weight	https://www.flypyka.com/pelican-2 Pyka Pellican empty weight to payload weight ratio	ON10, RN3	13.Feb	Khayaam	02.Apr	Luca
	Operational dimensions	W : 3,500 L : 3,000	mm	Low	NR10	Values presented are estimation values	The U7AG is a very similar drone to this UAV in terms of mass and payload, therefore a similar footprint is expected	https://www.forwardrobotics.com/u7ag	None	13.Feb	Aman		
	Transporational dimentions	2946 x 1784 x 1886	mm (LxWxH)	High	NR11	These are the minimum dimentions in each direction of the sourcevan sizes	This is to allow the drone to be transported inside of a van when in its transportational configuration	***In Indian Markets:*** https://truckguru.co.in/truck-size Indian Truck Sizes - Ford Transit (very popular transport van) 2946 x 1784 x 1886 https://www.vanguide.co.uk/guides/ford-transit-dimensions/ - Large Truck class (Hard limit) 4000 x 2200 x 2200 https://www.loadsofvans.com/blog/van-sizes-uk	DN4, DN5	13.Feb	Aman	14.Apr	Sam
Durability	Operational life	1'000	cycles	Low	NR12	Battery limited	This is in line with competitor drones	Specification reports for: Agras T0/40/50/100 & XAG P100 Pro	None	13.Feb	Sam		
	IP rating	IP55	IP rating	Medium	NR13	Target: IP55 Ideal: IP67	This is in line with competitor drones	DJI Agras T50 user manual	ON5	13.Feb	Khayaam	02.Apr	Luca
Down Time	Down time	3	minutes	Low	NR14	Includes battery and payload change / reload	Down time must be limited in order to serve more farms per day	https://dronespraypro.com/blogs/news/how-many-acres-in-1-day	O9, IN2, IN3, IN4	13.Feb	Sam	24.Apr	Luca
	Charging Time	8	minutes	Medium	NR15	15 min max	Charging time must be limited in order to serve more farms per day	https://agrispraydrones.com/blogs/news/how-long-can-a-sprayer-drone-fly-on-one-battery-charge & Specification reports for: Agras T0/40/50/100 & XAG P100 Pro	O9, IN2, IN3, IN4	13.Feb	Sam		

Appendix L: Morphological Chart

Functions	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5	Idea 6	Idea 7	Idea 8	Idea 9	Idea 10	Idea 11	Idea 12	Idea 13	Idea 14	Idea 15
Configuration			medium-sized Hybrid VTol: 	Picture	Quadcopter										
Propulsion (Horizontal)	Vertical Propeller As seen in Configuration				none	Rotor tilt - rotors tilt forward to provide thrust									
Propulsion (Vertical)	Horizontal Propeller As seen in Configuration				4 Stabilizers	Vertical propellor (see configuration)			In-wing rotors						
Control surfaces	Elevators and ailerons, V-tail -> no rudder.		Ailerons at wing and rudders at tail, no elevators required	Ailerons, Elevators, Rudder, Rotor control during vertical flight	Stabilizers	Rotor speed control - differential thrust manages pitch, roll and yaw control									
Spraying mechanism				Boom across wing trailing edge as seen in idea 2	nozzles on each 4 arms										
Payload location					Centre tank				Held in body moulded tank						
Payload loading				Payload can be removed using handles and swapped. The payload container will a lso use baffles to avoid sloshing					Body moulded tank						
Power System				Positioned on the front of fuselage and can be removed from the top by sliding out using handles		Battery pack on top of drone fuselage 									
Charging					Same as idea 1, mechanical button to descend battery module bay. Remove used battery to recharge externally, replace with new battery, resume route.			Docking station- high power system							
Landing gear		Picture	Fixed angled landing gear, two rigidly mounted to fuselage, rear location of the wing (in front of wing) and one at the tail end. See top down view.	Fixed skis	VTOL setup, simple landing station coupler for convenient docking (without recharging and refilling capabilities)										
Wing	Uses highly cambered aerofoil for high lift at slow speed and to have a lot of downwash for the spray to get caught in it and move down (like the propwash pushes the spray down on a SIR drone). Rectangular in shape (no sweep as cruise velocity will be small)		High swept wing with winglets at each wing end, No camber - aerofoil profile to be CLARKY for wing and NACA0012 for tail	Rectangular wing mounted on the top of the fuselage with high AR using Clark Y aerofoil with slight dihedral	No wing (multirotor)	No wing (multirotor)		may be possible to experiment with a less cambered aerofoil but allow capability to adjust wing inclination to allow higher wing AOA while keeping							
Transportation configuration							Folding wings	Telescopic wings							
Fuselage shape	Drop-shaped As seen in Configuration		Closely described to be a 'tear drop' shape, not dissimilar to other fixed wing VTOLs	Tubular fuselage	N/A	Aerodynamic hexagonal central hub		generic bluff body, first important to fit everything before streamlining, may be possible to make fuselage in shape of aerofoil to maximise body lift							
Tail	V-tail As seen in Configuration		V-tail design, 90 degree separation between each tail. Alternative - Traditional tail design		N/A	No tail (multirotor)									
															
															

	Category	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea	Idea
Idea number	Landing gear 3 votes each	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Description		Fixed Skids	Tripod	Wheels	Retractable	Launch & Catch	No landing gear	Tail sitter								
Votes		8	2	4	5	3	1	1								
Idea number	Propulsion 3 votes each	1	3	7	10	11	9									
Description		Fixed	Tilt	Ducted	In Wing	Tilt Wing	Like the Off-spray									
Votes		8	5	4	3	0	4									
Idea number	Tail 3 votes each	1	2	3	4	7	8	9	10							
Description		V-tail	Inverted V	Twin tail	U-tail	Tail sitter	H-tail	Vertical fin	No tail							
Votes		2	2	3	8	1	5	3	0							
Idea number	Spraying Mechanism 3 votes each	1	2	3	6	7	8	9	10							
Description		Pods	Boom	3 sprayers at the back	Under the vertical rotor	Central Spraying	Two central sprayers	Retractable	Non-wing mounted (on landing gear)							
Votes		3	6	1	3	3	2	1	5							
Idea number	Payload Tank 2 votes each	1	2	3												
Description		In wing	Internal (non removable)	Internal (removable)												
Votes		8	3	5												
Idea number	Power / Fuel 2 votes each	1	2	3	4											
Description		Battery	2-stroke engine	Hydrogen fuel cell	Hybrid system											
Votes		7	2	4	3											
Idea number	Configuration 3 votes each	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Description																
Votes		6	3	0	5	0	2	2	2	0	0	1	0	3	0	0
Idea number	Transport 1 vote each	1	2	3												
Description		Dis-assembly	Folding wings	Telescopic wings												
Votes		6	2	0												

Appendix M: Technical Accounts

Category	Attribute	Criteria	Unit	Iteration 1	Verification Method	Iteration 2	Verification Method	Iteration 3	Verification Method
Operation	Operational Productivity	21 (ideal case) & 18 (normal)	ha / h			55.728	Calculation	26 (ideal case) & 19-15 (normal)	Calculation and Simulation
	Max operational radius	4,000	m	6,000	Calculation	4,000.5	Calculation	4547.25	Calculation
	Flight time	30	m	30	Calculation	15	Battery Spec	15	Battery Spec
	Payload Capacity	47	kg	47	Calculation	47	Payload Spec	47	Payload Spec
	Operational Ceiling	1,500	m					2,000	Calculation
	Communication range	5 & Compatible with 4G and satellite	km		COTS Component Integration		COTS Component Integration	Yes	COTS Component Integration
Spraying Mech	Fluid flowrate	9	L/min	19.872	Calculation	18.48	Calculation	16.72	Calculation
Physical	MTOW	130	kg	110	Calculation and Estimation (previous target was 100 kg)	124	Calculation and Estimation	140.8329	Calculation
	Max MTOW	150	kg	110	Estimation	124		140.8329	
	Empty weight	83	kg	63	Calculation	77	Calculation	93.8329	Calculation
	Operational dimentions	L: 3,500 W: 3,000	mm	4500 x 4800	CAD model	4739 x 5180	CAD model	4701 x 5460	CAD model
	Transporational dimentions	2946 x 1784 x 1886 (L x W x H)	mm	2300 x 600 x 600	Estimation	2,700 x 1,810 x 720	CAD model Old target: 2,765 x 1,640 x 1900	2730 x 1,729 x 840	CAD model
Durability	Operational life	1,000	cycles			500	Battery Spec	500 for the battery, more for the rest	Battery Spec
	IP rating	IP55	IP__					IP35 (VTOL motors), rest is inside sealed fuselage	COTS Component Spec
Power	Down time (battery & payload change)	3	min					3 (5 worst case)	Simulation
	Charging time	8	min					Undefineable	
Other	Regulations compliance				Regulation Checking		Regulation Checking		Regulation Checking

Appendix N: FMEA, DFMEA and PFMEA

Full Aircraft Failure Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Mitigation through Analysis and Design	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Testing	Resp.	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN = S X O X D	IMPLEMENTATION	EFFECTIVENESS	MRPN (Mitigated RPN) = S X O X (I+E / 10)
When could the failure mode occur?	What sub-system is the failure mode part of	In what ways could the step change or failure go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or failure to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate this failure mode?			What test can be performed to mitigate this failure mode?	Who is responsible for making sure the actions are completed?							
During flight or during manoeuvres & gusts	Wing	Critical wing structure failure	Aircraft, field or crop damage and service not delivered	10	Assembly / manufacturing error, fatigue, inadequate pre-flight checks and unpredictable weather	1	Pre-flight, pre and post assembly / manufacture and QC checks, weather forecasts, inspections and certification documentation	7	70	Safety factor Structural redundancy in the wings FEA analysis g sensors VTOL system	7	4	Simulated failure test (wings are detached mid flight and the aircraft must recover in VTOL by itself)	Design, manufacturing and operations teams	10	1	7	70	7	4	28
During flight or during manoeuvres & gusts	Propulsion	Complete propulsion loss	Aircraft, field or crop damage and service not delivered	8	Electrical failure / power loss, faulty component, propeller / motor damage and structural damage	3	Pre-flight, pre and post assembly / manufacture and QC checks, inspections and certification documentation	4	96	Effective COTS component selection Motor redundancy Structural analysis of propeller booms CFD analysis for lift without power at cruise speed Waterproofing Battery sensors Fixed wings	8	5	Glide test (propulsion shut off once cruise speed reached). Sub-system propulsion and circuitry testing.	Design, manufacturing and operations teams	8	3	4	96	8	5	48
During flight or during manoeuvres & gusts	Spray Mechanism	Leakage of payload	Wastage of spraying product, loss of drone effectiveness and electrical issues	6	Puncture within payload system, worn connections, faulty seals, loss of valve control and incorrect payload loading	5	Pre-flight checks, QC checks pre and post assembly / manufacture, inspections and maintenance plan	3	90	Safety factor Proper o-ring selection Sensors Removable & swappable spraybooom	8	7	Full spray system test Cycle testing	Design, manufacturing and operations teams	6	5	3	90	8	7	63
During flight or during manoeuvres & gusts	Payload System and Spray Mechanism	Payload tubing blockage	Delayed service	6	Contamination in payload tank, tubing deformation, worn tubing causing debris to build up	3	Pre-flight checks, QC checks pre and post assembly / manufacture, inspections and maintenance plan	3	54	Liquid pressure analysis Tube sizing Sensors Removable & swappable spraybooom	5	7	Full spray system test Cycle testing	Design, manufacturing and operations teams	6	3	3	54	5	7	38
During landing	Landing gear	Structural failure of landing gear	Aircraft, field or crop damage and service delays	3	Fatigue, hard impact and manufacturing defects	2	Pre-flight, pre and post assembly / manufacture and QC checks, inspections and certification documentation	2	12	Safety factor Structural redundancy in fuselage FEA analysis g sensors Removable & swappable spraybooom	7	4	Hard landing test Fatigue testing	Design, manufacturing and operations teams	3	2	2	12	7	4	5
During flight or during manoeuvres & gusts and landing	Fuselage and Tail	Structural failure of fuselage or Tail	Aircraft, field or crop damage and service not delivered	10	Assembly / manufacturing error, fatigue, inadequate pre-flight checks and unpredictable weather	1	Pre-flight, pre and post assembly / manufacture and QC checks, weather forecasts, inspections and certification documentation	4	40	Safety factor Structural redundancy in the tail FEA analysis g sensors VTOL system	7	4	Simulated failure test (tail is detached mid flight and the aircraft must recover in VTOL by itself)	Design, manufacturing and operations teams	10	1	4	40	7	4	16
During flight or during manoeuvres & gusts and landing	Navigation	Faulty flight plan	Wastage of spraying product, loss of drone effectiveness and flight in illegal areas	6	Software issues, GPS location failure and faulty inputs (human error)	6	Software debugging, pre-flight check, GPS location check, flight plan input check, constant software patches and operator training	4	144	Thorough navigation model analysis Software strucutral robustness	9	6	Full operation field testing	Design and operations teams	6	6	4	144	9	6	86
During flight or during manoeuvres & gusts and landing	Battery	Power loss	Crop damage, electrical fire and service not delivered	10	Thermal degredation, overcharging, incorrect battery usage and handling and short circuiting	4	Flight log per battery, cell voltage checks, smart charging / discharging, proper storage, correct handling procedures and pre-flight checks	3	120	Effective COTS component selection Battery management / protection systems (BMS) Thermal simulation for battery degredation Battery lifecycle analysis Smart battery system Circuit redundancy Waterproofing	9	4	Accelerate mission testing Lifecycle testing	Design and operations teams	10	4	3	120	9	4	48
During flight or during manoeuvres & gusts and landing	Sensors	Faulty obstacle detection	Obstacle collision, damage to crop, aircraft and property damage, harm to client, unfulfilled service	10	Faulty hardware, power loss to system, dirty/covered sensors, Flight controller software issues	3	Pre flight checks, redundancy within sensor system	6	180	Effective COTS component selection Simulation of obstacle detection code accuracy Robustness through software Rendundancy within system	8	4	Sub-system bench testing Full navigation testing System validation testing	Design team	10	3	6	180	8	4	72
During flight or during manoeuvres & gusts and landing	Control Surfaces	Faulty / loss of actuation	Service not delivered, low efficiency in service	5	Faulty hardware, linkage failure, power loss. structural failure	5	Pre flight checks, QC at factory, documentation and inspection logs	1	25	Redundancy VTOL system	7	2	Fatigue testing Simulated failure test (placing the aircraft in an unstable manoeuvre where it must recover by itself)	Design, manufacturing and operations teams	5	5	1	25	7	2	5
During flight or during manoeuvres & gusts and landing	Circuitry	Circuitry failure	Aircraft, field or crop damage and service not delivered	10	Faulty wiring, short circuit, leakage, disconnected interfaces and overspec power inputs	3	Manufacturing regulation, QC checks, pre-flight checks, documentation and inspection logs	6	180	Effective COTS component selection Thorough PCB model and circuitry mode. Appropriate wire and component selection Built-in failsafes and circuit redundancies	8	4	Segmented and full sub-system bench testing	Design, manufacturing and operations teams	10	3	6	180	8	4	72
During flight or during manoeuvres & gusts and landing	Communication	Radio signal loss or signal interference	Undelivered service	5	Faulty hardware, weather location and geographical litiations	7	Radio signal detection, location scouting and pre-flight checks	1	35	Effective COTS component selection Powerful radio antennae RTH system if connection is lost Ground-based antenna on ground station	7	5	System bench test RTH system test	Design and operations teams	5	7	1	35	7	5	18
During flight or during manoeuvres & gusts	Fuselage	Disassembly connections failure	Aircraft, field or crop damage and service not delivered	10	Incorrect assembly from transport configuration to flight configuration, manufacturing error, fatigue and unpredictable weather	1	Pre-flight checks, assembly procedures, QC checks pre and post assembly / manufacture, weather forecasts and inspections	8	80	Safety factor Locking mechanisms Sensors Redundancies	7	5	Fatigue testing Simulated failure test (wings and tail are detached mid flight and the aircraft must recover in VTOL by itself)	Design, manufacturing and operations teams	10	1	8	80	7	5	40

VTOL Boom & Tail Interface Design Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Mitigation through Analysis and Design	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Testing	Resp.	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN = S x O x D	IMPLEMENTATION	EFFECTIVENESS	MRPN (Mitigated RPN) = S x O x E / 10
When could the failure mode occur?	What sub-system is the failure mode part of	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate this failure mode?			What test can be performed to mitigate this failure mode?	Who is responsible for making sure the actions are completed?							
During flight or during manoeuvres & gusts	VTOL Boom, Tail Boom and Tail	Epoxy adhesive failing	Structural failure in the Tail (the Tail disassembles)	9	Incorrect application of epoxy during manufacturing, incorrect adhesive used, insufficient surface area for epoxy to contact, product used after its rated service life	3	QC checks, manufacturing instructions, adhesive check, pre-flight checks	2	54	Gap added between interfaces Correct epoxy spec-ed	10	9	Full structures test	Design, manufacturing and operations teams	9	3	2	54	10	9	49
During flight or during manoeuvres & gusts	VTOL Boom	VTOL Boom (1) cracking or failing at interface with VTOL Boom Tail Attachment (4)	Structural failure in the Tail (the Tail detaches from the rest of the aircraft)	8	Incorrect manufacturing (of CFRP component and / or adaptors), prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor Deep insertion length in CFRP component	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78
During flight or during manoeuvres & gusts	VTOL Boom and Tail Boom	Threaded connection between Tightener (8) and VTOL Boom Tail Attachment (4) failing	Tail is no toger secured to the VTOL boom with a threaded connection and may fall out	5	Fatigue, prior damage, rough handling, poor manufacturing	2	Pre-flight checks, QC checks, inspection schedule & logs	4	40	Over-spec-ed threads and interference fit (H7 - k6) between the VTOL Boom Tail Attachment and the Tail Boom Connector	6	7	Full structures test and full assembly test	Design, manufacturing and operations teams	5	2	4	40	6	7	28
During flight or during manoeuvres & gusts	VTOL Boom and Tail Boom	Tail Boom Connector (6) sliding out of the VTOL Boom Tail Attachment (4)	The Tail detaches from the rest of the aircraft	7	Incorrect tolerances between two interfaces, threaded connection between Tightener and VTOL Boom Tail Attachment failing	2	Pre-flight checks, QC checks	2	28	H7 - k6 tolerances between the two components	6	9	Full structures test and full assembly test	Design, manufacturing and operations teams	7	2	2	28	6	9	25
During drone assembly	Tail Boom	Tightener (8) gets jammed and no longer slides over the Tail Boom Connector (6)	The Tail cannot be properly secured onto the VTOL Boom.	2	Incorrect tolerances between two interfaces	3	Pre-flight checks, QC checks, inspection schedule & logs	1	6	H8 tolerance on Tightener's inner hole Lubrification	8	9	Full assembly test	Design, manufacturing and operations teams	2	3	1	6	8	9	5
During flight or during manoeuvres & gusts	Tail Boom	Threaded connection between Tail Boom Connector (6) and Boom Screw Adaptor (9) failing	The Tail detaches from the rest of the aircraft	7	Fatigue, prior damage, rough handling, poor manufacturing	2	Pre-flight checks, QC checks, inspection schedule & logs	4	56	Over-spec-ed threads	7	7	Full structures test and full assembly test	Design, manufacturing and operations teams	7	2	4	56	7	7	39
During flight or during manoeuvres & gusts	Tail Boom	Tail Boom (5) cracking or failing at interface with Boom Screw Adaptors (9)	Structural failure in the Tail (the Tail disassembles or detaches from the rest of the aircraft)	8	Incorrect manufacturing (of CFRP component and / or adaptors), prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor Deep insertion length in CFRP component	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78
During flight or during manoeuvres & gusts	Tail Boom	Tail Boom (5) cracking and / or failing	Structural failure in the Tail	8	Incorrect manufacturing, prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78
During flight or during manoeuvres & gusts	Tail Boom and Tail	Threaded connection between Vertical Stabilizer Screw Adaptor (7) and Boom Screw Adaptor (9) failing	The Tail detaches from the rest of the aircraft	7	Fatigue, prior damage, rough handling, poor manufacturing	2	Pre-flight checks, QC checks, inspection schedule & logs	4	56	Over-spec-ed threads	7	7	Full structures test and full assembly test	Design, manufacturing and operations teams	7	2	4	56	7	7	39
During flight or during manoeuvres & gusts	Tail	Vertical Stabilizer (13) cracking or failing at interface with Vertical Stabilizer Screw Adaptor (7)	Structural failure in the Tail	8	Incorrect manufacturing (of CFRP component and / or adaptors), prior damage, fatigue, rough handling	2	Pre-flight checks, QC checks, inspection schedule & logs	7	112	Safety factor Deep insertion length in CFRP component	8	7	Full structures test	Design, manufacturing and operations teams	8	2	7	112	8	7	78
During drone assembly	Electrical Systems	Wires in Tail Boom (5) over-stretching, leading to quick-connect plugs disconnecting	Servos no longer recieve power to actuate the control surfaces	7	Incomcet handling during assembly, incomcet wire lengths installed, tngled wires inside the boom	7	Pre-flight checks	1	49	Wire length considered to maintain slack in connection	10	9	Full assembly test and Electrical systems test	Design, manufacturing and operations teams	7	7	1	49	10	9	44
During transportation, assembly and flight	Electrical Systems	Quick-connect plugs failing	Servos no longer recieve power to actuate the control surfaces and an electrical fire may start	8	Incomcet handling during assembly, fatigue, product used after its rated service life	3	Pre-flight checks, inspections	3	72	Correct spec-ing of quick-connect plugs	7	7	Electrical systems test	Design, manufacturing and operations teams	8	3	3	72	7	7	50
During transportation, assembly and flight	Electrical Systems	Damage to electrical wires	Servos no longer recieve power to actuate the control surfaces and an electrical fire may start	8	Incomcet handling during assembly, product used after its rated service life, incomcet wire used, incomcet wire length used	3	Pre-flight checks, inspections	4	96	Wire insulation and protection Sensors	7	7	Electrical systems test	Design, manufacturing and operations teams	8	3	4	96	7	7	67

Drone Assembly Process Failure Mode Effects Analysis

Process Step	Sub-system	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Process Controls	DETECTION (1 - 10)	RPN (Risk Priority Number)	Process Mitigation	IMPLEMENTATION (1 - 10)	EFFECTIVENESS (1 - 10)	Mitigation through Analysis and Design	Resp.	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN = S x O x D	IMPLEMENTATION	EFFECTIVENESS	MRPN (Mitigated RPN) = S x O x E / 10
When could the failure mode occur?	What sub-system is the failure mode part of	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What design feature can be implemented to mitigate this failure mode?			What design feature can be implemented to mitigate this failure mode?	Who is responsible for making sure the actions are completed?							
During drone assembly	Entire drone assembly	Incorrect handling of drone components	Broken components, higher likelyhood of structural failure modes occurring	8	Operator & Technician carelessness, Poor training	4	Component handling instructions and procedures, Technician training, strict hiring processes	4	128	Technician training	10	7	Higher structural redundancy and component sturdiness to avoid damage	Design team, Technicians and Operators	8	4	4	128	10	7	90
During drone assembly	Entire drone assembly	Over-tightened screws	Cracks in CFRP components, damage to screw or mounting	7	Incorrect tightening of screws	5	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	6	210	Specific screw torque value set for every screw	8	7	N/A	Design team, Technicians and Operators	7	5	6	210	8	7	147
During drone assembly	Entire drone assembly	Under-tightened screws	Higher likelyhood of components detaching	8	Incorrect tightening of screws	3	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	4	96	Specific screw torque value set for every screw	8	7	N/A	Design team, Technicians and Operators	8	3	4	96	8	7	67
During drone assembly	Entire drone assembly	Not all screws are installed	Higher likelyhood of components detaching	9	Operator & Technician carelessness, Poor training	3	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	27	Detailed assembly instructions	10	7	N/A	Design team, Technicians and Operators	9	3	1	27	10	7	19
During drone assembly	Entire drone assembly	Un-mounted components (not all components are mounted)	Drone cannot fly	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	N/A	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Entire drone assembly	Components mounted in the wrong location	Drone cannot fly	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	Specific interfaces for each component	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Entire drone assembly	Components mounted in the wrong orientation	Drone cannot fly or inability to carry out the mission	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions	10	7	Specific interfaces for each component	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Entire drone assembly	Power and electrical connections not connected (quick connect plugs not connected)	Drone cannot fly	10	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	10	Detailed assembly instructions Pre-flight systems test	10	7	N/A	Design team, Technicians and Operators	10	1	1	10	10	7	7
During drone assembly	Fuselage and Spray Booms	Tubing and hose connections not connected	Inability to carry out the mission, payload leak	7	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	7	Detailed assembly instructions Pre-flight systems test	10	7	N/A	Design team, Technicians and Operators	7	1	1	7	10	7	5
During drone assembly	Fuselage	Incorrect install of battery	Drone cannot fly, potential damage to the battery	8	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	1	8	Detailed assembly instructions Pre-flight systems test	10	7	Specific battery interface that highlights orientation	Design team, Technicians and Operators	8	1	1	8	10	7	6
During drone assembly	Fuselage	Hatches not properly closed	Aerodynamic inefficiencies, components falling out during manoeuvres	6	Operator & Technician carelessness, Poor training	3	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	4	72	Detailed assembly instructions	10	7	Simplified hatch system	Design team, Technicians and Operators	6	3	4	72	10	7	50
During drone assembly	Wings and Fuselage	Latches on wings not closed	Wings falling off of the drone during a manaouvre	9	Operator & Technician carelessness, Poor training	1	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	3	27	Detailed assembly instructions	10	7	Simplified wing connection system	Design team, Technicians and Operators	9	1	3	27	10	7	19
During drone assembly	Tail	Tail Booms not screwed correctly to tail	Tail falling off of the drone during a manaouvre	10	Operator & Technician carelessness, Poor training	2	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	3	60	Detailed assembly instructions	10	7	Simplified tail connection system	Design team, Technicians and Operators	10	2	3	60	10	7	42
During drone assembly	VTOL Booms and Tail	Connection between VTOL Booms and Tail Assembly not secured properly (Tightnener not fully tightened)	Tail falling off of the drone during a manaouvre	10	Operator & Technician carelessness, Poor training	2	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	3	60	Detailed assembly instructions	10	7	Simplified tail connection system	Design team, Technicians and Operators	10	2	3	60	10	7	42
During drone assembly	Fuselage	Payload tank not sealed	Payload leak	5	Operator & Technician carelessness, Poor training	6	Assembly procedures, Technician training, strict hiring processes, pre-flight checks	2	60	Detailed assembly instructions	10	7	Simplified hatch system	Design team, Technicians and Operators	5	6	2	60	10	7	42
During drone assembly	Entire drone assembly	Drone tipped over	Structural damage to the drone	7	Operator & Technician carelessness, Poor training	6	Component handling instructions and procedures, Technician training, strict hiring processes	1	42	Technician training Design: landing gear positioning	10	7	COG with / without payload and with / without battery considered in design regarding landing gear placement and spacing	Design team, Technicians and Operators	7	6	1	42	10	7	29
During drone assembly	Entire drone assembly	Pre-flight checks not followed or carried out	Higher likelyhood of failure modes occurring	9	Operator & Technician carelessness, Poor training	4	Technician training, strict hiring processes	5	180	Technician training	10	7	Reduced pre-flight operations needed to achieve UAV flight readiness	Design team, Technicians and Operators	9	4	5	180	10	7	126

Severity Scale

Effect	Criteria: Severity of Effect	Ranking
Hazardous - Without Warning	May expose client to loss, harm or major disruption - failure will occur **without** warning.	10
Hazardous - With Warning	May expose client to loss, harm or major disruption - failure **will occur with** warning.	9
Very High	Major disruption of service involving client interaction, resulting in either associate re-work or inconvenience to clients	8
High	Minor disruption of service involving client interaction and resulting in either associate re-work or inconvenience to clients	7
Moderate	Major disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	6
Low	Minor disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	5
Very Low	Minor disruption of service involving client interaction that does not result in either associate re-work or inconvenience to clients	4
Minor	Minor disruption of service not involving client interaction and does not result in either associate re-work or inconvenience to clients.	3
Very Minor	No disruption of service noticed by the client in any capacity and does not result in either associate re-work or inconvenience to clients.	2
None	No Effect	1

Occurance Scale

Probability of Failure	Time Period	Per Item Failure Rates	Ranking
Very High: Failure is almost inevitable	More than once per day	>= 1 in 2	10
	Once every 3-4 days	1 in 3	9
High: Generally associated with processes similar to previous processes that have often failed	Once every week	1 in 8	8
	Once every month	1in 20	7
Moderate: Generally associated with processes similar to previous processes which have experienced occasional failures, but not in major proportions	Once every 3 months	1 in 80	6
	Once every 6 months	1 in 400	5
	Once a year	1 in 800	4
Low: Isolated failures associated with similar processes	Once every 1 - 3 years	1 in 1,500	3
Very Low: Only isolated failures associated with almost identical processes	Once every 3 - 6 years	1 in 3,000	2
Remote: Failure is unlikely. No failures associated with almost identical processes	Once Every 7+ Years	1 in 6000	1

Detection Scale

Detection		Criteria: Likelihood the existence of a defect will be detected by process controls before next or subsequent process, -OR- before exposure to a client	Ranking
Almost Impossible	Can only be detected in flight	No known controls available to detect failure mode	10
Very Remote		Very remote likelihood current controls will detect failure mode	9
Remote		Remote likelihood current controls will detect failure mode	8
Very Low		Very low likelihood current controls will detect failure mode	7
Low	Can be detected during takeoff	Low likelihood current controls will detect failure mode	6
Moderate		Moderate likelihood current controls will detect failure mode	5
Moderately High	Can be detected on the ground / during setup	Moderately high likelihood current controls will detect failure mode	4
High		High likelihood current controls will detect failure mode	3
Very High		Very high likelihood current controls will detect failure mode	2
Almost Certain		Current controls almost certain to detect the failure mode. Reliable detection controls are known with similar processes.	1

Implementation Scale

Implementation	Criteria: Likelihood the implementation of the design feature or testing process as a form of mitigation for the failure mode	Ranking
Certain	Will be implemented	10
Very High	Has a very high chance of implementation	9
High	Has a high chance of implementation	8
Moderately High	Has a moderately high chance of implementation	7
Moderate	Has a moderate chance of being implemented	6
Low	Has a low chance of implementation	5
Very Low	Has a very low chance of implementation	4
Remote	Has a remote chance of implementation	3
Very Remote	Has a very remote chance of implementation	2
Impossible	Will not be implemented	1

Effectiveness Scale

Effectiveness	Criteria: Effectiveness of the design feature or testing process in solving or mitigating the failure mode	Ranking
Completely Ineffective	The mitigation will have no effect on the failure mode	10
Partially Ineffective	The mitigation will have very little effect on the failure mode	9
Very Low	The mitigation will reduces the chances of the failure mode occuring substantially	8
Low	The mitigation will moderately reduce the chances of the failure mode occuring	7
Moderately Low	The mitigation will reduce the chances of the failure mode occuring	6
Moderate	The mitigation will partly solve the problem	5
Moderately High	The mitigation will greatly reduce the chances of the failure mode occuring	4
High	The mitigation will reduces the chances of the failure mode occuring substantially	3
Very High	The mitigation will solve the problem almost in its entirety	2
Complete	The mitigation will solve the problem	1

The FMEA and DFMEA follow the same construction method. Each failure mode was fully defined in terms of process step (when the failure mode occurrs), the involved sub-system and its failure mode, the potential failure mode effects or consequences, potential causes, current process controls (processes put in place to detect or predict the failure mode's occurrence) and who is responsible for the oversight of this failure mode. A severity, occurrence and detection score was given, and those were multiplied to calculate the Risk Priority Number (RPN). Mitigations through design were outlined and given an implementation (probability of getting implemented and the willingness and ability for a manufacturer to manufacture that design feature*) and effectiveness (how well this design feature mitigates the failure mode) score. Mitigation through testing was also outlined. The effectiveness score was then used to calculate the Mitigated Risk Priority Number (MRPN).

The PFMEA follows a similar process, with the failure mode being defined in an identical way. However, a focus was put on process mitigations rather than design mitigations, with implementation and effectiveness scores being stated for the process mitigation. Analysis and design mitigations were also outlined. The MRPN was calculated in the identical way.

*A manufacturer may have a tolerance limit on their equipment and tools, thus the implementation score of a design feature that requires precise tolerancing is reduced.

Appendix O: Team Bath Drones CDR Weight Breakdown

Team Bath Drones CDR Weight Breakdown				
Sub-system	**Component**	**Material**	**Weight (kg)**	**% of MEW**
Fuselage	Internal Structure	Birch Faced Poplar Core Plywood	0.55	9.52
	Carbon Skin	Carbon Fibre Pre-Preg (MTC 510)	0.2	3.46
	Hatch	Glass Fibre Pre-Preg (XC130)	0.09	1.56
	Landing Structure	Plywood	0.08	1.38
	Wheels	COTS	0.1	1.73
	Tail Connection	PLA	0.05	0.87
	Spars	6061-T8 Aluminium	0.18	3.11
	Total		1.25	21.63
Wing Structures	Skin	Carbon Fibre Pre-Preg (XC 130)	0.8	13.84
	Ribs	Balsa Wood	0.4	6.92
	Stiffeners	ROHACELL	0.02	0.35
	Carbon Spars	Carbon Fibre Pre-Preg (MTC 510)	0.35	6.06
	Servo Housings	PLA	0.04	0.69
	Total		1.61	27.85
Avionics/Powertrain	Motor	COTS	0.45	7.79
	VESC	COTS	0.18	3.11
	Servos	COTS	0.12	2.08
	Avionics	COTS	0.35	6.06
	Battery	COTS (Lithium Polymer)	0.67	11.59
	Propeller	COTS (APC)	0.2	3.46
	Tracker and GPS	COTS	0.06	1.04
	Total		2.03	35.12
Payload	Water	-	4	
	Container	COTS (Polypropylene)	0.09	1.56
	Total		4.09	1.56
Tail Structure	Boom	Carbon Fibre Pre-Preg (MTC 510)	0.17	2.94
	Spars	Sandwich Nomex Core 3mm Carbon Fibre	0.03	0.52
	Surfaces	XPS Extruded Polystyrene Foam	0.06	1.04
	Servo Housings	PLA	0.04	0.69
	Total		0.3	5.19
Reserve	(Fasteners, Adhesives, Cables, Baffles)	Various	0.5	8.65
Total	**MTOW**		**9.78**	
Total	**MEW**		**5.78**	**100.00**

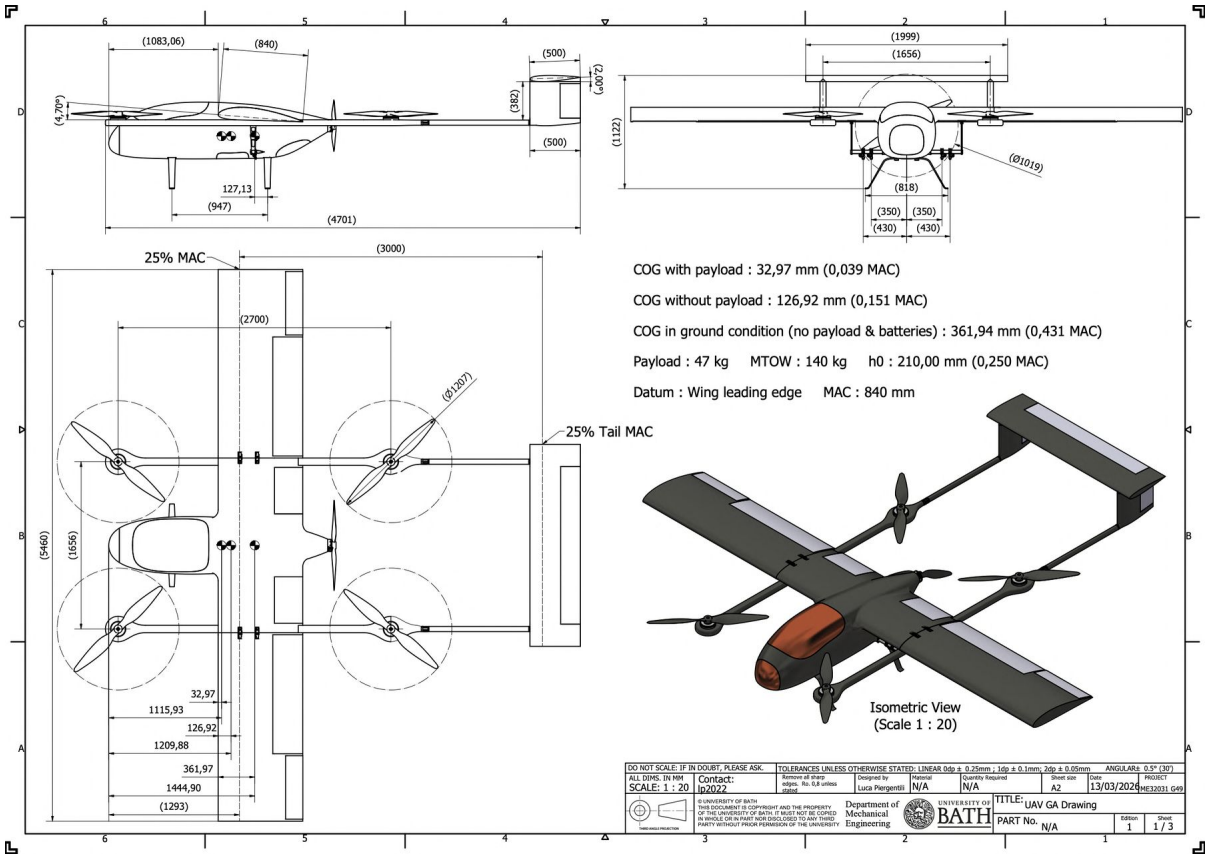
Appendix P: COTS Components Table

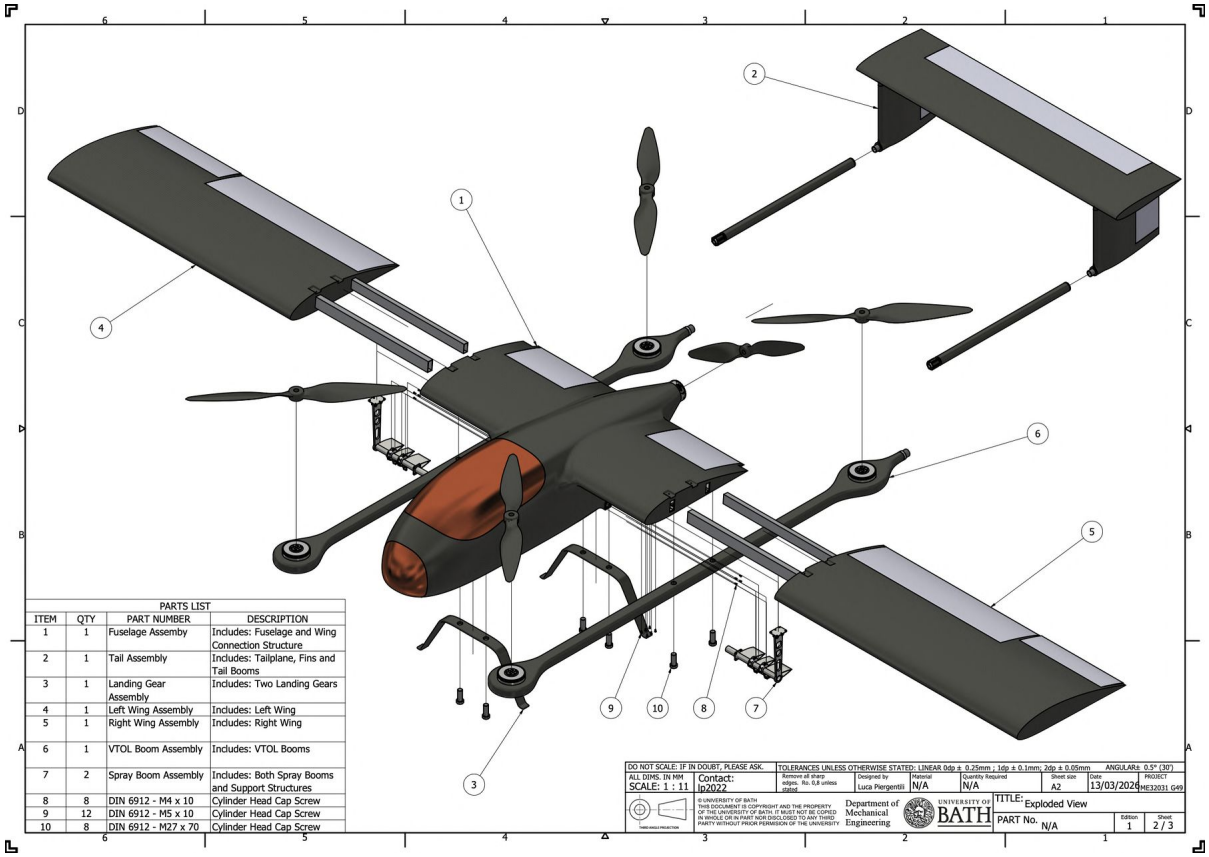
Component Identification			Geometrical Dimensions				Quantity and Weight			Location of component			Additional Information			
Sub System	Component	Use	Length (mm)	Width (mm)	Height (mm)	Diameter Ø	Quantity	Weight (kg)	Tot Weight (kg)	Positional Preference & Requirements	Mounting Preference	Connected components	Cost	Link to product	CAD model	Notes	Name
Propulsion	Vertical Motors	Vertical Propulsion	85			r152	6	2.789	11.136	Mounted on the VTOL boom structure. Required to orient airflow for cooling and far away from the fuselage or any other body to allow for a propeller clearance.	On ends of boom	VTOL ESC	2957.66	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Fuselage -> Parts -> COTS Components -> VTOL Motor		Jairam
	Vertical Propellers	Vertical Propulsion	~700		25.7	r13.06.5	6	0.45	1.8	Mounted on motor	None	VTOL Motor	1399.66	https://store.tmotor.com/goods-12-inch-carbon-fiber-propeller-for-vtol-drone-cw-ccw-1-pair-set.html	Sub-Assemblies -> Fuselage -> Parts -> COTS Components -> VTOL Propeller		Jairam
	Propulsion (VTOL Motor ESC)	Afuker Pro H2O53.2-65-B12-DC-FC -6 in total	159	68	32.5		6	0.42	1.68	Inside the structure of the aircraft, must be as close as possible to the motor/motor mount	Screwed onto using a u-shaped mounted bracket near the motors/disconnect point of the VTOL/Wing hang	VTOL motors	3552.86	https://www.tmotor.com/h55a-120a-esc.html		N/A	Jairam
	Horizontal Motor	Horizontal Propulsion	63.9		N/A	r120.1	1	1.319	1.319	Mounted inside the fuselage/close to the earth center as/as close to airflow for cooling.	On ends of Elevon	Cruise ESC	998.61	https://www.tmotor.com/u15-kv100.html	Sub-Assemblies -> Fuselage -> Parts -> COTS Components -> Cruise Motor		Jairam
	Horizontal Propeller	Horizontal Propulsion	~612.71		21.85	r1018.54	1	0.395	0.395	Mounted on motor	None	Cruise Motor	378.89	https://store.tmotor.com/goods-12-inch-carbon-fiber-propeller-for-vtol-drone-cw-ccw-1-pair-set.html	Sub-Assemblies -> Fuselage -> Parts -> COTS Components -> Cruise Propeller	Price not available online so will may be needed but a custom propeller may be used instead (TBD)	Jairam
	Propulsion (Cruise Motor ESC)	Thunder 1.2" 2ch 200A High Power 12S/14S ESC - 1 in total	148	69.6	36.1		1	0.62	0.62	Inside the structure of the aircraft, must be as close as possible to the motor/motor mount	Screwed onto using a u-shaped mounted bracket near the rear of the fuselage	Cruise motor	1009.65	https://store.tmotor.com/goods-thunder-1-2-2ch-200a-high-power-12s-14s-esc-1-piece.html	Sub-Assemblies -> Fuselage -> Parts -> COTS Components	N/A	Jairam
		Total							16.77				9695.27				
Payload (7 estimates)	Payload tank	Storing payload	330	n/a	n/a	564.9911582	1	0.2	0.2	Inside the fuselage, preferably on the fuselage's centroid/gravity	Rigid mounts to fuselage/some sorta/or vibration-dampening/rubber (possibly)	Barbed fitting adapter to hose to pump inlet	Unknown (Replicate/ custom built)	custom built	Sub-Assemblies -> Fuselage -> User's Structure Test		Sam
	1/4" BSP female/male connector / bulkhead	CUP-1C-S6BSP					1	0.03	0.03		Ziptie	1/4" BSP F -> 1/4" BSP F	6.25	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	3/8" BSP male x 19mm hose barb adapter	B0012203					1	0.018	0.018		Ziptie	3/8" BSP M -> 19mm barb	0.92	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	3/4" ID (19mm) reinforced braided PVC suction hose/1 m length cut)	BWR-19-SG					1	0.028	0.028		Ziptie	~	2.2	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Bag/Hepa-vac 1/2" in-line suction filter, 50 mesh	20982250					1	0.001	0.001		Ziptie	1/2" BSP F -> 1/2" BSP F	14.68	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Pump	Remco FATBOY Pump - 1 in total	198.12	127	115.5		1	4.5	4.5	Inside the fuselage, behind the payload tank	Pump sit's the central tank which should reduce the chance of air ingestion during spraying because the aircraft will in real likely fly at an angle of inclination so gravity will help	To central tank	181.65	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Components		Jonathan
	1/2" NPT male x 13mm hose barb adapter — pump outlet side	B0011111-NPT					1	0.015	0.015		Ziptie	1/2" NPT M -> 13mm barb	11.99	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" ID black reinforced rubber pressure hose, 25 bar (~3.6 psi)	APR-13					1	0.096	0.096		Ziptie	~	9.08	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" BSP male x 13mm hose barb adapter	B0011111					2	0.015	0.03		Ziptie	1/2" BSP M -> 13mm barb	9.86	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" BSP equal female threaded tee	B0210151					1	0.02	0.02		Ziptie	1/2" BSP F -> all 3 ports	2.8	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Pressure sensor	Leak checks, droplet size gauge and payload system health check		~17.9 (hex side length)	~40	~30	1	0.01	0.01	Inside the fuselage, close to the pump	Should be braced to fuselage if possible	Pump outlet, short hose to solenoid valve inlet	20.95	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	Requires small adapters and hoses for integration	Jonathan
	M12 A-coded 5-pin IP67 field-assembly connector	M12-5A-IP67					1	0.02	0.02			M12 A-coded 5-pin	35.2	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Solenoid Valve	Leak check and flow control	65	33	33		1	0.06	0.06	Inside the fuselage	Small bracket mount on fuselage	Pressure sensor T-piece outlet, short hose to flow splitter inlet	105.3	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	Requires small adapters and hoses for integration	Jonathan
	DIN 43650 Form A IP65 cable plug	DIN43650A-IP65					1	0.01	0.01			DIN 43650A socket	1.29	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" BSP female x 1/2" NPT male adapter	BSP12-NPT34					1	0.03	0.03			1/2" BSP F -> 3/4" NPT M		Any plumbing supplier	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Banjo 8-station manifold 7M/7F/6.5TAD	TM07F/6TAD					1	0.25	0.25			3/8" NPT M inlet / 6×1/2" NPT F outlets	56.66	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Jonathan
	1/2" NPT plug — bladed outlets	NPT12-PLG					4	0.005	0.02			1/2" NPT M -> sealed end		Any plumbing supplier	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" NPT x 13mm hose barb adapter — active section	B0011113-NPT					6	0.015	0.09			1/2" NPT M -> 13mm barb	47.96	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	1/2" ID clear PVC hose — main field to nozzle feed (~2 runs * 5 m each)	CUP-13-1					1	0.18	0.18			~	2.33	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Stainless hose clips 13-20 mm	SS-HC-13-20					2	0.005	0.01			13-20 mm clamp range	0.6	Any hardware store	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Tee Jet DJ7TS02, 20mm/3/8 nozzle body with Diaphragm check valve	Mounting					6	0.005	0.18	Mounted on the spray boom	Flat surface mount via integral mounting hole in upper clamp, bolt directly to spray boom tube M6 stainless fasteners	Flow splitter outlet/hose & Spray tip	19.84	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Spray Boom -> Parts -> COTS Components -> Nozzle Body		Jonathan
	Teejet AIXR8002VK extended range flat fan spray tip, 80°, stainless steel insert, white visiflo	XR8002VK					6	0.001	0.02	1.8	7.2	Teejet bayonet -fit flat fan spray outlet		Any retail / supplier	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	PTFE tape						1	0.01	0.01	41.00	41.00		1	Any hardware store			
Spray boom	6mm x 50mm .02 stainless quick-release ball-lock detent pin with pull ring	QRDP0502 (Voltage connection — power through single hole in main rod and bushing/flange to retain hose assembly in airframe)					2	0.014	0.028	Inside spray boom			2.95	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)	24 bought for £2.95, only needed	Jonathan
	6mm x 100mm .02 stainless quick-release ball-lock detent pin with pull ring	QRDP0210 (Wiring connections — one pin wing rod outer hole, each pin through single wing rod and wiring flight bus)					4	0.014	0.056	Inside spray boom			0	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	2mm x 150mm stainless steel lanyard with loop ends	LSMD-2-150-SS (Tether pin lock pin — tether each pin to its flange to prevent loss during field removal or in-flight vibration)					4	0.005	0.02	Inside spray boom			Any rigging supplier	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	25mm bore single-split shaft collar, .62 diameter, 304 stainless, 10mm ID, 15mm wide, M6 clamp screw	SC-25-SS (Two per wing rod position — one collar each side of sleeve to prevent wing rod sliding axially along its rotation/ leading/ turns and lateral acceleration)					10	0.1765	1.765	Inside spray boom			28.95	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Loctite 243 medium-strength threadlocker (blue), 10ml	LOCTITE-243-10ML (Applied to M6 shaft collar clamp screws — prevents collar loosening under flight vibration while allowing hex-key removal for repositioning)					2	0.012	0.024	Inside spray boom			15.97	https://www.ebay.co.uk/itm/115	N/A(Adapter/hose/electrical component DO NOT CAD)		Jonathan
	Spray boom (estimate)						1	1	1	Under Structure	Mounted on the fuselage				Sub-Assemblies -> Spray Boom -> Parts		
		Total							8.665				579.76				
Power/Electrical Electronics	Battery	Semi-solid State 21700Ah,5g 51800mAh 10C 6S 4V 12S1P LiPO Battery	215	91	118	N/A	4	0.686	18.764	Inside the fuselage, can be/used as ballast for the COG but it would be/very beneficial to be/as close to the/ and closest to the/ front of the aircraft	Mounted flat within fuselage for better cabling (reduce cable stretching)	To power module	5016.2	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	5V Step-Down Voltage Module	Responsible for converting battery voltage and power to suitable values for flight controller, working in tandem with power distribution blocks. Applicable to FC, RGB lamps, Telemetry and ESCs.	113	60	25	N/A	2	0.25	1	Inside the fuselage, in close proximity to power distribution blocks	screwed down (with a mounting bracket)	power distribution blocks	93.84	~	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	12V Step-Down Voltage Module (for Servo UBEC's) - RSD-1000-12	Step-down 1000v to 12V for each of the both tail and wing-aileron servo UBEC's	161	68	96	N/A	3	0.6	1.8	Inside the fuselage, in close proximity to power distribution blocks	screwed down (with a mounting bracket)	power distribution blocks	169.53	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	12V Step-Down Voltage Module (for Servo UBEC's) - RSD-1000-12	Step-down 1000v to 12V for each of both/each flap servo UBEC	191	86	60	N/A	1	1	1	Inside the fuselage, near where the wing attach/detach	screwed down (with a mounting bracket)	power distribution blocks	80.61	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	12V Step-Down Voltage Module (for Spray Pump) - RSD-1000-12-250 TP	Step-down 1000V to 12V for spray pump	81	57.9	12.7	N/A	1	0.105	0.105	Inside the fuselage, in close proximity to pump motor controller	Screwed down with a mounting bracket	pump motor controller	207.35	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	To be noted that the version in the COTS link is the standard version. The RSD-1000-12-250-12-TP, the/ effective overall wire termination for connecting wires to it, which is the actual component required but that is to be/ directly requested from supplier.	Aman
	7.4V Step-Down Voltage Module - Holdsporting UBEC 2S-6V	Step-down 12V to 7.4V for each of/ the servos	55	20.2	17.6	N/A	2	0.074	0.148	Mounting bracket	Mounting bracket	Each wire connected to a step-down 12V converter with wire connecting to the wing servos, and the other connected to all four tail servos	116.98	~	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	Power distribution block	Responsible for power diversion to all sub-components	100.8	67	66.3	N/A	4	0.44	1.76	Within fuselage, 2 near the flight controller and 2 towards the rear of the fuselage/ close to the payload system , cruise motor and wing attachment points	Mounted flat within fuselage with sufficient clearance for cabling	2 out of distribution blocks for connecting batteries to all components, 1 out of distribution block for each wing that either 2 blocks to be connected to the first two blocks to act as a hub to connect all other auxiliary components electrically	782.4	https://www.tmotor.co.uk/Products/Detail/Power%20Distribution%20Board	Sub-Assemblies -> Wings -> Parts -> Aileron Servo		Aman
	Servo(Aileron)	Aileron Control Surface Actuation - MD80T35NBP-DroneCAN	41	21	26.5	N/A	2	0.07	0.14	Integrated into the wing Unit & RHS Aileron	Within a Holder Bracket to be screwed into the wing, provide around 10 mm gap from servo to aileron	To Flight Controller through a CANBus splitter & connected to UBEC	387.04	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Wings -> Parts -> Aileron Servo		Aman
	Servo(Elevator)	Elevator Control Surface Actuation - MD90S70M-DroneCAN	60	30	38	N/A	2	0.0868	0.1328	Integrated into the Unit & RHS Elevator on Tail	Within a Holder Bracket to be screwed into the Tail, provide around 10 mm gap from servo to elevator	To Flight Controller through a CANBus splitter & connected to UBEC	314.76	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Tail -> Parts -> Elevator Servo		Aman
	Servo(Rudder) (same as elevator)	Rudder Control Surface Actuation - MD90S70M-DroneCAN	60	30	38	N/A	2	0.0868	0.1328	Integrated into the rudder on the Tail	Within a Holder Bracket to be screwed into the Tail, provide around 10 mm gap from servo to rudder	To Flight Controller through a CANBus splitter & connected to UBEC	314.76	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Tail -> Parts -> Elevator Servo		Aman
	Servo (Flaps on wing)	Large Control Surface Actuation - MD80T60-DAP-CAN-DroneCAN	60	30	39	N/A	2	0.09	0.18	Integrated into the wing Unit & RHS Flaps	Within a Holder Bracket to be screwed into the Flat, provide around 10 mm gap from servo to elevator	To Flight Controller through a CANBus splitter & connected to UBEC	408.78	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Tail -> Parts -> Elevator Servo		Aman
	Servo (Flaps on fuselage)	Small Flap Control Surface Actuation - MD80T190DP-DroneCAN	41	21	26.5	N/A	2	0.07	0.14	Integrated into the wing Unit & RHS Flaps	Within a Holder Bracket to be screwed into the Flat, provide around 10 mm gap from servo to elevator	Between servo ESC for the ailerons and each of the ailerons (as the ESC powers both ailerons servos)	867.04	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Tail -> Parts -> Elevator Servo		Aman
	Wing-rod connectors	The connectors are electrical/mechanical junctions located where/ critical/ MEG wing and fuselage meet	N/A	N/A	N/A	N/A	2	0.04	0.08	Where wing-section and fuselage meet, as central/ within the airfoil cross-section as possible	To mount to	Between servo ESC for the ailerons and each of the ailerons (as the ESC powers both ailerons servos)	113.98	https://www.tmotor.com/u8-kv100.html	Sub-Assemblies -> Tail -> Parts -> Elevator Servo		Aman
	VTOL Motor Bullet Connectors - 4mm	To allow for a quick connect between the VTOL ESC to the individual VTOL motors	19	N/A	N/A	4	24	0.009	0.216	Located within Booms	Soldered to power and ground wiring where the female male connectors meet.	For the wiring connecting each VTOL motor to its individual ESC	48	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Tail -> Parts -> Elevator Servo	3 Male and 3 female connectors between each VTOL motor -> ESC pair.	Aman
	Tail-cuddle/elevator servo Bullet Connectors - 4mm	To allow for a quick connect between the Servo ESC to all the servos for the tail control surfaces.	19	N/A	N/A	4	12	0.0019	0.0168	Between rear booms and tail	Soldered to power and ground wiring where the female male connectors meet.	For the wiring connecting each servo to the Hurricane ESC	2	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Tail -> Parts -> Elevator Servo	2 Male and 2 female connectors between each servo -> ESC pair.	Aman
	CANbus Splitter	Extender for CANbus ports on FC carrier board	12	54	5 (assumed)	N/A	3	0.003	0.009	Close proximity to flight controller	Holder bracket held in place within fuselage using velcro	CANG and CANG portion Orange Cube, outputs to/ tie signal wires for all/ servos. Position one splitter somewhat/ close to/ longitudinal location of wing and the other splitter towards the rear of the drones.	35.7	https://docs.cubepilot.org/user-guides/carrier-boards/can-hub-and-splitter	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	Sensor Hub - PL Sensor Hub 42 V2	Sends the info to the flight controller from the/ dual sensor boards.	42	54	10	N/A	1	0.007	0.007	Close proximity to batteries, close to where the battery connector wires come out from.	Screwed down on a flat surface (or small mounting bracket)	both sensor boards. Output of sensor hub is connected directly to/ flight controller.	36.43	https://docs.cubepilot.org/user-guides/carrier-boards/can-hub-and-splitter	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	Sensor Board	Measuring current output from both serial strings from the battery packs	63.5	23	15	N/A	2	0.05	0.1	Close proximity to batteries, close to where the battery connector wires come out from.	Sensor board comes attached with soldered on 0.5000 wires, so allocate 1 board to each battery pack pair and place each 4 short of these	Input for each board comes from/ serial connector output for each battery pack pair. Output for each board is/ connected to the sensor hub	75.08	https://docs.cubepilot.org/user-guides/carrier-boards/can-hub-and-splitter	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Aman
	Pump Motor Controller - Syren 25	Control over spray pump	61	56	21	N/A	1	0.055	0.055	Close proximity to payload pump	Mounted flat near spray pump	To 12V Step-down as an input and to the pump as an output, while also connected to the arrange rules	66.25	https://www.dimensionengineering.com/products/syren25	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	Looking into controllers for pump	Aman
		Total							21.7642				8720.53				
Navigation and Sensors	CubeOrange Flight controller	To control flight	94	35	33	N/A	1	0.073	0.073	Anywhere in the fuselage, preferably close to the main	None	Any electrical component	375	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	GPS		80	80	20	N/A	2	0.05	0.1	Placed near the top of the fuselage, second one must be at least 10 cm further from the main than the first module, both in/ the longitudinal/ fuselage and same height. Needs an RF transparent material above it. Placed away from other electronics to avoid disturbances.	Mounted on fiberglass or plastic not metal	Flight controller	225	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Radar sensor	To detect altitude	110	80	20	N/A	1	0.11	0.11	Mounted underneath the fuselage. Requires a clear view of the ground, not through any material		Flight controller	662.81	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Crop health sensor body	To sense crop health	60	60	20	N/A	1	0.072	0.072	Mounted under the drone facing towards the ground, nearer, at flow or nozzle/ to take a view of the ground. Not through any material. Vibration reduced mounting	For these purposes it also is semi-required/ beneficial to/ put on a channel that allows airflow into this section. Rubber mounting/ gasket/ pliable for vibration isolation.	Flight controller	7300	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Crop health sunshine sensor	To sense sunshine	47	40	20	N/A	1	0.055	0.055	Mounted near/ access to the top. Could be mounted on top of the fuselage but allow for routing of cables.	None	Flight controller		https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Radio Receiver	To receive RC control	65	30	15	N/A	1	0.017	0.017	Placed near the top of the fuselage, second one must be at least 10 cm further from the main than the first module, both in/ the longitudinal/ fuselage and same height. Needs an RF transparent material above it. Placed away from other electronics to avoid disturbances.	None	Flight controller	50	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Telemetry Radio	To receive mission parameters	30	50	10	N/A	1	0.03	0.03	Located near/ the wing joints , between second to be an RF transparent area . Potential idea to route coax cables through the wings and have antennas in RF transparent wing tips. Otherwise antennas will be routed to the outside of the wing / fuselage	None	Flight controller	50	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	Depends on region	Peter
	Airspeed sensor	To sense airspeed	27	30	31.4	N/A	1	0.01	0.01	Anywhere in the fuselage, preferably close to the pitot-tube	None	Flight controller		https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Pitot tube		102	N/A	N/A	5	1	0.01	0.01	Located at front or other suitable forward facing location	None	Airspeed sensor	65	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	N/A	Peter
	Camera	Provide vision to pilot	36	36	2	17.5 (cameralens)	1	0.01	0.01	Located at front or other suitable forward facing location	None	Hawkeye	227.9	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents	Radius is given for the lens, other dimensions are for the circuit board	Peter
	Obstacle avoidance radar	Avoid obstacles	115	90	25	N/A	1	0.3	0.3	Located in RF transparent location	Make sure the covers facing forwards, 45° x 45°, refer to data sheet	Flight controller		https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Peter
	Hawkeye system	Communicate with drone	78.5	30	15	N/A	1	0.095	0.095	Located in RF transparent location	1.5 deg from propeller/axial , please see https://docs.cubepilot.org/user-guides/telemetry/herelink-user-guide/beam-fly-and-connection	Flight controller	1100	https://www.ebay.co.uk/itm/115	Sub-Assemblies -> Fuselage -> Parts -> COTS Core parents		Peter
		Total							0.853				4055.71				
		Total							52.0614				25308.27				

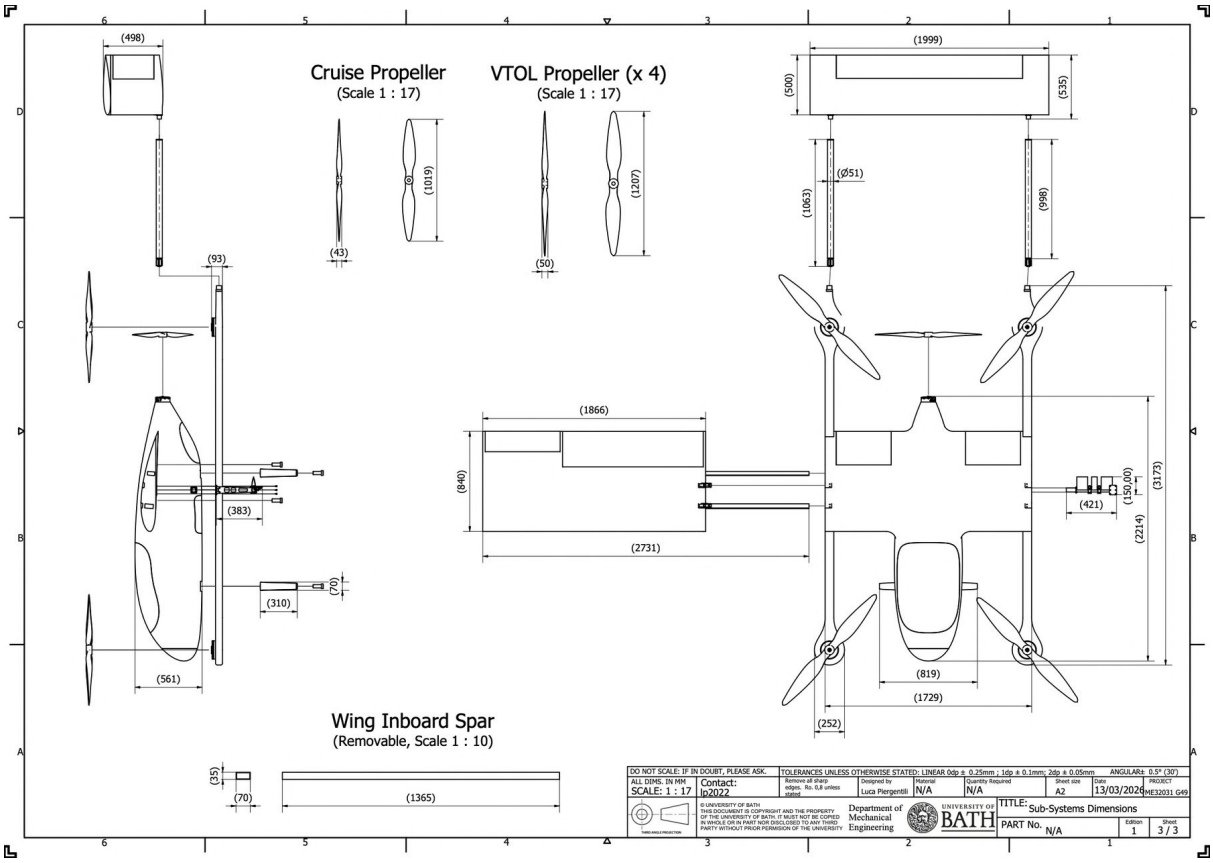
Appendix Q: COG Management Sheet

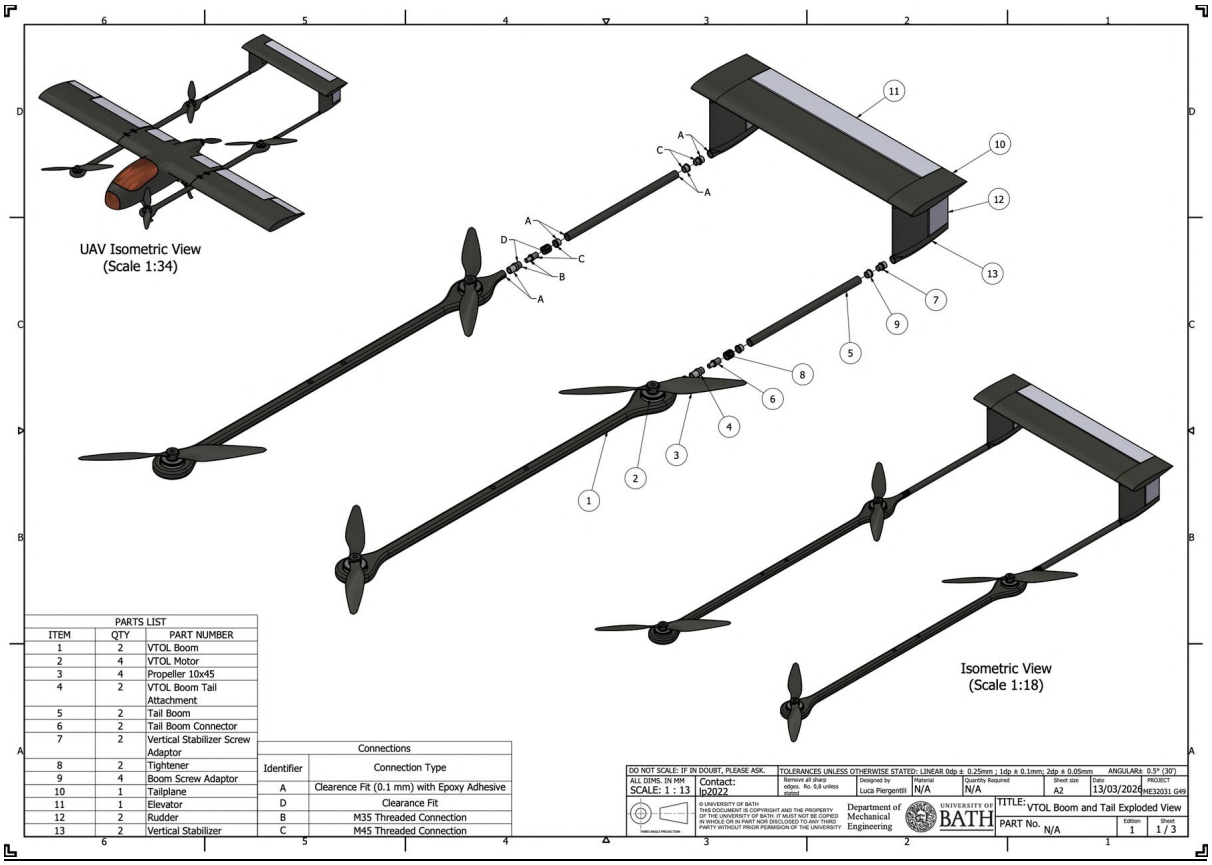
Component Identification			***Datum: Wing Leading Edge***		
Sub-system	**Item**	**Weight (kg)**	**Distance from DATUM (mm)**	**h**	**Moment**
Propulsion	Starboard rear VTOL motor	2.789	**1709.338**	2.035	5.675
	Port rear VTOL motor	2.789	1709.338	2.035	5.675
	Starboard forward VTOL motor	2.789	**-990.662**	-1.179	-3.289
	Port forward VTOL motor	2.789	-990.662	-1.179	-3.289
	Starboard rear VTOL propeller	0.450	1709.338	2.035	0.916
	Port rear VTOL propeller	0.450	1709.338	2.035	0.916
	Starboard forward VTOL propeller	0.450	-990.662	-1.179	-0.531
	Port forward VTOL propller	0.450	-990.662	-1.179	-0.531
	Starboard rear VTOL motor ESC	0.420	302.320	0.360	0.151
	Port rear VTOL motor ESC	0.420	302.320	0.360	0.151
	Starboard forward VTOL motor ESC	0.420	230.320	0.274	0.115
	Port forward VTOL motor ESC	0.420	230.320	0.274	0.115
	Horizontal motor	1.319	**1095.195**	1.304	1.720
	Horizontal propeller	0.195	1146.120	1.364	0.266
16.77	Horizontal motor ESC	0.620	993.070	1.182	0.733
Payload	Payload tank	0.200	-154.609	-0.184	-0.037
	3/4" BSP female tank connector / bulkhead	0.030	370.669	0.441	0.013
	3/4" BSP male × 19mm hose barb adapter	0.018	370.669	0.441	0.008
	3/4" ID (19mm) reinforced braided PVC suction hose (~0.3m single run)	0.028	370.669	0.441	0.012
	Arag/Hypro 1/2" inline suction filter, 50 mesh	0.045	370.669	0.441	0.020
	Pump	4.500	**146.238**	0.174	0.783
	1/2" NPT male × 13mm hose barb adapter — pump outlet side	0.015	370.669	0.441	0.007
	1/2" ID black reinforced rubber pressure hose, 25 bar (~0.3 m)	0.036	370.669	0.441	0.016
	1/2" BSP male × 13mm hose barb adapter	0.030	370.669	0.441	0.013
	1/2" BSP equal female threaded tee	0.020	370.669	0.441	0.009
	Pressure sensor	0.010	273.980	0.326	0.003
	M12 A-coded 3-pin IP67 field-assembly connector	0.020	370.669	0.441	0.009
	Solenoid Valve	0.040	281.189	0.335	0.013
	DIN 43650 Form A IP65 cable plug	0.010	370.669	0.441	0.004
	1/2" BSP female × 3/4" NPT male adapter	0.030	370.669	0.441	0.013
	Banjo 8-station manifold TM075/050	0.250	370.669	0.441	0.110
	1/2" NPT plug — blocked outlets	0.020	370.669	0.441	0.009
	1/2" NPT × 13mm hose barb adapter — active outlets	0.060	370.669	0.441	0.026
	1/2" ID clear PVC hose — manifold to nozzle bodies (4 runs ~1 m each)	0.180	370.669	0.441	0.079
	Stainless hose clips 13–20 mm	0.010	370.669	0.441	0.004
	TeeJet QJ17560A-25mm-NYB nozzle body with ChemSaver check valve	0.180	370.669	0.441	0.079
	TeeJet XR80008VS extended range flat fan spray tip, 80°, stainless steel insert, white VisiFlo	0.020	370.669	0.441	0.009
5.762	PTFE tape	0.010	370.669	0.441	0.004
8.645	**Payload**	47.000	**-154.609**	-0.184	-8.651
Spray Boom	6mm × 30mm A2 stainless quick-release ball-lock detent pin with pull ring	0.028	347.070	0.413	0.012
	6mm × 30mm A2 stainless quick-release ball-lock detent pin with pull ring	0.056	347.070	0.413	0.023
	2mm × 150mm stainless steel lanyard with loop ends	0.030	347.070	0.413	0.012
	25mm bore single-split shaft collar, A2 stainless, 45mm OD, 15mm wide, M6 clamp screw	1.745	347.070	0.413	0.721
	Loctite 243 medium-strength threadlocker (blue), 10ml	0.024	347.070	0.413	0.010
2.883	Spay boom (estimte)	1.000	**347.070**	0.413	0.413
Power / General Electronics	Battery pack	18.744	**-814.583**	-0.970	-18.177
	5V Step-Down Voltage Module	1.000	-814.333	-0.969	-0.969
	12V Step-Down Voltage Module (For Servo UBECs) - RSD-100D-12	1.800	-814.333	-0.969	-1.745
	12V Step-Down Voltage Module (For Servo UBECs) - RSD-200D-12	1.000	-814.333	-0.969	-0.969
	12V Step-Down Voltage Module (For Spray Pump) - HAE200-110S12W-TF	0.105	-814.333	-0.969	-0.102
	7.4V Step-Down Voltage Module	0.148	-814.333	-0.969	-0.143
	Power distribution unit	1.760	-814.333	-0.969	-1.706
	Servo (Aileron)	0.140	576.754	0.687	0.096
	Servo (Elevator)	0.133	3232.145	3.848	0.511
	Servo(Rudder)	0.133	3232.145	3.848	0.511
	Servo (Flaps on wings)	0.180	449.707	0.535	0.096
	Servo (Flaps on fuselage)	0.140	449.707	0.535	0.075
	Wing-root connectors	0.080	307.070	0.366	0.029
	VTOL Motor Bullet Connectors - 8mm	0.216	307.070	0.366	0.079
	Tail rudder/elevator servo Bullet Connectors - 4mm	0.035	3139.019	3.737	0.130
	CANbus Splitter	0.009	-814.333	-0.969	-0.009
	Sensor Hub - PL Sensor Hub X2-V2	0.007	-814.333	-0.969	-0.007
	Sensor Board	0.100	-814.333	-0.969	-0.097
25.7844	Pump Motor Controller - Syren 25	0.055	146.238	0.174	0.010
Navigation and Sensors	Cube Orange flight controller	0.073	-431.955	-0.514	-0.038
	GPS	0.100	**-431.955**	-0.514	-0.051
	Radar sensor	0.110	-431.955	-0.514	-0.057
	Crop health sensor body	0.072	**-1034.777**	-1.232	-0.089
	Crop health sunshine sensor	0.035	-1034.777	-1.232	-0.043
	Radio Receiver	0.017	-431.955	-0.514	-0.009
	Telemetry Radio	0.030	-431.955	-0.514	-0.015
	Airspeed sensor	0.010	-431.955	-0.514	-0.005
	Pitot tube	0.010	**-1034.777**	-1.232	-0.012
	Camera	0.010	-1034.777	-1.232	-0.012
	Obstacle avoidence system	0.300	-1034.777	-1.232	-0.370
0.862	Herelink system	0.095	-431.955	-0.514	-0.049
Structures	Fuselage structure	14.00	-154.609	-0.184	-2.577
	Wing attatchment points and VTOL booms structure	5.23	210.000	0.250	1.308
	Wing structure	10.47	210.000	0.250	2.617
	Tailplane Structure	5.23	3209.958	3.821	19.999
	VTOL booms	1.70	359.338	0.428	0.728
	Horizontal tail booms	1.31	2552.062	3.038	3.975
41.7715	Landing gear	3.83	15.570	0.019	0.071
	Sum of Weights	**140.83**	**Sum of Moments**		**5.527**
	Centre of Gravity	**0.039**	**Centre of Gravity (mm)**	**32.97**	
	Centre of Gravity (empty payload)	**0.151**	**Centre of Gravity (no payload) (mm)**	**126.92**	
	Centre of Gravity (ground condition: no battery & payload)	**0.431**	**Centre of Gravity (ground condition: no battery & payload) (mm)**	**361.94**	
	h0	**0.25**	**h0 (mm)**	**210.00**	

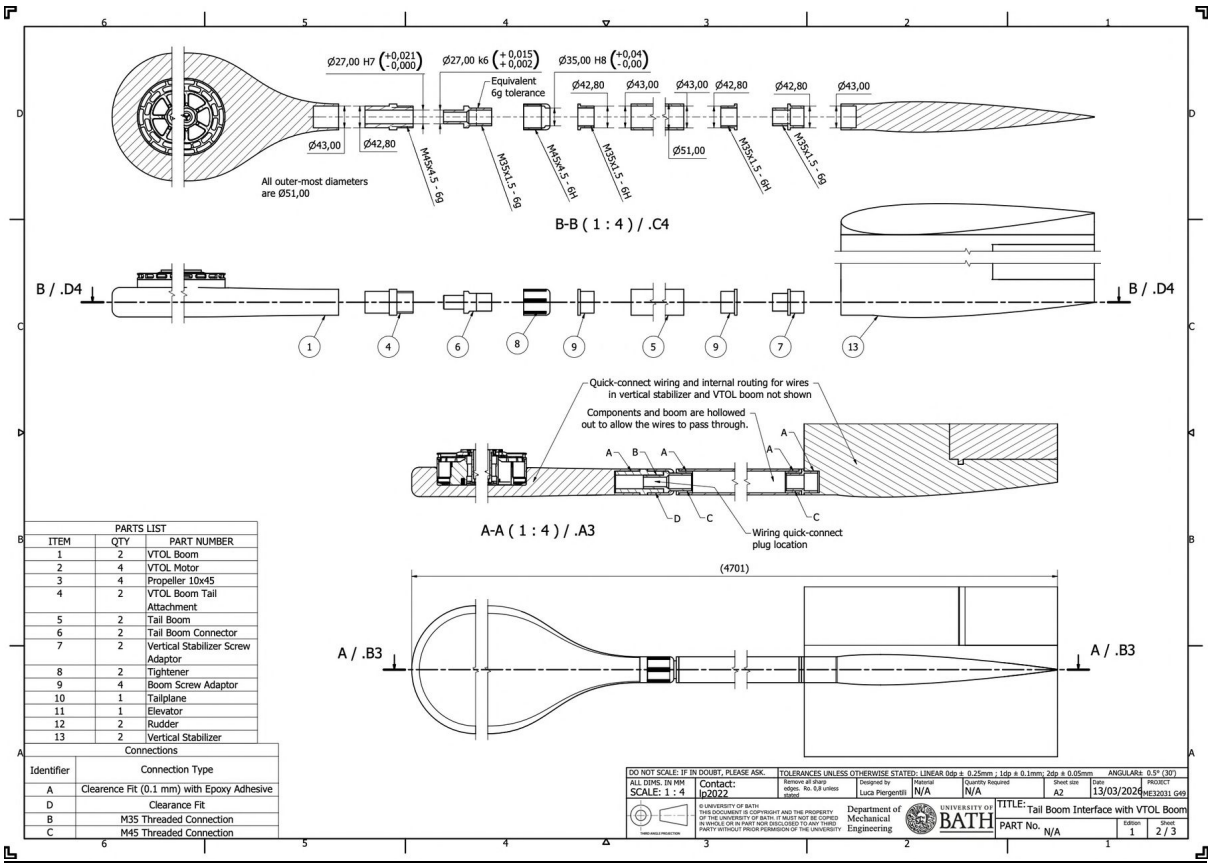
Appendix R: Technical Drawings






Cruise Propeller
(Scale 1 : 17)
VTOL Propeller (x 4)
(Scale 1 : 17)
Wing Inboard Spar
(Removable, Scale 1 : 10)
(498)
(1019)
(43)
(1207)
(50)
(1999)
(500)
(535)
(Ø51)
(1063)
(998)
(93)
(383)
(310)
(70)
(561)
(1866)
(840)
(2731)
(421)
(150.00)
(3172)
(2214)
(819)
(1729)
(252)
(35)
(70)
(1365)
DO NOT SCALE: IF IN DOUBT, PLEASE ASK.
TOLERANCES UNLESS OTHERWISE STATED: LINEAR 0dp ± 0.25mm ; 1dp ± 0.1mm; 2dp ± 0.05mm ANGULAR± 0.5° (30')
ALL DIMS. IN MM
SCALE: 1 : 17
Contact: lp2022
Remove all sharp edges. Ro. 0,8 unless stated
Designed by Luca Piergentili
Material N/A
Quantity Required N/A
Sheet size A2
Date 13/03/2026
PROJECT ME32031 G49
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TITLE: Sub-Systems Dimensions
PART No. N/A
Edition 1
Sheet 3 / 3

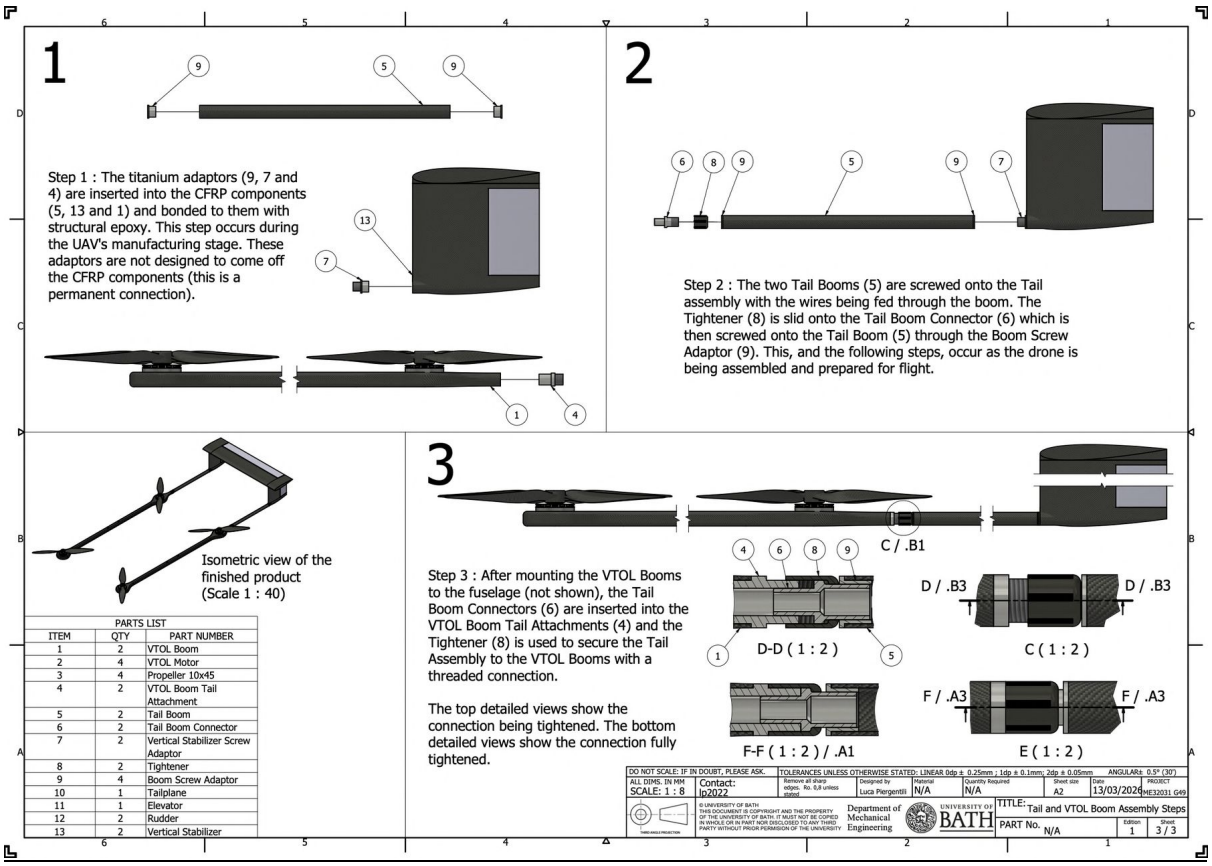

UAV Isometric View
(Scale 1:34)
Isometric View
(Scale 1:18)
PARTS LIST
ITEM | QTY | PART NUMBER
1 | 2 | VTOL Boom
2 | 4 | VTOL Motor
3 | 4 | Propeller 10x45
4 | 2 | VTOL Boom Tail Attachment
5 | 2 | Tail Boom
6 | 2 | Tail Boom Connector
7 | 2 | Vertical Stabilizer Screw Adaptor
8 | 2 | Tightener
9 | 4 | Boom Screw Adaptor
10 | 1 | Tailplane
11 | 1 | Elevator
12 | 2 | Rudder
13 | 2 | Vertical Stabilizer
Connections
Identifier | Connection Type
A | Clearence Fit (0.1 mm) with Epoxy Adhesive
D | Clearance Fit
B | M35 Threaded Connection
C | M45 Threaded Connection
DO NOT SCALE: IF IN DOUBT, PLEASE ASK.
TOLERANCES UNLESS OTHERWISE STATED: LINEAR 0dp ± 0.25mm ; 1dp ± 0.1mm; 2dp ± 0.05mm ANGULAR± 0.5° (30')
ALL DIMS. IN MM
SCALE: 1 : 13
Contact: lp2022
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Designed by Luca Piergentili
Material N/A
Quantity Required N/A
Sheet size A2
Date 13/03/2026
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TITLE: VTOL Boom and Tail Exploded View
PART No. N/A
Edition 1
Sheet 1 / 3



PARTS LIST		
ITEM	QTY	PART NUMBER
1	2	VTOL Boom
2	4	VTOL Motor
3	4	Propeller 10x45
4	2	VTOL Boom Tail Attachment
5	2	Tail Boom
6	2	Tail Boom Connector
7	2	Vertical Stabilizer Screw Adaptor
8	2	Tightener
9	4	Boom Screw Adaptor
10	1	Tailplane
11	1	Elevator
12	2	Rudder
13	2	Vertical Stabilizer

Connections	
Identifier	Connection Type
A	Clearance Fit (0.1 mm) with Epoxy Adhesive
D	Clearance Fit
B	M35 Threaded Connection
C	M45 Threaded Connection

DO NOT SCALE: IF IN DOUBT, PLEASE ASK. TOLERANCES UNLESS OTHERWISE STATED: LINEAR 0dp ± 0.25mm ; 1dp ± 0.1mm; 2dp ± 0.05mm ANGULAR± 0.5° (30')
ALL DIMS. IN MM SCALE: 1 : 4 Contact: lp2022 Remove all sharp edges. Ro. 0,8 unless stated Designed by Luca Piergentili Material N/A Quantity Required N/A Sheet size A2 Date 13/03/2026 PROJECT ME32031 G49
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TITLE: Tail Boom Interface with VTOL Boom
PART No. N/A Edition 1 Sheet 2 / 3



PARTS LIST		
ITEM	QTY	PART NUMBER
1	2	VTOL Boom
2	4	VTOL Motor
3	4	Propeller 10x45
4	2	VTOL Boom Tail Attachment
5	2	Tail Boom
6	2	Tail Boom Connector
7	2	Vertical Stabilizer Screw Adaptor
8	2	Tightener
9	4	Boom Screw Adaptor
10	1	Tailplane
11	1	Elevator
12	2	Rudder
13	2	Vertical Stabilizer

DO NOT SCALE: IF IN DOUBT, PLEASE ASK. TOLERANCES UNLESS OTHERWISE STATED: LINEAR 0dp ± 0.25mm ; 1dp ± 0.1mm; 2dp ± 0.05mm ANGULAR± 0.5° (30')
ALL DIMS. IN MM SCALE: 1 : 8 Contact: lp2022 Remove all sharp edges. Ro. 0,8 unless stated Designed by Luca Piergentili Material N/A Quantity Required N/A Sheet size A2 Date 13/03/2026 PROJECT ME32031 G49
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TITLE: Tail and VTOL Boom Assembly Steps
PART No. N/A Edition 1 Sheet 3 / 3

Appendix S: Compliance Matrix

Requirement Identification			*Complience and Verification*							
Category	**Requirements**	**ID**	**Design Phase Verification Technique**	**Expected outcome**	**Rationale**	**Development / Test Phase Verification Technique**	**Expected outcome**	**Rationale**	**Compliance Status**	**Rationale for Non-Complience**
Operation	The drone's spraying must cover the entire area that has been marked for spraying	ON1	Spraying and Navigation model	Theoretical mission profile and operational productivity	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Partial Compliance	Irregular fields were not considered during this design analysis, future work will involve consideration of irregular and curved field shapes
	The drone must be capable of autonomous flight	ON2	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be able to perform its spraying function autonomously	ON3	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of VTOL	ON4	Propulsion analysis and calculation	Theoretical drone performace metrics	Only available verification technique at this stage	Propulsion test and Full operations test	Real-life drone performance capabilities	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of flying under rain or perturbed (non sunny and with wind) weather conditions	ON5	Structural analysis and calculation	Theoretical structures capabilities	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Full waterproofing analysis was not completed, future work would involve waterproofing design for the fuselage and o-ring design for relevant interphases
	The drone must be able to withstand the structural loads outlined in the flight envelope specific to its category	ON6	Structural analysis and calculation	Theoretical structures capabilities	Only available verification technique at this stage	Structural testing	Real-life drone structures behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The payload system must be accessible in order for leak checks to be performed	ON7	Payload design	Compliance or non-compliance depending on payload design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Full waterproofing and leak analysis was not completed, future work would involve waterproofing design for the fuselage, water-tight payload connections and o-ring design for relevant interphases
	If signal / communication is lost between the drone and the operator, the drone must automatically reposition itself to regain the connection or have a RTH (return-to-home) function	ON8	COTS components spec and Navigation model	System compliance or non-compliance depending on the COTS component specs and built-in function in navigation code	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Beyond GDBP scope	N/A
	The drone operation must have a higher operational productivity than 0.4 to 0.7 hectares per day --> 0.04 to 0.07 hectares / hour	ON9	Spraying and Navigation model and theoretical operational productivity calculations	Theoretical mission profile and operational productivity	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must carry enough payload to spray 2 hectares in one mission	ON10	Structural analysis and calculation	Theoretical structures capabilities	Only available verification technique at this stage	Payload tank design and Full operations test	Real-life payload system behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be capable of spraying the field or area outlined in the mission plan autonomously	ON11	Spraying and Navigation model with emphasis on spray drift	Theoretical mission profile and operational productivity	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must have accurate obstacle detection	ON12	COTS components spec and Navigation model	System compliance or non-compliance depending on the COTS component specs and built-in function in navigation code	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must follow a flight path that avoids obstacles, no-fly zones and terrain with autonomous operation	ON13	6-degree of freedom model simulation	Simulated behaviour of the drone.	Only available verification technique at this stage	Flight navigation test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Navigation model only includes a field drawn by the user, no real life mapping, and therefore no-fly zones, were included in the model. Future work would be to import the model into a world map and include no-fly zones in the map, coding the software to avoid them
	The drone must have an operational radius of 4km	ON14	Propulsion and battery analysis and calculation	Theoretical drone performace metrics and range	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must have 4G, GPS and sattellite connectivity	ON15	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Off-the-shelf component search and testing	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone spraying systems must be compatible with existing spraying products	ON16	Payload design and material selection	Compliance or non-compliance depending on payload design and material selection from theoretical materials analysis	Only available verification technique at this stage	Sprayer test and Full operations test	Real-life payload system behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone's battery system must be rechargeable on site (at the farm)	ON17	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Battery recharge test	Real-life battery behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone battery system must be rechargable using existing infrastructure	ON18	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Off-the-shelf component search and testing	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone must be capable of servicing farms located at 4921 ft (1,500 m) above sea level	ON19	Propulsion analysis and calculation	Theoretical drone performace metrics	Only available verification technique at this stage	High altitude Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone's structure must be capable of surviving a crash landing with the drone travelling at the its cruise velocity	ON20	Structural analysis and calculation	Theoretical structures capabilities	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Crash analysis was not performed on the drone, future work would involve full crash analysis after every phase of flight
Design	The payload tank must be removable	DN1	Payload design	Compliance or non-compliance depending on payload design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Non Compliance	The payload tank became a water-proof payload bay to allow solid and liquid payload to be transported (meeting need DN3). The payload tank was then integrated into the fuselage to save weight
	The payload tank must be shaped in such a way that fascilitates cleaning	DN2	Payload design	Compliance or non-compliance depending on payload design and engineering intuition	Only available verification technique at this stage	Cleaning test	Cleaning effectiveness with current design	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must have a modular design that allows for different mission profiles to be performed	DN3	Overall design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Alternative missions test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must be disassemblable or have given configuration for transport	DN4	Interface design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Assembly and transport test	Compliance or non-compliance based on real-life drone capabilities	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	In its transportational configuration, the drone must fit inside of a medium sized van	DN5	CAD model measurements	Estimated physical measurements (without taking tolerances and manufacturing imperfections into account)	Only available verification technique at this stage	Assembly and transport test	Compliance or non-compliance based on real-life drone capabilities	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	Internal components should be easily accessable	DN6	Fuselage design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The disassembled drone (or in its transport configuration) must have designated carrying points	DN7	Overall design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Assembly and transport test	Compliance or non-compliance based on real-life drone capabilities	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
Business	The drone service must be cheaper than 725 ₹ / hectare	BN1	Cost analysis and estimation	Estimated cost of the operation	Only available verification technique at this stage	Full operations test and business analysis	Real-life drone behaviour and costs, revenue and profits data	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	It must be easy for the farmer to contact the company to order the service	BN2	N/A	N/A	Only available verification technique at this stage	Infrastucture setup and surveys	Business capabilities and customers spraying timetable	This is the best way to get reliable data on the business's real-life behaviour and customer needs	Beyond GDBP scope	N/A
	The company must have enough drone capacity to service multiple farmers	BN3	N/A	N/A	Only available verification technique at this stage	Infrastucture setup and purchasing analysis	Business capabilities	This is the best way to get reliable data on the business's real-life behaviour	Beyond GDBP scope	N/A
	The company must be able to provide the service within 1 week of the farmer contacting the company	BN4	N/A	N/A	Only available verification technique at this stage	Infrastucture setup and purchasing analysis	Business capabilities	This is the best way to get reliable data on the business's real-life behaviour	Beyond GDBP scope	N/A
	The drone must have documentation that proves its spraying effectiveness	BN5	N/A	N/A	Only available verification technique at this stage	Final documentation audit	Compliance or non-compliance assessed by external auditors	Mandatory for all companies	Beyond GDBP scope	N/A
Regulation	The drone must reduce the farmer's exposure to harmful chemicals during spraying	RN1	Spraying and Navigation model and drift analysis	Theoretical mission profile and spray particulate behaviour	Only available verification technique at this stage	Full operations test and drift analysis	Real-life drone and spray particulate behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Spay drift analysis was not performed and would be performed during future work
	The drone and its use case must respect the regional laws of the area where it is operating	RN2	Overall design and regulations review	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Regulations review by an external body	Compliance or non-compliance based on real-life drone capabilities and design	This is the only way regulation compliance can be checked	Compliance	N/A
	The drone must not be above 150 kg MTOW	RN3	Weight estimations	Theoretical drone weight	Only available verification technique at this stage	Regulations review by an external body	Compliance or non-compliance based on real-life drone capabilities and design	This is the only way regulation compliance can be checked	Compliance	N/A
	The drone must respect every regulation associated with airworthiness in its category according to EASA	RN4	Overall design and regulations review	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Regulations review by an external body	Compliance or non-compliance based on real-life drone capabilities and design	This is the only way regulation compliance can be checked	Compliance	N/A
	The drone's V-n envelope must respect guidelines set by EASA	RN5	Structural analysis and calculation and regulations review	Theoretical structures capabilities	Only available verification technique at this stage	Regulations review by an external body	Compliance or non-compliance based on real-life drone capabilities and design	This is the only way regulation compliance can be checked	Compliance	N/A
	The drone must adheare to the agricultural regulations set by the local government	RN6	Overall design and regulations review	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Regulations review by an external body	Compliance or non-compliance based on real-life drone capabilities and design	This is the only way regulation compliance can be checked	Compliance	N/A
	When operting, the drone should respect the noise constraints established by the local government	RN7	Propeller noise analysis and regulations review	Theoretical propeller noise analysis	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Full in-depth propeller noise analysis was not performed, future work would involve in-depth noise analysis for all propellers in all phases of flight
	The drone must be safe for any person located within a 30m radius of the area of operation	RN8	Overall design and regulations review	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The spraying system must comply with the relevant regulatory bodies that regulate pesticide, herbicide and fertilizer dispersion over crops or a field	RN9	Payload design and regulations review	Compliance or non-compliance depending on payload design outcomes and decisions	Only available verification technique at this stage	Full operations test and drift analysis	Real-life drone and spray particulate behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Spay drift analysis was not performed and would be performed during future work
Drone-Operator Interaction	The drone must be intuitive to pilot / control	IN1	Navigation and control system inputs model	Compliance or non-compliance depending on control system design decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Beyond GDBP scope	N/A
	The mission setup process must be straight forward	IN2	GUI design	Compliance or non-compliance depending on GUI design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Beyond GDBP scope	N/A
	Down-time between two drone missions must be limited	IN3	Spraying and Navigation model	Theoretical mission profile and operational productivity	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Partial Compliance	Drone architecture favored a small down-time between missions, but the target of 2 minutes was not achieved.
	The battery swap and payload loading must be tool-less	IN4	Overall design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Further analysis and design required (Future Work)	Fuselage access hatches were considered and located but not fully designed and modelled. Future work would involve the design of hatch systems to gain access to internal components
	Transition operation from drone transport configuration to flight configuration must be tool-less	IN5	Interface design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Partial Compliance	To provide a better and sturdier connection, bolts were used in the interface between the VTOL boom, landing gear and spray boom and the fuselage. This was also done to minimise the drone's wetted area for better aerodynamic efficiency
	The drone must have the necessary equipment to be remote controlled by a pilot	IN6	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone's operations must be monitorable from a distance by the operator	IN7	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
	The drone must possess a camera that is able to give the operator the drone's POV	IN8	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Full operations test	Real-life drone behaviour	This is the best way to get reliable data on the drone or system's real-life behaviour	Compliance	N/A
Manufacturing and End-of-life	Material selection, contruction / assembly method and battery technology must be compatible with end-of-life processes	MN1	Material selection and lifecycle assessment	Compliance or non-compliance depending on design outcomes and decisions and theoretical LCA	Only available verification technique at this stage	Liaison with service	Detailed specifications or requirements for end-of-life processes	To understand whether the drone is suitable for end-of-life processes	Further analysis and design required (Future Work)	An in-depth life cycle assessment was not conducted and materials were not chosen with end-of-life recycling in mind, future work would consider these elements during design
	Off-the-shelf components must be used in the design with little to no alterations	MN2	COTS components spec and overall design	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Liaison with suppliers	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone's avionics and sensor systems need to be made from COTS components	MN3	COTS components spec	System compliance or non-compliance depending on the COTS component specs	Only available verification technique at this stage	Liaison with suppliers	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone's power system needs to be made from COTS components	MN4	COTS components spec and electrical systems analysis	System compliance or non-compliance depending on the COTS component specs, chosen using the results of a power systems model	Only available verification technique at this stage	Liaison with suppliers	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	The drone's propulsion system needs to be made from COTS components	MN5	COTS components spec and propulsion analysis	System compliance or non-compliance depending on the COTS component specs, chosen using the results of a propulsion model	Only available verification technique at this stage	Liaison with suppliers	Full understanding of COTS component capabilities	To understand whether the COTS component is suitable for the sub-system	Compliance	N/A
	Bespoke components must be manufacturable with existing industrial manufacturing methods	MN6	Manufacturing assessment and overall design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Liaison with suppliers	Detailed specifications or requirements for component assembly and manufacturing	To understand whether the drone is suitable for manufacturing and assembly	Compliance	N/A
	The drone must be built using commercially available and industrially produced materials	MN7	Material selection	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Liaison with suppliers	Detailed specifications or requirements for component assembly and manufacturing	To understand whether the drone is suitable for manufacturing and assembly	Compliance	N/A
	The drone must be assemblable using industry standard tooling and current assembling and manufacturing techniques	MN8	Manufacturing assessment and overall design	Compliance or non-compliance depending on design outcomes and decisions	Only available verification technique at this stage	Liaison with suppliers	Detailed specifications or requirements for component assembly and manufacturing	To understand whether the drone is suitable for manufacturing and assembly	Compliance	N/A

Requirement Definition						Complience and Verification								
Category	Attribute	Requirement	Units	Importance	ID	Design Phase Verification Technique	Rationale	Expected outcome	Development / Test Phase Verification Technique	Rationale	Expected outcome	Iteration 3 Compliance	Rationale For Non-Compliance
Operation	Operational Productivity (OP)	21 (ideal case) & 18 (normal)	ha/h	High	NR1	Calculation	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Operational Radius	4,000	meters	High	NR2	Calculation	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Flight time	30	minutes	High	NR3	Calculation and battery spec	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Non-Compliance	A flight time of 30 minutes was extremely over-spec-ed for the mission. See Flight Time Change Trade-Off Summary for more info
	Payload capacity	47	kg	High	NR4	Defined from requirement	This is the only verification method available at this stage	Theoretical value	Physical drone construction	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone specifications	Compliance	N/A
	Operational ceiling	1,500	meters	Medium	NR5	Motors spec and Calculation	This is the only verification method available at this stage	Theoretical value	High altitude full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
	Communication range	5 & Compatible with 4G and satellite	km	High	NR6	Avionics spec	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
Spraying mechanism	Fluid flowrate	9	L/min	High	NR7	Calculation	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone behaviour	Compliance	N/A
Physical	MTOW	130 Absolute Max: 150 kg	kg	High	NR8	Calculation	This is the only verification method available at this stage	Theoretical value	Physical drone construction	This is the only way to get accurate drone physical data	Real-life drone physical attributes	Non-Compliance	The empty weight fraction with the target payload and target MTOW was unachievable. To reach an MTOW of 130 kg, payload capacity would have had to reduce
	Empty weight	83 Absolute Max: 103 kg	kg	Medium	NR9	Calculation	This is the only verification method available at this stage	Theoretical value	Physical drone construction	This is the only way to get accurate drone physical data	Real-life drone physical attributes	Compliance	N/A
	Operational dimensions	W : 3,500 L : 3,000	mm	Low	NR10	CAD model	This is the only verification method available at this stage	Theoretical value	Physical drone construction	This is the only way to get accurate drone physical data	Real-life drone physical attributes	Non-Compliance	The initial size estimates based on competitor aircraft were inaccurate. Non-compliance did not lead to any issues
	Transportational dimentions	2946 x 1784 x 1886	mm (LxWxH)	High	NR11	CAD model	This is the only verification method available at this stage	Theoretical value	Physical drone construction	This is the only way to get accurate drone physical data	Real-life drone physical attributes	Compliance	N/A
Durability	Operational life	1'000	cycles	Low	NR12	Calculation and battery spec	This is the only verification method available at this stage	Theoretical value	Full operations and cycle testing	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone service life	Partial Compliance	Current battery technology has a limited lifespan, however, the structural elements of the aircraft have a longer service life
	IP rating	IP55	IP rating	Medium	NR13	CAD model design	This is the only verification method available at this stage	Theoretical value	Full operations and IP test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone IP rating	Partial Compliance	The VTOL motors are IP35 rated, however, the rest of the aircraft would have a IP67 rating
Down Time	Down time	3	minutes	Low	NR14	Calculation	This is the only verification method available at this stage	Theoretical value	Full operations test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life mission down time	Compliance	N/A
	Charging Time	8	minutes	Medium	NR15	Calculation and battery spec	This is the only verification method available at this stage	Theoretical value	Full operations and charging test	This is the best way to get reliable data on the drone or system's real-life behaviour	Real-life drone charging time	Further analysis and design required (Future Work)	Charging time analysis could not be performed during this project and together with real-life battery tests, are part of Future Work

Integrated Set of Needs Compliance Summary		
Number of Needs	**Percentage of Total Needs**	**Explenation**
37	**65**	**Percent of needs complied with**
3	**5**	**Percent of needs partially complied with**
9	**16**	**Percent of needs require future work to be complied with**
1	**2**	**Percent of needs not complied with**
7	**12**	**Percent of needs are beyond the scope of GDBP**
57	**100**	**57 Needs total, divided in 6 categories**

Numerical Requirements Compliance Summary		
Number of Requirements	**Percentage of Total Needs**	**Explenation**
9	**60**	**Percent of needs complied with**
2	**13**	**Percent of needs partially complied with**
1	**7**	**Percent of needs require future work to be complied with**
3	**20**	**Percent of needs not complied with**
15	**100**	**15 Needs total, divided in 5 categories**

Appendix T: CAD Model Renders

